Mathematics

Class 4 (English Medium)

Solved Exercises — Sindh Textbook Board

8 lessons — 1074 questions

1. Numbers and Arithmetic Operations

Exercise 1.1: Numbers in Words and Figures

Question: Exercise 1.1, Question 1: Write the number in figures: Four thousand six hundred and ninety two.

Answer: Step 1: Break the words into parts: four thousand = 4,000; six hundred = 600; ninety two = 92; 4,000 + 600 + 92 = 4,692. | Step 2: Add the parts together to make the full number. | Final Answer: 4,692.

Question: Exercise 1.1, Question 1: Write the number in figures: Sixty nine thousand four hundred and seven.

Answer: Step 1: Break the words into parts: sixty nine thousand = 69,000; four hundred = 400; seven = 7; 69,000 + 400 + 7 = 69,407. | Step 2: Add the parts together to make the full number. | Final Answer: 69,407.

Question: Exercise 1.1, Question 1: Write the number in figures: Four hundred fifty six thousand and nine hundred twenty six.

Answer: Step 1: Break the words into parts: four hundred fifty six thousand = 456,000; nine hundred twenty six = 926; 456,000 + 926 = 456,926. | Step 2: Add the parts together to make the full number. | Final Answer: 456,926.

Question: Exercise 1.1, Question 1: Write the number in figures: Seven hundred thousand.

Answer: Step 1: Break the words into parts: seven hundred thousand = 700,000. | Step 2: Add the parts together to make the full number. | Final Answer: 700,000.

Question: Exercise 1.1, Question 1: Write the number in figures: Nine hundred two thousand and forty two.

Answer: Step 1: Break the words into parts: nine hundred two thousand = 902,000; forty two = 42; 902,000 + 42 = 902,042. | Step 2: Add the parts together to make the full number. | Final Answer: 902,042.

Question: Exercise 1.1, Question 1: Write the number in figures: Seven hundred twenty nine thousand and six.

Answer: Step 1: Break the words into parts: seven hundred twenty nine thousand = 729,000; six = 6; 729,000 + 6 = 729,006. | Step 2: Add the parts together to make the full number. | Final Answer: 729,006.

Question: Exercise 1.1, Question 2 (i): Write the number in words: 2,561.

Answer: Step 1: Split the number into thousands and the last three digits. | Step 2: Read the thousands part first, then the hundreds, tens and ones. | Final Answer: Two thousand five hundred and sixty one.

Question: Exercise 1.1, Question 2 (ii): Write the number in words: 34,561.

Answer: Step 1: Split the number into thousands and the last three digits. | Step 2: Read the thousands part first, then the hundreds, tens and ones. | Final Answer: Thirty four thousand five hundred and sixty one.

Question: Exercise 1.1, Question 2 (iii): Write the number in words: 56,081.

Answer: Step 1: Split the number into thousands and the last three digits. | Step 2: Read the thousands part first, then the hundreds, tens and ones. | Final Answer: Fifty six thousand eighty one.

Question: Exercise 1.1, Question 2 (iv): Write the number in words: 92,000.

Answer: Step 1: Split the number into thousands and the last three digits. | Step 2: Read the thousands part first, then the hundreds, tens and ones. | Final Answer: Ninety two thousand.

Question: Exercise 1.1, Question 2 (v): Write the number in words: 245,612.

Answer: Step 1: Split the number into thousands and the last three digits. | Step 2: Read the thousands part first, then the hundreds, tens and ones. |

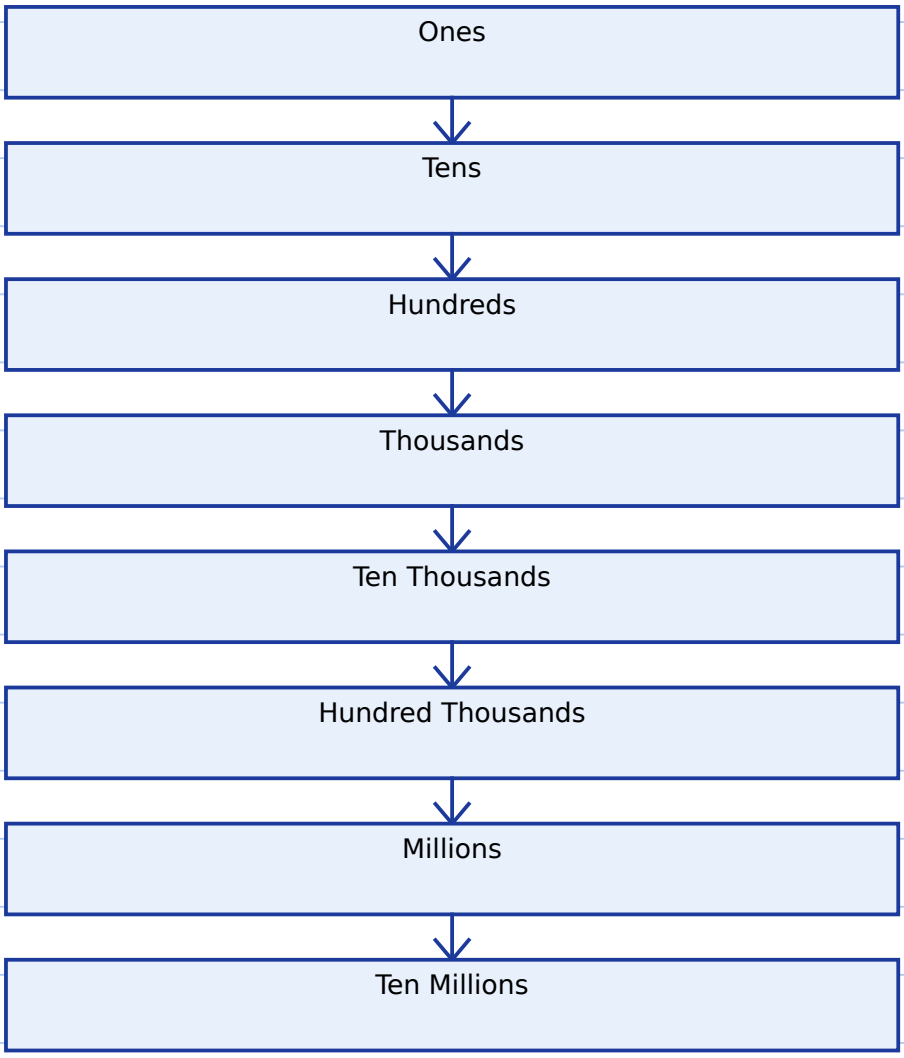
Final Answer: Two hundred and forty five thousand six hundred and twelve.

Question: Exercise 1.1, Question 2 (vi): Write the number in words: 349,650.

Answer: Step 1: Split the number into thousands and the last three digits. | Step 2: Read the thousands part first, then the hundreds, tens and ones. | Final Answer: Three hundred and forty nine thousand six hundred and fifty.

Exercise 1.2: Place Value and Value

Place Value Chart



Question: Exercise 1.2, Question 1 (i): Write the place value and value of the coloured digit(s) in 12,345,678: digit 2 (in the millions place).

Answer: Step 1: Write 12,345,678 and count the places from the right:

ones, tens, hundreds, thousands and so on. | Step 2: The digit 2 stands in the millions place, so its place value is millions and its value is 2 x 1,000,000 = 2,000,000. | Step 3: Final Answer: 2 in the millions place = 2,000,000.

Question: Exercise 1.2, Question 1 (ii): Write the place value and value of the coloured digit(s) in 58,923,107: digit 8 (in the millions place).

Answer: Step 1: Write 58,923,107 and count the places from the right: ones, tens, hundreds, thousands and so on. | Step 2: The digit 8 stands in the millions place, so its place value is millions and its value is 8 x 1,000,000 = 8,000,000. | Step 3: Final Answer: 8 in the millions place = 8,000,000.

Question: Exercise 1.2, Question 1 (iii): Write the place value and value of the coloured digit(s) in 100,000,000: digit 1 (in the hundred millions place).

Answer: Step 1: Write 100,000,000 and count the places from the right: ones, tens, hundreds, thousands and so on. | Step 2: The digit 1 stands in the hundred millions place, so its place value is hundred millions and its value is 1 x 100,000,000 = 100,000,000. | Step 3: Final Answer: 1 in the hundred millions place = 100,000,000.

Question: Exercise 1.2, Question 1 (iv): Write the place value and value of the coloured digit(s) in 23,964,579: digit 9 (in the hundred thousands place).

Answer: Step 1: Write 23,964,579 and count the places from the right: ones, tens, hundreds, thousands and so on. | Step 2: The digit 9 stands in the hundred thousands place, so its place value is hundred thousands and its value is 9 x 100,000 = 900,000. | Step 3: Final Answer: 9 in the hundred thousands place = 900,000.

Question: Exercise 1.2, Question 1 (v): Write the place value and value of

the coloured digit(s) in 25,960,238: digit 5 (in the millions place), digit 2 (in the hundreds place).

Answer: Step 1: Write 25,960,238 and count the places from the right: ones, tens, hundreds, thousands and so on. | Step 2: The digit 5 stands in the millions place, so its place value is millions and its value is 5 x 1,000,000 = 5,000,000. | Step 3: The digit 2 stands in the hundreds place, so its place value is hundreds and its value is 2 x 100 = 200. | Step 4: Final Answer: 5 in the millions place = 5,000,000; 2 in the hundreds place = 200.

Question: Exercise 1.2, Question 1 (vi): Write the place value and value of the coloured digit(s) in 9,621,382: digit 9 (in the millions place), digit 1 (in the thousands place).

Answer: Step 1: Write 9,621,382 and count the places from the right: ones, tens, hundreds, thousands and so on. | Step 2: The digit 9 stands in the millions place, so its place value is millions and its value is 9 x 1,000,000 = 9,000,000. | Step 3: The digit 1 stands in the thousands place, so its place value is thousands and its value is 1 x 1,000 = 1,000. | Step 4: Final Answer: 9 in the millions place = 9,000,000; 1 in the thousands place = 1,000.

Question: Exercise 1.2, Question 1 (vii): Write the place value and value of the coloured digit(s) in 8,998,776: digit 9 (in the hundred thousands place), digit 8 (in the thousands place).

Answer: Step 1: Write 8,998,776 and count the places from the right: ones, tens, hundreds, thousands and so on. | Step 2: The digit 9 stands in the hundred thousands place, so its place value is hundred thousands and its value is 9 x 100,000 = 900,000. | Step 3: The digit 8 stands in the thousands place, so its place value is thousands and its value is 8 x 1,000 = 8,000. | Step 4: Final Answer: 9 in the hundred thousands place =

900,000; 8 in the thousands place = 8,000.

Question: Exercise 1.2, Question 1 (viii): Write the place value and value of the coloured digit(s) in 76,905,851: digit 7 (in the ten millions place), digit 6 (in the millions place).

Answer: Step 1: Write 76,905,851 and count the places from the right: ones, tens, hundreds, thousands and so on. | Step 2: The digit 7 stands in the ten millions place, so its place value is ten millions and its value is 7 x 10,000,000 = 70,000,000. | Step 3: The digit 6 stands in the millions place, so its place value is millions and its value is 6 x 1,000,000 = 6,000,000. | Step 4: Final Answer: 7 in the ten millions place = 70,000,000; 6 in the millions place = 6,000,000.

Question: Exercise 1.2, Question 2 (i): Write the value of every digit in 9,234,513.

Answer: Step 1: Read 9,234,513 place by place from the left. | Step 2: The digit 9 is in the millions place; its value is 9,000,000. | Step 3: The digit 2 is in the hundred thousands place; its value is 200,000. | Step 4: The digit 3 is in the ten thousands place; its value is 30,000. | Step 5: The digit 4 is in the thousands place; its value is 4,000. | Step 6: The digit 5 is in the hundreds place; its value is 500. | Step 7: The digit 1 is in the tens place; its value is 10. | Step 8: The digit 3 is in the ones place; its value is 3. | Step 9: Final Answer: 9 = 9,000,000; 2 = 200,000; 3 = 30,000; 4 = 4,000; 5 = 500; 1 = 10; 3 = 3.

Question: Exercise 1.2, Question 2 (ii): Write the value of every digit in 50,120,306.

Answer: Step 1: Read 50,120,306 place by place from the left. | Step 2: The digit 5 is in the ten millions place; its value is 50,000,000. | Step 3: The digit 0 is in the millions place; its value is 0. | Step 4: The digit 1 is in the hundred thousands place; its value is 100,000. | Step 5: The digit 2 is

in the ten thousands place; its value is 20,000. | Step 6: The digit 0 is in the thousands place; its value is 0. | Step 7: The digit 3 is in the hundreds place; its value is 300. | Step 8: The digit 0 is in the tens place; its value is 0. | Step 9: The digit 6 is in the ones place; its value is 6. | Step 10: Final Answer: 5 = 50,000,000; 0 = 0; 1 = 100,000; 2 = 20,000; 0 = 0; 3 = 300; 0 = 0; 6 = 6.

Question: Exercise 1.2, Question 2 (iii): Write the value of every digit in 3,567,899.

Answer: Step 1: Read 3,567,899 place by place from the left. | Step 2: The digit 3 is in the millions place; its value is 3,000,000. | Step 3: The digit 5 is in the hundred thousands place; its value is 500,000. | Step 4: The digit 6 is in the ten thousands place; its value is 60,000. | Step 5: The digit 7 is in the thousands place; its value is 7,000. | Step 6: The digit 8 is in the hundreds place; its value is 800. | Step 7: The digit 9 is in the tens place; its value is 90. | Step 8: The digit 9 is in the ones place; its value is 9. | Step 9: Final Answer: 3 = 3,000,000; 5 = 500,000; 6 = 60,000; 7 = 7,000; 8 = 800; 9 = 90; 9 = 9.

Question: Exercise 1.2, Question 2 (iv): Write the value of every digit in 36,564,396.

Answer: Step 1: Read 36,564,396 place by place from the left. | Step 2: The digit 3 is in the ten millions place; its value is 30,000,000. | Step 3: The digit 6 is in the millions place; its value is 6,000,000. | Step 4: The digit 5 is in the hundred thousands place; its value is 500,000. | Step 5: The digit 6 is in the ten thousands place; its value is 60,000. | Step 6: The digit 4 is in the thousands place; its value is 4,000. | Step 7: The digit 3 is in the hundreds place; its value is 300. | Step 8: The digit 9 is in the tens place; its value is 90. | Step 9: The digit 6 is in the ones place; its value is 6. | Step 10: Final Answer: 3 = 30,000,000; 6 = 6,000,000; 5 = 500,000; 6

= 60,000; 4 = 4,000; 3 = 300; 9 = 90; 6 = 6.

Exercise 1.3: Numbers up to Millions

Question: Exercise 1.3, Question 1 (i): Read and write the number in words: 241,935.

Answer: Step 1: Group the digits: the millions part, the thousands part and the last three digits. | Step 2: Read each group in order. | Final Answer: Two hundred and forty one thousand nine hundred and thirty five.

Question: Exercise 1.3, Question 1 (ii): Read and write the number in words: 4,312,687.

Answer: Step 1: Group the digits: the millions part, the thousands part and the last three digits. | Step 2: Read each group in order. | Final Answer: Four million three hundred and twelve thousand six hundred and eighty seven.

Question: Exercise 1.3, Question 1 (iii): Read and write the number in words: 5,000,000.

Answer: Step 1: Group the digits: the millions part, the thousands part and the last three digits. | Step 2: Read each group in order. | Final Answer: Five million.

Question: Exercise 1.3, Question 1 (iv): Read and write the number in words: 25,134,564.

Answer: Step 1: Group the digits: the millions part, the thousands part and the last three digits. | Step 2: Read each group in order. | Final Answer: Twenty five million one hundred and thirty four thousand five hundred and sixty four.

Question: Exercise 1.3, Question 1 (v): Read and write the number in words: 100,000,000.

Answer: Step 1: Group the digits: the millions part, the thousands part and the last three digits. | Step 2: Read each group in order. | Final Answer:

One hundred million.

Question: Exercise 1.3, Question 1 (vi): Read and write the number in words: 9,264,387.

Answer: Step 1: Group the digits: the millions part, the thousands part and the last three digits. | Step 2: Read each group in order. | Final Answer: Nine million two hundred and sixty four thousand three hundred and eighty seven.

Question: Exercise 1.3, Question 1 (vii): Read and write the number in words: 50,001,000.

Answer: Step 1: Group the digits: the millions part, the thousands part and the last three digits. | Step 2: Read each group in order. | Final Answer: Fifty million one thousand.

Question: Exercise 1.3, Question 1 (viii): Read and write the number in words: 4,109,200.

Answer: Step 1: Group the digits: the millions part, the thousands part and the last three digits. | Step 2: Read each group in order. | Final Answer: Four million one hundred and nine thousand two hundred.

Question: Exercise 1.3, Question 1 (ix): Read and write the number in words: 99,990,090.

Answer: Step 1: Group the digits: the millions part, the thousands part and the last three digits. | Step 2: Read each group in order. | Final Answer: Ninety nine million nine hundred and ninety thousand ninety.

Question: Exercise 1.3, Question 2 (i): Write in figures: Two million, ninety thousand and sixty eight.

Answer: Step 1: Note the millions part and the thousands part in the words. | Step 2: Write each part in its correct places, filling empty places with zeros. | Final Answer: 2,090,068.

Question: Exercise 1.3, Question 2 (ii): Write in figures: Thirty million, six

hundred thousand and forty five.

Answer: Step 1: Note the millions part and the thousands part in the words. | Step 2: Write each part in its correct places, filling empty places with zeros. | Final Answer: 30,600,045.

Question: Exercise 1.3, Question 2 (iii): Write in figures: One hundred million.

Answer: Step 1: Note the millions part and the thousands part in the words. | Step 2: Write each part in its correct places, filling empty places with zeros. | Final Answer: 100,000,000.

Question: Exercise 1.3, Question 2 (iv): Write in figures: Twenty million and twenty.

Answer: Step 1: Note the millions part and the thousands part in the words. | Step 2: Write each part in its correct places, filling empty places with zeros. | Final Answer: 20,000,020.

Question: Exercise 1.3, Question 2 (v): Write in figures: Ninety million and sixty seven thousand.

Answer: Step 1: Note the millions part and the thousands part in the words. | Step 2: Write each part in its correct places, filling empty places with zeros. | Final Answer: 90,067,000.

Question: Exercise 1.3, Question 3: Write the numbers from two million to two million twenty in figures.

Answer: Step 1: Write two million in figures: 2,000,000. | Step 2: Keep counting one by one, adding 1 each time. | Final Answer: 2,000,000, 2,000,001, 2,000,002, 2,000,003, 2,000,004, 2,000,005, 2,000,006, 2,000,007, 2,000,008, 2,000,009, 2,000,010, 2,000,011, 2,000,012, 2,000,013, 2,000,014, 2,000,015, 2,000,016, 2,000,017, 2,000,018, 2,000,019, 2,000,020.

Question: Exercise 1.3, Question 4 (i): Write the missing numbers:

2,450,761, 2,451,761, 2,452,761, ..., ..., ...

Answer: Step 1: Find the gap between two neighbours: 2,451,761 - 2,450,761 = 1,000. | Step 2: Add 1,000 each time: 2,452,761 + 1,000 = 2,453,761, then 2,454,761, then 2,455,761. | Final Answer: 2,453,761, 2,454,761, 2,455,761.

Question: Exercise 1.3, Question 4 (ii): Write the missing numbers: 7,000,300, 7,000,400, 7,000,500, ..., ..., ...

Answer: Step 1: Find the gap between two neighbours: 7,000,400 - 7,000,300 = 100. | Step 2: Add 100 each time: 7,000,500 + 100 = 7,000,600, then 7,000,700, then 7,000,800. | Final Answer: 7,000,600, 7,000,700, 7,000,800.

Question: Exercise 1.3, Question 4 (iii): Write the missing numbers: 67,213,415, 67,223,415, 67,233,415, ..., ..., ...

Answer: Step 1: Find the gap between two neighbours: 67,223,415 - 67,213,415 = 10,000. | Step 2: Add 10,000 each time: 67,233,415 + 10,000 = 67,243,415, then 67,253,415, then 67,263,415. | Final Answer: 67,243,415, 67,253,415, 67,263,415.

Question: Exercise 1.3, Question 5: Separate the millions, ten millions and hundred millions in the following numbers: 2,456,178; 22,233,341; 1,000,000; 10,000,000; 100,000,000; 2,561,000; 20,001,010.

Answer: Step 1: Split each number into hundred millions, ten millions, millions and the remaining part. | Step 2: 2,456,178 = 0 hundred millions, 0 ten millions, 2 millions and 456,178. | Step 3: 22,233,341 = 0 hundred millions, 2 ten millions, 2 millions and 233,341. | Step 4: 1,000,000 = 0 hundred millions, 0 ten millions, 1 millions and 0. | Step 5: 10,000,000 = 0 hundred millions, 1 ten millions, 0 millions and 0. | Step 6: 100,000,000 = 1 hundred millions, 0 ten millions, 0 millions and 0. | Step 7: 2,561,000 = 0 hundred millions, 0 ten millions, 2 millions and 561,000. | Step 8:

20,001,010 = 0 hundred millions, 2 ten millions, 0 millions and 1,010. | Step 9: Final Answer: the splits are written in the steps above.

Exercise 1.4: Comparing and Ordering Numbers

Question: Exercise 1.4, Question 1 (i): Compare using <, > or =: 52,001,000 and 345,912.

Answer: Step 1: Count the digits: 52,001,000 has 8 digits and 345,912 has 6 digits. | Step 2: The number with more digits is the bigger number. | Final Answer: 52,001,000 > 345,912; 52,001,000 is bigger and 345,912 is smaller.

Question: Exercise 1.4, Question 1 (ii): Compare using <, > or =: 2,456,123 and 24,345,611.

Answer: Step 1: Count the digits: 2,456,123 has 7 digits and 24,345,611 has 8 digits. | Step 2: The number with more digits is the bigger number. | Final Answer: 2,456,123 < 24,345,611; 24,345,611 is bigger and 2,456,123 is smaller.

Question: Exercise 1.4, Question 1 (iii): Compare using <, > or =: 7,123,400 and 8,567,001.

Answer: Step 1: Both numbers have 7 digits, so compare from the leftmost place. | Step 2: The digits agree until the millions place, where 7,123,400 has 7 and 8,567,001 has 8. | Step 3: 8 is bigger than 7, so the number with 8 in this place is bigger. | Final Answer: 7,123,400 < 8,567,001.

Question: Exercise 1.4, Question 1 (iv): Compare using <, > or =: 92,333,444 and 92,315,617.

Answer: Step 1: Both numbers have 8 digits, so compare from the leftmost place. | Step 2: The digits agree until the ten thousands place, where 92,333,444 has 3 and 92,315,617 has 1. | Step 3: 3 is bigger than 1, so the number with 3 in this place is bigger. | Final Answer: 92,333,444 >

92,315,617.

Question: Exercise 1.4, Question 1 (v): Compare using <, > or =: 24,000,008 and 24,000,005.

Answer: Step 1: Both numbers have 8 digits, so compare from the leftmost place. | Step 2: The digits agree until the ones place, where 24,000,008 has 8 and 24,000,005 has 5. | Step 3: 8 is bigger than 5, so the number with 8 in this place is bigger. | Final Answer: 24,000,008 > 24,000,005.

Question: Exercise 1.4, Question 2 (i): Arrange in ascending and descending order: 3,174,215; 3,741,512; 3,076,005.

Answer: Step 1: All the numbers have the same number of digits, so compare them place by place from the left. | Step 2: Arrange them from smallest to biggest. | Final Answer: Ascending order: 3,076,005, 3,174,215, 3,741,512. Descending order: 3,741,512, 3,174,215, 3,076,005.

Question: Exercise 1.4, Question 2 (ii): Arrange in ascending and descending order: 95,123,415; 95,312,415; 95,113,417.

Answer: Step 1: All the numbers have the same number of digits, so compare them place by place from the left. | Step 2: Arrange them from smallest to biggest. | Final Answer: Ascending order: 95,113,417, 95,123,415, 95,312,415. Descending order: 95,312,415, 95,123,415, 95,113,417.

Question: Exercise 1.4, Question 2 (iii): Arrange in ascending and descending order: 59,178,215; 59,296,712; 52,111,222; 58,110,176.

Answer: Step 1: All the numbers have the same number of digits, so compare them place by place from the left. | Step 2: Arrange them from smallest to biggest. | Final Answer: Ascending order: 52,111,222, 58,110,176, 59,178,215, 59,296,712. Descending order: 59,296,712, 59,178,215, 58,110,176, 52,111,222.

Question: Exercise 1.4, Question 2 (iv): Arrange in ascending and

descending order: 14,111,920; 14,160,000; 13,200,415; 13,100,219.

Answer: Step 1: All the numbers have the same number of digits, so compare them place by place from the left. | Step 2: Arrange them from smallest to biggest. | Final Answer: Ascending order: 13,100,219, 13,200,415, 14,111,920, 14,160,000. Descending order: 14,160,000, 14,111,920, 13,200,415, 13,100,219.

Exercise 1.5: Addition

Question: Exercise 1.5, Question 1 (i): Solve in vertical form: 25,431 + 41,245.

Answer: Step 1: Write the numbers one below the other, ones under ones, tens under tens. | Step 2: Add the ones column: 1 + 5 = 6; write 6. | Step 3: Add the tens column: 3 + 4 = 7; write 7. | Step 4: Add the hundreds column: 4 + 2 = 6; write 6. | Step 5: Add the thousands column: 5 + 1 = 6; write 6. | Step 6: Add the ten thousands column: 2 + 4 = 6; write 6. | Step 7: Final Answer: 25,431 + 41,245 = 66,676.

Question: Exercise 1.5, Question 1 (ii): Solve in vertical form: 92,345 + 50,178.

Answer: Step 1: Write the numbers one below the other, ones under ones, tens under tens. | Step 2: Add the ones column: 5 + 8 = 13; write 3 and carry 1 to the next column. | Step 3: Add the tens column: 4 + 7 + 1 carried over = 12; write 2 and carry 1 to the next column. | Step 4: Add the hundreds column: 3 + 1 + 1 carried over = 5; write 5. | Step 5: Add the thousands column: 2 + 0 = 2; write 2. | Step 6: Add the ten thousands column: 9 + 5 = 14; write 4 and carry 1 to the next column. | Step 7: Write the carried 1 in the next column. | Step 8: Final Answer: 92,345 + 50,178 = 142,523.

Question: Exercise 1.5, Question 1 (iii): Solve in vertical form: 107,236 +

298,017.

Answer: Step 1: Write the numbers one below the other, ones under ones, tens under tens. | Step 2: Add the ones column: 6 + 7 = 13; write 3 and carry 1 to the next column. | Step 3: Add the tens column: 3 + 1 + 1 carried over = 5; write 5. | Step 4: Add the hundreds column: 2 + 0 = 2; write 2. | Step 5: Add the thousands column: 7 + 8 = 15; write 5 and carry 1 to the next column. | Step 6: Add the ten thousands column: 0 + 9 + 1 carried over = 10; write 0 and carry 1 to the next column. | Step 7: Add the hundred thousands column: 1 + 2 + 1 carried over = 4; write 4. | Step 8: Final Answer: 107,236 + 298,017 = 405,253.

Question: Exercise 1.5, Question 1 (iv): Solve in vertical form: 349,629 + 201,548.

Answer: Step 1: Write the numbers one below the other, ones under ones, tens under tens. | Step 2: Add the ones column: 9 + 8 = 17; write 7 and carry 1 to the next column. | Step 3: Add the tens column: 2 + 4 + 1 carried over = 7; write 7. | Step 4: Add the hundreds column: 6 + 5 = 11; write 1 and carry 1 to the next column. | Step 5: Add the thousands column: 9 + 1 + 1 carried over = 11; write 1 and carry 1 to the next column. | Step 6: Add the ten thousands column: 4 + 0 + 1 carried over = 5; write 5. | Step 7: Add the hundred thousands column: 3 + 2 = 5; write 5. | Step 8: Final Answer: 349,629 + 201,548 = 551,177.

Question: Exercise 1.5, Question 1 (v): Solve in vertical form: 964,328 + 428,961.

Answer: Step 1: Write the numbers one below the other, ones under ones, tens under tens. | Step 2: Add the ones column: 8 + 1 = 9; write 9. | Step 3: Add the tens column: 2 + 6 = 8; write 8. | Step 4: Add the hundreds column: 3 + 9 = 12; write 2 and carry 1 to the next column. | Step 5: Add the thousands column: 4 + 8 + 1 carried over = 13; write 3 and carry 1 to

the next column. | Step 6: Add the ten thousands column: 6 + 2 + 1 carried over = 9; write 9. | Step 7: Add the hundred thousands column: 9 + 4 = 13; write 3 and carry 1 to the next column. | Step 8: Write the carried 1 in the next column. | Step 9: Final Answer: 964,328 + 428,961 = 1,393,289.

Question: Exercise 1.5, Question 1 (vi): Solve in vertical form: 999,555 + 312,016.

Answer: Step 1: Write the numbers one below the other, ones under ones, tens under tens. | Step 2: Add the ones column: 5 + 6 = 11; write 1 and carry 1 to the next column. | Step 3: Add the tens column: 5 + 1 + 1 carried over = 7; write 7. | Step 4: Add the hundreds column: 5 + 0 = 5; write 5. | Step 5: Add the thousands column: 9 + 2 = 11; write 1 and carry 1 to the next column. | Step 6: Add the ten thousands column: 9 + 1 + 1 carried over = 11; write 1 and carry 1 to the next column. | Step 7: Add the hundred thousands column: 9 + 3 + 1 carried over = 13; write 3 and carry 1 to the next column. | Step 8: Write the carried 1 in the next column. | Step 9: Final Answer: 999,555 + 312,016 = 1,311,571.

Question: Exercise 1.5, Question 2 (i): Add 24,317 and 90,235.

Answer: Step 1: Write the numbers one below the other, ones under ones, tens under tens. | Step 2: Add the ones column: 7 + 5 = 12; write 2 and carry 1 to the next column. | Step 3: Add the tens column: 1 + 3 + 1 carried over = 5; write 5. | Step 4: Add the hundreds column: 3 + 2 = 5; write 5. | Step 5: Add the thousands column: 4 + 0 = 4; write 4. | Step 6: Add the ten thousands column: 2 + 9 = 11; write 1 and carry 1 to the next column. | Step 7: Write the carried 1 in the next column. | Step 8: Final Answer: 24,317 + 90,235 = 114,552.

Question: Exercise 1.5, Question 2 (ii): Add 67,999 and 249,982.

Answer: Step 1: Write the numbers one below the other, ones under ones,

tens under tens. | Step 2: Add the ones column: 9 + 2 = 11; write 1 and carry 1 to the next column. | Step 3: Add the tens column: 9 + 8 + 1 carried over = 18; write 8 and carry 1 to the next column. | Step 4: Add the hundreds column: 9 + 9 + 1 carried over = 19; write 9 and carry 1 to the next column. | Step 5: Add the thousands column: 7 + 9 + 1 carried over = 17; write 7 and carry 1 to the next column. | Step 6: Add the ten thousands column: 6 + 4 + 1 carried over = 11; write 1 and carry 1 to the next column. | Step 7: Add the hundred thousands column: 0 + 2 + 1 carried over = 3; write 3. | Step 8: Final Answer: 67,999 + 249,982 = 317,981.

Question: Exercise 1.5, Question 2 (iii): Add 392,612 and 722,334.

Answer: Step 1: Write the numbers one below the other, ones under ones, tens under tens. | Step 2: Add the ones column: 2 + 4 = 6; write 6. | Step 3: Add the tens column: 1 + 3 = 4; write 4. | Step 4: Add the hundreds column: 6 + 3 = 9; write 9. | Step 5: Add the thousands column: 2 + 2 = 4; write 4. | Step 6: Add the ten thousands column: 9 + 2 = 11; write 1 and carry 1 to the next column. | Step 7: Add the hundred thousands column: 3 + 7 + 1 carried over = 11; write 1 and carry 1 to the next column. | Step 8: Write the carried 1 in the next column. | Step 9: Final Answer: 392,612 + 722,334 = 1,114,946.

Question: Exercise 1.5, Question 2 (iv): Add 591,023 and 942,589.

Answer: Step 1: Write the numbers one below the other, ones under ones, tens under tens. | Step 2: Add the ones column: 3 + 9 = 12; write 2 and carry 1 to the next column. | Step 3: Add the tens column: 2 + 8 + 1 carried over = 11; write 1 and carry 1 to the next column. | Step 4: Add the hundreds column: 0 + 5 + 1 carried over = 6; write 6. | Step 5: Add the thousands column: 1 + 2 = 3; write 3. | Step 6: Add the ten thousands column: 9 + 4 = 13; write 3 and carry 1 to the next column. | Step 7: Add

the hundred thousands column: 5 + 9 + 1 carried over = 15; write 5 and carry 1 to the next column. | Step 8: Write the carried 1 in the next column. | Step 9: Final Answer: 591,023 + 942,589 = 1,533,612.

Question: Exercise 1.5, Question 2 (v): Add 217,640 and 921,079.

Answer: Step 1: Write the numbers one below the other, ones under ones, tens under tens. | Step 2: Add the ones column: 0 + 9 = 9; write 9. | Step 3: Add the tens column: 4 + 7 = 11; write 1 and carry 1 to the next column. | Step 4: Add the hundreds column: 6 + 0 + 1 carried over = 7; write 7. | Step 5: Add the thousands column: 7 + 1 = 8; write 8. | Step 6: Add the ten thousands column: 1 + 2 = 3; write 3. | Step 7: Add the hundred thousands column: 2 + 9 = 11; write 1 and carry 1 to the next column. | Step 8: Write the carried 1 in the next column. | Step 9: Final Answer: 217,640 + 921,079 = 1,138,719.

Question: Exercise 1.5, Question 2 (vi): Add 555,566 and 778,896.

Answer: Step 1: Write the numbers one below the other, ones under ones, tens under tens. | Step 2: Add the ones column: 6 + 6 = 12; write 2 and carry 1 to the next column. | Step 3: Add the tens column: 6 + 9 + 1 carried over = 16; write 6 and carry 1 to the next column. | Step 4: Add the hundreds column: 5 + 8 + 1 carried over = 14; write 4 and carry 1 to the next column. | Step 5: Add the thousands column: 5 + 8 + 1 carried over = 14; write 4 and carry 1 to the next column. | Step 6: Add the ten thousands column: 5 + 7 + 1 carried over = 13; write 3 and carry 1 to the next column. | Step 7: Add the hundred thousands column: 5 + 7 + 1 carried over = 13; write 3 and carry 1 to the next column. | Step 8: Write the carried 1 in the next column. | Step 9: Final Answer: 555,566 + 778,896 = 1,334,462.

Question: Exercise 1.5, Question 2 (vii): Add 500,983 and 645,008.

Answer: Step 1: Write the numbers one below the other, ones under ones,

tens under tens. | Step 2: Add the ones column: 3 + 8 = 11; write 1 and carry 1 to the next column. | Step 3: Add the tens column: 8 + 0 + 1 carried over = 9; write 9. | Step 4: Add the hundreds column: 9 + 0 = 9; write 9. | Step 5: Add the thousands column: 0 + 5 = 5; write 5. | Step 6: Add the ten thousands column: 0 + 4 = 4; write 4. | Step 7: Add the hundred thousands column: 5 + 6 = 11; write 1 and carry 1 to the next column. | Step 8: Write the carried 1 in the next column. | Step 9: Final Answer: 500,983 + 645,008 = 1,145,991.

Question: Exercise 1.5, Question 2 (viii): Add 910,052 and 881,223.

Answer: Step 1: Write the numbers one below the other, ones under ones, tens under tens. | Step 2: Add the ones column: 2 + 3 = 5; write 5. | Step 3: Add the tens column: 5 + 2 = 7; write 7. | Step 4: Add the hundreds column: 0 + 2 = 2; write 2. | Step 5: Add the thousands column: 0 + 1 = 1; write 1. | Step 6: Add the ten thousands column: 1 + 8 = 9; write 9. | Step 7: Add the hundred thousands column: 9 + 8 = 17; write 7 and carry 1 to the next column. | Step 8: Write the carried 1 in the next column. | Step 9: Final Answer: 910,052 + 881,223 = 1,791,275.

Exercise 1.6: Addition Word Problems

Question: Exercise 1.6, Question 1: The Government of Sindh spent Rs 581,034 on one road and Rs 347,083 on another road. How much is it in total?

Answer: Step 1: To find the total spending, add the two amounts. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 4 + 3 = 7; write 7. | Step 4: Add the tens column: 3 + 8 = 11; write 1 and carry 1 to the next column. | Step 5: Add the hundreds column: 0 + 0 + 1 carried over = 1; write 1. | Step 6: Add the thousands column: 1 + 7 = 8; write 8. | Step 7: Add the ten thousands

column: 8 + 4 = 12; write 2 and carry 1 to the next column. | Step 8: Add the hundred thousands column: 5 + 3 + 1 carried over = 9; write 9. | Step 9: Final Answer: The total amount spent is Rs 928,117.

Question: Exercise 1.6, Question 2: In an examination, 27,514 girls and 20,328 boys appeared. How much is it in total?

Answer: Step 1: To find the total students, add the girls and the boys. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 4 + 8 = 12; write 2 and carry 1 to the next column. | Step 4: Add the tens column: 1 + 2 + 1 carried over = 4; write 4. | Step 5: Add the hundreds column: 5 + 3 = 8; write 8. | Step 6: Add the thousands column: 7 + 0 = 7; write 7. | Step 7: Add the ten thousands column: 2 + 2 = 4; write 4. | Step 8: Final Answer: 47,842 students appeared in the examination in total.

Question: Exercise 1.6, Question 3: Bismah spent Rs 857,600 on computers and Aleesha spent Rs 641,200. How much is it in total?

Answer: Step 1: Add both amounts to find the total spent by both. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 0 + 0 = 0; write 0. | Step 4: Add the tens column: 0 + 0 = 0; write 0. | Step 5: Add the hundreds column: 6 + 2 = 8; write 8. | Step 6: Add the thousands column: 7 + 1 = 8; write 8. | Step 7: Add the ten thousands column: 5 + 4 = 9; write 9. | Step 8: Add the hundred thousands column: 8 + 6 = 14; write 4 and carry 1 to the next column. | Step 9: Write the carried 1 in the next column. | Step 10: Final Answer: Together they spent Rs 1,498,800.

Question: Exercise 1.6, Question 4: The first Pakistan vs India match at the National Stadium Karachi was watched by 55,384 people and the second match at Qaddafi Stadium Lahore by 43,298 people. How much is it in total?

Answer: Step 1: Add the people of both matches to find the total. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 4 + 8 = 12; write 2 and carry 1 to the next column. | Step 4: Add the tens column: 8 + 9 + 1 carried over = 18; write 8 and carry 1 to the next column. | Step 5: Add the hundreds column: 3 + 2 + 1 carried over = 6; write 6. | Step 6: Add the thousands column: 5 + 3 = 8; write 8. | Step 7: Add the ten thousands column: 5 + 4 = 9; write 9. | Step 8: Final Answer: 98,682 people watched the matches in total.

Question: Exercise 1.6, Question 5: One company makes 865,271 bicycles in a year and another company makes 725,059 bicycles. How much is it in total?

Answer: Step 1: Add the bicycles of both companies for the total production. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 1 + 9 = 10; write 0 and carry 1 to the next column. | Step 4: Add the tens column: 7 + 5 + 1 carried over = 13; write 3 and carry 1 to the next column. | Step 5: Add the hundreds column: 2 + 0 + 1 carried over = 3; write 3. | Step 6: Add the thousands column: 5 + 5 = 10; write 0 and carry 1 to the next column. | Step 7: Add the ten thousands column: 6 + 2 + 1 carried over = 9; write 9. | Step 8: Add the hundred thousands column: 8 + 7 = 15; write 5 and carry 1 to the next column. | Step 9: Write the carried 1 in the next column. | Step 10: Final Answer: The total production is 1,590,330 bicycles.

Question: Exercise 1.6, Question 6: Pakistan Railways carried 347,180 kg of mangoes in the first week and 449,130 kg in the next week. How much is it in total?

Answer: Step 1: Add the weights of both weeks for the total weight. | Step 2: Write the numbers one below the other, ones under ones, tens under

tens. | Step 3: Add the ones column: 0 + 0 = 0; write 0. | Step 4: Add the tens column: 8 + 3 = 11; write 1 and carry 1 to the next column. | Step 5: Add the hundreds column: 1 + 1 + 1 carried over = 3; write 3. | Step 6: Add the thousands column: 7 + 9 = 16; write 6 and carry 1 to the next column. | Step 7: Add the ten thousands column: 4 + 4 + 1 carried over = 9; write 9. | Step 8: Add the hundred thousands column: 3 + 4 = 7; write 7. | Step 9: Final Answer: 796,310 kg of mangoes were carried altogether.

Exercise 1.7: Subtraction

Question: Exercise 1.7, Question 1 (i): Solve in vertical form: 32,164 - 20,053.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Subtract the ones column: 4 - 3 = 1; write 1. | Step 3: Subtract the tens column: 6 - 5 = 1; write 1. | Step 4: Subtract the hundreds column: 1 - 0 = 1; write 1. | Step 5: Subtract the thousands column: 2 - 0 = 2; write 2. | Step 6: Subtract the ten thousands column: 3 - 2 = 1; write 1. | Step 7: Final Answer: 32,164 - 20,053 = 12,111.

Question: Exercise 1.7, Question 1 (ii): Solve in vertical form: 583,729 - 21,678.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Subtract the ones column: 9 - 8 = 1; write 1. | Step 3: Tens column: after borrowing, the top digit becomes 12; 12 - 7 = 5; write 5. | Step 4: Hundreds column: after borrowing, the top digit becomes 6; 6 - 6 = 0; write 0. | Step 5: Subtract the thousands column: 3 - 1 = 2; write 2. | Step 6: Subtract the ten thousands column: 8 - 2 = 6; write 6. | Step 7: Subtract the hundred thousands column: 5 - 0 = 5; write 5. | Step 8: Final Answer: 583,729 - 21,678 = 562,051.

Question: Exercise 1.7, Question 1 (iii): Solve in vertical form: 627,948 -

16,328.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Subtract the ones column: 8 - 8 = 0; write 0. | Step 3: Subtract the tens column: 4 - 2 = 2; write 2. | Step 4: Subtract the hundreds column: 9 - 3 = 6; write 6. | Step 5: Subtract the thousands column: 7 - 6 = 1; write 1. | Step 6: Subtract the ten thousands column: 2 - 1 = 1; write 1. | Step 7: Subtract the hundred thousands column: 6 - 0 = 6; write 6. | Step 8: Final Answer: 627,948 - 16,328 = 611,620.

Question: Exercise 1.7, Question 1 (iv): Solve in vertical form: 793,854 - 185,035.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Ones column: after borrowing, the top digit becomes 14; 14 - 5 = 9; write 9. | Step 3: Tens column: after borrowing, the top digit becomes 4; 4 - 3 = 1; write 1. | Step 4: Subtract the hundreds column: 8 - 0 = 8; write 8. | Step 5: Thousands column: after borrowing, the top digit becomes 13; 13 - 5 = 8; write 8. | Step 6: Ten thousands column: after borrowing, the top digit becomes 8; 8 - 8 = 0; write 0. | Step 7: Subtract the hundred thousands column: 7 - 1 = 6; write 6. | Step 8: Final Answer: 793,854 - 185,035 = 608,819.

Question: Exercise 1.7, Question 1 (v): Solve in vertical form: 790,000 - 628,453.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Ones column: after borrowing, the top digit becomes 10; 10 - 3 = 7; write 7. | Step 3: Tens column: after borrowing, the top digit becomes 9; 9 - 5 = 4; write 4. | Step 4: Hundreds column: after borrowing, the top digit becomes 9; 9 - 4 = 5; write 5. | Step 5: Thousands column: after borrowing, the top digit becomes 9; 9 - 8 = 1; write 1. | Step 6: Ten thousands column: after borrowing, the top digit

becomes 8; 8 - 2 = 6; write 6. | Step 7: Subtract the hundred thousands column: 7 - 6 = 1; write 1. | Step 8: Final Answer: 790,000 - 628,453 = 161,547.

Question: Exercise 1.7, Question 1 (vi): Solve in vertical form: 894,093 - 847,141.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Subtract the ones column: 3 - 1 = 2; write 2. | Step 3: Subtract the tens column: 9 - 4 = 5; write 5. | Step 4: Hundreds column: after borrowing, the top digit becomes 10; 10 - 1 = 9; write 9. | Step 5: Thousands column: after borrowing, the top digit becomes 13; 13 - 7 = 6; write 6. | Step 6: Ten thousands column: after borrowing, the top digit becomes 8; 8 - 4 = 4; write 4. | Step 7: Subtract the hundred thousands column: 8 - 8 = 0; write 0. | Step 8: Final Answer: 894,093 - 847,141 = 46,952.

Question: Exercise 1.7, Question 2 (i): Subtract 43,210 from 98,765.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Subtract the ones column: 5 - 0 = 5; write 5. | Step 3: Subtract the tens column: 6 - 1 = 5; write 5. | Step 4: Subtract the hundreds column: 7 - 2 = 5; write 5. | Step 5: Subtract the thousands column: 8 - 3 = 5; write 5. | Step 6: Subtract the ten thousands column: 9 - 4 = 5; write 5. | Step 7: Final Answer: 98,765 - 43,210 = 55,555.

Question: Exercise 1.7, Question 2 (ii): Subtract 39,072 from 273,194.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Subtract the ones column: 4 - 2 = 2; write 2. | Step 3: Subtract the tens column: 9 - 7 = 2; write 2. | Step 4: Subtract the hundreds column: 1 - 0 = 1; write 1. | Step 5: Thousands column: after borrowing, the top digit becomes 13; 13 - 9 = 4; write 4. | Step 6: Ten thousands column: after borrowing, the top digit becomes 6; 6 - 3 = 3;

write 3. | Step 7: Subtract the hundred thousands column: 2 - 0 = 2; write 2. | Step 8: Final Answer: 273,194 - 39,072 = 234,122.

Question: Exercise 1.7, Question 2 (iii): Subtract 99,999 from 100,000.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Ones column: after borrowing, the top digit becomes 10; 10 - 9 = 1; write 1. | Step 3: Tens column: after borrowing, the top digit becomes 9; 9 - 9 = 0; write 0. | Step 4: Hundreds column: after borrowing, the top digit becomes 9; 9 - 9 = 0; write 0. | Step 5: Thousands column: after borrowing, the top digit becomes 9; 9 - 9 = 0; write 0. | Step 6: Ten thousands column: after borrowing, the top digit becomes 9; 9 - 9 = 0; write 0. | Step 7: Hundred thousands column: after borrowing, the top digit becomes 0; 0 - 0 = 0; write 0. | Step 8: Final Answer: 100,000 - 99,999 = 1.

Question: Exercise 1.7, Question 2 (iv): Subtract 537,864 from 700,000.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Ones column: after borrowing, the top digit becomes 10; 10 - 4 = 6; write 6. | Step 3: Tens column: after borrowing, the top digit becomes 9; 9 - 6 = 3; write 3. | Step 4: Hundreds column: after borrowing, the top digit becomes 9; 9 - 8 = 1; write 1. | Step 5: Thousands column: after borrowing, the top digit becomes 9; 9 - 7 = 2; write 2. | Step 6: Ten thousands column: after borrowing, the top digit becomes 9; 9 - 3 = 6; write 6. | Step 7: Hundred thousands column: after borrowing, the top digit becomes 6; 6 - 5 = 1; write 1. | Step 8: Final Answer: 700,000 - 537,864 = 162,136.

Question: Exercise 1.7, Question 2 (v): Subtract 685,439 from 874,189.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Subtract the ones column: 9 - 9 = 0; write 0. | Step 3: Subtract the tens column: 8 - 3 = 5; write 5. | Step 4: Hundreds

column: after borrowing, the top digit becomes 11; 11 - 4 = 7; write 7. | Step 5: Thousands column: after borrowing, the top digit becomes 13; 13 - 5 = 8; write 8. | Step 6: Ten thousands column: after borrowing, the top digit becomes 16; 16 - 8 = 8; write 8. | Step 7: Hundred thousands column: after borrowing, the top digit becomes 7; 7 - 6 = 1; write 1. | Step 8: Final Answer: 874,189 - 685,439 = 188,750.

Question: Exercise 1.7, Question 2 (vi): Subtract 721,059 from 751,342.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Ones column: after borrowing, the top digit becomes 12; 12 - 9 = 3; write 3. | Step 3: Tens column: after borrowing, the top digit becomes 13; 13 - 5 = 8; write 8. | Step 4: Hundreds column: after borrowing, the top digit becomes 2; 2 - 0 = 2; write 2. | Step 5: Subtract the thousands column: 1 - 1 = 0; write 0. | Step 6: Subtract the ten thousands column: 5 - 2 = 3; write 3. | Step 7: Subtract the hundred thousands column: 7 - 7 = 0; write 0. | Step 8: Final Answer: 751,342 - 721,059 = 30,283.

Question: Exercise 1.7, Question 2 (vii): Subtract 781,500 from 871,600.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Subtract the ones column: 0 - 0 = 0; write 0. | Step 3: Subtract the tens column: 0 - 0 = 0; write 0. | Step 4: Subtract the hundreds column: 6 - 5 = 1; write 1. | Step 5: Subtract the thousands column: 1 - 1 = 0; write 0. | Step 6: Ten thousands column: after borrowing, the top digit becomes 17; 17 - 8 = 9; write 9. | Step 7: Hundred thousands column: after borrowing, the top digit becomes 7; 7 - 7 = 0; write 0. | Step 8: Final Answer: 871,600 - 781,500 = 90,100.

Question: Exercise 1.7, Question 2 (viii): Subtract 894,354 from 994,354.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Subtract the ones column: 4 - 4 = 0; write 0.

| Step 3: Subtract the tens column: 5 - 5 = 0; write 0. | Step 4: Subtract the hundreds column: 3 - 3 = 0; write 0. | Step 5: Subtract the thousands column: 4 - 4 = 0; write 0. | Step 6: Subtract the ten thousands column: 9 - 9 = 0; write 0. | Step 7: Subtract the hundred thousands column: 9 - 8 = 1; write 1. | Step 8: Final Answer: 994,354 - 894,354 = 100,000.

Question: Exercise 1.7, Question 2 (ix): Subtract 990,001 from 991,000.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Ones column: after borrowing, the top digit becomes 10; 10 - 1 = 9; write 9. | Step 3: Tens column: after borrowing, the top digit becomes 9; 9 - 0 = 9; write 9. | Step 4: Hundreds column: after borrowing, the top digit becomes 9; 9 - 0 = 9; write 9. | Step 5: Thousands column: after borrowing, the top digit becomes 0; 0 - 0 = 0; write 0. | Step 6: Subtract the ten thousands column: 9 - 9 = 0; write 0. | Step 7: Subtract the hundred thousands column: 9 - 9 = 0; write 0. | Step 8: Final Answer: 991,000 - 990,001 = 999.

Question: Exercise 1.7, Question 2 (x): Subtract 184,019 from 765,129.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Subtract the ones column: 9 - 9 = 0; write 0. | Step 3: Subtract the tens column: 2 - 1 = 1; write 1. | Step 4: Subtract the hundreds column: 1 - 0 = 1; write 1. | Step 5: Subtract the thousands column: 5 - 4 = 1; write 1. | Step 6: Ten thousands column: after borrowing, the top digit becomes 16; 16 - 8 = 8; write 8. | Step 7: Hundred thousands column: after borrowing, the top digit becomes 6; 6 - 1 = 5; write 5. | Step 8: Final Answer: 765,129 - 184,019 = 581,110.

Question: Exercise 1.7, Question 2 (xi): Subtract 853,492 from 891,400.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Ones column: after borrowing, the top digit becomes 10; 10 - 2 = 8; write 8. | Step 3: Tens column: after borrowing,

the top digit becomes 9; 9 - 9 = 0; write 0. | Step 4: Hundreds column: after borrowing, the top digit becomes 13; 13 - 4 = 9; write 9. | Step 5: Thousands column: after borrowing, the top digit becomes 10; 10 - 3 = 7; write 7. | Step 6: Ten thousands column: after borrowing, the top digit becomes 8; 8 - 5 = 3; write 3. | Step 7: Subtract the hundred thousands column: 8 - 8 = 0; write 0. | Step 8: Final Answer: 891,400 - 853,492 = 37,908.

Question: Exercise 1.7, Question 2 (xii): Subtract 493,994 from 943,002.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Ones column: after borrowing, the top digit becomes 12; 12 - 4 = 8; write 8. | Step 3: Tens column: after borrowing, the top digit becomes 9; 9 - 9 = 0; write 0. | Step 4: Hundreds column: after borrowing, the top digit becomes 9; 9 - 9 = 0; write 0. | Step 5: Thousands column: after borrowing, the top digit becomes 12; 12 - 3 = 9; write 9. | Step 6: Ten thousands column: after borrowing, the top digit becomes 13; 13 - 9 = 4; write 4. | Step 7: Hundred thousands column: after borrowing, the top digit becomes 8; 8 - 4 = 4; write 4. | Step 8: Final Answer: 943,002 - 493,994 = 449,008.

Exercise 1.8: Subtraction Word Problems

Question: Exercise 1.8, Question 1: The government spent Rs 985,000 on two schools in total. Rs 539,450 were spent on one school. How much was spent on the other school?

Answer: Step 1: To find the other school's spending, subtract one school's amount from the total. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 0 - 0 = 0; write 0. | Step 4: Tens column: after borrowing, the top digit becomes 10; 10 - 5 = 5; write 5. | Step 5: Hundreds column: after

borrowing, the top digit becomes 9; 9 - 4 = 5; write 5. | Step 6: Thousands column: after borrowing, the top digit becomes 14; 14 - 9 = 5; write 5. | Step 7: Ten thousands column: after borrowing, the top digit becomes 7; 7 - 3 = 4; write 4. | Step 8: Subtract the hundred thousands column: 9 - 5 = 4; write 4. | Step 9: Final Answer: Rs 445,550 were spent on the other school.

Question: Exercise 1.8, Question 2: 57,986 students of grade IV appeared in the examination and 43,985 of them passed. How many students did not pass?

Answer: Step 1: To find those who did not pass, subtract the passed students from the total. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 6 - 5 = 1; write 1. | Step 4: Subtract the tens column: 8 - 8 = 0; write 0. | Step 5: Subtract the hundreds column: 9 - 9 = 0; write 0. | Step 6: Subtract the thousands column: 7 - 3 = 4; write 4. | Step 7: Subtract the ten thousands column: 5 - 4 = 1; write 1. | Step 8: Final Answer: 14,001 students did not pass.

Question: Exercise 1.8, Question 3: Aslam and Atif together invested Rs 658,700 in a cattle farm. Aslam's share is Rs 385,780. What is Atif's share?

Answer: Step 1: To find Atif's share, subtract Aslam's share from the total. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 0 - 0 = 0; write 0. | Step 4: Tens column: after borrowing, the top digit becomes 10; 10 - 8 = 2; write 2. | Step 5: Hundreds column: after borrowing, the top digit becomes 16; 16 - 7 = 9; write 9. | Step 6: Thousands column: after borrowing, the top digit becomes 7; 7 - 5 = 2; write 2. | Step 7: Ten thousands column: after borrowing, the top digit becomes 15; 15 - 8 = 7; write 7. | Step 8: Hundred thousands column: after borrowing, the top digit becomes 5; 5 - 3 = 2;

write 2. | Step 9: Final Answer: Atif's share is Rs 272,920.

Question: Exercise 1.8, Question 4: A cotton company's profit for two months is Rs 320,000. The profit of one month is Rs 139,998. What is the other month's profit?

Answer: Step 1: To find the other month's profit, subtract one month's profit from the total. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Ones column: after borrowing, the top digit becomes 10; 10 - 8 = 2; write 2. | Step 4: Tens column: after borrowing, the top digit becomes 9; 9 - 9 = 0; write 0. | Step 5: Hundreds column: after borrowing, the top digit becomes 9; 9 - 9 = 0; write 0. | Step 6: Thousands column: after borrowing, the top digit becomes 9; 9 - 9 = 0; write 0. | Step 7: Ten thousands column: after borrowing, the top digit becomes 11; 11 - 3 = 8; write 8. | Step 8: Hundred thousands column: after borrowing, the top digit becomes 2; 2 - 1 = 1; write 1. | Step 9: Final Answer: The other month's profit is Rs 180,002.

Question: Exercise 1.8, Question 5: On Eid days, 3,955 people visited the zoo on the first day and 3,843 people on the second day. How many more people came on the first day?

Answer: Step 1: To find the difference, subtract the second day's number from the first day's. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 5 - 3 = 2; write 2. | Step 4: Subtract the tens column: 5 - 4 = 1; write 1. | Step 5: Subtract the hundreds column: 9 - 8 = 1; write 1. | Step 6: Subtract the thousands column: 3 - 3 = 0; write 0. | Step 7: Final Answer: 112 more people came on the first day.

Question: Exercise 1.8, Question 6: A poultry farm had 89,534 hens. 43,294 hens died of bird flu. How many hens were left?

Answer: Step 1: To find the hens left, subtract the dead hens from the

total. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 4 - 4 = 0; write 0. | Step 4: Tens column: after borrowing, the top digit becomes 13; 13 - 9 = 4; write 4. | Step 5: Hundreds column: after borrowing, the top digit becomes 4; 4 - 2 = 2; write 2. | Step 6: Subtract the thousands column: 9 - 3 = 6; write 6. | Step 7: Subtract the ten thousands column: 8 - 4 = 4; write 4. | Step 8: Final Answer: 46,240 hens were left.

Exercise 1.9: Multiplication

Question: Exercise 1.9, Question 1 (i): Solve: 1,632 x 23.

Answer: Step 1: Break 23 into 3 + 20. | Step 2: Multiply by the ones: 1,632 x 3 = 1000 x 3 = 3000 + 600 x 3 = 1800 + 30 x 3 = 90 + 2 x 3 = 6; total 4896. | Step 3: Multiply by the tens: first 1,632 x 2 = 1000 x 2 = 2000 + 600 x 2 = 1200 + 30 x 2 = 60 + 2 x 2 = 4; total 3264; then x 10 gives 32,640. | Step 4: Add the partial products: 4,896 + 32,640 = 37,536. | Step 5: Final Answer: 1,632 x 23 = 37,536.

Question: Exercise 1.9, Question 1 (ii): Solve: 2,341 x 70.

Answer: Step 1: Break 70 into 70. | Step 2: Multiply by the tens: first 2,341 x 7 = 2000 x 7 = 14000 + 300 x 7 = 2100 + 40 x 7 = 280 + 1 x 7 = 7; total 16387; then x 10 gives 163,870. | Step 3: Final Answer: 2,341 x 70 = 163,870.

Question: Exercise 1.9, Question 1 (iii): Solve: 6,314 x 52.

Answer: Step 1: Break 52 into 2 + 50. | Step 2: Multiply by the ones: 6,314 x 2 = 6000 x 2 = 12000 + 300 x 2 = 600 + 10 x 2 = 20 + 4 x 2 = 8; total 12628. | Step 3: Multiply by the tens: first 6,314 x 5 = 6000 x 5 = 30000 + 300 x 5 = 1500 + 10 x 5 = 50 + 4 x 5 = 20; total 31570; then x 10 gives 315,700. | Step 4: Add the partial products: 12,628 + 315,700 = 328,328. | Step 5: Final Answer: 6,314 x 52 = 328,328.

Question: Exercise 1.9, Question 1 (iv): Solve: 2,109 x 84.

Answer: Step 1: Break 84 into 4 + 80. | Step 2: Multiply by the ones: 2,109 x 4 = 2000 x 4 = 8000 + 100 x 4 = 400 + 9 x 4 = 36; total 8436. | Step 3: Multiply by the tens: first 2,109 x 8 = 2000 x 8 = 16000 + 100 x 8 = 800 + 9 x 8 = 72; total 16872; then x 10 gives 168,720. | Step 4: Add the partial products: 8,436 + 168,720 = 177,156. | Step 5: Final Answer: 2,109 x 84 = 177,156.

Question: Exercise 1.9, Question 1 (v): Solve: 51,389 x 562.

Answer: Step 1: Break 562 into 2 + 60 + 500. | Step 2: Multiply by the ones: 51,389 x 2 = 50000 x 2 = 100000 + 1000 x 2 = 2000 + 300 x 2 = 600 + 80 x 2 = 160 + 9 x 2 = 18; total 102778. | Step 3: Multiply by the tens: first 51,389 x 6 = 50000 x 6 = 300000 + 1000 x 6 = 6000 + 300 x 6 = 1800 + 80 x 6 = 480 + 9 x 6 = 54; total 308334; then x 10 gives 3,083,340. | Step 4: Multiply by the hundreds: first 51,389 x 5 = 50000 x 5 = 250000 + 1000 x 5 = 5000 + 300 x 5 = 1500 + 80 x 5 = 400 + 9 x 5 = 45; total 256945; then x 100 gives 25,694,500. | Step 5: Add the partial products: 102,778 + 3,083,340 + 25,694,500 = 28,880,618. | Step 6: Final Answer: 51,389 x 562 = 28,880,618.

Question: Exercise 1.9, Question 1 (vi): Solve: 43,851 x 725.

Answer: Step 1: Break 725 into 5 + 20 + 700. | Step 2: Multiply by the ones: 43,851 x 5 = 40000 x 5 = 200000 + 3000 x 5 = 15000 + 800 x 5 = 4000 + 50 x 5 = 250 + 1 x 5 = 5; total 219255. | Step 3: Multiply by the tens: first 43,851 x 2 = 40000 x 2 = 80000 + 3000 x 2 = 6000 + 800 x 2 = 1600 + 50 x 2 = 100 + 1 x 2 = 2; total 87702; then x 10 gives 877,020. | Step 4: Multiply by the hundreds: first 43,851 x 7 = 40000 x 7 = 280000 + 3000 x 7 = 21000 + 800 x 7 = 5600 + 50 x 7 = 350 + 1 x 7 = 7; total 306957; then x 100 gives 30,695,700. | Step 5: Add the partial products: 219,255 + 877,020 + 30,695,700 = 31,791,975. | Step 6: Final Answer:

43,851 x 725 = 31,791,975.

Question: Exercise 1.9, Question 1 (vii): Solve: 65,123 x 316.

Answer: Step 1: Break 316 into 6 + 10 + 300. | Step 2: Multiply by the ones: 65,123 x 6 = 60000 x 6 = 360000 + 5000 x 6 = 30000 + 100 x 6 = 600 + 20 x 6 = 120 + 3 x 6 = 18; total 390738. | Step 3: Multiply by the tens: first 65,123 x 1 = 60000 x 1 = 60000 + 5000 x 1 = 5000 + 100 x 1 = 100 + 20 x 1 = 20 + 3 x 1 = 3; total 65123; then x 10 gives 651,230. | Step 4: Multiply by the hundreds: first 65,123 x 3 = 60000 x 3 = 180000 + 5000 x 3 = 15000 + 100 x 3 = 300 + 20 x 3 = 60 + 3 x 3 = 9; total 195369; then x 100 gives 19,536,900. | Step 5: Add the partial products: 390,738 + 651,230 + 19,536,900 = 20,578,868. | Step 6: Final Answer: 65,123 x 316 = 20,578,868.

Question: Exercise 1.9, Question 1 (viii): Solve: 74,156 x 163.

Answer: Step 1: Break 163 into 3 + 60 + 100. | Step 2: Multiply by the ones: 74,156 x 3 = 70000 x 3 = 210000 + 4000 x 3 = 12000 + 100 x 3 = 300 + 50 x 3 = 150 + 6 x 3 = 18; total 222468. | Step 3: Multiply by the tens: first 74,156 x 6 = 70000 x 6 = 420000 + 4000 x 6 = 24000 + 100 x 6 = 600 + 50 x 6 = 300 + 6 x 6 = 36; total 444936; then x 10 gives 4,449,360. | Step 4: Multiply by the hundreds: first 74,156 x 1 = 70000 x 1 = 70000 + 4000 x 1 = 4000 + 100 x 1 = 100 + 50 x 1 = 50 + 6 x 1 = 6; total 74156; then x 100 gives 7,415,600. | Step 5: Add the partial products: 222,468 + 4,449,360 + 7,415,600 = 12,087,428. | Step 6: Final Answer: 74,156 x 163 = 12,087,428.

Question: Exercise 1.9, Question 2 (i): Multiply 11,689 by 100.

Answer: Step 1: Break 100 into 100. | Step 2: Multiply by the hundreds: first 11,689 x 1 = 10000 x 1 = 10000 + 1000 x 1 = 1000 + 600 x 1 = 600 + 80 x 1 = 80 + 9 x 1 = 9; total 11689; then x 100 gives 1,168,900. | Step 3: Final Answer: 11,689 x 100 = 1,168,900.

Question: Exercise 1.9, Question 2 (ii): Multiply 21,499 by 120.

Answer: Step 1: Break 120 into 20 + 100. | Step 2: Multiply by the tens: first 21,499 x 2 = 20000 x 2 = 40000 + 1000 x 2 = 2000 + 400 x 2 = 800 + 90 x 2 = 180 + 9 x 2 = 18; total 42998; then x 10 gives 429,980. | Step 3: Multiply by the hundreds: first 21,499 x 1 = 20000 x 1 = 20000 + 1000 x 1 = 1000 + 400 x 1 = 400 + 90 x 1 = 90 + 9 x 1 = 9; total 21499; then x 100 gives 2,149,900. | Step 4: Add the partial products: 429,980 + 2,149,900 = 2,579,880. | Step 5: Final Answer: 21,499 x 120 = 2,579,880.

Question: Exercise 1.9, Question 2 (iii): Multiply 25,701 by 553.

Answer: Step 1: Break 553 into 3 + 50 + 500. | Step 2: Multiply by the ones: 25,701 x 3 = 20000 x 3 = 60000 + 5000 x 3 = 15000 + 700 x 3 = 2100 + 1 x 3 = 3; total 77103. | Step 3: Multiply by the tens: first 25,701 x 5 = 20000 x 5 = 100000 + 5000 x 5 = 25000 + 700 x 5 = 3500 + 1 x 5 = 5; total 128505; then x 10 gives 1,285,050. | Step 4: Multiply by the hundreds: first 25,701 x 5 = 20000 x 5 = 100000 + 5000 x 5 = 25000 + 700 x 5 = 3500 + 1 x 5 = 5; total 128505; then x 100 gives 12,850,500. | Step 5: Add the partial products: 77,103 + 1,285,050 + 12,850,500 = 14,212,653. | Step 6: Final Answer: 25,701 x 553 = 14,212,653.

Question: Exercise 1.9, Question 2 (iv): Multiply 32,145 by 152.

Answer: Step 1: Break 152 into 2 + 50 + 100. | Step 2: Multiply by the ones: 32,145 x 2 = 30000 x 2 = 60000 + 2000 x 2 = 4000 + 100 x 2 = 200 + 40 x 2 = 80 + 5 x 2 = 10; total 64290. | Step 3: Multiply by the tens: first 32,145 x 5 = 30000 x 5 = 150000 + 2000 x 5 = 10000 + 100 x 5 = 500 + 40 x 5 = 200 + 5 x 5 = 25; total 160725; then x 10 gives 1,607,250. | Step 4: Multiply by the hundreds: first 32,145 x 1 = 30000 x 1 = 30000 + 2000 x 1 = 2000 + 100 x 1 = 100 + 40 x 1 = 40 + 5 x 1 = 5; total 32145; then x 100 gives 3,214,500. | Step 5: Add the partial products: 64,290 + 1,607,250 + 3,214,500 = 4,886,040. | Step 6: Final

Answer: 32,145 x 152 = 4,886,040.

Question: Exercise 1.9, Question 2 (v): Multiply 41,078 by 203.

Answer: Step 1: Break 203 into 3 + 200. | Step 2: Multiply by the ones: 41,078 x 3 = 40000 x 3 = 120000 + 1000 x 3 = 3000 + 70 x 3 = 210 + 8 x 3 = 24; total 123234. | Step 3: Multiply by the hundreds: first 41,078 x 2 = 40000 x 2 = 80000 + 1000 x 2 = 2000 + 70 x 2 = 140 + 8 x 2 = 16; total 82156; then x 100 gives 8,215,600. | Step 4: Add the partial products: 123,234 + 8,215,600 = 8,338,834. | Step 5: Final Answer: 41,078 x 203 = 8,338,834.

Question: Exercise 1.9, Question 2 (vi): Multiply 12,345 by 123.

Answer: Step 1: Break 123 into 3 + 20 + 100. | Step 2: Multiply by the ones: 12,345 x 3 = 10000 x 3 = 30000 + 2000 x 3 = 6000 + 300 x 3 = 900 + 40 x 3 = 120 + 5 x 3 = 15; total 37035. | Step 3: Multiply by the tens: first 12,345 x 2 = 10000 x 2 = 20000 + 2000 x 2 = 4000 + 300 x 2 = 600 + 40 x 2 = 80 + 5 x 2 = 10; total 24690; then x 10 gives 246,900. | Step 4: Multiply by the hundreds: first 12,345 x 1 = 10000 x 1 = 10000 + 2000 x 1 = 2000 + 300 x 1 = 300 + 40 x 1 = 40 + 5 x 1 = 5; total 12345; then x 100 gives 1,234,500. | Step 5: Add the partial products: 37,035 + 246,900 + 1,234,500 = 1,518,435. | Step 6: Final Answer: 12,345 x 123 = 1,518,435.

Question: Exercise 1.9, Question 2 (vii): Multiply 54,321 by 321.

Answer: Step 1: Break 321 into 1 + 20 + 300. | Step 2: Multiply by the ones: 54,321 x 1 = 50000 x 1 = 50000 + 4000 x 1 = 4000 + 300 x 1 = 300 + 20 x 1 = 20 + 1 x 1 = 1; total 54321. | Step 3: Multiply by the tens: first 54,321 x 2 = 50000 x 2 = 100000 + 4000 x 2 = 8000 + 300 x 2 = 600 + 20 x 2 = 40 + 1 x 2 = 2; total 108642; then x 10 gives 1,086,420. | Step 4: Multiply by the hundreds: first 54,321 x 3 = 50000 x 3 = 150000 + 4000 x 3 = 12000 + 300 x 3 = 900 + 20 x 3 = 60 + 1 x 3 = 3; total

162963; then x 100 gives 16,296,300. | Step 5: Add the partial products: 54,321 + 1,086,420 + 16,296,300 = 17,437,041. | Step 6: Final Answer: 54,321 x 321 = 17,437,041.

Question: Exercise 1.9, Question 2 (viii): Multiply 89,713 by 401.

Answer: Step 1: Break 401 into 1 + 400. | Step 2: Multiply by the ones: 89,713 x 1 = 80000 x 1 = 80000 + 9000 x 1 = 9000 + 700 x 1 = 700 + 10 x 1 = 10 + 3 x 1 = 3; total 89713. | Step 3: Multiply by the hundreds: first 89,713 x 4 = 80000 x 4 = 320000 + 9000 x 4 = 36000 + 700 x 4 = 2800 + 10 x 4 = 40 + 3 x 4 = 12; total 358852; then x 100 gives 35,885,200. | Step 4: Add the partial products: 89,713 + 35,885,200 = 35,974,913. | Step 5: Final Answer: 89,713 x 401 = 35,974,913.

Exercise 1.10: Multiplication Word Problems

Question: Exercise 1.10, Question 1: One truck carries 5,324 bottles. How many bottles can be carried in 132 trucks?

Answer: Step 1: For 132 trucks, multiply 5,324 by 132. | Step 2: Break 132 into 2 + 30 + 100. | Step 3: Multiply by the ones: 5,324 x 2 = 5000 x 2 = 10000 + 300 x 2 = 600 + 20 x 2 = 40 + 4 x 2 = 8; total 10648. | Step 4: Multiply by the tens: first 5,324 x 3 = 5000 x 3 = 15000 + 300 x 3 = 900 + 20 x 3 = 60 + 4 x 3 = 12; total 15972; then x 10 gives 159,720. | Step 5: Multiply by the hundreds: first 5,324 x 1 = 5000 x 1 = 5000 + 300 x 1 = 300 + 20 x 1 = 20 + 4 x 1 = 4; total 5324; then x 100 gives 532,400. | Step 6: Add the partial products: 10,648 + 159,720 + 532,400 = 702,768. | Step 7: Final Answer: 702,768 bottles can be carried in 132 trucks.

Question: Exercise 1.10, Question 2: A school has 630 students and each student pays a monthly fee of Rs 1,200. How much fee is collected in a month?

Answer: Step 1: For one month's fee, multiply 1,200 by 630. | Step 2:

Break 630 into 30 + 600. | Step 3: Multiply by the tens: first 1,200 x 3 = 1000 x 3 = 3000 + 200 x 3 = 600; total 3600; then x 10 gives 36,000. | Step 4: Multiply by the hundreds: first 1,200 x 6 = 1000 x 6 = 6000 + 200 x 6 = 1200; total 7200; then x 100 gives 720,000. | Step 5: Add the partial products: 36,000 + 720,000 = 756,000. | Step 6: Final Answer: Rs 756,000 fee is collected in a month.

Question: Exercise 1.10, Question 3: The price of one motorcycle is Rs 35,800. What is the price of 325 motorcycles?

Answer: Step 1: For 325 motorcycles, multiply 35,800 by 325. | Step 2: Break 325 into 5 + 20 + 300. | Step 3: Multiply by the ones: 35,800 x 5 = 30000 x 5 = 150000 + 5000 x 5 = 25000 + 800 x 5 = 4000; total 179000. | Step 4: Multiply by the tens: first 35,800 x 2 = 30000 x 2 = 60000 + 5000 x 2 = 10000 + 800 x 2 = 1600; total 71600; then x 10 gives 716,000. | Step 5: Multiply by the hundreds: first 35,800 x 3 = 30000 x 3 = 90000 + 5000 x 3 = 15000 + 800 x 3 = 2400; total 107400; then x 100 gives 10,740,000. | Step 6: Add the partial products: 179,000 + 716,000 + 10,740,000 = 11,635,000. | Step 7: Final Answer: The price of 325 motorcycles is Rs 11,635,000.

Question: Exercise 1.10, Question 4: A poultry farm produces 43,290 eggs in one day. How many eggs are produced in 400 days?

Answer: Step 1: For 400 days, multiply 43,290 by 400. | Step 2: Break 400 into 400. | Step 3: Multiply by the hundreds: first 43,290 x 4 = 40000 x 4 = 160000 + 3000 x 4 = 12000 + 200 x 4 = 800 + 90 x 4 = 360; total 173160; then x 100 gives 17,316,000. | Step 4: Final Answer: 17,316,000 eggs are produced in 400 days.

Question: Exercise 1.10, Question 5: A town uses 45,038 litres of water in one week. How much water is used in 890 weeks?

Answer: Step 1: For 890 weeks, multiply 45,038 by 890. | Step 2: Break

890 into 90 + 800. | Step 3: Multiply by the tens: first 45,038 x 9 = 40000 x 9 = 360000 + 5000 x 9 = 45000 + 30 x 9 = 270 + 8 x 9 = 72; total 405342; then x 10 gives 4,053,420. | Step 4: Multiply by the hundreds: first 45,038 x 8 = 40000 x 8 = 320000 + 5000 x 8 = 40000 + 30 x 8 = 240 + 8 x 8 = 64; total 360304; then x 100 gives 36,030,400. | Step 5: Add the partial products: 4,053,420 + 36,030,400 = 40,083,820. | Step 6: Final Answer: 40,083,820 litres of water are used in 890 weeks.

Question: Exercise 1.10, Question 6: One family spends Rs 15,980 on food in a month. How much do 580 such families spend?

Answer: Step 1: For 580 such families, multiply 15,980 by 580. | Step 2: Break 580 into 80 + 500. | Step 3: Multiply by the tens: first 15,980 x 8 = 10000 x 8 = 80000 + 5000 x 8 = 40000 + 900 x 8 = 7200 + 80 x 8 = 640; total 127840; then x 10 gives 1,278,400. | Step 4: Multiply by the hundreds: first 15,980 x 5 = 10000 x 5 = 50000 + 5000 x 5 = 25000 + 900 x 5 = 4500 + 80 x 5 = 400; total 79900; then x 100 gives 7,990,000. | Step 5: Add the partial products: 1,278,400 + 7,990,000 = 9,268,400. | Step 6: Final Answer: 580 families spend Rs 9,268,400.

Exercise 1.11: Division

Question: Exercise 1.11, Question 1 (i): Solve: 6,744 / 12.

Answer: Step 1: We divide 6,744 by 12, digit by digit. | Step 2: 67 / 12: 12 x 5 = 60; remainder 67 - 60 = 7. | Step 3: 74 / 12: 12 x 6 = 72; remainder 74 - 72 = 2. | Step 4: 24 / 12: 12 x 2 = 24; remainder 24 - 24 = 0. | Step 5: Final Answer: 6,744 / 12 = 562.

Question: Exercise 1.11, Question 1 (ii): Solve: 3,795 / 15.

Answer: Step 1: We divide 3,795 by 15, digit by digit. | Step 2: 37 / 15: 15 x 2 = 30; remainder 37 - 30 = 7. | Step 3: 79 / 15: 15 x 5 = 75; remainder 79 - 75 = 4. | Step 4: 45 / 15: 15 x 3 = 45; remainder 45 - 45 = 0. | Step 5:

Final Answer: 3,795 / 15 = 253.

Question: Exercise 1.11, Question 1 (iii): Solve: 7,293 / 13.

Answer: Step 1: We divide 7,293 by 13, digit by digit. | Step 2: 72 / 13: 13 x 5 = 65; remainder 72 - 65 = 7. | Step 3: 79 / 13: 13 x 6 = 78; remainder 79 - 78 = 1. | Step 4: 13 / 13: 13 x 1 = 13; remainder 13 - 13 = 0. | Step 5: Final Answer: 7,293 / 13 = 561.

Question: Exercise 1.11, Question 1 (iv): Solve: 9,384 / 12.

Answer: Step 1: We divide 9,384 by 12, digit by digit. | Step 2: 93 / 12: 12 x 7 = 84; remainder 93 - 84 = 9. | Step 3: 98 / 12: 12 x 8 = 96; remainder 98 - 96 = 2. | Step 4: 24 / 12: 12 x 2 = 24; remainder 24 - 24 = 0. | Step 5: Final Answer: 9,384 / 12 = 782.

Question: Exercise 1.11, Question 1 (v): Solve: 2,214 / 18.

Answer: Step 1: We divide 2,214 by 18, digit by digit. | Step 2: 22 / 18: 18 x 1 = 18; remainder 22 - 18 = 4. | Step 3: 41 / 18: 18 x 2 = 36; remainder 41 - 36 = 5. | Step 4: 54 / 18: 18 x 3 = 54; remainder 54 - 54 = 0. | Step 5: Final Answer: 2,214 / 18 = 123.

Question: Exercise 1.11, Question 1 (vi): Solve: 9,944 / 22.

Answer: Step 1: We divide 9,944 by 22, digit by digit. | Step 2: 99 / 22: 22 x 4 = 88; remainder 99 - 88 = 11. | Step 3: 114 / 22: 22 x 5 = 110; remainder 114 - 110 = 4. | Step 4: 44 / 22: 22 x 2 = 44; remainder 44 - 44 = 0. | Step 5: Final Answer: 9,944 / 22 = 452.

Question: Exercise 1.11, Question 2 (i): Divide and find the quotient: divide 4,368 by 28.

Answer: Step 1: We divide 4,368 by 28, digit by digit. | Step 2: 43 / 28: 28 x 1 = 28; remainder 43 - 28 = 15. | Step 3: 156 / 28: 28 x 5 = 140; remainder 156 - 140 = 16. | Step 4: 168 / 28: 28 x 6 = 168; remainder 168 - 168 = 0. | Step 5: Final Answer: 4,368 / 28 = 156.

Question: Exercise 1.11, Question 2 (ii): Divide and find the quotient:

divide 8,890 by 35.

Answer: Step 1: We divide 8,890 by 35, digit by digit. | Step 2: 88 / 35: 35 x 2 = 70; remainder 88 - 70 = 18. | Step 3: 189 / 35: 35 x 5 = 175; remainder 189 - 175 = 14. | Step 4: 140 / 35: 35 x 4 = 140; remainder 140 - 140 = 0. | Step 5: Final Answer: 8,890 / 35 = 254.

Question: Exercise 1.11, Question 2 (iii): Divide and find the quotient: divide 5,056 by 32.

Answer: Step 1: We divide 5,056 by 32, digit by digit. | Step 2: 50 / 32: 32 x 1 = 32; remainder 50 - 32 = 18. | Step 3: 185 / 32: 32 x 5 = 160; remainder 185 - 160 = 25. | Step 4: 256 / 32: 32 x 8 = 256; remainder 256 - 256 = 0. | Step 5: Final Answer: 5,056 / 32 = 158.

Question: Exercise 1.11, Question 2 (iv): Divide and find the quotient: divide 6,300 by 25.

Answer: Step 1: We divide 6,300 by 25, digit by digit. | Step 2: 63 / 25: 25 x 2 = 50; remainder 63 - 50 = 13. | Step 3: 130 / 25: 25 x 5 = 125; remainder 130 - 125 = 5. | Step 4: 50 / 25: 25 x 2 = 50; remainder 50 - 50 = 0. | Step 5: Final Answer: 6,300 / 25 = 252.

Question: Exercise 1.11, Question 2 (v): Divide and find the quotient: divide 3,920 by 16.

Answer: Step 1: We divide 3,920 by 16, digit by digit. | Step 2: 39 / 16: 16 x 2 = 32; remainder 39 - 32 = 7. | Step 3: 72 / 16: 16 x 4 = 64; remainder 72 - 64 = 8. | Step 4: 80 / 16: 16 x 5 = 80; remainder 80 - 80 = 0. | Step 5: Final Answer: 3,920 / 16 = 245.

Question: Exercise 1.11, Question 2 (vi): Divide and find the quotient: divide 6,642 by 18.

Answer: Step 1: We divide 6,642 by 18, digit by digit. | Step 2: 66 / 18: 18 x 3 = 54; remainder 66 - 54 = 12. | Step 3: 124 / 18: 18 x 6 = 108; remainder 124 - 108 = 16. | Step 4: 162 / 18: 18 x 9 = 162; remainder 162

- 162 = 0. | Step 5: Final Answer: 6,642 / 18 = 369.

Question: Exercise 1.11, Question 3: Find the quotient and remainder when the dividend is 5,075 and the divisor is 35.

Answer: Step 1: We divide 5,075 by 35, digit by digit. | Step 2: 50 / 35: 35 x 1 = 35; remainder 50 - 35 = 15. | Step 3: 157 / 35: 35 x 4 = 140; remainder 157 - 140 = 17. | Step 4: 175 / 35: 35 x 5 = 175; remainder 175 - 175 = 0. | Step 5: Final Answer: 5,075 / 35 = 145.

Question: Exercise 1.11, Question 4: Find the quotient and remainder when the dividend is 5,696 and the divisor is 16.

Answer: Step 1: We divide 5,696 by 16, digit by digit. | Step 2: 56 / 16: 16 x 3 = 48; remainder 56 - 48 = 8. | Step 3: 89 / 16: 16 x 5 = 80; remainder 89 - 80 = 9. | Step 4: 96 / 16: 16 x 6 = 96; remainder 96 - 96 = 0. | Step 5: Final Answer: 5,696 / 16 = 356.

Question: Exercise 1.11, Question 5: Find the quotient and remainder when the dividend is 31,035 and the divisor is 12.

Answer: Step 1: We divide 31,035 by 12, digit by digit. | Step 2: 31 / 12: 12 x 2 = 24; remainder 31 - 24 = 7. | Step 3: 70 / 12: 12 x 5 = 60; remainder 70 - 60 = 10. | Step 4: 103 / 12: 12 x 8 = 96; remainder 103 - 96 = 7. | Step 5: 75 / 12: 12 x 6 = 72; remainder 75 - 72 = 3. | Step 6: Final Answer: Quotient = 2,586, Remainder = 3.

Question: Exercise 1.11, Question 6: Find the quotient and remainder when the dividend is 9,267 and the divisor is 15.

Answer: Step 1: We divide 9,267 by 15, digit by digit. | Step 2: 92 / 15: 15 x 6 = 90; remainder 92 - 90 = 2. | Step 3: 26 / 15: 15 x 1 = 15; remainder 26 - 15 = 11. | Step 4: 117 / 15: 15 x 7 = 105; remainder 117 - 105 = 12. | Step 5: Final Answer: Quotient = 617, Remainder = 12.

Exercise 1.12: Division Word Problems

Question: Exercise 1.12, Question 1: A piece of cloth is 6,272 metres long. One suit needs 7 metres of cloth. How many suits can be made?

Answer: Step 1: To find the number of suits, divide 6,272 by 7. | Step 2: We divide 6,272 by 7, digit by digit. | Step 3: 62 / 7: 7 x 8 = 56; remainder 62 - 56 = 6. | Step 4: 67 / 7: 7 x 9 = 63; remainder 67 - 63 = 4. | Step 5: 42 / 7: 7 x 6 = 42; remainder 42 - 42 = 0. | Step 6: Final Answer: 896 suits can be made.

Question: Exercise 1.12, Question 2: The monthly salary of 12 workers together is Rs 7,032 and all get the same salary. What is the salary of one worker?

Answer: Step 1: For one worker's salary, divide 7,032 by 12. | Step 2: We divide 7,032 by 12, digit by digit. | Step 3: 70 / 12: 12 x 5 = 60; remainder 70 - 60 = 10. | Step 4: 103 / 12: 12 x 8 = 96; remainder 103 - 96 = 7. | Step 5: 72 / 12: 12 x 6 = 72; remainder 72 - 72 = 0. | Step 6: Final Answer: The salary of one worker is Rs 586.

Question: Exercise 1.12, Question 3: Asif spent Rs 2,925 to buy 13 shirts of the same price. What is the price of each shirt?

Answer: Step 1: For the price of one shirt, divide 2,925 by 13. | Step 2: We divide 2,925 by 13, digit by digit. | Step 3: 29 / 13: 13 x 2 = 26; remainder 29 - 26 = 3. | Step 4: 32 / 13: 13 x 2 = 26; remainder 32 - 26 = 6. | Step 5: 65 / 13: 13 x 5 = 65; remainder 65 - 65 = 0. | Step 6: Final Answer: The price of one shirt is Rs 225.

Question: Exercise 1.12, Question 4: The weight of 16 sacks of flour is 4,496 kg and all sacks weigh the same. What is the weight of one sack?

Answer: Step 1: For the weight of one sack, divide 4,496 by 16. | Step 2: We divide 4,496 by 16, digit by digit. | Step 3: 44 / 16: 16 x 2 = 32; remainder 44 - 32 = 12. | Step 4: 129 / 16: 16 x 8 = 128; remainder 129 - 128 = 1. | Step 5: 16 / 16: 16 x 1 = 16; remainder 16 - 16 = 0. | Step 6:

Final Answer: The weight of one sack is 281 kg.

Question: Exercise 1.12, Question 5: The price of 25 cricket bats is Rs 3,075. What is the price of one bat?

Answer: Step 1: For the price of one bat, divide 3,075 by 25. | Step 2: We divide 3,075 by 25, digit by digit. | Step 3: 30 / 25: 25 x 1 = 25; remainder 30 - 25 = 5. | Step 4: 57 / 25: 25 x 2 = 50; remainder 57 - 50 = 7. | Step 5: 75 / 25: 25 x 3 = 75; remainder 75 - 75 = 0. | Step 6: Final Answer: The price of one bat is Rs 123.

Question: Exercise 1.12, Question 6: One crate holds 36 bottles. There are 5,616 bottles in all. How many crates are required?

Answer: Step 1: To find the number of crates, divide 5,616 by 36. | Step 2: We divide 5,616 by 36, digit by digit. | Step 3: 56 / 36: 36 x 1 = 36; remainder 56 - 36 = 20. | Step 4: 201 / 36: 36 x 5 = 180; remainder 201 - 180 = 21. | Step 5: 216 / 36: 36 x 6 = 216; remainder 216 - 216 = 0. | Step 6: Final Answer: 156 crates are required.

Exercise 1.13: Mixed Operations

Question: Exercise 1.13, Question 1: Solve: 46 - 23 + 17.

Answer: Step 1: Only addition and subtraction are here, so work from left to right. | Step 2: 46 - 23 = 23. | Step 3: 23 + 17 = 40. | Step 4: Final Answer: 46 - 23 + 17 = 40.

Question: Exercise 1.13, Question 2: Solve: 99 - 77 + 33.

Answer: Step 1: Only addition and subtraction are here, so work from left to right. | Step 2: 99 - 77 = 22. | Step 3: 22 + 33 = 55. | Step 4: Final Answer: 99 - 77 + 33 = 55.

Question: Exercise 1.13, Question 3: Solve: 98 - 46 + 24.

Answer: Step 1: Only addition and subtraction are here, so work from left to right. | Step 2: 98 - 46 = 52. | Step 3: 52 + 24 = 76. | Step 4: Final

Answer: 98 - 46 + 24 = 76.

Question: Exercise 1.13, Question 4: Solve: 48 - 21 + 31.

Answer: Step 1: Only addition and subtraction are here, so work from left to right. | Step 2: 48 - 21 = 27. | Step 3: 27 + 31 = 58. | Step 4: Final Answer: 48 - 21 + 31 = 58.

Question: Exercise 1.13, Question 5: Solve: 324 - 152 + 182.

Answer: Step 1: Only addition and subtraction are here, so work from left to right. | Step 2: 324 - 152 = 172. | Step 3: 172 + 182 = 354. | Step 4: Final Answer: 324 - 152 + 182 = 354.

Question: Exercise 1.13, Question 6: Solve: 582 - 325 + 154.

Answer: Step 1: Only addition and subtraction are here, so work from left to right. | Step 2: 582 - 325 = 257. | Step 3: 257 + 154 = 411. | Step 4: Final Answer: 582 - 325 + 154 = 411.

Question: Exercise 1.13, Question 7: Solve: 682 + 329 - 159.

Answer: Step 1: Only addition and subtraction are here, so work from left to right. | Step 2: 682 + 329 = 1,011. | Step 3: 1,011 - 159 = 852. | Step 4: Final Answer: 682 + 329 - 159 = 852.

Question: Exercise 1.13, Question 8: Solve: 489 / 5 + 393.

Answer: Step 1: Division and multiplication are done before addition and subtraction. | Step 2: First do the division: 489 / 5 = 97 remainder 4; we take the quotient 97. | Step 3: Now add: 97 + 393 = 490. | Step 4: Final Answer: 489 / 5 + 393 = 490.

Question: Exercise 1.13, Question 9: Solve: 253 / 11 x 5.

Answer: Step 1: Only addition and subtraction are here, so work from left to right. | Step 2: 253 / 11 = 23. | Step 3: 23 x 5 = 115. | Step 4: Final Answer: 253 / 11 x 5 = 115.

Question: Exercise 1.13, Question 10: Solve: 540 / 15 x 8.

Answer: Step 1: Only addition and subtraction are here, so work from left

to right. | Step 2: 540 / 15 = 36. | Step 3: 36 x 8 = 288. | Step 4: Final Answer: 540 / 15 x 8 = 288.

Question: Exercise 1.13, Question 11: Solve: 992 / 16 x 4.

Answer: Step 1: Only addition and subtraction are here, so work from left to right. | Step 2: 992 / 16 = 62. | Step 3: 62 x 4 = 248. | Step 4: Final Answer: 992 / 16 x 4 = 248.

Question: Exercise 1.13, Question 12: Solve: 7 x 375 / 15.

Answer: Step 1: Only addition and subtraction are here, so work from left to right. | Step 2: 7 x 375 = 2,625. | Step 3: 2,625 / 15 = 175. | Step 4: Final Answer: 7 x 375 / 15 = 175.

Question: Exercise 1.13, Question 13: Solve: 12 x 114 / 19 + 10.

Answer: Step 1: Division and multiplication are done before addition and subtraction. | Step 2: 12 x 114 = 1,368. | Step 3: 1,368 / 19 = 72. | Step 4: 72 + 10 = 82. | Step 5: Final Answer: 12 x 114 / 19 + 10 = 82.

Question: Exercise 1.13, Question 14: Solve: 23 + 800 / 20 x 2.

Answer: Step 1: Division and multiplication are done before addition and subtraction. | Step 2: 800 / 20 = 40. | Step 3: 40 x 2 = 80. | Step 4: 23 + 80 = 103. | Step 5: Final Answer: 23 + 800 / 20 x 2 = 103.

Question: Exercise 1.13, Question 15: Solve: 32 x 400 / 16 + 23.

Answer: Step 1: Division and multiplication are done before addition and subtraction. | Step 2: 32 x 400 = 12,800. | Step 3: 12,800 / 16 = 800. | Step 4: 800 + 23 = 823. | Step 5: Final Answer: 32 x 400 / 16 + 23 = 823.

Question: Exercise 1.13, Question 16: Solve: 451 x 690 / 30 - 15.

Answer: Step 1: Division and multiplication are done before addition and subtraction. | Step 2: 451 x 690 = 311,190. | Step 3: 311,190 / 30 = 10,373. | Step 4: 10,373 - 15 = 10,358. | Step 5: Final Answer: 451 x 690 / 30 - 15 = 10,358.

Exercise 1.14: Mixed Word Problems

Question: Exercise 1.14, Question 1: One car costs Rs 748,630 and another car costs Rs 630,010. What is the total cost of both?

Answer: Step 1: Add the prices of both cars. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 0 + 0 = 0; write 0. | Step 4: Add the tens column: 3 + 1 = 4; write 4. | Step 5: Add the hundreds column: 6 + 0 = 6; write 6. | Step 6: Add the thousands column: 8 + 0 = 8; write 8. | Step 7: Add the ten thousands column: 4 + 3 = 7; write 7. | Step 8: Add the hundred thousands column: 7 + 6 = 13; write 3 and carry 1 to the next column. | Step 9: Write the carried 1 in the next column. | Step 10: Final Answer: The total cost of both cars is Rs 1,378,640.

Question: Exercise 1.14, Question 2: Shagufta had Rs 389,000 in the bank. She spent Rs 183,499 on house repair. How much money is left with her?

Answer: Step 1: For the money left, subtract the spending from the total. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Ones column: after borrowing, the top digit becomes 10; 10 - 9 = 1; write 1. | Step 4: Tens column: after borrowing, the top digit becomes 9; 9 - 9 = 0; write 0. | Step 5: Hundreds column: after borrowing, the top digit becomes 9; 9 - 4 = 5; write 5. | Step 6: Thousands column: after borrowing, the top digit becomes 8; 8 - 3 = 5; write 5. | Step 7: Subtract the ten thousands column: 8 - 8 = 0; write 0. | Step 8: Subtract the hundred thousands column: 3 - 1 = 2; write 2. | Step 9: Final Answer: Rs 205,501 were left with Shagufta.

Question: Exercise 1.14, Question 3: A poultry farm sold 143,860 chickens in one month and 354,180 in the next month. What is the difference in sale?

Answer: Step 1: For the difference, subtract the smaller sale from the

bigger sale. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 0 - 0 = 0; write 0. | Step 4: Subtract the tens column: 8 - 6 = 2; write 2. | Step 5: Hundreds column: after borrowing, the top digit becomes 11; 11 - 8 = 3; write 3. | Step 6: Thousands column: after borrowing, the top digit becomes 3; 3 - 3 = 0; write 0. | Step 7: Subtract the ten thousands column: 5 - 4 = 1; write 1. | Step 8: Subtract the hundred thousands column: 3 - 1 = 2; write 2. | Step 9: Final Answer: The difference in sale is 210,320 chickens.

Question: Exercise 1.14, Question 4: The cost of one TV is Rs 95,400. What is the cost of 150 TVs?

Answer: Step 1: For 150 TVs, multiply 95,400 by 150. | Step 2: Break 150 into 50 + 100. | Step 3: Multiply by the tens: first 95,400 x 5 = 90000 x 5 = 450000 + 5000 x 5 = 25000 + 400 x 5 = 2000; total 477000; then x 10 gives 4,770,000. | Step 4: Multiply by the hundreds: first 95,400 x 1 = 90000 x 1 = 90000 + 5000 x 1 = 5000 + 400 x 1 = 400; total 95400; then x 100 gives 9,540,000. | Step 5: Add the partial products: 4,770,000 + 9,540,000 = 14,310,000. | Step 6: Final Answer: The cost of 150 TVs is Rs 14,310,000.

Question: Exercise 1.14, Question 5: One farm contains 56,321 date trees. How many trees are there in 835 such farms?

Answer: Step 1: For 835 farms, multiply 56,321 by 835. | Step 2: Break 835 into 5 + 30 + 800. | Step 3: Multiply by the ones: 56,321 x 5 = 50000 x 5 = 250000 + 6000 x 5 = 30000 + 300 x 5 = 1500 + 20 x 5 = 100 + 1 x 5 = 5; total 281605. | Step 4: Multiply by the tens: first 56,321 x 3 = 50000 x 3 = 150000 + 6000 x 3 = 18000 + 300 x 3 = 900 + 20 x 3 = 60 + 1 x 3 = 3; total 168963; then x 10 gives 1,689,630. | Step 5: Multiply by the hundreds: first 56,321 x 8 = 50000 x 8 = 400000 + 6000 x 8 = 48000

+ 300 x 8 = 2400 + 20 x 8 = 160 + 1 x 8 = 8; total 450568; then x 100 gives 45,056,800. | Step 6: Add the partial products: 281,605 + 1,689,630 + 45,056,800 = 47,028,035. | Step 7: Final Answer: There are 47,028,035 date trees in 835 farms.

Question: Exercise 1.14, Question 6: A school spent Rs 4,375 on a picnic and only 35 students went. How much did each student pay?

Answer: Step 1: To find each student's share, divide 4,375 by 35. | Step 2: We divide 4,375 by 35, digit by digit. | Step 3: 43 / 35: 35 x 1 = 35; remainder 43 - 35 = 8. | Step 4: 87 / 35: 35 x 2 = 70; remainder 87 - 70 = 17. | Step 5: 175 / 35: 35 x 5 = 175; remainder 175 - 175 = 0. | Step 6: Final Answer: Each student paid Rs 125.

Question: Exercise 1.14, Question 7: Najeeb is distributing 3,290 food packs among 235 families. How many packs does each family get?

Answer: Step 1: To find the packs per family, divide 3,290 by 235. | Step 2: We divide 3,290 by 235, digit by digit. | Step 3: 329 / 235: 235 x 1 = 235; remainder 329 - 235 = 94. | Step 4: 940 / 235: 235 x 4 = 940; remainder 940 - 940 = 0. | Step 5: Final Answer: Each family gets 14 food packs.

Review Exercise 1

Review Exercise 1, Question 1 (i): Tick the correct option. The place value of 8 in 2,485,612 is _____

(a) 800

(b) 8,000



(c) 80,000

(d) 800,000

Review Exercise 1, Question 1 (ii): Tick the correct option. Seven digit numbers represent _____

(a) thousand

✓ (b) millions

(c) ten millions

(d) hundred millions

Review Exercise 1, Question 1 (iii): Tick the correct option. 20 + 8 / 4 = _____

(a) 4

(b) 7

(c) 8

✓ (d) 22

Question: Review Exercise 1, Question 2 (i): Write in words: 2,412,316.

Answer: Step 1: Group the digits into millions, thousands and the last three digits. | Step 2: Read each group in order. | Final Answer: Two million four hundred and twelve thousand three hundred and sixteen.

Question: Review Exercise 1, Question 2 (ii): Write in words: 36,123,101.

Answer: Step 1: Group the digits into millions, thousands and the last three digits. | Step 2: Read each group in order. | Final Answer: Thirty six million one hundred and twenty three thousand one hundred and one.

Question: Review Exercise 1, Question 2 (iii): Write in words: 600,216.

Answer: Step 1: Group the digits into millions, thousands and the last three digits. | Step 2: Read each group in order. | Final Answer: Six hundred thousand two hundred and sixteen.

Question: Review Exercise 1, Question 3 (i): Write in figures: Twenty million.

Answer: Step 1: Identify the place value named in the words. | Step 2: Write it in figures in the correct places. | Final Answer: 20,000,000.

Question: Review Exercise 1, Question 3 (ii): Write in figures: One hundred million.

Answer: Step 1: Identify the place value named in the words. | Step 2:

Write it in figures in the correct places. | Final Answer: 100,000,000.

Question: Review Exercise 1, Question 4 (i): Add 416,712 and 712,145.

Answer: Step 1: Write the numbers one below the other, ones under ones, tens under tens. | Step 2: Add the ones column: 2 + 5 = 7; write 7. | Step 3: Add the tens column: 1 + 4 = 5; write 5. | Step 4: Add the hundreds column: 7 + 1 = 8; write 8. | Step 5: Add the thousands column: 6 + 2 = 8; write 8. | Step 6: Add the ten thousands column: 1 + 1 = 2; write 2. | Step 7: Add the hundred thousands column: 4 + 7 = 11; write 1 and carry 1 to the next column. | Step 8: Write the carried 1 in the next column. | Step 9: Final Answer: 416,712 + 712,145 = 1,128,857.

Question: Review Exercise 1, Question 4 (ii): Add 900,102 and 812,156.

Answer: Step 1: Write the numbers one below the other, ones under ones, tens under tens. | Step 2: Add the ones column: 2 + 6 = 8; write 8. | Step 3: Add the tens column: 0 + 5 = 5; write 5. | Step 4: Add the hundreds column: 1 + 1 = 2; write 2. | Step 5: Add the thousands column: 0 + 2 = 2; write 2. | Step 6: Add the ten thousands column: 0 + 1 = 1; write 1. | Step 7: Add the hundred thousands column: 9 + 8 = 17; write 7 and carry 1 to the next column. | Step 8: Write the carried 1 in the next column. | Step 9: Final Answer: 900,102 + 812,156 = 1,712,258.

Question: Review Exercise 1, Question 5 (i): Subtract 218,822 from 967,829.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Subtract the ones column: 9 - 2 = 7; write 7. | Step 3: Subtract the tens column: 2 - 2 = 0; write 0. | Step 4: Subtract the hundreds column: 8 - 8 = 0; write 0. | Step 5: Thousands column: after borrowing, the top digit becomes 17; 17 - 8 = 9; write 9. | Step 6: Ten thousands column: after borrowing, the top digit becomes 5; 5 - 1 = 4; write 4. | Step 7: Subtract the hundred thousands column: 9 - 2 = 7; write

7. | Step 8: Final Answer: 967,829 - 218,822 = 749,007.

Question: Review Exercise 1, Question 5 (ii): Subtract 100,512 from 200,603.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Subtract the ones column: 3 - 2 = 1; write 1. | Step 3: Tens column: after borrowing, the top digit becomes 10; 10 - 1 = 9; write 9. | Step 4: Hundreds column: after borrowing, the top digit becomes 5; 5 - 5 = 0; write 0. | Step 5: Subtract the thousands column: 0 - 0 = 0; write 0. | Step 6: Subtract the ten thousands column: 0 - 0 = 0; write 0. | Step 7: Subtract the hundred thousands column: 2 - 1 = 1; write 1. | Step 8: Final Answer: 200,603 - 100,512 = 100,091.

Question: Review Exercise 1, Question 5 (iii): Subtract 555,666 from 723,444.

Answer: Step 1: Write the bigger number on top and the smaller number below it, in columns. | Step 2: Ones column: after borrowing, the top digit becomes 14; 14 - 6 = 8; write 8. | Step 3: Tens column: after borrowing, the top digit becomes 13; 13 - 6 = 7; write 7. | Step 4: Hundreds column: after borrowing, the top digit becomes 13; 13 - 6 = 7; write 7. | Step 5: Thousands column: after borrowing, the top digit becomes 12; 12 - 5 = 7; write 7. | Step 6: Ten thousands column: after borrowing, the top digit becomes 11; 11 - 5 = 6; write 6. | Step 7: Hundred thousands column: after borrowing, the top digit becomes 6; 6 - 5 = 1; write 1. | Step 8: Final Answer: 723,444 - 555,666 = 167,778.

Question: Review Exercise 1, Question 6 (i): Solve: 61,243 x 261.

Answer: Step 1: Break 261 into 1 + 60 + 200. | Step 2: Multiply by the ones: 61,243 x 1 = 60000 x 1 = 60000 + 1000 x 1 = 1000 + 200 x 1 = 200 + 40 x 1 = 40 + 3 x 1 = 3; total 61243. | Step 3: Multiply by the tens: first 61,243 x 6 = 60000 x 6 = 360000 + 1000 x 6 = 6000 + 200 x 6 =

1200 + 40 x 6 = 240 + 3 x 6 = 18; total 367458; then x 10 gives 3,674,580. | Step 4: Multiply by the hundreds: first 61,243 x 2 = 60000 x 2 = 120000 + 1000 x 2 = 2000 + 200 x 2 = 400 + 40 x 2 = 80 + 3 x 2 = 6; total 122486; then x 100 gives 12,248,600. | Step 5: Add the partial products: 61,243 + 3,674,580 + 12,248,600 = 15,984,423. | Step 6: Final Answer: 61,243 x 261 = 15,984,423.

Question: Review Exercise 1, Question 6 (ii): Solve: 21,588 x 120.

Answer: Step 1: Break 120 into 20 + 100. | Step 2: Multiply by the tens: first 21,588 x 2 = 20000 x 2 = 40000 + 1000 x 2 = 2000 + 500 x 2 = 1000 + 80 x 2 = 160 + 8 x 2 = 16; total 43176; then x 10 gives 431,760. | Step 3: Multiply by the hundreds: first 21,588 x 1 = 20000 x 1 = 20000 + 1000 x 1 = 1000 + 500 x 1 = 500 + 80 x 1 = 80 + 8 x 1 = 8; total 21588; then x 100 gives 2,158,800. | Step 4: Add the partial products: 431,760 + 2,158,800 = 2,590,560. | Step 5: Final Answer: 21,588 x 120 = 2,590,560.

Question: Review Exercise 1, Question 6 (iii): Solve: 3,810 / 15.

Answer: Step 1: We divide 3,810 by 15, digit by digit. | Step 2: 38 / 15: 15 x 2 = 30; remainder 38 - 30 = 8. | Step 3: 81 / 15: 15 x 5 = 75; remainder 81 - 75 = 6. | Step 4: 60 / 15: 15 x 4 = 60; remainder 60 - 60 = 0. | Step 5: Final Answer: 3,810 / 15 = 254.

Question: Review Exercise 1, Question 6 (iv): Solve: 5,088 / 32.

Answer: Step 1: We divide 5,088 by 32, digit by digit. | Step 2: 50 / 32: 32 x 1 = 32; remainder 50 - 32 = 18. | Step 3: 188 / 32: 32 x 5 = 160; remainder 188 - 160 = 28. | Step 4: 288 / 32: 32 x 9 = 288; remainder 288 - 288 = 0. | Step 5: Final Answer: 5,088 / 32 = 159.

Question: Review Exercise 1, Question 6 (v): Solve: 555 / 15 x 2.

Answer: Step 1: Do the division and multiplication from left to right. | Step 2: 555 / 15 = 37. | Step 3: 37 x 2 = 74. | Step 4: Final Answer: 555 / 15 x 2 = 74.

Question: Review Exercise 1, Question 7: The annual saving of Raheel is Rs 89,560. He purchases a TV for Rs 35,000. How much money is left with him?

Answer: Step 1: For the saving left, subtract the TV price from the annual saving. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 0 - 0 = 0; write 0. | Step 4: Subtract the tens column: 6 - 0 = 6; write 6. | Step 5: Subtract the hundreds column: 5 - 0 = 5; write 5. | Step 6: Subtract the thousands column: 9 - 5 = 4; write 4. | Step 7: Subtract the ten thousands column: 8 - 3 = 5; write 5. | Step 8: Final Answer: Rs 54,560 were left with Raheel.

2. Factors and Multiples

Activity: Finding Factors

Question: Activity: Find the factors of 2.

Answer: Step 1: Divide 2 by every number starting from 1 and note the exact divisions. | Step 2: Exact divisions: 2 = 1 x 2. | Step 3: Final Answer: The factors of 2 are: 1, 2.

Question: Activity: Find the factors of 4.

Answer: Step 1: Divide 4 by every number starting from 1 and note the exact divisions. | Step 2: Exact divisions: 4 = 1 x 4; 4 = 2 x 2. | Step 3: Final Answer: The factors of 4 are: 1, 2, 4.

Question: Activity: Find the factors of 5.

Answer: Step 1: Divide 5 by every number starting from 1 and note the exact divisions. | Step 2: Exact divisions: 5 = 1 x 5. | Step 3: Final Answer: The factors of 5 are: 1, 5.

Question: Activity: Find the factors of 7.

Answer: Step 1: Divide 7 by every number starting from 1 and note the exact divisions. | Step 2: Exact divisions: 7 = 1 x 7. | Step 3: Final Answer: The factors of 7 are: 1, 7.

Question: Activity: Find the factors of 9.

Answer: Step 1: Divide 9 by every number starting from 1 and note the exact divisions. | Step 2: Exact divisions: 9 = 1 x 9; 9 = 3 x 3. | Step 3: Final Answer: The factors of 9 are: 1, 3, 9.

Question: Activity: Find the factors of 12.

Answer: Step 1: Divide 12 by every number starting from 1 and note the exact divisions. | Step 2: Exact divisions: 12 = 1 x 12; 12 = 2 x 6; 12 = 3 x

4. | Step 3: Final Answer: The factors of 12 are: 1, 2, 3, 4, 6, 12.

Question: Activity: Find the factors of 18.

Answer: Step 1: Divide 18 by every number starting from 1 and note the exact divisions. | Step 2: Exact divisions: 18 = 1 x 18; 18 = 2 x 9; 18 = 3 x 6. | Step 3: Final Answer: The factors of 18 are: 1, 2, 3, 6, 9, 18.

Question: Activity: Find the factors of 24.

Answer: Step 1: Divide 24 by every number starting from 1 and note the exact divisions. | Step 2: Exact divisions: 24 = 1 x 24; 24 = 2 x 12; 24 = 3 x 8; 24 = 4 x 6. | Step 3: Final Answer: The factors of 24 are: 1, 2, 3, 4, 6, 8, 12, 24.

Question: Activity: Find the factors of 32.

Answer: Step 1: Divide 32 by every number starting from 1 and note the exact divisions. | Step 2: Exact divisions: 32 = 1 x 32; 32 = 2 x 16; 32 = 4 x 8. | Step 3: Final Answer: The factors of 32 are: 1, 2, 4, 8, 16, 32.

Question: Activity: Find the factors of 48.

Answer: Step 1: Divide 48 by every number starting from 1 and note the exact divisions. | Step 2: Exact divisions: 48 = 1 x 48; 48 = 2 x 24; 48 = 3 x 16; 48 = 4 x 12; 48 = 6 x 8. | Step 3: Final Answer: The factors of 48 are: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48.

Question: Activity: Check the divisibility of 15,381 by 2, 3, 5 and 10.

Answer: Step 1: Test for 2: the last digit is 1, which is odd, so 15,381 is NOT divisible by 2. | Step 2: Test for 3: the sum of the digits is 1 + 5 + 3 + 8 + 1 = 18. 18 is divisible by 3, so 15,381 is divisible by 3. Check: 15,381 / 3 = 5,127. | Step 3: Test for 5: the last digit is neither 0 nor 5, so it is NOT divisible by 5. | Step 4: Test for 10: the last digit is not 0, so it is NOT divisible by 10. | Step 5: Final Answer: 15,381 is divisible by 3 only.

Exercise 2.1: Divisibility by 2, 3, 5 and 10

Question: Exercise 2.1, Question 1 (i): Is 120 divisible by 2?

Answer: Step 1: Rule: a number is divisible by 2 if its last digit is even. | Step 2: The last digit of 120 is 0, which is even. | Step 3: Final Answer: 120 is divisible by 2.

Question: Exercise 2.1, Question 1 (ii): Is 1,001 divisible by 2?

Answer: Step 1: Rule: a number is divisible by 2 if its last digit is even. | Step 2: The last digit of 1,001 is 1, which is odd. | Step 3: Final Answer: 1,001 is NOT divisible by 2.

Question: Exercise 2.1, Question 1 (iii): Is 1,434 divisible by 2?

Answer: Step 1: Rule: a number is divisible by 2 if its last digit is even. | Step 2: The last digit of 1,434 is 4, which is even. | Step 3: Final Answer: 1,434 is divisible by 2.

Question: Exercise 2.1, Question 1 (iv): Is 2,221 divisible by 2?

Answer: Step 1: Rule: a number is divisible by 2 if its last digit is even. | Step 2: The last digit of 2,221 is 1, which is odd. | Step 3: Final Answer: 2,221 is NOT divisible by 2.

Question: Exercise 2.1, Question 1 (v): Is 13,574 divisible by 2?

Answer: Step 1: Rule: a number is divisible by 2 if its last digit is even. | Step 2: The last digit of 13,574 is 4, which is even. | Step 3: Final Answer: 13,574 is divisible by 2.

Question: Exercise 2.1, Question 2 (i): Test the divisibility of 135 by 3.

Answer: Step 1: Rule: a number is divisible by 3 if the sum of its digits is divisible by 3. | Step 2: Sum of the digits of 135: 1 + 3 + 5 = 9. | Step 3: 9 is divisible by 3. | Step 4: Final Answer: 135 is divisible by 3.

Question: Exercise 2.1, Question 2 (ii): Test the divisibility of 1,471 by 3.

Answer: Step 1: Rule: a number is divisible by 3 if the sum of its digits is divisible by 3. | Step 2: Sum of the digits of 1,471: 1 + 4 + 7 + 1 = 13. | Step 3: 13 is NOT divisible by 3. | Step 4: Final Answer: 1,471 is NOT

divisible by 3.

Question: Exercise 2.1, Question 2 (iii): Test the divisibility of 2,100 by 3.

Answer: Step 1: Rule: a number is divisible by 3 if the sum of its digits is divisible by 3. | Step 2: Sum of the digits of 2,100: 2 + 1 + 0 + 0 = 3. | Step 3: 3 is divisible by 3. | Step 4: Final Answer: 2,100 is divisible by 3.

Question: Exercise 2.1, Question 2 (iv): Test the divisibility of 3,331 by 3.

Answer: Step 1: Rule: a number is divisible by 3 if the sum of its digits is divisible by 3. | Step 2: Sum of the digits of 3,331: 3 + 3 + 3 + 1 = 10. | Step 3: 10 is NOT divisible by 3. | Step 4: Final Answer: 3,331 is NOT divisible by 3.

Question: Exercise 2.1, Question 2 (v): Test the divisibility of 31,242 by 3.

Answer: Step 1: Rule: a number is divisible by 3 if the sum of its digits is divisible by 3. | Step 2: Sum of the digits of 31,242: 3 + 1 + 2 + 4 + 2 = 12. | Step 3: 12 is divisible by 3. | Step 4: Final Answer: 31,242 is divisible by 3.

Question: Exercise 2.1, Question 3 (i): Is 1,235 divisible by 5?

Answer: Step 1: Rule: a number is divisible by 5 if its last digit is 0 or 5. | Step 2: The last digit of 1,235 is 5. | Step 3: Final Answer: 1,235 is divisible by 5.

Question: Exercise 2.1, Question 3 (ii): Is 5,552 divisible by 5?

Answer: Step 1: Rule: a number is divisible by 5 if its last digit is 0 or 5. | Step 2: The last digit of 5,552 is 2. | Step 3: Final Answer: 5,552 is NOT divisible by 5.

Question: Exercise 2.1, Question 3 (iii): Is 6,035 divisible by 5?

Answer: Step 1: Rule: a number is divisible by 5 if its last digit is 0 or 5. | Step 2: The last digit of 6,035 is 5. | Step 3: Final Answer: 6,035 is divisible by 5.

Question: Exercise 2.1, Question 3 (iv): Is 10,001 divisible by 5?

Answer: Step 1: Rule: a number is divisible by 5 if its last digit is 0 or 5. | Step 2: The last digit of 10,001 is 1. | Step 3: Final Answer: 10,001 is NOT divisible by 5.

Question: Exercise 2.1, Question 3 (v): Is 53,550 divisible by 5?

Answer: Step 1: Rule: a number is divisible by 5 if its last digit is 0 or 5. | Step 2: The last digit of 53,550 is 0. | Step 3: Final Answer: 53,550 is divisible by 5.

Question: Exercise 2.1, Question 4 (i): Test the divisibility of 1,350 by 10.

Answer: Step 1: Rule: a number is divisible by 10 if its last digit is 0. | Step 2: The last digit of 1,350 is 0. | Step 3: Final Answer: 1,350 is divisible by 10.

Question: Exercise 2.1, Question 4 (ii): Test the divisibility of 2,225 by 10.

Answer: Step 1: Rule: a number is divisible by 10 if its last digit is 0. | Step 2: The last digit of 2,225 is 5. | Step 3: Final Answer: 2,225 is NOT divisible by 10.

Question: Exercise 2.1, Question 4 (iii): Test the divisibility of 30,500 by 10.

Answer: Step 1: Rule: a number is divisible by 10 if its last digit is 0. | Step 2: The last digit of 30,500 is 0. | Step 3: Final Answer: 30,500 is divisible by 10.

Question: Exercise 2.1, Question 4 (iv): Test the divisibility of 13,575 by 10.

Answer: Step 1: Rule: a number is divisible by 10 if its last digit is 0. | Step 2: The last digit of 13,575 is 5. | Step 3: Final Answer: 13,575 is NOT divisible by 10.

Question: Exercise 2.1, Question 4 (v): Test the divisibility of 20,050 by 10.

Answer: Step 1: Rule: a number is divisible by 10 if its last digit is 0. | Step 2: The last digit of 20,050 is 0. | Step 3: Final Answer: 20,050 is divisible by 10.

Question: Exercise 2.1, Question 5 (i): Is 12,000 divisible by both 5 and

10?

Answer: Step 1: Test for 5: the last digit is 0, so 12,000 is divisible by 5. | Step 2: Test for 10: the last digit is 0, so 12,000 is divisible by 10. | Step 3: Final Answer: 12,000 is divisible by both 5 and 10.

Question: Exercise 2.1, Question 5 (ii): Is 2,145 divisible by both 5 and 10?

Answer: Step 1: Test for 5: the last digit is 5, so 2,145 is divisible by 5. | Step 2: Test for 10: the last digit is not 0, so 2,145 is NOT divisible by 10. | Step 3: Final Answer: 2,145 is NOT divisible by both 5 and 10.

Question: Exercise 2.1, Question 5 (iii): Is 4,040 divisible by both 5 and 10?

Answer: Step 1: Test for 5: the last digit is 0, so 4,040 is divisible by 5. | Step 2: Test for 10: the last digit is 0, so 4,040 is divisible by 10. | Step 3: Final Answer: 4,040 is divisible by both 5 and 10.

Question: Exercise 2.1, Question 5 (iv): Is 12,345 divisible by both 5 and 10?

Answer: Step 1: Test for 5: the last digit is 5, so 12,345 is divisible by 5. | Step 2: Test for 10: the last digit is not 0, so 12,345 is NOT divisible by 10. | Step 3: Final Answer: 12,345 is NOT divisible by both 5 and 10.

Question: Exercise 2.1, Question 5 (v): Is 7,270 divisible by both 5 and 10?

Answer: Step 1: Test for 5: the last digit is 0, so 7,270 is divisible by 5. | Step 2: Test for 10: the last digit is 0, so 7,270 is divisible by 10. | Step 3: Final Answer: 7,270 is divisible by both 5 and 10.

Question: Exercise 2.1, Question 6 (i): Check the divisibility of 405 by 2, 3, 5 and 10.

Answer: Step 1: Last digit of 405 is 5 and the sum of its digits is 4 + 0 + 5 = 9. | Step 2: By 2: No. | Step 3: By 3: Yes. | Step 4: By 5: Yes. | Step 5: By 10: No. | Step 6: Final Answer: 405 is divisible by 3, 5.

Question: Exercise 2.1, Question 6 (ii): Check the divisibility of 3,354 by 2, 3, 5 and 10.

Answer: Step 1: Last digit of 3,354 is 4 and the sum of its digits is 3 + 3 + 5 + 4 = 15. | Step 2: By 2: Yes. | Step 3: By 3: Yes. | Step 4: By 5: No. | Step 5: By 10: No. | Step 6: Final Answer: 3,354 is divisible by 2, 3.

Question: Exercise 2.1, Question 6 (iii): Check the divisibility of 2,340 by 2, 3, 5 and 10.

Answer: Step 1: Last digit of 2,340 is 0 and the sum of its digits is 2 + 3 + 4 + 0 = 9. | Step 2: By 2: Yes. | Step 3: By 3: Yes. | Step 4: By 5: Yes. | Step 5: By 10: Yes. | Step 6: Final Answer: 2,340 is divisible by 2, 3, 5, 10.

Question: Exercise 2.1, Question 6 (iv): Check the divisibility of 41,220 by 2, 3, 5 and 10.

Answer: Step 1: Last digit of 41,220 is 0 and the sum of its digits is 4 + 1 + 2 + 2 + 0 = 9. | Step 2: By 2: Yes. | Step 3: By 3: Yes. | Step 4: By 5: Yes. | Step 5: By 10: Yes. | Step 6: Final Answer: 41,220 is divisible by 2, 3, 5, 10.

Question: Exercise 2.1, Question 6 (v): Check the divisibility of 34,329 by 2, 3, 5 and 10.

Answer: Step 1: Last digit of 34,329 is 9 and the sum of its digits is 3 + 4 + 3 + 2 + 9 = 21. | Step 2: By 2: No. | Step 3: By 3: Yes. | Step 4: By 5: No. | Step 5: By 10: No. | Step 6: Final Answer: 34,329 is divisible by 3.

Exercise 2.2: Prime and Composite Numbers

Question: Exercise 2.2, Question 1 (i): Is 22 a prime number?

Answer: Step 1: Find the factors of 22: 1, 2, 11, 22. | Step 2: 22 has 4 factors in total. | Step 3: Final Answer: 22 is a composite number.

Question: Exercise 2.2, Question 1 (ii): Is 41 a prime number?

Answer: Step 1: Find the factors of 41: 1, 41. | Step 2: 41 has 2 factors in total. | Step 3: Final Answer: 41 is a prime number.

Question: Exercise 2.2, Question 1 (iii): Is 63 a prime number?

Answer: Step 1: Find the factors of 63: 1, 3, 7, 9, 21, 63. | Step 2: 63 has 6 factors in total. | Step 3: Final Answer: 63 is a composite number.

Question: Exercise 2.2, Question 1 (iv): Is 51 a prime number?

Answer: Step 1: Find the factors of 51: 1, 3, 17, 51. | Step 2: 51 has 4 factors in total. | Step 3: Final Answer: 51 is a composite number.

Question: Exercise 2.2, Question 1 (v): Is 81 a prime number?

Answer: Step 1: Find the factors of 81: 1, 3, 9, 27, 81. | Step 2: 81 has 5 factors in total. | Step 3: Final Answer: 81 is a composite number.

Question: Exercise 2.2, Question 1 (vi): Is 119 a prime number?

Answer: Step 1: Find the factors of 119: 1, 7, 17, 119. | Step 2: 119 has 4 factors in total. | Step 3: Final Answer: 119 is a composite number.

Question: Exercise 2.2, Question 1 (vii): Is 223 a prime number?

Answer: Step 1: Find the factors of 223: 1, 223. | Step 2: 223 has 2 factors in total. | Step 3: Final Answer: 223 is a prime number.

Question: Exercise 2.2, Question 1 (viii): Is 1 a prime number?

Answer: Step 1: Find the factors of 1: 1. | Step 2: 1 has only one factor: 1. | Step 3: Final Answer: 1 is neither prime nor composite.

Question: Exercise 2.2, Question 1 (ix): Is 101 a prime number?

Answer: Step 1: Find the factors of 101: 1, 101. | Step 2: 101 has 2 factors in total. | Step 3: Final Answer: 101 is a prime number.

Question: Exercise 2.2, Question 1 (x): Is 222 a prime number?

Answer: Step 1: Find the factors of 222: 1, 2, 3, 6, 37, 74, 111, 222. | Step 2: 222 has 8 factors in total. | Step 3: Final Answer: 222 is a composite number.

Question: Exercise 2.2, Question 2 (i): Is 34 a composite number?

Answer: Step 1: Find the factors of 34: 1, 2, 17, 34. | Step 2: 34 has 4 factors in total. | Step 3: Final Answer: 34 is a composite number.

Question: Exercise 2.2, Question 2 (ii): Is 71 a composite number?

Answer: Step 1: Find the factors of 71: 1, 71. | Step 2: 71 has 2 factors in total. | Step 3: Final Answer: 71 is a prime number.

Question: Exercise 2.2, Question 2 (iii): Is 163 a composite number?

Answer: Step 1: Find the factors of 163: 1, 163. | Step 2: 163 has 2 factors in total. | Step 3: Final Answer: 163 is a prime number.

Question: Exercise 2.2, Question 2 (iv): Is 351 a composite number?

Answer: Step 1: Find the factors of 351: 1, 3, 9, 13, 27, 39, 117, 351. | Step 2: 351 has 8 factors in total. | Step 3: Final Answer: 351 is a composite number.

Question: Exercise 2.2, Question 2 (v): Is 81 a composite number?

Answer: Step 1: Find the factors of 81: 1, 3, 9, 27, 81. | Step 2: 81 has 5 factors in total. | Step 3: Final Answer: 81 is a composite number.

Question: Exercise 2.2, Question 2 (vi): Is 19 a composite number?

Answer: Step 1: Find the factors of 19: 1, 19. | Step 2: 19 has 2 factors in total. | Step 3: Final Answer: 19 is a prime number.

Question: Exercise 2.2, Question 2 (vii): Is 23 a composite number?

Answer: Step 1: Find the factors of 23: 1, 23. | Step 2: 23 has 2 factors in total. | Step 3: Final Answer: 23 is a prime number.

Question: Exercise 2.2, Question 2 (viii): Is 100 a composite number?

Answer: Step 1: Find the factors of 100: 1, 2, 4, 5, 10, 20, 25, 50, 100. | Step 2: 100 has 9 factors in total. | Step 3: Final Answer: 100 is a composite number.

Question: Exercise 2.2, Question 2 (ix): Is 18 a composite number?

Answer: Step 1: Find the factors of 18: 1, 2, 3, 6, 9, 18. | Step 2: 18 has 6 factors in total. | Step 3: Final Answer: 18 is a composite number.

Question: Exercise 2.2, Question 2 (x): Is 135 a composite number?

Answer: Step 1: Find the factors of 135: 1, 3, 5, 9, 15, 27, 45, 135. | Step 2: 135 has 8 factors in total. | Step 3: Final Answer: 135 is a composite

number.

Question: Exercise 2.2, Question 3: Separate the prime and composite numbers from 41 to 60.

Answer: Step 1: Test the factors of every number from 41 to 60: a number with only two factors (1 and itself) is prime, the rest are composite. | Step 2: Final Answer: Prime numbers: 41, 43, 47, 53, 59; Composite numbers: 42, 44, 45, 46, 48, 49, 50, 51, 52, 54, 55, 56, 57, 58, 60.

Question: Exercise 2.2, Question 4: Write down all the prime numbers between 1 and 20.

Answer: Step 1: 1 is neither prime nor composite. | Step 2: Test every number from 2 to 20: a number with exactly two factors is prime. | Step 3: Final Answer: The prime numbers between 1 and 20 are: 2, 3, 5, 7, 11, 13, 17, 19.

Question: Exercise 2.2, Question 5: Write down all the composite numbers between 10 and 30.

Answer: Step 1: Test the factors of every number from 10 to 30: a number with more than two factors is composite. | Step 2: Final Answer: The composite numbers between 10 and 30 are: 10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25, 26, 27, 28, 30.

Question: Exercise 2.2, Question 6: Write down all the prime and composite numbers between 20 and 40.

Answer: Step 1: Test every number from 20 to 40 and split it into prime and composite. | Step 2: Final Answer: Prime numbers: 23, 29, 31, 37; Composite numbers: 20, 21, 22, 24, 25, 26, 27, 28, 30, 32, 33, 34, 35, 36, 38, 39, 40.

Exercise 2.3: Multiples and Factors

Question: Exercise 2.3, Question 1: List the first twelve multiples of 4.

Answer: Step 1: Multiply 4 by 1 to 12: 4 x 1 = 4; 4 x 2 = 8; 4 x 3 = 12; 4 x 4 = 16; 4 x 5 = 20; 4 x 6 = 24; 4 x 7 = 28; 4 x 8 = 32; 4 x 9 = 36; 4 x 10 = 40; 4 x 11 = 44; 4 x 12 = 48. | Step 2: Final Answer: The first twelve multiples of 4 are: 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48.

Question: Exercise 2.3, Question 1: List the first twelve multiples of 7.

Answer: Step 1: Multiply 7 by 1 to 12: 7 x 1 = 7; 7 x 2 = 14; 7 x 3 = 21; 7 x 4 = 28; 7 x 5 = 35; 7 x 6 = 42; 7 x 7 = 49; 7 x 8 = 56; 7 x 9 = 63; 7 x 10 = 70; 7 x 11 = 77; 7 x 12 = 84. | Step 2: Final Answer: The first twelve multiples of 7 are: 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84.

Question: Exercise 2.3, Question 1: List the first twelve multiples of 9.

Answer: Step 1: Multiply 9 by 1 to 12: 9 x 1 = 9; 9 x 2 = 18; 9 x 3 = 27; 9 x 4 = 36; 9 x 5 = 45; 9 x 6 = 54; 9 x 7 = 63; 9 x 8 = 72; 9 x 9 = 81; 9 x 10 = 90; 9 x 11 = 99; 9 x 12 = 108. | Step 2: Final Answer: The first twelve multiples of 9 are: 9, 18, 27, 36, 45, 54, 63, 72, 81, 90, 99, 108.

Question: Exercise 2.3, Question 2: Write down all the factors of 16.

Answer: Step 1: Divide 16 by every number starting from 1 and note the exact divisions. | Step 2: Exact divisions: 16 = 1 x 16; 16 = 2 x 8; 16 = 4 x 4. | Step 3: Final Answer: The factors of 16 are: 1, 2, 4, 8, 16.

Question: Exercise 2.3, Question 2: Write down all the factors of 26.

Answer: Step 1: Divide 26 by every number starting from 1 and note the exact divisions. | Step 2: Exact divisions: 26 = 1 x 26; 26 = 2 x 13. | Step 3: Final Answer: The factors of 26 are: 1, 2, 13, 26.

Question: Exercise 2.3, Question 2: Write down all the factors of 45.

Answer: Step 1: Divide 45 by every number starting from 1 and note the exact divisions. | Step 2: Exact divisions: 45 = 1 x 45; 45 = 3 x 15; 45 = 5 x 9. | Step 3: Final Answer: The factors of 45 are: 1, 3, 5, 9, 15, 45.

Question: Exercise 2.3, Question 3: Circle all the multiples of 7 from: 14, 24, 28, 35, 45, 56, 62, 84, 69.

Answer: Step 1: 14 / 7 = 2 exactly, so 14 is a multiple of 7. | Step 2: 24 / 7 = 3 remainder 3, so 24 is NOT a multiple of 7. | Step 3: 28 / 7 = 4 exactly, so 28 is a multiple of 7. | Step 4: 35 / 7 = 5 exactly, so 35 is a multiple of 7. | Step 5: 45 / 7 = 6 remainder 3, so 45 is NOT a multiple of 7. | Step 6: 56 / 7 = 8 exactly, so 56 is a multiple of 7. | Step 7: 62 / 7 = 8 remainder 6, so 62 is NOT a multiple of 7. | Step 8: 84 / 7 = 12 exactly, so 84 is a multiple of 7. | Step 9: 69 / 7 = 9 remainder 6, so 69 is NOT a multiple of 7. | Step 10: Final Answer: The multiples of 7 are: 14, 28, 35, 56, 84.

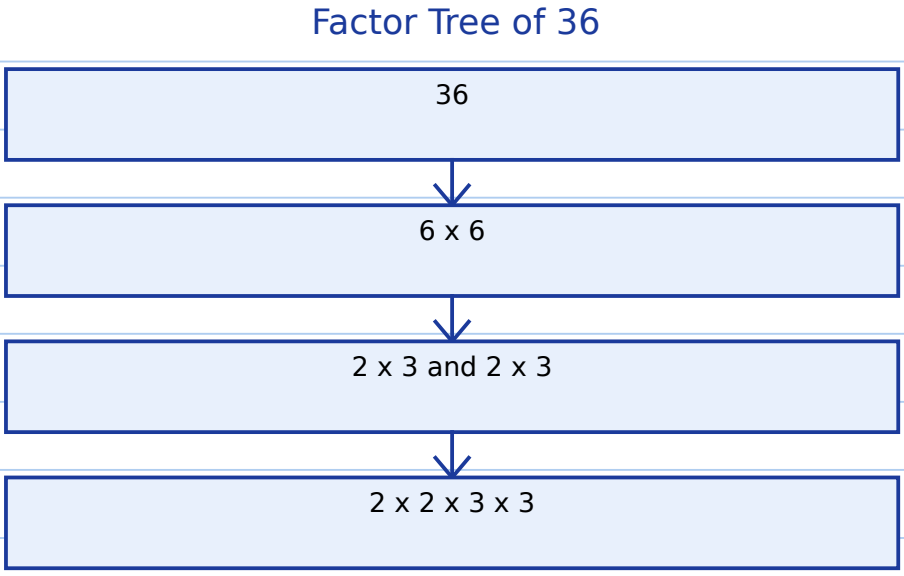
Question: Exercise 2.3, Question 4: Write down all the multiples of 8 between 20 and 90.

Answer: Step 1: Write the multiples of 8: 8, 16, 24, 32, 40, 48, 56, 64, 72, 80, 88. | Step 2: Pick those that are greater than 20 and less than 90. | Step 3: Final Answer: The multiples of 8 between 20 and 90 are: 24, 32, 40, 48, 56, 64, 72, 80, 88.

Question: Exercise 2.3, Question 5: Write all the factors of 50.

Answer: Step 1: Divide 50 by every number starting from 1 and note the exact divisions. | Step 2: Exact divisions: 50 = 1 x 50; 50 = 2 x 25; 50 = 5 x 10. | Step 3: Final Answer: The factors of 50 are: 1, 2, 5, 10, 25, 50.

Exercise 2.4: Prime Factorization



Question: Exercise 2.4, Question 1 (i): Write down all the factors of 12 and underline the prime factors.

Answer: Step 1: All the factors of 12: 1, 2, 3, 4, 6, 12. | Step 2: Among them, the prime factors are: 2, 3 (those with exactly two factors). | Step 3: Final Answer: Factors of 12: 1, 2, 3, 4, 6, 12; prime factors (underlined): 2, 3.

Question: Exercise 2.4, Question 1 (ii): Write down all the factors of 20 and underline the prime factors.

Answer: Step 1: All the factors of 20: 1, 2, 4, 5, 10, 20. | Step 2: Among them, the prime factors are: 2, 5 (those with exactly two factors). | Step 3: Final Answer: Factors of 20: 1, 2, 4, 5, 10, 20; prime factors (underlined): 2, 5.

Question: Exercise 2.4, Question 1 (iii): Write down all the factors of 25 and underline the prime factors.

Answer: Step 1: All the factors of 25: 1, 5, 25. | Step 2: Among them, the prime factors are: 5 (those with exactly two factors). | Step 3: Final Answer: Factors of 25: 1, 5, 25; prime factors (underlined): 5.

Question: Exercise 2.4, Question 1 (iv): Write down all the factors of 44

and underline the prime factors.

Answer: Step 1: All the factors of 44: 1, 2, 4, 11, 22, 44. | Step 2: Among them, the prime factors are: 2, 11 (those with exactly two factors). | Step 3: Final Answer: Factors of 44: 1, 2, 4, 11, 22, 44; prime factors (underlined): 2, 11.

Question: Exercise 2.4, Question 1 (v): Write down all the factors of 64 and underline the prime factors.

Answer: Step 1: All the factors of 64: 1, 2, 4, 8, 16, 32, 64. | Step 2: Among them, the prime factors are: 2 (those with exactly two factors). | Step 3: Final Answer: Factors of 64: 1, 2, 4, 8, 16, 32, 64; prime factors (underlined): 2.

Question: Exercise 2.4, Question 1 (vi): Write down all the factors of 28 and underline the prime factors.

Answer: Step 1: All the factors of 28: 1, 2, 4, 7, 14, 28. | Step 2: Among them, the prime factors are: 2, 7 (those with exactly two factors). | Step 3: Final Answer: Factors of 28: 1, 2, 4, 7, 14, 28; prime factors (underlined): 2, 7.

Question: Exercise 2.4, Question 2 (i): Factorize 36 using the factor tree method.

Answer: Step 1: Start with 36. | Step 2: Split 36 = 2 x 18. 2 is prime, so keep it; split 18 further. | Step 3: Split 18 = 2 x 9. 2 is prime, so keep it; split 9 further. | Step 4: Split 9 = 3 x 3. Both are prime. | Step 5: Collect all the prime factors: 36 = 2 x 2 x 3 x 3. | Step 6: Final Answer: 36 = 2 x 2 x 3 x 3.

Question: Exercise 2.4, Question 2 (ii): Factorize 54 using the factor tree method.

Answer: Step 1: Start with 54. | Step 2: Split 54 = 2 x 27. 2 is prime, so keep it; split 27 further. | Step 3: Split 27 = 3 x 9. 3 is prime, so keep it;

split 9 further. | Step 4: Split 9 = 3 x 3. Both are prime. | Step 5: Collect all the prime factors: 54 = 2 x 3 x 3 x 3. | Step 6: Final Answer: 54 = 2 x 3 x 3 x 3.

Question: Exercise 2.4, Question 2 (iii): Factorize 27 using the factor tree method.

Answer: Step 1: Start with 27. | Step 2: Split 27 = 3 x 9. 3 is prime, so keep it; split 9 further. | Step 3: Split 9 = 3 x 3. Both are prime. | Step 4: Collect all the prime factors: 27 = 3 x 3 x 3. | Step 5: Final Answer: 27 = 3 x 3 x 3.

Question: Exercise 2.4, Question 2 (iv): Factorize 45 using the factor tree method.

Answer: Step 1: Start with 45. | Step 2: Split 45 = 3 x 15. 3 is prime, so keep it; split 15 further. | Step 3: Split 15 = 3 x 5. Both are prime. | Step 4: Collect all the prime factors: 45 = 3 x 3 x 5. | Step 5: Final Answer: 45 = 3 x 3 x 5.

Question: Exercise 2.4, Question 2 (v): Factorize 32 using the factor tree method.

Answer: Step 1: Start with 32. | Step 2: Split 32 = 2 x 16. 2 is prime, so keep it; split 16 further. | Step 3: Split 16 = 2 x 8. 2 is prime, so keep it; split 8 further. | Step 4: Split 8 = 2 x 4. 2 is prime, so keep it; split 4 further. | Step 5: Split 4 = 2 x 2. Both are prime. | Step 6: Collect all the prime factors: 32 = 2 x 2 x 2 x 2 x 2. | Step 7: Final Answer: 32 = 2 x 2 x 2 x 2 x 2.

Question: Exercise 2.4, Question 2 (vi): Factorize 82 using the factor tree method.

Answer: Step 1: Start with 82. | Step 2: Split 82 = 2 x 41. Both are prime. | Step 3: Collect all the prime factors: 82 = 2 x 41. | Step 4: Final Answer: 82 = 2 x 41.

Question: Exercise 2.4, Question 3 (i): Factorize 63 using the division method.

Answer: Step 1: Start dividing by the smallest prime number. | Step 2: 63 / 3 = 21. | Step 3: 21 / 3 = 7. | Step 4: 7 / 7 = 1. | Step 5: Final Answer: 63 = 3 x 3 x 7.

Question: Exercise 2.4, Question 3 (ii): Factorize 45 using the division method.

Answer: Step 1: Start dividing by the smallest prime number. | Step 2: 45 / 3 = 15. | Step 3: 15 / 3 = 5. | Step 4: 5 / 5 = 1. | Step 5: Final Answer: 45 = 3 x 3 x 5.

Question: Exercise 2.4, Question 3 (iii): Factorize 72 using the division method.

Answer: Step 1: Start dividing by the smallest prime number. | Step 2: 72 / 2 = 36. | Step 3: 36 / 2 = 18. | Step 4: 18 / 2 = 9. | Step 5: 9 / 3 = 3. | Step 6: 3 / 3 = 1. | Step 7: Final Answer: 72 = 2 x 2 x 2 x 3 x 3.

Question: Exercise 2.4, Question 3 (iv): Factorize 54 using the division method.

Answer: Step 1: Start dividing by the smallest prime number. | Step 2: 54 / 2 = 27. | Step 3: 27 / 3 = 9. | Step 4: 9 / 3 = 3. | Step 5: 3 / 3 = 1. | Step 6: Final Answer: 54 = 2 x 3 x 3 x 3.

Question: Exercise 2.4, Question 3 (v): Factorize 38 using the division method.

Answer: Step 1: Start dividing by the smallest prime number. | Step 2: 38 / 2 = 19. | Step 3: 19 / 19 = 1. | Step 4: Final Answer: 38 = 2 x 19.

Question: Exercise 2.4, Question 3 (vi): Factorize 90 using the division method.

Answer: Step 1: Start dividing by the smallest prime number. | Step 2: 90 / 2 = 45. | Step 3: 45 / 3 = 15. | Step 4: 15 / 3 = 5. | Step 5: 5 / 5 = 1. | Step

6: Final Answer: 90 = 2 x 3 x 3 x 5.

Exercise 2.5: Highest Common Factor (HCF)

Question: Exercise 2.5, Question 1 (i): Write down all the factors of 12 and 18 and loop the common factors.

Answer: Step 1: Factors of 12: 1, 2, 3, 4, 6, 12. | Step 2: Factors of 18: 1, 2, 3, 6, 9, 18. | Step 3: The common factors are: 1, 2, 3, 6. | Step 4: Final Answer: The common factors are: 1, 2, 3, 6.

Question: Exercise 2.5, Question 1 (ii): Write down all the factors of 10 and 15 and loop the common factors.

Answer: Step 1: Factors of 10: 1, 2, 5, 10. | Step 2: Factors of 15: 1, 3, 5, 15. | Step 3: The common factors are: 1, 5. | Step 4: Final Answer: The common factors are: 1, 5.

Question: Exercise 2.5, Question 1 (iii): Write down all the factors of 22 and 44 and loop the common factors.

Answer: Step 1: Factors of 22: 1, 2, 11, 22. | Step 2: Factors of 44: 1, 2, 4, 11, 22, 44. | Step 3: The common factors are: 1, 2, 11, 22. | Step 4: Final Answer: The common factors are: 1, 2, 11, 22.

Question: Exercise 2.5, Question 1 (iv): Write down all the factors of 8 and 32 and loop the common factors.

Answer: Step 1: Factors of 8: 1, 2, 4, 8. | Step 2: Factors of 32: 1, 2, 4, 8, 16, 32. | Step 3: The common factors are: 1, 2, 4, 8. | Step 4: Final Answer: The common factors are: 1, 2, 4, 8.

Question: Exercise 2.5, Question 1 (v): Write down all the factors of 24 and 30 and 36 and loop the common factors.

Answer: Step 1: Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. | Step 2: Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30. | Step 3: Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36. | Step 4: The common factors are: 1, 2, 3, 6. | Step 5: Final Answer: The common factors are: 1, 2, 3, 6.

Question: Exercise 2.5, Question 2 (i): Find the HCF of 36 and 42 by the Prime Factorization Method.

Answer: Step 1: Prime factors of 36: 36 = 2 x 2 x 3 x 3. | Step 2: Prime factors of 42: 42 = 2 x 3 x 7. | Step 3: Take the common prime factors: 2 x 3 = 6. | Step 4: Final Answer: HCF = 6.

Question: Exercise 2.5, Question 2 (ii): Find the HCF of 28 and 42 by the Prime Factorization Method.

Answer: Step 1: Prime factors of 28: 28 = 2 x 2 x 7. | Step 2: Prime factors of 42: 42 = 2 x 3 x 7. | Step 3: Take the common prime factors: 2 x 7 = 14. | Step 4: Final Answer: HCF = 14.

Question: Exercise 2.5, Question 2 (iii): Find the HCF of 45 and 75 by the Prime Factorization Method.

Answer: Step 1: Prime factors of 45: 45 = 3 x 3 x 5. | Step 2: Prime factors of 75: 75 = 3 x 5 x 5. | Step 3: Take the common prime factors: 3 x 5 = 15. | Step 4: Final Answer: HCF = 15.

Question: Exercise 2.5, Question 2 (iv): Find the HCF of 25 and 75 by the Prime Factorization Method.

Answer: Step 1: Prime factors of 25: 25 = 5 x 5. | Step 2: Prime factors of 75: 75 = 3 x 5 x 5. | Step 3: Take the common prime factors: 5 x 5 = 25. | Step 4: Final Answer: HCF = 25.

Question: Exercise 2.5, Question 2 (v): Find the HCF of 18 and 72 by the Prime Factorization Method.

Answer: Step 1: Prime factors of 18: 18 = 2 x 3 x 3. | Step 2: Prime factors of 72: 72 = 2 x 2 x 2 x 3 x 3. | Step 3: Take the common prime factors: 2 x 3 x 3 = 18. | Step 4: Final Answer: HCF = 18.

Question: Exercise 2.5, Question 2 (vi): Find the HCF of 32 and 64 by the Prime Factorization Method.

Answer: Step 1: Prime factors of 32: 32 = 2 x 2 x 2 x 2 x 2. | Step 2: Prime

factors of 64: 64 = 2 x 2 x 2 x 2 x 2 x 2. | Step 3: Take the common prime factors: 2 x 2 x 2 x 2 x 2 = 32. | Step 4: Final Answer: HCF = 32.

Question: Exercise 2.5, Question 2 (vii): Find the HCF of 60 and 90 by the Prime Factorization Method.

Answer: Step 1: Prime factors of 60: 60 = 2 x 2 x 3 x 5. | Step 2: Prime factors of 90: 90 = 2 x 3 x 3 x 5. | Step 3: Take the common prime factors: 2 x 3 x 5 = 30. | Step 4: Final Answer: HCF = 30.

Question: Exercise 2.5, Question 2 (viii): Find the HCF of 25 and 40 and 50 by the Prime Factorization Method.

Answer: Step 1: Prime factors of 25: 25 = 5 x 5. | Step 2: Prime factors of 40: 40 = 2 x 2 x 2 x 5. | Step 3: Prime factors of 50: 50 = 2 x 5 x 5. | Step 4: Take the common prime factors: 5 = 5. | Step 5: Final Answer: HCF = 5.

Question: Exercise 2.5, Question 3 (i): Find the HCF of 24 and 36 by a Venn Diagram.

Answer: Step 1: Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. | Step 2: Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36. | Step 3: The common factors are: 1, 2, 3, 4, 6, 12. | Step 4: Write the common factors in the overlapping middle part of the Venn diagram. | Step 5: The greatest factor in the middle part is 12. | Step 6: Final Answer: HCF = 12.

Question: Exercise 2.5, Question 3 (ii): Find the HCF of 63 and 54 by a Venn Diagram.

Answer: Step 1: Factors of 63: 1, 3, 7, 9, 21, 63. | Step 2: Factors of 54: 1, 2, 3, 6, 9, 18, 27, 54. | Step 3: The common factors are: 1, 3, 9. | Step 4: Write the common factors in the overlapping middle part of the Venn diagram. | Step 5: The greatest factor in the middle part is 9. | Step 6: Final Answer: HCF = 9.

Question: Exercise 2.5, Question 3 (iii): Find the HCF of 40 and 44 by a Venn Diagram.

Answer: Step 1: Factors of 40: 1, 2, 4, 5, 8, 10, 20, 40. | Step 2: Factors of 44: 1, 2, 4, 11, 22, 44. | Step 3: The common factors are: 1, 2, 4. | Step 4: Write the common factors in the overlapping middle part of the Venn diagram. | Step 5: The greatest factor in the middle part is 4. | Step 6: Final Answer: HCF = 4.

Question: Exercise 2.5, Question 3 (iv): Find the HCF of 48 and 84 by a Venn Diagram.

Answer: Step 1: Factors of 48: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48. | Step 2: Factors of 84: 1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84. | Step 3: The common factors are: 1, 2, 3, 4, 6, 12. | Step 4: Write the common factors in the overlapping middle part of the Venn diagram. | Step 5: The greatest factor in the middle part is 12. | Step 6: Final Answer: HCF = 12.

Question: Exercise 2.5, Question 3 (v): Find the HCF of 22 and 24 by a Venn Diagram.

Answer: Step 1: Factors of 22: 1, 2, 11, 22. | Step 2: Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. | Step 3: The common factors are: 1, 2. | Step 4: Write the common factors in the overlapping middle part of the Venn diagram. | Step 5: The greatest factor in the middle part is 2. | Step 6: Final Answer: HCF = 2.

Question: Exercise 2.5, Question 3 (vi): Find the HCF of 14 and 21 and 49 by a Venn Diagram.

Answer: Step 1: Factors of 14: 1, 2, 7, 14. | Step 2: Factors of 21: 1, 3, 7, 21. | Step 3: Factors of 49: 1, 7, 49. | Step 4: The common factors are: 1, 7. | Step 5: Write the common factors in the overlapping middle part of the Venn diagram. | Step 6: The greatest factor in the middle part is 7. | Step 7: Final Answer: HCF = 7.

Exercise 2.6: HCF Word Problems

Question: Exercise 2.6, Question 1: Haider wants to distribute 36 sweets and 48 jellies equally among students. What is the maximum number of students among whom he can distribute them equally?

Answer: Step 1: For an equal distribution we need the HCF of 36, 48. | Step 2: Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36. | Step 3: Factors of 48: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48. | Step 4: Common factors: 1, 2, 3, 4, 6, 12. | Step 5: The greatest common factor is 12. | Step 6: Final Answer: The maximum number of students is 12.

Question: Exercise 2.6, Question 2: 18 Science books, 24 Maths books and 36 English books are to be arranged in equal bundles. What is the greatest number of bundles that can be made?

Answer: Step 1: For an equal distribution we need the HCF of 18, 24, 36. | Step 2: Factors of 18: 1, 2, 3, 6, 9, 18. | Step 3: Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. | Step 4: Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36. | Step 5: Common factors: 1, 2, 3, 6. | Step 6: The greatest common factor is 6. | Step 7: Final Answer: The greatest number of bundles is 6.

Question: Exercise 2.6, Question 3: Two ribbons are 44 m and 66 m long and are to be cut into pieces of the same length. What will be the greatest length of each piece?

Answer: Step 1: For an equal distribution we need the HCF of 44, 66. | Step 2: Factors of 44: 1, 2, 4, 11, 22, 44. | Step 3: Factors of 66: 1, 2, 3, 6, 11, 22, 33, 66. | Step 4: Common factors: 1, 2, 11, 22. | Step 5: The greatest common factor is 22. | Step 6: Final Answer: The greatest length of each piece is 22 m.

Question: Exercise 2.6, Question 4: Mona has two pieces of tablecloth, one 48 cm wide and the other 90 cm wide. Both are to be cut into strips of equal width. What will be the maximum width of each strip?

Answer: Step 1: For an equal distribution we need the HCF of 48, 90. | Step

2: Factors of 48: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48. | Step 3: Factors of 90: 1, 2, 3, 5, 6, 9, 10, 15, 18, 30, 45, 90. | Step 4: Common factors: 1, 2, 3, 6. | Step 5: The greatest common factor is 6. | Step 6: Final Answer: The maximum width of each strip is 6 cm.

Question: Exercise 2.6, Question 5: A class teacher has 32 red balloons, 28 blue balloons and 20 green balloons and wants to distribute them equally among students. What is the maximum number of balloons which could be distributed equally between students?

Answer: Step 1: For an equal distribution we need the HCF of 32, 28, 20. | Step 2: Factors of 32: 1, 2, 4, 8, 16, 32. | Step 3: Factors of 28: 1, 2, 4, 7, 14, 28. | Step 4: Factors of 20: 1, 2, 4, 5, 10, 20. | Step 5: Common factors: 1, 2, 4. | Step 6: The greatest common factor is 4. | Step 7: Final Answer: The greatest number that divides all of them equally is 4, so the balloons can be distributed equally in 4 shares.

Question: Exercise 2.6, Question 6: Zulfiqar has three containers containing 144, 176 and 256 litres of coconut oil. Find the capacity of the largest tin by which he can measure the oil exactly.

Answer: Step 1: For an equal distribution we need the HCF of 144, 176, 256. | Step 2: Factors of 144: 1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 36, 48, 72, 144. | Step 3: Factors of 176: 1, 2, 4, 8, 11, 16, 22, 44, 88, 176. | Step 4: Factors of 256: 1, 2, 4, 8, 16, 32, 64, 128, 256. | Step 5: Common factors: 1, 2, 4, 8, 16. | Step 6: The greatest common factor is 16. | Step 7: Final Answer: The capacity of the largest tin is 16 litres.

Exercise 2.7: Lowest Common Multiple (LCM)

Question: Exercise 2.7, Question 1 (i): Find the LCM of 25 and 15 by the common multiples method.

Answer: Step 1: Multiples of 25: 25, 50, 75. | Step 2: Multiples of 15: 15,

30, 45, 60, 75. | Step 3: The smallest multiple common to all is 75. | Step 4: Final Answer: LCM = 75.

Question: Exercise 2.7, Question 1 (ii): Find the LCM of 12 and 14 by the common multiples method.

Answer: Step 1: Multiples of 12: 12, 24, 36, 48, 60, 72, 84. | Step 2: Multiples of 14: 14, 28, 42, 56, 70, 84. | Step 3: The smallest multiple common to all is 84. | Step 4: Final Answer: LCM = 84.

Question: Exercise 2.7, Question 1 (iii): Find the LCM of 10 and 20 by the common multiples method.

Answer: Step 1: Multiples of 10: 10, 20. | Step 2: Multiples of 20: 20. | Step 3: The smallest multiple common to all is 20. | Step 4: Final Answer: LCM = 20.

Question: Exercise 2.7, Question 1 (iv): Find the LCM of 6 and 9 by the common multiples method.

Answer: Step 1: Multiples of 6: 6, 12, 18. | Step 2: Multiples of 9: 9, 18. | Step 3: The smallest multiple common to all is 18. | Step 4: Final Answer: LCM = 18.

Question: Exercise 2.7, Question 1 (v): Find the LCM of 7 and 5 by the common multiples method.

Answer: Step 1: Multiples of 7: 7, 14, 21, 28, 35. | Step 2: Multiples of 5: 5, 10, 15, 20, 25, 30, 35. | Step 3: The smallest multiple common to all is 35. | Step 4: Final Answer: LCM = 35.

Question: Exercise 2.7, Question 2 (i): Find the LCM of 42 and 18 by the prime factorization method.

Answer: Step 1: Prime factors of 42: 42 = 2 x 3 x 7. | Step 2: Prime factors of 18: 18 = 2 x 3 x 3. | Step 3: Take every prime factor the greatest number of times it appears: 2 x 3 x 3 x 7 = 126. | Step 4: Final Answer: LCM = 126.

Question: Exercise 2.7, Question 2 (ii): Find the LCM of 15 and 36 by the prime factorization method.

Answer: Step 1: Prime factors of 15: 15 = 3 x 5. | Step 2: Prime factors of 36: 36 = 2 x 2 x 3 x 3. | Step 3: Take every prime factor the greatest number of times it appears: 2 x 2 x 3 x 3 x 5 = 180. | Step 4: Final Answer: LCM = 180.

Question: Exercise 2.7, Question 2 (iii): Find the LCM of 12 and 45 by the prime factorization method.

Answer: Step 1: Prime factors of 12: 12 = 2 x 2 x 3. | Step 2: Prime factors of 45: 45 = 3 x 3 x 5. | Step 3: Take every prime factor the greatest number of times it appears: 2 x 2 x 3 x 3 x 5 = 180. | Step 4: Final Answer: LCM = 180.

Question: Exercise 2.7, Question 2 (iv): Find the LCM of 35 and 15 by the prime factorization method.

Answer: Step 1: Prime factors of 35: 35 = 5 x 7. | Step 2: Prime factors of 15: 15 = 3 x 5. | Step 3: Take every prime factor the greatest number of times it appears: 3 x 5 x 7 = 105. | Step 4: Final Answer: LCM = 105.

Question: Exercise 2.7, Question 2 (v): Find the LCM of 20 and 48 by the prime factorization method.

Answer: Step 1: Prime factors of 20: 20 = 2 x 2 x 5. | Step 2: Prime factors of 48: 48 = 2 x 2 x 2 x 2 x 3. | Step 3: Take every prime factor the greatest number of times it appears: 2 x 2 x 2 x 2 x 3 x 5 = 240. | Step 4: Final Answer: LCM = 240.

Question: Exercise 2.7, Question 2 (vi): Find the LCM of 27 and 24 by the prime factorization method.

Answer: Step 1: Prime factors of 27: 27 = 3 x 3 x 3. | Step 2: Prime factors of 24: 24 = 2 x 2 x 2 x 3. | Step 3: Take every prime factor the greatest number of times it appears: 2 x 2 x 2 x 3 x 3 x 3 = 216. | Step 4: Final

Answer: LCM = 216.

Question: Exercise 2.7, Question 2 (vii): Find the LCM of 45 and 63 by the prime factorization method.

Answer: Step 1: Prime factors of 45: 45 = 3 x 3 x 5. | Step 2: Prime factors of 63: 63 = 3 x 3 x 7. | Step 3: Take every prime factor the greatest number of times it appears: 3 x 3 x 5 x 7 = 315. | Step 4: Final Answer: LCM = 315.

Question: Exercise 2.7, Question 2 (viii): Find the LCM of 30 and 18 by the prime factorization method.

Answer: Step 1: Prime factors of 30: 30 = 2 x 3 x 5. | Step 2: Prime factors of 18: 18 = 2 x 3 x 3. | Step 3: Take every prime factor the greatest number of times it appears: 2 x 3 x 3 x 5 = 90. | Step 4: Final Answer: LCM = 90.

Question: Exercise 2.7, Question 2 (ix): Find the LCM of 11 and 22 and 33 by the prime factorization method.

Answer: Step 1: Prime factors of 11: 11 = 11. | Step 2: Prime factors of 22: 22 = 2 x 11. | Step 3: Prime factors of 33: 33 = 3 x 11. | Step 4: Take every prime factor the greatest number of times it appears: 2 x 3 x 11 = 66. | Step 5: Final Answer: LCM = 66.

Question: Exercise 2.7, Question 2 (x): Find the LCM of 24 and 30 and 36 by the prime factorization method.

Answer: Step 1: Prime factors of 24: 24 = 2 x 2 x 2 x 3. | Step 2: Prime factors of 30: 30 = 2 x 3 x 5. | Step 3: Prime factors of 36: 36 = 2 x 2 x 3 x 3. | Step 4: Take every prime factor the greatest number of times it appears: 2 x 2 x 2 x 3 x 3 x 5 = 360. | Step 5: Final Answer: LCM = 360.

Exercise 2.8: LCM Word Problems

Question: Exercise 2.8, Question 1: Ali saves Rs 9 every day while Maria

saves Rs 12 every day. What is the least number of days it will take each person to save the same amount of money?

Answer: Step 1: Find the LCM for the day when both have saved the same amount. | Step 2: Multiples of 9: 9, 18, 27, 36. | Step 3: Multiples of 12: 12, 24, 36. | Step 4: The smallest multiple common to all is 36. | Step 5: Final Answer: In the least 36 days each will have saved the same amount (Rs 324).

Question: Exercise 2.8, Question 2: A child smiles at another child every 3 seconds while the other child smiles back every 6 seconds. When will both children smile at each other at the same time?

Answer: Step 1: Find the LCM for the time when both smile together. | Step 2: Multiples of 3: 3, 6. | Step 3: Multiples of 6: 6. | Step 4: The smallest multiple common to all is 6. | Step 5: Final Answer: Both children will smile at each other at the same time every 6 seconds.

Question: Exercise 2.8, Question 3: Boxes that are 12 inches tall are being piled next to boxes that are 10 inches tall. What is the least height in feet at which the two piles will be the same height?

Answer: Step 1: The equal height of both piles is the LCM. | Step 2: Multiples of 12: 12, 24, 36, 48, 60. | Step 3: Multiples of 10: 10, 20, 30, 40, 50, 60. | Step 4: The smallest multiple common to all is 60. | Step 5: Final Answer: The least height is 60 inches, and 60 / 12 = 5, so it is 5 feet.

Question: Exercise 2.8, Question 4: Two men running a marathon took a sip of water at the same time 72 minutes after they started. If the first man took a sip every 9 minutes, how often did the other man take a sip?

Answer: Step 1: Both runners took a sip together after 72 minutes, so 72 is the LCM of the two gaps. | Step 2: The first man sips every 9 minutes and 72 = 9 x 8. | Step 3: If the other man sips every 8 minutes, then LCM(9, 8) = 72, which matches the given time. | Step 4: Final Answer: The other man

took a sip every 8 minutes.

Question: Exercise 2.8, Question 5: A radio station plays “yesterday” by the Beatles once every 2 days; another station plays the same song once every 3 days. How many times in 30 days will both stations play the same song on the same day?

Answer: Step 1: Find the LCM for the days when both stations play the song together. | Step 2: Multiples of 2: 2, 4, 6. | Step 3: Multiples of 3: 3, 6. | Step 4: The smallest multiple common to all is 6. | Step 5: Final Answer: Both play together every 6 days. In 30 days, 30 / 6 = 5 times both stations will play the same song on the same day.

Question: Exercise 2.8, Question 6: Ali runs every 12 days, Anees every 8 days and Ameen every 6 days. All of them ran today. After how many days will they all run together again?

Answer: Step 1: Find the LCM to get the day they come together. | Step 2: Multiples of 12: 12, 24. | Step 3: Multiples of 8: 8, 16, 24. | Step 4: Multiples of 6: 6, 12, 18, 24. | Step 5: The smallest multiple common to all is 24. | Step 6: Final Answer: They will all run together again after 24 days.

Question: Exercise 2.8, Question 7: Ali visits the garden every 15 days and Azhar every 20 days. Both met in the garden today. After how many days will they meet in the garden again?

Answer: Step 1: Find the LCM to get the day they come together. | Step 2: Multiples of 15: 15, 30, 45, 60. | Step 3: Multiples of 20: 20, 40, 60. | Step 4: The smallest multiple common to all is 60. | Step 5: Final Answer: They will meet in the garden again after 60 days.

Review Exercise 2

Review Exercise 2, Question 1 (i): Choose the correct answer. Which number is divisible by 2?

(a) 567

✓ (b) 484

(c) 257

(d) 193

Review Exercise 2, Question 1 (ii): Choose the correct answer. Which number is divisible by 3?

(a) 143

✓ (b) 483

(c) 367

(d) 941

Review Exercise 2, Question 1 (iii): Choose the correct answer. Which number is divisible by both 2 and 10?

(a) 4579

(b) 3921

✓ (c) 4050

(d) 2108

Question: Review Exercise 2, Question 1 (iv): Choose the correct answer. What is the HCF of 39 and 52? (a) 39 (b) 26 (c) 13 (d) 1

Answer: Step 1: Factors of 39: 1, 3, 13, 39. | Step 2: Factors of 52: 1, 2, 4, 13, 26, 52. | Step 3: The common factors are: 1, 13. | Step 4: Final Answer: (c) 13 — the HCF of 39 and 52 is 13.

Question: Review Exercise 2, Question 1 (v): Choose the correct answer. What is the LCM of 48 and 66? (a) 528 (b) 185 (c) 246 (d) 114

Answer: Step 1: Prime factors of 48: 48 = 2 x 2 x 2 x 2 x 3. | Step 2: Prime factors of 66: 66 = 2 x 3 x 11. | Step 3: Take every prime factor the greatest number of times: 2 x 2 x 2 x 2 x 3 x 11 = 528. | Step 4: Final Answer: (a) 528.

Question: Review Exercise 2, Question 2: Find the HCF and LCM of 48, 69

and 36.

Answer: Step 1: Factors of 48: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48. | Step 2: Factors of 69: 1, 3, 23, 69. | Step 3: Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36. | Step 4: The common factors are: 1, 3. | Step 5: The greatest common factor is 3, so HCF = 3. | Step 6: Prime factors of 48: 48 = 2 x 2 x 2 x 2 x 3. | Step 7: Prime factors of 69: 69 = 3 x 23. | Step 8: Prime factors of 36: 36 = 2 x 2 x 3 x 3. | Step 9: Take every prime factor the greatest number of times: 2 x 2 x 2 x 2 x 3 x 3 x 23 = 3,312. | Step 10: Final Answer: HCF = 3 and LCM = 3,312.

Question: Review Exercise 2, Question 3: What is the HCF and LCM of 37 and 41? Explain how you got your answer.

Answer: Step 1: Factors of 37: 1, 37. It has only two factors, so 37 is prime. | Step 2: Factors of 41: 1, 41. It has only two factors, so 41 is also prime. | Step 3: Two different prime numbers have only one common factor: 1. So HCF = 1. | Step 4: The LCM of two prime numbers is their product: 37 x 41 = 1,517. | Step 5: Final Answer: HCF = 1 and LCM = 1,517.

Question: Review Exercise 2, Question 4: Mrs. Sayem wants to distribute 45 patties, 60 samosas and 75 juices equally. Find the greatest number of students among which these can be distributed equally.

Answer: Step 1: For an equal distribution, find the HCF of 45, 60 and 75. | Step 2: Factors of 45: 1, 3, 5, 9, 15, 45. | Step 3: Factors of 60: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60. | Step 4: Factors of 75: 1, 3, 5, 15, 25, 75. | Step 5: Common factors: 1, 3, 5, 15. | Step 6: The greatest common factor is 15. | Step 7: Final Answer: The greatest number of students is 15.

Question: Review Exercise 2, Question 5: Rida is arranging her birthday party. She wants to share 15 sandwiches or 30 slices of pizza equally to each friend. Find the number of friends that could be invited.

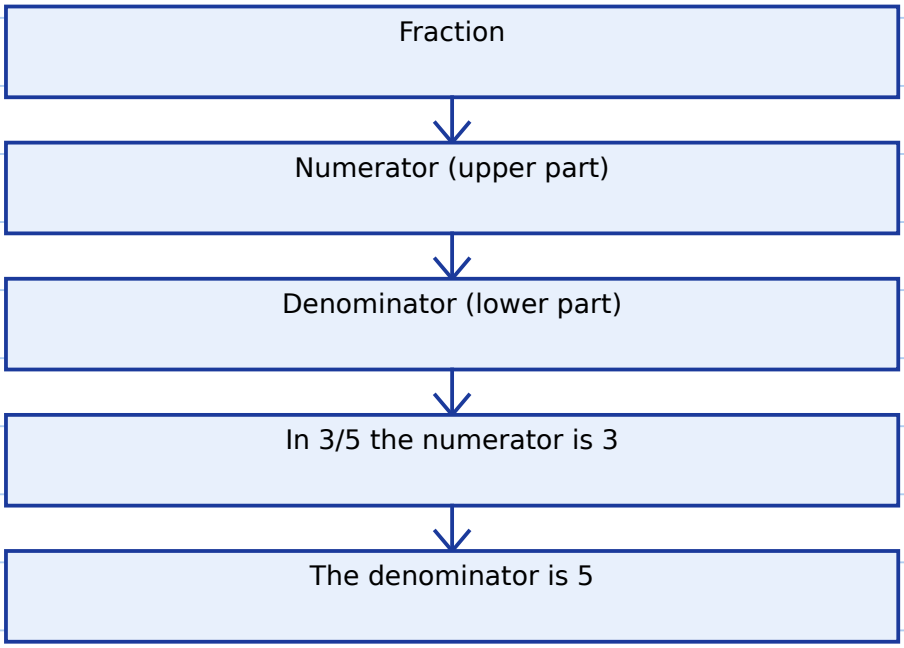
Answer: Step 1: Each friend gets either half of a sandwich or one slice of

pizza. | Step 2: Halves from 15 sandwiches: 15 x 2 = 30 portions. | Step 3: The pizza already has 30 slices. | Step 4: Both items divide equally among 30 friends: LCM(15, 30) = 30. | Step 5: Final Answer: Rida can invite 30 friends.

3. Fractions

Exercise 3.1: Like and Unlike Fractions and Their Comparison

Parts of a Fraction



Question: Exercise 3.1, Question 1 (i): Identify whether 2/7 and 6/7 are like or unlike fractions.

Answer: Step 1: First look at the denominator of 2/7: it is 7. | Step 2: Then look at the denominator of 6/7: it is 7. | Step 3: The two denominators are the same. | Step 4: Final Answer: 2/7 and 6/7 are like fractions.

Question: Exercise 3.1, Question 1 (ii): Identify whether 2/9 and 2/15 are like or unlike fractions.

Answer: Step 1: First look at the denominator of 2/9: it is 9. | Step 2: Then look at the denominator of 2/15: it is 15. | Step 3: The two denominators are different. | Step 4: Final Answer: 2/9 and 2/15 are unlike fractions.

Question: Exercise 3.1, Question 1 (iii): Identify whether 8/17 and 7/17 are like or unlike fractions.

Answer: Step 1: First look at the denominator of 8/17: it is 17. | Step 2: Then look at the denominator of 7/17: it is 17. | Step 3: The two denominators are the same. | Step 4: Final Answer: 8/17 and 7/17 are like fractions.

Question: Exercise 3.1, Question 1 (iv): Identify whether 3/10 and 6/11 are like or unlike fractions.

Answer: Step 1: First look at the denominator of 3/10: it is 10. | Step 2: Then look at the denominator of 6/11: it is 11. | Step 3: The two denominators are different. | Step 4: Final Answer: 3/10 and 6/11 are unlike fractions.

Question: Exercise 3.1, Question 1 (v): Identify whether 7/11 and 7/10 are like or unlike fractions.

Answer: Step 1: First look at the denominator of 7/11: it is 11. | Step 2: Then look at the denominator of 7/10: it is 10. | Step 3: The two denominators are different. | Step 4: Final Answer: 7/11 and 7/10 are unlike fractions.

Question: Exercise 3.1, Question 1 (vi): Identify whether 1/4 and 3/4 are like or unlike fractions.

Answer: Step 1: First look at the denominator of 1/4: it is 4. | Step 2: Then look at the denominator of 3/4: it is 4. | Step 3: The two denominators are the same. | Step 4: Final Answer: 1/4 and 3/4 are like fractions.

Question: Exercise 3.1, Question 1 (vii): Identify whether 4/13 and 13/15 are like or unlike fractions.

Answer: Step 1: First look at the denominator of 4/13: it is 13. | Step 2: Then look at the denominator of 13/15: it is 15. | Step 3: The two denominators are different. | Step 4: Final Answer: 4/13 and 13/15 are unlike fractions.

Question: Exercise 3.1, Question 1 (viii): Identify whether 11/19 and 16/19

are like or unlike fractions.

Answer: Step 1: First look at the denominator of 11/19: it is 19. | Step 2: Then look at the denominator of 16/19: it is 19. | Step 3: The two denominators are the same. | Step 4: Final Answer: 11/19 and 16/19 are like fractions.

Question: Exercise 3.1, Question 1 (ix): Identify whether 77/27 and 4/27 are like or unlike fractions.

Answer: Step 1: First look at the denominator of 77/27: it is 27. | Step 2: Then look at the denominator of 4/27: it is 27. | Step 3: The two denominators are the same. | Step 4: Final Answer: 77/27 and 4/27 are like fractions.

Question: Exercise 3.1, Question 2 (i): Convert into like fractions first and then compare with < or > or =: 3/8 and 3/8.

Answer: Step 1: The denominators are 8 and 8; their LCM is 8. | Step 2: Convert 3/8: multiply the numerator by 1 to get 3; the new fraction is 3/8. | Step 3: Convert 3/8: multiply the numerator by 1 to get 3; the new fraction is 3/8. | Step 4: Now the denominators are equal, so compare the numerators: 3 and 3. | Step 5: Final Answer: 3/8 = 3/8.

Question: Exercise 3.1, Question 2 (ii): Convert into like fractions first and then compare with < or > or =: 5/6 and 4/9.

Answer: Step 1: The denominators are 6 and 9; their LCM is 18. | Step 2: Convert 5/6: multiply the numerator by 3 to get 15; the new fraction is 15/18. | Step 3: Convert 4/9: multiply the numerator by 2 to get 8; the new fraction is 8/18. | Step 4: Now the denominators are equal, so compare the numerators: 15 and 8. | Step 5: Final Answer: 5/6 > 4/9.

Question: Exercise 3.1, Question 2 (iii): Convert into like fractions first and then compare with < or > or =: 7/12 and 1/4.

Answer: Step 1: The denominators are 12 and 4; their LCM is 12. | Step 2:

Convert 7/12: multiply the numerator by 1 to get 7; the new fraction is 7/12. | Step 3: Convert 1/4: multiply the numerator by 3 to get 3; the new fraction is 3/12. | Step 4: Now the denominators are equal, so compare the numerators: 7 and 3. | Step 5: Final Answer: 7/12 > 1/4.

Question: Exercise 3.1, Question 2 (iv): Convert into like fractions first and then compare with < or > or =: 2/7 and 13/14.

Answer: Step 1: The denominators are 7 and 14; their LCM is 14. | Step 2: Convert 2/7: multiply the numerator by 2 to get 4; the new fraction is 4/14. | Step 3: Convert 13/14: multiply the numerator by 1 to get 13; the new fraction is 13/14. | Step 4: Now the denominators are equal, so compare the numerators: 4 and 13. | Step 5: Final Answer: 2/7 < 13/14.

Question: Exercise 3.1, Question 2 (v): Convert into like fractions first and then compare with < or > or =: 6/15 and 3/10.

Answer: Step 1: The denominators are 15 and 10; their LCM is 30. | Step 2: Convert 6/15: multiply the numerator by 2 to get 12; the new fraction is 12/30. | Step 3: Convert 3/10: multiply the numerator by 3 to get 9; the new fraction is 9/30. | Step 4: Now the denominators are equal, so compare the numerators: 12 and 9. | Step 5: Final Answer: 6/15 > 3/10.

Question: Exercise 3.1, Question 2 (vi): Convert into like fractions first and then compare with < or > or =: 9/11 and 16/22.

Answer: Step 1: The denominators are 11 and 22; their LCM is 22. | Step 2: Convert 9/11: multiply the numerator by 2 to get 18; the new fraction is 18/22. | Step 3: Convert 16/22: multiply the numerator by 1 to get 16; the new fraction is 16/22. | Step 4: Now the denominators are equal, so compare the numerators: 18 and 16. | Step 5: Final Answer: 9/11 > 16/22.

Question: Exercise 3.1, Question 3 (i): Arrange in ascending order: 6/7, 9/7, 5/7.

Answer: Step 1: The LCM of all the denominators is 7. | Step 2: Convert

every fraction to denominator 7: 6/7, 9/7, 5/7. | Step 3: Now arrange by looking at the numerators (ascending order). | Step 4: Final Answer: 5/7, 6/7, 9/7.

Question: Exercise 3.1, Question 3 (ii): Arrange in ascending order: 5/6, 2/3, 5/9.

Answer: Step 1: The LCM of all the denominators is 18. | Step 2: Convert every fraction to denominator 18: 15/18, 12/18, 10/18. | Step 3: Now arrange by looking at the numerators (ascending order). | Step 4: Final Answer: 5/9, 2/3, 5/6.

Question: Exercise 3.1, Question 3 (iii): Arrange in ascending order: 2/5, 1/3, 4/15.

Answer: Step 1: The LCM of all the denominators is 15. | Step 2: Convert every fraction to denominator 15: 6/15, 5/15, 4/15. | Step 3: Now arrange by looking at the numerators (ascending order). | Step 4: Final Answer: 4/15, 1/3, 2/5.

Question: Exercise 3.1, Question 3 (iv): Arrange in ascending order: 5/12, 4/8, 3/4, 1/6.

Answer: Step 1: The LCM of all the denominators is 24. | Step 2: Convert every fraction to denominator 24: 10/24, 12/24, 18/24, 4/24. | Step 3: Now arrange by looking at the numerators (ascending order). | Step 4: Final Answer: 1/6, 5/12, 4/8, 3/4.

Question: Exercise 3.1, Question 3 (v): Arrange in ascending order: 9/8, 7/2, 13/6, 5/4.

Answer: Step 1: The LCM of all the denominators is 24. | Step 2: Convert every fraction to denominator 24: 27/24, 84/24, 52/24, 30/24. | Step 3: Now arrange by looking at the numerators (ascending order). | Step 4: Final Answer: 9/8, 5/4, 13/6, 7/2.

Question: Exercise 3.1, Question 3 (vi): Arrange in ascending order: 5/12,

7/6, 7/4, 5/2.

Answer: Step 1: The LCM of all the denominators is 12. | Step 2: Convert every fraction to denominator 12: 5/12, 14/12, 21/12, 30/12. | Step 3: Now arrange by looking at the numerators (ascending order). | Step 4: Final Answer: 5/12, 7/6, 7/4, 5/2.

Question: Exercise 3.1, Question 4 (i): Simplify to the lowest form: 24/32.

Answer: Step 1: Find the common factors of 24 and 32. | Step 2: Divide the numerator and denominator by 2: 24/32 = 12/16. | Step 3: Divide the numerator and denominator by 2: 12/16 = 6/8. | Step 4: Divide the numerator and denominator by 2: 6/8 = 3/4. | Step 5: Final Answer: 3/4.

Question: Exercise 3.1, Question 4 (ii): Simplify to the lowest form: 18/27.

Answer: Step 1: Find the common factors of 18 and 27. | Step 2: Divide the numerator and denominator by 3: 18/27 = 6/9. | Step 3: Divide the numerator and denominator by 3: 6/9 = 2/3. | Step 4: Final Answer: 2/3.

Question: Exercise 3.1, Question 4 (iii): Simplify to the lowest form: 30/50.

Answer: Step 1: Find the common factors of 30 and 50. | Step 2: Divide the numerator and denominator by 2: 30/50 = 15/25. | Step 3: Divide the numerator and denominator by 5: 15/25 = 3/5. | Step 4: Final Answer: 3/5.

Question: Exercise 3.1, Question 4 (iv): Simplify to the lowest form: 14/42.

Answer: Step 1: Find the common factors of 14 and 42. | Step 2: Divide the numerator and denominator by 2: 14/42 = 7/21. | Step 3: Divide the numerator and denominator by 7: 7/21 = 1/3. | Step 4: Final Answer: 1/3.

Question: Exercise 3.1, Question 4 (v): Simplify to the lowest form: 33/66.

Answer: Step 1: Find the common factors of 33 and 66. | Step 2: Divide the numerator and denominator by 3: 33/66 = 11/22. | Step 3: Divide the numerator and denominator by 11: 11/22 = 1/2. | Step 4: Final Answer: 1/2.

Exercise 3.2: Types of Fractions and Conversion

Question: Exercise 3.2, Question 1 (i): Identify whether 1/3 is a unit, proper, improper or mixed fraction.

Answer: Step 1: The numerator of 1/3 is 1. | Step 2: Final Answer: a unit fraction.

Question: Exercise 3.2, Question 1 (ii): Identify whether 1/4 is a unit, proper, improper or mixed fraction.

Answer: Step 1: The numerator of 1/4 is 1. | Step 2: Final Answer: a unit fraction.

Question: Exercise 3.2, Question 1 (iii): Identify whether 4 3/4 is a unit, proper, improper or mixed fraction.

Answer: Step 1: In 4 3/4 the numerator is greater than the denominator. | Step 2: Final Answer: an improper fraction.

Question: Exercise 3.2, Question 1 (iv): Identify whether 1/7 is a unit, proper, improper or mixed fraction.

Answer: Step 1: The numerator of 1/7 is 1. | Step 2: Final Answer: a unit fraction.

Question: Exercise 3.2, Question 1 (v): Identify whether 5/9 is a unit, proper, improper or mixed fraction.

Answer: Step 1: In 5/9 the numerator 5 is smaller than the denominator 9. | Step 2: Final Answer: a proper fraction.

Question: Exercise 3.2, Question 1 (vi): Identify whether 2 1/8 is a unit, proper, improper or mixed fraction.

Answer: Step 1: In 2 1/8 the numerator is greater than the denominator. | Step 2: Final Answer: a mixed fraction.

Question: Exercise 3.2, Question 1 (vii): Identify whether 1 1/3 is a unit, proper, improper or mixed fraction.

Answer: Step 1: In 1 1/3 the numerator is greater than the denominator. |

Step 2: Final Answer: an improper fraction.

Question: Exercise 3.2, Question 1 (viii): Identify whether the fraction shown in the figure (a circle in 5 equal parts, 1 part shaded) is a unit, proper, improper or mixed fraction.

Answer: Step 1: The figure shows a circle divided into 5 equal parts with 1 part shaded, which is the fraction 1/5. | Step 2: Final Answer: a unit fraction.

Question: Exercise 3.2, Question 1 (ix): Identify whether the fraction shown in the figure (a triangle in 3 equal parts, 1 part shaded) is a unit, proper, improper or mixed fraction.

Answer: Step 1: The figure shows a triangle divided into 3 equal parts with 1 part shaded, which is the fraction 1/3. | Step 2: Final Answer: a unit fraction.

Question: Exercise 3.2, Question 1 (x): Identify whether the fraction shown in the figure (two parallelograms, one fully shaded and one half shaded) is a unit, proper, improper or mixed fraction.

Answer: Step 1: The figure shows two parallelograms: one fully shaded (1 whole) and one half shaded (1/2), together 1 1/2. | Step 2: Final Answer: a mixed fraction.

Question: Exercise 3.2, Question 2 (i): Convert the improper fraction 43/7 into a mixed fraction.

Answer: Step 1: Divide the numerator 43 by the denominator 7. | Step 2: 43 / 7 = 6 remainder 1. | Step 3: The quotient 6 becomes the whole part and the remainder 1 becomes the new numerator; the denominator stays 7. | Step 4: Final Answer: 6 1/7.

Question: Exercise 3.2, Question 2 (ii): Convert the improper fraction 29/4 into a mixed fraction.

Answer: Step 1: Divide the numerator 29 by the denominator 4. | Step 2:

29 / 4 = 7 remainder 1. | Step 3: The quotient 7 becomes the whole part and the remainder 1 becomes the new numerator; the denominator stays 4. | Step 4: Final Answer: 7 1/4.

Question: Exercise 3.2, Question 2 (iii): Convert the improper fraction 74/9 into a mixed fraction.

Answer: Step 1: Divide the numerator 74 by the denominator 9. | Step 2: 74 / 9 = 8 remainder 2. | Step 3: The quotient 8 becomes the whole part and the remainder 2 becomes the new numerator; the denominator stays 9. | Step 4: Final Answer: 8 2/9.

Question: Exercise 3.2, Question 2 (iv): Convert the improper fraction 14/6 into a mixed fraction.

Answer: Step 1: Divide the numerator 14 by the denominator 6. | Step 2: 14 / 6 = 2 remainder 2. | Step 3: The quotient 2 becomes the whole part and the remainder 2 becomes the new numerator; the denominator stays 6. | Step 4: Simplify the fraction part: 2/6 = 1/3. | Step 5: Final Answer: 2 1/3.

Question: Exercise 3.2, Question 2 (v): Convert the improper fraction 28/5 into a mixed fraction.

Answer: Step 1: Divide the numerator 28 by the denominator 5. | Step 2: 28 / 5 = 5 remainder 3. | Step 3: The quotient 5 becomes the whole part and the remainder 3 becomes the new numerator; the denominator stays 5. | Step 4: Final Answer: 5 3/5.

Question: Exercise 3.2, Question 2 (vi): Convert the improper fraction 67/3 into a mixed fraction.

Answer: Step 1: Divide the numerator 67 by the denominator 3. | Step 2: 67 / 3 = 22 remainder 1. | Step 3: The quotient 22 becomes the whole part and the remainder 1 becomes the new numerator; the denominator stays 3. | Step 4: Final Answer: 22 1/3.

Question: Exercise 3.2, Question 2 (vii): Convert the improper fraction 85/11 into a mixed fraction.

Answer: Step 1: Divide the numerator 85 by the denominator 11. | Step 2: 85 / 11 = 7 remainder 8. | Step 3: The quotient 7 becomes the whole part and the remainder 8 becomes the new numerator; the denominator stays 11. | Step 4: Final Answer: 7 8/11.

Question: Exercise 3.2, Question 2 (viii): Convert the improper fraction 86/5 into a mixed fraction.

Answer: Step 1: Divide the numerator 86 by the denominator 5. | Step 2: 86 / 5 = 17 remainder 1. | Step 3: The quotient 17 becomes the whole part and the remainder 1 becomes the new numerator; the denominator stays 5. | Step 4: Final Answer: 17 1/5.

Question: Exercise 3.2, Question 3 (i): Express the mixed fraction 4 3/10 as an improper fraction.

Answer: Step 1: Multiply the whole part 4 by the denominator 10: 4 x 10 = 40. | Step 2: Add the numerator 3: 40 + 3 = 43; this is the new numerator. | Step 3: The denominator stays 10. | Step 4: Final Answer: 43/10.

Question: Exercise 3.2, Question 3 (ii): Express the mixed fraction 5 2/3 as an improper fraction.

Answer: Step 1: Multiply the whole part 5 by the denominator 3: 5 x 3 = 15. | Step 2: Add the numerator 2: 15 + 2 = 17; this is the new numerator. | Step 3: The denominator stays 3. | Step 4: Final Answer: 17/3.

Question: Exercise 3.2, Question 3 (iii): Express the mixed fraction 6 1/2 as an improper fraction.

Answer: Step 1: Multiply the whole part 6 by the denominator 2: 6 x 2 = 12. | Step 2: Add the numerator 1: 12 + 1 = 13; this is the new numerator. | Step 3: The denominator stays 2. | Step 4: Final Answer: 13/2.

Question: Exercise 3.2, Question 3 (iv): Express the mixed fraction 3 2/7 as

an improper fraction.

Answer: Step 1: Multiply the whole part 3 by the denominator 7: 3 x 7 = 21. | Step 2: Add the numerator 2: 21 + 2 = 23; this is the new numerator. | Step 3: The denominator stays 7. | Step 4: Final Answer: 23/7.

Question: Exercise 3.2, Question 3 (v): Express the mixed fraction 8 1/4 as an improper fraction.

Answer: Step 1: Multiply the whole part 8 by the denominator 4: 8 x 4 = 32. | Step 2: Add the numerator 1: 32 + 1 = 33; this is the new numerator. | Step 3: The denominator stays 4. | Step 4: Final Answer: 33/4.

Question: Exercise 3.2, Question 3 (vi): Express the mixed fraction 1 11/13 as an improper fraction.

Answer: Step 1: Multiply the whole part 1 by the denominator 13: 1 x 13 = 13. | Step 2: Add the numerator 11: 13 + 11 = 24; this is the new numerator. | Step 3: The denominator stays 13. | Step 4: Final Answer: 24/13.

Question: Exercise 3.2, Question 3 (vii): Express the mixed fraction 7 7/8 as an improper fraction.

Answer: Step 1: Multiply the whole part 7 by the denominator 8: 7 x 8 = 56. | Step 2: Add the numerator 7: 56 + 7 = 63; this is the new numerator. | Step 3: The denominator stays 8. | Step 4: Final Answer: 63/8.

Question: Exercise 3.2, Question 3 (viii): Express the mixed fraction 2 6/7 as an improper fraction.

Answer: Step 1: Multiply the whole part 2 by the denominator 7: 2 x 7 = 14. | Step 2: Add the numerator 6: 14 + 6 = 20; this is the new numerator. | Step 3: The denominator stays 7. | Step 4: Final Answer: 20/7.

Activity: Adding and Subtracting Like Fractions

Question: Activity, Add (i): Add 3/7 and 5/7.

Answer: Step 1: The denominators are 7, 7. Their LCM is 7. | Step 2: Convert 3/7: 3 x 1 = 3, so it becomes 3/7. | Step 3: Convert 5/7: 5 x 1 = 5, so it becomes 5/7. | Step 4: Now add the numerators: 3 + 5 = 8; the result is 8/7. | Step 5: Final Answer: 1 1/7.

Question: Activity, Add (ii): Add 2/9 and 3/9.

Answer: Step 1: The denominators are 9, 9. Their LCM is 9. | Step 2: Convert 2/9: 2 x 1 = 2, so it becomes 2/9. | Step 3: Convert 3/9: 3 x 1 = 3, so it becomes 3/9. | Step 4: Now add the numerators: 2 + 3 = 5; the result is 5/9. | Step 5: Final Answer: 5/9.

Question: Activity, Add (iii): Add 5/7 and 1/7.

Answer: Step 1: The denominators are 7, 7. Their LCM is 7. | Step 2: Convert 5/7: 5 x 1 = 5, so it becomes 5/7. | Step 3: Convert 1/7: 1 x 1 = 1, so it becomes 1/7. | Step 4: Now add the numerators: 5 + 1 = 6; the result is 6/7. | Step 5: Final Answer: 6/7.

Question: Activity, Add (iv): Add 6/10 and 3/10.

Answer: Step 1: The denominators are 10, 10. Their LCM is 10. | Step 2: Convert 6/10: 6 x 1 = 6, so it becomes 6/10. | Step 3: Convert 3/10: 3 x 1 = 3, so it becomes 3/10. | Step 4: Now add the numerators: 6 + 3 = 9; the result is 9/10. | Step 5: Final Answer: 9/10.

Question: Activity, Add (v): Add 5/12 and 2/12.

Answer: Step 1: The denominators are 12, 12. Their LCM is 12. | Step 2: Convert 5/12: 5 x 1 = 5, so it becomes 5/12. | Step 3: Convert 2/12: 2 x 1 = 2, so it becomes 2/12. | Step 4: Now add the numerators: 5 + 2 = 7; the result is 7/12. | Step 5: Final Answer: 7/12.

Question: Activity, Add (vi): Add 7/11 and 2/11.

Answer: Step 1: The denominators are 11, 11. Their LCM is 11. | Step 2: Convert 7/11: 7 x 1 = 7, so it becomes 7/11. | Step 3: Convert 2/11: 2 x 1 = 2, so it becomes 2/11. | Step 4: Now add the numerators: 7 + 2 = 9; the

result is 9/11. | Step 5: Final Answer: 9/11.

Question: Activity, Subtract (i): Subtract 2/5 from 4/5.

Answer: Step 1: The denominators are 5, 5. Their LCM is 5. | Step 2: Convert 4/5: 4 x 1 = 4, so it becomes 4/5. | Step 3: Convert 2/5: 2 x 1 = 2, so it becomes 2/5. | Step 4: Now subtract the numerators: 4 - 2 = 2; the result is 2/5. | Step 5: Final Answer: 2/5.

Question: Activity, Subtract (ii): Subtract 8/15 from 9/15.

Answer: Step 1: The denominators are 15, 15. Their LCM is 15. | Step 2: Convert 9/15: 9 x 1 = 9, so it becomes 9/15. | Step 3: Convert 8/15: 8 x 1 = 8, so it becomes 8/15. | Step 4: Now subtract the numerators: 9 - 8 = 1; the result is 1/15. | Step 5: Final Answer: 1/15.

Question: Activity, Subtract (iii): Subtract 4/15 from 7/15.

Answer: Step 1: The denominators are 15, 15. Their LCM is 15. | Step 2: Convert 7/15: 7 x 1 = 7, so it becomes 7/15. | Step 3: Convert 4/15: 4 x 1 = 4, so it becomes 4/15. | Step 4: Now subtract the numerators: 7 - 4 = 3; the result is 3/15. | Step 5: Divide the numerator and denominator by 3: 3/15 = 1/5. | Step 6: Final Answer: 1/5.

Question: Activity, Subtract (iv): Subtract 6/12 from 7/12.

Answer: Step 1: The denominators are 12, 12. Their LCM is 12. | Step 2: Convert 7/12: 7 x 1 = 7, so it becomes 7/12. | Step 3: Convert 6/12: 6 x 1 = 6, so it becomes 6/12. | Step 4: Now subtract the numerators: 7 - 6 = 1; the result is 1/12. | Step 5: Final Answer: 1/12.

Question: Activity, Subtract (v): Subtract 5/14 from 8/14.

Answer: Step 1: The denominators are 14, 14. Their LCM is 14. | Step 2: Convert 8/14: 8 x 1 = 8, so it becomes 8/14. | Step 3: Convert 5/14: 5 x 1 = 5, so it becomes 5/14. | Step 4: Now subtract the numerators: 8 - 5 = 3; the result is 3/14. | Step 5: Final Answer: 3/14.

Question: Activity, Subtract (vi): Subtract 4/17 from 13/17.

Answer: Step 1: The denominators are 17, 17. Their LCM is 17. | Step 2: Convert 13/17: 13 x 1 = 13, so it becomes 13/17. | Step 3: Convert 4/17: 4 x 1 = 4, so it becomes 4/17. | Step 4: Now subtract the numerators: 13 - 4 = 9; the result is 9/17. | Step 5: Final Answer: 9/17.

Exercise 3.3: Addition and Subtraction of Fractions

Question: Exercise 3.3, Question 1 (i): Add: 2/3 + 1 2/9.

Answer: Step 1: The denominators are 3, 9. Their LCM is 9. | Step 2: Convert 2/3: 2 x 3 = 6, so it becomes 6/9. | Step 3: Convert 1 2/9: 11 x 1 = 11, so it becomes 11/9. | Step 4: Now add the numerators: 6 + 11 = 17; the result is 17/9. | Step 5: Final Answer: 1 8/9.

Question: Exercise 3.3, Question 1 (ii): Add: 1 1/6 + 1/2.

Answer: Step 1: The denominators are 6, 2. Their LCM is 6. | Step 2: Convert 1 1/6: 7 x 1 = 7, so it becomes 7/6. | Step 3: Convert 1/2: 1 x 3 = 3, so it becomes 3/6. | Step 4: Now add the numerators: 7 + 3 = 10; the result is 10/6. | Step 5: Divide the numerator and denominator by 2: 10/6 = 5/3. | Step 6: Final Answer: 1 2/3.

Question: Exercise 3.3, Question 1 (iii): Add: 8/10 + 1/2 + 2/4.

Answer: Step 1: The denominators are 10, 2, 4. Their LCM is 20. | Step 2: Convert 8/10: 8 x 2 = 16, so it becomes 16/20. | Step 3: Convert 1/2: 1 x 10 = 10, so it becomes 10/20. | Step 4: Convert 2/4: 2 x 5 = 10, so it becomes 10/20. | Step 5: Now add the numerators: 16 + 10 + 10 = 36; the result is 36/20. | Step 6: Divide the numerator and denominator by 2: 36/20 = 18/10. | Step 7: Divide the numerator and denominator by 2: 18/10 = 9/5. | Step 8: Final Answer: 1 4/5.

Question: Exercise 3.3, Question 1 (iv): Add: 3/4 + 1/2 + 4/6.

Answer: Step 1: The denominators are 4, 2, 6. Their LCM is 12. | Step 2: Convert 3/4: 3 x 3 = 9, so it becomes 9/12. | Step 3: Convert 1/2: 1 x 6 =

6, so it becomes 6/12. | Step 4: Convert 4/6: 4 x 2 = 8, so it becomes 8/12. | Step 5: Now add the numerators: 9 + 6 + 8 = 23; the result is 23/12. | Step 6: Final Answer: 1 11/12.

Question: Exercise 3.3, Question 1 (v): Add: 3 1/2 + 5 1/3.

Answer: Step 1: The denominators are 2, 3. Their LCM is 6. | Step 2: Convert 3 1/2: 7 x 3 = 21, so it becomes 21/6. | Step 3: Convert 5 1/3: 16 x 2 = 32, so it becomes 32/6. | Step 4: Now add the numerators: 21 + 32 = 53; the result is 53/6. | Step 5: Final Answer: 8 5/6.

Question: Exercise 3.3, Question 1 (vi): Add: 5 1/3 + 2 3/4.

Answer: Step 1: The denominators are 3, 4. Their LCM is 12. | Step 2: Convert 5 1/3: 16 x 4 = 64, so it becomes 64/12. | Step 3: Convert 2 3/4: 11 x 3 = 33, so it becomes 33/12. | Step 4: Now add the numerators: 64 + 33 = 97; the result is 97/12. | Step 5: Final Answer: 8 1/12.

Question: Exercise 3.3, Question 1 (vii): Add: 2/3 + 1 1/5 + 1 1/2.

Answer: Step 1: The denominators are 3, 5, 2. Their LCM is 30. | Step 2: Convert 2/3: 2 x 10 = 20, so it becomes 20/30. | Step 3: Convert 1 1/5: 6 x 6 = 36, so it becomes 36/30. | Step 4: Convert 1 1/2: 3 x 15 = 45, so it becomes 45/30. | Step 5: Now add the numerators: 20 + 36 + 45 = 101; the result is 101/30. | Step 6: Final Answer: 3 11/30.

Question: Exercise 3.3, Question 1 (viii): Add: 2 1/7 + 2/5 + 1 1/7.

Answer: Step 1: The denominators are 7, 5, 7. Their LCM is 35. | Step 2: Convert 2 1/7: 15 x 5 = 75, so it becomes 75/35. | Step 3: Convert 2/5: 2 x 7 = 14, so it becomes 14/35. | Step 4: Convert 1 1/7: 8 x 5 = 40, so it becomes 40/35. | Step 5: Now add the numerators: 75 + 14 + 40 = 129; the result is 129/35. | Step 6: Final Answer: 3 24/35.

Question: Exercise 3.3, Question 2 (i): Subtract: 5/6 - 1/2.

Answer: Step 1: The denominators are 6, 2. Their LCM is 6. | Step 2: Convert 5/6: 5 x 1 = 5, so it becomes 5/6. | Step 3: Convert 1/2: 1 x 3 = 3,

so it becomes 3/6. | Step 4: Now subtract the numerators: 5 - 3 = 2; the result is 2/6. | Step 5: Divide the numerator and denominator by 2: 2/6 = 1/3. | Step 6: Final Answer: 1/3.

Question: Exercise 3.3, Question 2 (ii): Subtract: 7/8 - 3/4.

Answer: Step 1: The denominators are 8, 4. Their LCM is 8. | Step 2: Convert 7/8: 7 x 1 = 7, so it becomes 7/8. | Step 3: Convert 3/4: 3 x 2 = 6, so it becomes 6/8. | Step 4: Now subtract the numerators: 7 - 6 = 1; the result is 1/8. | Step 5: Final Answer: 1/8.

Question: Exercise 3.3, Question 2 (iii): Subtract: 4/5 - 1/3.

Answer: Step 1: The denominators are 5, 3. Their LCM is 15. | Step 2: Convert 4/5: 4 x 3 = 12, so it becomes 12/15. | Step 3: Convert 1/3: 1 x 5 = 5, so it becomes 5/15. | Step 4: Now subtract the numerators: 12 - 5 = 7; the result is 7/15. | Step 5: Final Answer: 7/15.

Question: Exercise 3.3, Question 2 (iv): Subtract: 3 3/10 - 1 1/4.

Answer: Step 1: The denominators are 10, 4. Their LCM is 20. | Step 2: Convert 3 3/10: 33 x 2 = 66, so it becomes 66/20. | Step 3: Convert 1 1/4: 5 x 5 = 25, so it becomes 25/20. | Step 4: Now subtract the numerators: 66 - 25 = 41; the result is 41/20. | Step 5: Final Answer: 2 1/20.

Question: Exercise 3.3, Question 2 (v): Subtract: 9 1/2 - 3 1/5.

Answer: Step 1: The denominators are 2, 5. Their LCM is 10. | Step 2: Convert 9 1/2: 19 x 5 = 95, so it becomes 95/10. | Step 3: Convert 3 1/5: 16 x 2 = 32, so it becomes 32/10. | Step 4: Now subtract the numerators: 95 - 32 = 63; the result is 63/10. | Step 5: Final Answer: 6 3/10.

Question: Exercise 3.3, Question 2 (vi): Subtract: 4/5 - 1/10 - 2/15.

Answer: Step 1: The denominators are 5, 10, 15. Their LCM is 30. | Step 2: Convert 4/5: 4 x 6 = 24, so it becomes 24/30. | Step 3: Convert 1/10: 1 x 3 = 3, so it becomes 3/30. | Step 4: Convert 2/15: 2 x 2 = 4, so it becomes 4/30. | Step 5: Now subtract the numerators: 24 - 3 - 4 = 17; the result is

17/30. | Step 6: Final Answer: 17/30.

Question: Exercise 3.3, Question 3 (i): Apply the commutative and associative property of addition to find the missing values: 3/5 + 7/4 = ? + 3/5.

Answer: Step 1: Match the fractions on both sides and see which fraction is missing. | Step 2: This is the commutative property of addition (the order of adding does not change the sum). | Step 3: Final Answer: 7/4

Question: Exercise 3.3, Question 3 (ii): Apply the commutative and associative property of addition to find the missing values: 2/9 + ? = 4/7 + ?.

Answer: Step 1: Match the fractions on both sides and see which fraction is missing. | Step 2: This is the commutative property of addition: the two fractions change places, so the first missing value is 4/7 and the second missing value is 2/9. | Step 3: Final Answer: the first missing value is 4/7 and the second missing value is 2/9

Question: Exercise 3.3, Question 3 (iii): Apply the commutative and associative property of addition to find the missing values: 1/3 + (1/5 + 1/4) = (1/3 + ?) + 1/4.

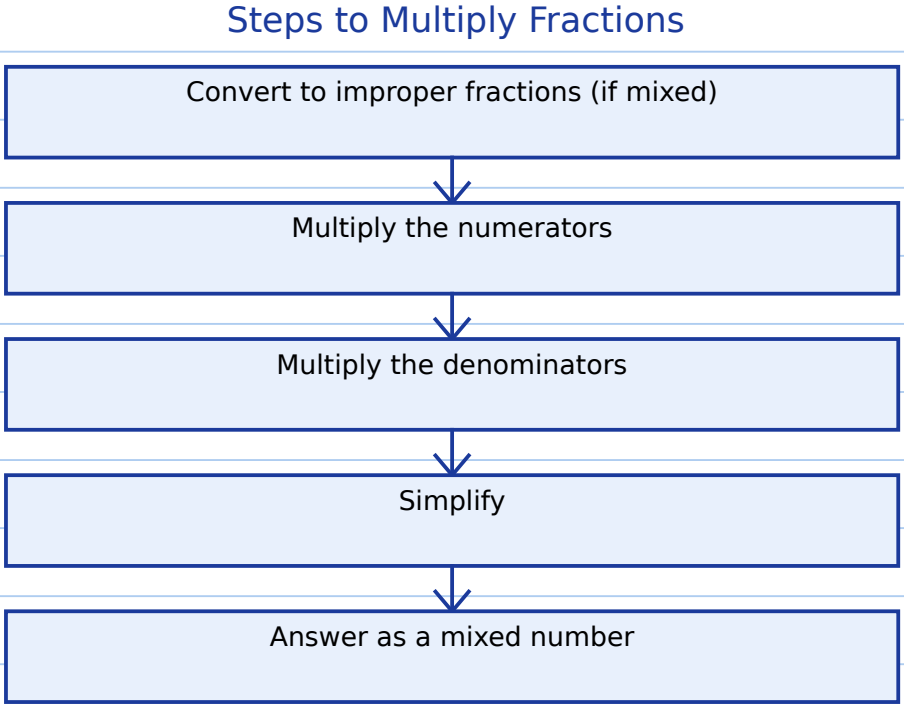
Answer: Step 1: Match the fractions on both sides and see which fraction is missing. | Step 2: This is the associative property of addition (the grouping of the fractions changes, the fractions stay the same). | Step 3: Final Answer: 1/5

Question: Exercise 3.3, Question 3 (iv): Apply the commutative and associative property of addition to find the missing values: 3/10 + (? + 4/5) = (? + 2/7) + 4/5.

Answer: Step 1: Match the fractions on both sides and see which fraction is missing. | Step 2: This is the associative property of addition: the first missing value is 2/7 and the second missing value is 3/10. | Step 3: Final

Answer: the first missing value is 2/7 and the second missing value is 3/10

Exercise 3.4: Multiplication of Fractions



Question: Exercise 3.4, Question 1 (i): Simplify: 5/6 of 9.

Answer: Step 1: Convert the mixed number 9 into an improper fraction: 9 x 1 + 0 = 9, so it is 9/1. | Step 2: Multiply the numerators: 5 x 9 = 45. | Step 3: Multiply the denominators: 6 x 1 = 6. | Step 4: So the product is 45/6. | Step 5: Divide the numerator and denominator by 3: 45/6 = 15/2. | Step 6: Final Answer: 7 1/2.

Question: Exercise 3.4, Question 1 (ii): Simplify: 4/8 of 6.

Answer: Step 1: Convert the mixed number 6 into an improper fraction: 6 x 1 + 0 = 6, so it is 6/1. | Step 2: Multiply the numerators: 4 x 6 = 24. | Step 3: Multiply the denominators: 8 x 1 = 8. | Step 4: So the product is 24/8. | Step 5: Divide the numerator and denominator by 2: 24/8 = 12/4. | Step 6: Divide the numerator and denominator by 2: 12/4 = 6/2. | Step 7: Divide the numerator and denominator by 2: 6/2 = 3/1. | Step 8: Final Answer: 3.

Question: Exercise 3.4, Question 1 (iii): Simplify: 3/14 of 45.

Answer: Step 1: Convert the mixed number 45 into an improper fraction: 45 x 1 + 0 = 45, so it is 45/1. | Step 2: Multiply the numerators: 3 x 45 = 135. | Step 3: Multiply the denominators: 14 x 1 = 14. | Step 4: So the product is 135/14. | Step 5: Final Answer: 9 9/14.

Question: Exercise 3.4, Question 2 (i): Find the product: 1/2 x 4/10.

Answer: Step 1: Multiply the numerators: 1 x 4 = 4. | Step 2: Multiply the denominators: 2 x 10 = 20. | Step 3: So the product is 4/20. | Step 4: Divide the numerator and denominator by 2: 4/20 = 2/10. | Step 5: Divide the numerator and denominator by 2: 2/10 = 1/5. | Step 6: Final Answer: 1/5.

Question: Exercise 3.4, Question 2 (ii): Find the product: 6/9 x 8/6.

Answer: Step 1: Convert the mixed number 1 2/6 into an improper fraction: 1 x 6 + 2 = 8, so it is 8/6. | Step 2: Multiply the numerators: 6 x 8 = 48. | Step 3: Multiply the denominators: 9 x 6 = 54. | Step 4: So the product is 48/54. | Step 5: Divide the numerator and denominator by 2: 48/54 = 24/27. | Step 6: Divide the numerator and denominator by 3: 24/27 = 8/9. | Step 7: Final Answer: 8/9.

Question: Exercise 3.4, Question 2 (iii): Find the product: 15/21 x 6/9 x 1/2.

Answer: Step 1: Multiply the numerators: 15 x 6 x 1 = 90. | Step 2: Multiply the denominators: 21 x 9 x 2 = 378. | Step 3: So the product is 90/378. | Step 4: Divide the numerator and denominator by 2: 90/378 = 45/189. | Step 5: Divide the numerator and denominator by 3: 45/189 = 15/63. | Step 6: Divide the numerator and denominator by 3: 15/63 = 5/21. | Step 7: Final Answer: 5/21.

Question: Exercise 3.4, Question 2 (iv): Find the product: 12/24 x 15/18.

Answer: Step 1: Multiply the numerators: 12 x 15 = 180. | Step 2: Multiply the denominators: 24 x 18 = 432. | Step 3: So the product is 180/432. | Step 4: Divide the numerator and denominator by 2: 180/432 = 90/216. |

Step 5: Divide the numerator and denominator by 2: 90/216 = 45/108. | Step 6: Divide the numerator and denominator by 3: 45/108 = 15/36. | Step 7: Divide the numerator and denominator by 3: 15/36 = 5/12. | Step 8: Final Answer: 5/12.

Question: Exercise 3.4, Question 2 (v): Find the product: 6/12 x 21/24.

Answer: Step 1: Multiply the numerators: 6 x 21 = 126. | Step 2: Multiply the denominators: 12 x 24 = 288. | Step 3: So the product is 126/288. | Step 4: Divide the numerator and denominator by 2: 126/288 = 63/144. | Step 5: Divide the numerator and denominator by 3: 63/144 = 21/48. | Step 6: Divide the numerator and denominator by 3: 21/48 = 7/16. | Step 7: Final Answer: 7/16.

Question: Exercise 3.4, Question 2 (vi): Find the product: 8/16 x 4/8 x 3/2.

Answer: Step 1: Convert the mixed number 1 1/2 into an improper fraction: 1 x 2 + 1 = 3, so it is 3/2. | Step 2: Multiply the numerators: 8 x 4 x 3 = 96. | Step 3: Multiply the denominators: 16 x 8 x 2 = 256. | Step 4: So the product is 96/256. | Step 5: Divide the numerator and denominator by 2: 96/256 = 48/128. | Step 6: Divide the numerator and denominator by 2: 48/128 = 24/64. | Step 7: Divide the numerator and denominator by 2: 24/64 = 12/32. | Step 8: Divide the numerator and denominator by 2: 12/32 = 6/16. | Step 9: Divide the numerator and denominator by 2: 6/16 = 3/8. | Step 10: Final Answer: 3/8.

Question: Exercise 3.4, Question 2 (vii): Find the product: 4 4/5 x 4 3/3.

Answer: Step 1: Convert the mixed number 4 4/5 into an improper fraction: 4 x 5 + 4 = 24, so it is 24/5. | Step 2: Convert the mixed number 5 into an improper fraction: 5 x 3 + 0 = 15, so it is 15/3. | Step 3: Multiply the numerators: 24 x 15 = 360. | Step 4: Multiply the denominators: 5 x 3 = 15. | Step 5: So the product is 360/15. | Step 6: Divide the numerator and denominator by 3: 360/15 = 120/5. | Step 7: Divide the numerator

and denominator by 5: 120/5 = 24/1. | Step 8: Final Answer: 24.

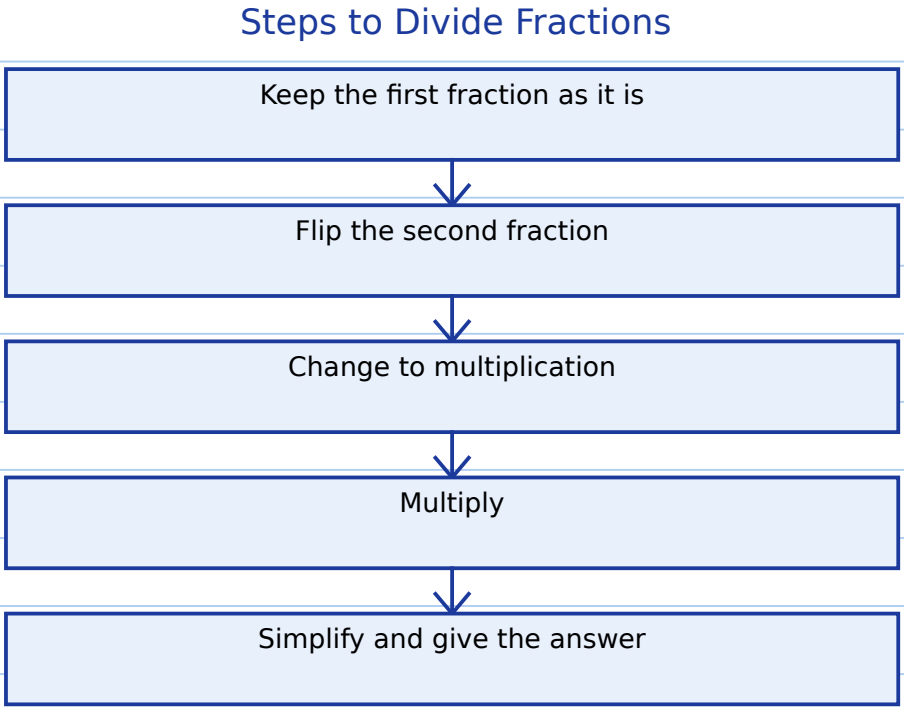
Question: Exercise 3.4, Question 2 (viii): Find the product: 8 1/4 x 3 2/6.

Answer: Step 1: Convert the mixed number 8 1/4 into an improper fraction: 8 x 4 + 1 = 33, so it is 33/4. | Step 2: Convert the mixed number 3 2/6 into an improper fraction: 3 x 6 + 2 = 20, so it is 20/6. | Step 3: Multiply the numerators: 33 x 20 = 660. | Step 4: Multiply the denominators: 4 x 6 = 24. | Step 5: So the product is 660/24. | Step 6: Divide the numerator and denominator by 2: 660/24 = 330/12. | Step 7: Divide the numerator and denominator by 2: 330/12 = 165/6. | Step 8: Divide the numerator and denominator by 3: 165/6 = 55/2. | Step 9: Final Answer: 27 1/2.

Question: Exercise 3.4, Question 2 (ix): Find the product: 8 4/5 x 3 6/8.

Answer: Step 1: Convert the mixed number 8 4/5 into an improper fraction: 8 x 5 + 4 = 44, so it is 44/5. | Step 2: Convert the mixed number 3 6/8 into an improper fraction: 3 x 8 + 6 = 30, so it is 30/8. | Step 3: Multiply the numerators: 44 x 30 = 1320. | Step 4: Multiply the denominators: 5 x 8 = 40. | Step 5: So the product is 1320/40. | Step 6: Divide the numerator and denominator by 2: 1320/40 = 660/20. | Step 7: Divide the numerator and denominator by 2: 660/20 = 330/10. | Step 8: Divide the numerator and denominator by 2: 330/10 = 165/5. | Step 9: Divide the numerator and denominator by 5: 165/5 = 33/1. | Step 10: Final Answer: 33.

Exercise 3.5: Division of Fractions



Question: Exercise 3.5, Question (i): Solve: 4/9 / 16/9.

Answer: Step 1: To divide, keep the first fraction 4/9 as it is. | Step 2: Flip the second fraction: 1 7/9 becomes 9/16. | Step 3: Now multiply: 4/9 x 9/16. | Step 4: Multiply the numerators: 4 x 9 = 36. | Step 5: Multiply the denominators: 9 x 16 = 144. | Step 6: So the product is 36/144. | Step 7: Divide the numerator and denominator by 2: 36/144 = 18/72. | Step 8: Divide the numerator and denominator by 2: 18/72 = 9/36. | Step 9: Divide the numerator and denominator by 3: 9/36 = 3/12. | Step 10: Divide the numerator and denominator by 3: 3/12 = 1/4. | Step 11: Final Answer: 1/4.

Question: Exercise 3.5, Question (ii): Solve: 4/8 / 2/12.

Answer: Step 1: To divide, keep the first fraction 4/8 as it is. | Step 2: Flip the second fraction: 2/12 becomes 12/2. | Step 3: Now multiply: 4/8 x 12/2. | Step 4: Convert the mixed number 6 into an improper fraction: 6 x 2 + 0 = 12, so it is 12/2. | Step 5: Multiply the numerators: 4 x 12 = 48. | Step 6: Multiply the denominators: 8 x 2 = 16. | Step 7: So the product is 48/16. |

Step 8: Divide the numerator and denominator by 2: 48/16 = 24/8. | Step 9: Divide the numerator and denominator by 2: 24/8 = 12/4. | Step 10: Divide the numerator and denominator by 2: 12/4 = 6/2. | Step 11: Divide the numerator and denominator by 2: 6/2 = 3/1. | Step 12: Final Answer: 3.

Question: Exercise 3.5, Question (iii): Solve: 15/20 / 3/12.

Answer: Step 1: To divide, keep the first fraction 15/20 as it is. | Step 2: Flip the second fraction: 3/12 becomes 12/3. | Step 3: Now multiply: 15/20 x 12/3. | Step 4: Convert the mixed number 4 into an improper fraction: 4 x 3 + 0 = 12, so it is 12/3. | Step 5: Multiply the numerators: 15 x 12 = 180. | Step 6: Multiply the denominators: 20 x 3 = 60. | Step 7: So the product is 180/60. | Step 8: Divide the numerator and denominator by 2: 180/60 = 90/30. | Step 9: Divide the numerator and denominator by 2: 90/30 = 45/15. | Step 10: Divide the numerator and denominator by 3: 45/15 = 15/5. | Step 11: Divide the numerator and denominator by 5: 15/5 = 3/1. | Step 12: Final Answer: 3.

Question: Exercise 3.5, Question (iv): Solve: 1/4 / 7/16.

Answer: Step 1: To divide, keep the first fraction 1/4 as it is. | Step 2: Flip the second fraction: 7/16 becomes 16/7. | Step 3: Now multiply: 1/4 x 16/7. | Step 4: Convert the mixed number 2 2/7 into an improper fraction: 2 x 7 + 2 = 16, so it is 16/7. | Step 5: Multiply the numerators: 1 x 16 = 16. | Step 6: Multiply the denominators: 4 x 7 = 28. | Step 7: So the product is 16/28. | Step 8: Divide the numerator and denominator by 2: 16/28 = 8/14. | Step 9: Divide the numerator and denominator by 2: 8/14 = 4/7. | Step 10: Final Answer: 4/7.

Question: Exercise 3.5, Question (v): Solve: 9/10 / 12/15.

Answer: Step 1: To divide, keep the first fraction 9/10 as it is. | Step 2: Flip the second fraction: 12/15 becomes 15/12. | Step 3: Now multiply: 9/10 x

15/12. | Step 4: Convert the mixed number 1 3/12 into an improper fraction: 1 x 12 + 3 = 15, so it is 15/12. | Step 5: Multiply the numerators: 9 x 15 = 135. | Step 6: Multiply the denominators: 10 x 12 = 120. | Step 7: So the product is 135/120. | Step 8: Divide the numerator and denominator by 3: 135/120 = 45/40. | Step 9: Divide the numerator and denominator by 5: 45/40 = 9/8. | Step 10: Final Answer: 1 1/8.

Question: Exercise 3.5, Question (vi): Solve: 9/30 / 6/12.

Answer: Step 1: To divide, keep the first fraction 9/30 as it is. | Step 2: Flip the second fraction: 6/12 becomes 12/6. | Step 3: Now multiply: 9/30 x 12/6. | Step 4: Convert the mixed number 2 into an improper fraction: 2 x 6 + 0 = 12, so it is 12/6. | Step 5: Multiply the numerators: 9 x 12 = 108. | Step 6: Multiply the denominators: 30 x 6 = 180. | Step 7: So the product is 108/180. | Step 8: Divide the numerator and denominator by 2: 108/180 = 54/90. | Step 9: Divide the numerator and denominator by 2: 54/90 = 27/45. | Step 10: Divide the numerator and denominator by 3: 27/45 = 9/15. | Step 11: Divide the numerator and denominator by 3: 9/15 = 3/5. | Step 12: Final Answer: 3/5.

Question: Exercise 3.5, Question (vii): Solve: 12/21 / 6/27.

Answer: Step 1: To divide, keep the first fraction 12/21 as it is. | Step 2: Flip the second fraction: 6/27 becomes 27/6. | Step 3: Now multiply: 12/21 x 27/6. | Step 4: Convert the mixed number 4 3/6 into an improper fraction: 4 x 6 + 3 = 27, so it is 27/6. | Step 5: Multiply the numerators: 12 x 27 = 324. | Step 6: Multiply the denominators: 21 x 6 = 126. | Step 7: So the product is 324/126. | Step 8: Divide the numerator and denominator by 2: 324/126 = 162/63. | Step 9: Divide the numerator and denominator by 3: 162/63 = 54/21. | Step 10: Divide the numerator and denominator by 3: 54/21 = 18/7. | Step 11: Final Answer: 2 4/7.

Question: Exercise 3.5, Question (viii): Solve: 5/25 / 20/30.

Answer: Step 1: To divide, keep the first fraction 5/25 as it is. | Step 2: Flip the second fraction: 20/30 becomes 30/20. | Step 3: Now multiply: 5/25 x 30/20. | Step 4: Convert the mixed number 1 10/20 into an improper fraction: 1 x 20 + 10 = 30, so it is 30/20. | Step 5: Multiply the numerators: 5 x 30 = 150. | Step 6: Multiply the denominators: 25 x 20 = 500. | Step 7: So the product is 150/500. | Step 8: Divide the numerator and denominator by 2: 150/500 = 75/250. | Step 9: Divide the numerator and denominator by 5: 75/250 = 15/50. | Step 10: Divide the numerator and denominator by 5: 15/50 = 3/10. | Step 11: Final Answer: 3/10.

Question: Exercise 3.5, Question (ix): Solve: 2 1/3 / 2/5.

Answer: Step 1: To divide, keep the first fraction 2 1/3 as it is. | Step 2: Flip the second fraction: 2/5 becomes 5/2. | Step 3: Now multiply: 2 1/3 x 5/2. | Step 4: Convert the mixed number 2 1/3 into an improper fraction: 2 x 3 + 1 = 7, so it is 7/3. | Step 5: Convert the mixed number 2 1/2 into an improper fraction: 2 x 2 + 1 = 5, so it is 5/2. | Step 6: Multiply the numerators: 7 x 5 = 35. | Step 7: Multiply the denominators: 3 x 2 = 6. | Step 8: So the product is 35/6. | Step 9: Final Answer: 5 5/6.

Question: Exercise 3.5, Question (x): Solve: 2 1/3 / 2/4.

Answer: Step 1: To divide, keep the first fraction 2 1/3 as it is. | Step 2: Flip the second fraction: 2/4 becomes 4/2. | Step 3: Now multiply: 2 1/3 x 4/2. | Step 4: Convert the mixed number 2 1/3 into an improper fraction: 2 x 3 + 1 = 7, so it is 7/3. | Step 5: Convert the mixed number 2 into an improper fraction: 2 x 2 + 0 = 4, so it is 4/2. | Step 6: Multiply the numerators: 7 x 4 = 28. | Step 7: Multiply the denominators: 3 x 2 = 6. | Step 8: So the product is 28/6. | Step 9: Divide the numerator and denominator by 2: 28/6 = 14/3. | Step 10: Final Answer: 4 2/3.

Question: Exercise 3.5, Question (xi): Solve: 2 4/5 / 2/3.

Answer: Step 1: To divide, keep the first fraction 2 4/5 as it is. | Step 2: Flip

the second fraction: 2/3 becomes 3/2. | Step 3: Now multiply: 2 4/5 x 3/2. | Step 4: Convert the mixed number 2 4/5 into an improper fraction: 2 x 5 + 4 = 14, so it is 14/5. | Step 5: Convert the mixed number 1 1/2 into an improper fraction: 1 x 2 + 1 = 3, so it is 3/2. | Step 6: Multiply the numerators: 14 x 3 = 42. | Step 7: Multiply the denominators: 5 x 2 = 10. | Step 8: So the product is 42/10. | Step 9: Divide the numerator and denominator by 2: 42/10 = 21/5. | Step 10: Final Answer: 4 1/5.

Question: Exercise 3.5, Question (xii): Solve: 3 6/83 / 4 2/4.

Answer: Step 1: To divide, keep the first fraction 3 6/83 as it is. | Step 2: Flip the second fraction: 4 1/2 becomes 2/9. | Step 3: Now multiply: 3 6/83 x 2/9. | Step 4: Convert the mixed number 3 6/83 into an improper fraction: 3 x 83 + 6 = 255, so it is 255/83. | Step 5: Multiply the numerators: 255 x 2 = 510. | Step 6: Multiply the denominators: 83 x 9 = 747. | Step 7: So the product is 510/747. | Step 8: Divide the numerator and denominator by 3: 510/747 = 170/249. | Step 9: Final Answer: 170/249.

Question: Exercise 3.5, Question (xiii): Solve: 6 / 2/3.

Answer: Step 1: To divide, keep the first fraction 6 as it is. | Step 2: Flip the second fraction: 2/3 becomes 3/2. | Step 3: Now multiply: 6 x 3/2. | Step 4: Convert the mixed number 6 into an improper fraction: 6 x 1 + 0 = 6, so it is 6/1. | Step 5: Convert the mixed number 1 1/2 into an improper fraction: 1 x 2 + 1 = 3, so it is 3/2. | Step 6: Multiply the numerators: 6 x 3 = 18. | Step 7: Multiply the denominators: 1 x 2 = 2. | Step 8: So the product is 18/2. | Step 9: Divide the numerator and denominator by 2: 18/2 = 9/1. | Step 10: Final Answer: 9.

Question: Exercise 3.5, Question (xiv): Solve: 2/3 / 8.

Answer: Step 1: To divide, keep the first fraction 2/3 as it is. | Step 2: Flip the second fraction: 8 becomes 1/8. | Step 3: Now multiply: 2/3 x 1/8. |

Step 4: Multiply the numerators: 2 x 1 = 2. | Step 5: Multiply the denominators: 3 x 8 = 24. | Step 6: So the product is 2/24. | Step 7: Divide the numerator and denominator by 2: 2/24 = 1/12. | Step 8: Final Answer: 1/12.

Question: Exercise 3.5, Question (xv): Solve: 12/5 / 9.

Answer: Step 1: To divide, keep the first fraction 2 2/5 as it is. | Step 2: Flip the second fraction: 9 becomes 1/9. | Step 3: Now multiply: 2 2/5 x 1/9. | Step 4: Convert the mixed number 2 2/5 into an improper fraction: 2 x 5 + 2 = 12, so it is 12/5. | Step 5: Multiply the numerators: 12 x 1 = 12. | Step 6: Multiply the denominators: 5 x 9 = 45. | Step 7: So the product is 12/45. | Step 8: Divide the numerator and denominator by 3: 12/45 = 4/15. | Step 9: Final Answer: 4/15.

Exercise 3.6: Fraction Word Problems

Question: Exercise 3.6, Question 1: A family used 4 1/4 litres of milk and another family used 7 1/2 litres in a day. How much milk was used by both families?

Answer: Step 1: The milk of both families is to be put together, so add the two amounts. | Step 2: The denominators are 4, 2. Their LCM is 4. | Step 3: Convert 4 1/4: 17 x 1 = 17, so it becomes 17/4. | Step 4: Convert 7 1/2: 15 x 2 = 30, so it becomes 30/4. | Step 5: Now add the numerators: 17 + 30 = 47; the result is 47/4. | Step 6: Final Answer: Both families used 11 3/4 litres of milk in all.

Question: Exercise 3.6, Question 2: My aunty buys a piece of ribbon 4 1/6 cm long; she buys another piece 4 2/5 cm long. How much longer was the second piece?

Answer: Step 1: To find how much longer the second piece was, subtract the first (smaller) length from the second (bigger) one. | Step 2: The

denominators are 5, 6. Their LCM is 30. | Step 3: Convert 4 2/5: 22 x 6 = 132, so it becomes 132/30. | Step 4: Convert 4 1/6: 25 x 5 = 125, so it becomes 125/30. | Step 5: Now subtract the numerators: 132 - 125 = 7; the result is 7/30. | Step 6: Final Answer: The second piece was 7/30 cm longer.

Question: Exercise 3.6, Question 3: I am 8 1/2 years old and my brother is 2 1/2 years younger than me. What is the age of my brother?

Answer: Step 1: My brother is 2 1/2 years younger than me, so subtract these years from my age. | Step 2: The denominators are 2, 2. Their LCM is 2. | Step 3: Convert 8 1/2: 17 x 1 = 17, so it becomes 17/2. | Step 4: Convert 2 1/2: 5 x 1 = 5, so it becomes 5/2. | Step 5: Now subtract the numerators: 17 - 5 = 12; the result is 12/2. | Step 6: Divide the numerator and denominator by 2: 12/2 = 6/1. | Step 7: Final Answer: My brother is 6 years old.

Question: Exercise 3.6, Question 4: Nazia has 4 4/5 metres of ribbon; she used 1/2 of this ribbon to tie a present. How many metres did she use?

Answer: Step 1: Nazia has 4 4/5 metres of ribbon and she used 1/2 of it, so multiply. | Step 2: Convert the mixed number 4 4/5 into an improper fraction: 4 x 5 + 4 = 24, so it is 24/5. | Step 3: Multiply the numerators: 24 x 1 = 24. | Step 4: Multiply the denominators: 5 x 2 = 10. | Step 5: So the product is 24/10. | Step 6: Divide the numerator and denominator by 2: 24/10 = 12/5. | Step 7: Final Answer: Nazia used 2 2/5 metres of ribbon.

Question: Exercise 3.6, Question 5: A tree is 18 metres tall; how many pieces of wood can be cut if each piece is 2/3 metre long?

Answer: Step 1: The tree is 18 metres long and each piece is 2/3 metre, so divide the total length by the length of one piece. | Step 2: To divide, keep the first fraction 18 as it is. | Step 3: Flip the second fraction: 2/3 becomes 3/2. | Step 4: Now multiply: 18 x 3/2. | Step 5: Convert the mixed number

18 into an improper fraction: 18 x 1 + 0 = 18, so it is 18/1. | Step 6: Convert the mixed number 1 1/2 into an improper fraction: 1 x 2 + 1 = 3, so it is 3/2. | Step 7: Multiply the numerators: 18 x 3 = 54. | Step 8: Multiply the denominators: 1 x 2 = 2. | Step 9: So the product is 54/2. | Step 10: Divide the numerator and denominator by 2: 54/2 = 27/1. | Step 11: Final Answer: 27 pieces of wood can be cut.

Question: Exercise 3.6, Question 6: A shopkeeper has 15 kg of peanuts; he places 2/5 kg in a bag. How many bags will he use?

Answer: Step 1: The shopkeeper has 15 kg of peanuts and each bag holds 2/5 kg, so divide. | Step 2: To divide, keep the first fraction 15 as it is. | Step 3: Flip the second fraction: 2/5 becomes 5/2. | Step 4: Now multiply: 15 x 5/2. | Step 5: Convert the mixed number 15 into an improper fraction: 15 x 1 + 0 = 15, so it is 15/1. | Step 6: Convert the mixed number 2 1/2 into an improper fraction: 2 x 2 + 1 = 5, so it is 5/2. | Step 7: Multiply the numerators: 15 x 5 = 75. | Step 8: Multiply the denominators: 1 x 2 = 2. | Step 9: So the product is 75/2. | Step 10: Final Answer: The shopkeeper will use 37 1/2 bags, that is, 38 bags will be needed (the last bag will be half filled).

Review Exercise 3

Question: Exercise Review 3, Question 1 (i): Convert the improper fraction 66/8 into a mixed fraction.

Answer: Step 1: Divide the numerator 66 by the denominator 8. | Step 2: 66 / 8 = 8 remainder 2. | Step 3: The quotient 8 becomes the whole part and the remainder 2 becomes the new numerator; the denominator stays 8. | Step 4: Simplify the fraction part: 2/8 = 1/4. | Step 5: Final Answer: 8 1/4.

Question: Exercise Review 3, Question 1 (ii): Convert the improper fraction

17/3 into a mixed fraction.

Answer: Step 1: Divide the numerator 17 by the denominator 3. | Step 2: 17 / 3 = 5 remainder 2. | Step 3: The quotient 5 becomes the whole part and the remainder 2 becomes the new numerator; the denominator stays 3. | Step 4: Final Answer: 5 2/3.

Question: Exercise Review 3, Question 1 (iii): Convert the improper fraction 23/5 into a mixed fraction.

Answer: Step 1: Divide the numerator 23 by the denominator 5. | Step 2: 23 / 5 = 4 remainder 3. | Step 3: The quotient 4 becomes the whole part and the remainder 3 becomes the new numerator; the denominator stays 5. | Step 4: Final Answer: 4 3/5.

Question: Exercise Review 3, Question 1 (iv): Convert the improper fraction 40/5 into a mixed fraction.

Answer: Step 1: Divide the numerator 40 by the denominator 5. | Step 2: 40 / 5 = 8 remainder 0. | Step 3: The quotient 8 becomes the whole part and the remainder 0 becomes the new numerator; the denominator stays 5. | Step 4: Final Answer: 8.

Question: Exercise Review 3, Question 2 (i): Express the mixed fraction 8 4/5 as an improper fraction.

Answer: Step 1: Multiply the whole part 8 by the denominator 5: 8 x 5 = 40. | Step 2: Add the numerator 4: 40 + 4 = 44; this is the new numerator. | Step 3: The denominator stays 5. | Step 4: Final Answer: 44/5.

Question: Exercise Review 3, Question 2 (ii): Express the mixed fraction 3 5/8 as an improper fraction.

Answer: Step 1: Multiply the whole part 3 by the denominator 8: 3 x 8 = 24. | Step 2: Add the numerator 5: 24 + 5 = 29; this is the new numerator. | Step 3: The denominator stays 8. | Step 4: Final Answer: 29/8.

Question: Exercise Review 3, Question 2 (iii): Express the mixed fraction 7

3/7 as an improper fraction.

Answer: Step 1: Multiply the whole part 7 by the denominator 7: 7 x 7 = 49. | Step 2: Add the numerator 3: 49 + 3 = 52; this is the new numerator. | Step 3: The denominator stays 7. | Step 4: Final Answer: 52/7.

Question: Exercise Review 3, Question 2 (iv): Express the mixed fraction 4 3/2 as an improper fraction.

Answer: Step 1: Multiply the whole part 4 by the denominator 2: 4 x 2 = 8. | Step 2: Add the numerator 3: 8 + 3 = 11; this is the new numerator. | Step 3: The denominator stays 2. | Step 4: Final Answer: 11/2.

Question: Exercise Review 3, Question 3 (i): Arrange in ascending order: 3/8, 9/8, 5/8.

Answer: Step 1: The LCM of all the denominators is 8. | Step 2: Convert every fraction to denominator 8: 3/8, 9/8, 5/8. | Step 3: Now arrange by looking at the numerators (ascending order). | Step 4: Final Answer: 3/8, 5/8, 9/8.

Question: Exercise Review 3, Question 3 (ii): Arrange in ascending order: 2/3, 1/4, 5/6.

Answer: Step 1: The LCM of all the denominators is 12. | Step 2: Convert every fraction to denominator 12: 8/12, 3/12, 10/12. | Step 3: Now arrange by looking at the numerators (ascending order). | Step 4: Final Answer: 1/4, 2/3, 5/6.

Question: Exercise Review 3, Question 3 (iii): Arrange in ascending order: 6/4, 8/3, 7/6, 5/2.

Answer: Step 1: The LCM of all the denominators is 12. | Step 2: Convert every fraction to denominator 12: 18/12, 32/12, 14/12, 30/12. | Step 3: Now arrange by looking at the numerators (ascending order). | Step 4: Final Answer: 7/6, 6/4, 5/2, 8/3.

Question: Exercise Review 3, Question 3 (iv): Arrange in ascending order:

2/7, 5/11, 4/5, 3/11.

Answer: Step 1: The LCM of all the denominators is 385. | Step 2: Convert every fraction to denominator 385: 110/385, 175/385, 308/385, 105/385. | Step 3: Now arrange by looking at the numerators (ascending order). | Step 4: Final Answer: 3/11, 2/7, 5/11, 4/5.

Question: Exercise Review 3, Question 4 (i): Arrange in descending order: 13/6, 11/6, 7/6.

Answer: Step 1: The LCM of all the denominators is 6. | Step 2: Convert every fraction to denominator 6: 13/6, 11/6, 7/6. | Step 3: Now arrange by looking at the numerators (descending order). | Step 4: Final Answer: 13/6, 11/6, 7/6.

Question: Exercise Review 3, Question 4 (ii): Arrange in descending order: 2/3, 3/5, 5/6.

Answer: Step 1: The LCM of all the denominators is 30. | Step 2: Convert every fraction to denominator 30: 20/30, 18/30, 25/30. | Step 3: Now arrange by looking at the numerators (descending order). | Step 4: Final Answer: 5/6, 2/3, 3/5.

Question: Exercise Review 3, Question 4 (iii): Arrange in descending order: 4/15, 3/10, 9/20, 4/5.

Answer: Step 1: The LCM of all the denominators is 60. | Step 2: Convert every fraction to denominator 60: 16/60, 18/60, 27/60, 48/60. | Step 3: Now arrange by looking at the numerators (descending order). | Step 4: Final Answer: 4/5, 9/20, 3/10, 4/15.

Question: Exercise Review 3, Question 4 (iv): Arrange in descending order: 4/3, 3/5, 5/7, 1/3.

Answer: Step 1: The LCM of all the denominators is 105. | Step 2: Convert every fraction to denominator 105: 140/105, 63/105, 75/105, 35/105. | Step 3: Now arrange by looking at the numerators (descending order). |

Step 4: Final Answer: 4/3, 5/7, 3/5, 1/3.

Question: Review Exercise 3, Question 5 (i): Reduce into the lowest terms: 30/42.

Answer: Step 1: Find the common factors of 30 and 42. | Step 2: Divide the numerator and denominator by 2: 30/42 = 15/21. | Step 3: Divide the numerator and denominator by 3: 15/21 = 5/7. | Step 4: Final Answer: 5/7.

Question: Review Exercise 3, Question 5 (ii): Reduce into the lowest terms: 24/44.

Answer: Step 1: Find the common factors of 24 and 44. | Step 2: Divide the numerator and denominator by 2: 24/44 = 12/22. | Step 3: Divide the numerator and denominator by 2: 12/22 = 6/11. | Step 4: Final Answer: 6/11.

Question: Review Exercise 3, Question 5 (iii): Reduce into the lowest terms: 2 4/8.

Answer: Step 1: First keep the whole part 2 aside; the fractional part is 4/8. | Step 2: Divide the numerator and denominator by 4: 4/8 = 1/2. | Step 3: Final Answer: 2 1/2.

Question: Review Exercise 3, Question 5 (iv): Reduce into the lowest terms: 3 3/6.

Answer: Step 1: First keep the whole part 3 aside; the fractional part is 3/6. | Step 2: Divide the numerator and denominator by 3: 3/6 = 1/2. | Step 3: Final Answer: 3 1/2.

Question: Review Exercise 3, Question 6 (i): Simplify: 3/5 + 2/4.

Answer: Step 1: The denominators are 5, 4. Their LCM is 20. | Step 2: Convert 3/5: 3 x 4 = 12, so it becomes 12/20. | Step 3: Convert 2/4: 2 x 5 = 10, so it becomes 10/20. | Step 4: Now add the numerators: 12 + 10 = 22; the result is 22/20. | Step 5: Divide the numerator and denominator by 2: 22/20 = 11/10. | Step 6: Final Answer: 1 1/10.

Question: Review Exercise 3, Question 6 (ii): Simplify: 7/9 + 4/6 + 2/3.

Answer: Step 1: The denominators are 9, 6, 3. Their LCM is 18. | Step 2: Convert 7/9: 7 x 2 = 14, so it becomes 14/18. | Step 3: Convert 4/6: 4 x 3 = 12, so it becomes 12/18. | Step 4: Convert 2/3: 2 x 6 = 12, so it becomes 12/18. | Step 5: Now add the numerators: 14 + 12 + 12 = 38; the result is 38/18. | Step 6: Divide the numerator and denominator by 2: 38/18 = 19/9. | Step 7: Final Answer: 2 1/9.

Question: Review Exercise 3, Question 6 (iii): Simplify: 4/5 + 6/10 + 1/2.

Answer: Step 1: The denominators are 5, 10, 2. Their LCM is 10. | Step 2: Convert 4/5: 4 x 2 = 8, so it becomes 8/10. | Step 3: Convert 6/10: 6 x 1 = 6, so it becomes 6/10. | Step 4: Convert 1/2: 1 x 5 = 5, so it becomes 5/10. | Step 5: Now add the numerators: 8 + 6 + 5 = 19; the result is 19/10. | Step 6: Final Answer: 1 9/10.

Question: Review Exercise 3, Question 6 (iv): Simplify: 2/3 + 4 1/2.

Answer: Step 1: The denominators are 3, 2. Their LCM is 6. | Step 2: Convert 2/3: 2 x 2 = 4, so it becomes 4/6. | Step 3: Convert 4 1/2: 9 x 3 = 27, so it becomes 27/6. | Step 4: Now add the numerators: 4 + 27 = 31; the result is 31/6. | Step 5: Final Answer: 5 1/6.

Question: Review Exercise 3, Question 7 (i): Simplify: 3/4 - 4/8.

Answer: Step 1: The denominators are 4, 8. Their LCM is 8. | Step 2: Convert 3/4: 3 x 2 = 6, so it becomes 6/8. | Step 3: Convert 4/8: 4 x 1 = 4, so it becomes 4/8. | Step 4: Now subtract the numerators: 6 - 4 = 2; the result is 2/8. | Step 5: Divide the numerator and denominator by 2: 2/8 = 1/4. | Step 6: Final Answer: 1/4.

Question: Review Exercise 3, Question 7 (ii): Simplify: 2 1/5 - 1 4/5.

Answer: Step 1: The denominators are 5, 5. Their LCM is 5. | Step 2: Convert 2 1/5: 11 x 1 = 11, so it becomes 11/5. | Step 3: Convert 1 4/5: 9 x 1 = 9, so it becomes 9/5. | Step 4: Now subtract the numerators: 11 - 9

= 2; the result is 2/5. | Step 5: Final Answer: 2/5.

Question: Review Exercise 3, Question 7 (iii): Simplify: 5 2/3 - 2 2/4.

Answer: Step 1: The denominators are 3, 4. Their LCM is 12. | Step 2: Convert 5 2/3: 17 x 4 = 68, so it becomes 68/12. | Step 3: Convert 2 2/4: 10 x 3 = 30, so it becomes 30/12. | Step 4: Now subtract the numerators: 68 - 30 = 38; the result is 38/12. | Step 5: Divide the numerator and denominator by 2: 38/12 = 19/6. | Step 6: Final Answer: 3 1/6.

Question: Review Exercise 3, Question 7 (iv): Simplify: 13/15 - 5/10 - 1/5.

Answer: Step 1: The denominators are 15, 10, 5. Their LCM is 30. | Step 2: Convert 13/15: 13 x 2 = 26, so it becomes 26/30. | Step 3: Convert 5/10: 5 x 3 = 15, so it becomes 15/30. | Step 4: Convert 1/5: 1 x 6 = 6, so it becomes 6/30. | Step 5: Now subtract the numerators: 26 - 15 - 6 = 5; the result is 5/30. | Step 6: Divide the numerator and denominator by 5: 5/30 = 1/6. | Step 7: Final Answer: 1/6.

Question: Review Exercise 3, Question 8 (i): Multiply, giving the answer in the lowest terms: 2 2/5 x 2 1/2.

Answer: Step 1: Convert the mixed number 2 2/5 into an improper fraction: 2 x 5 + 2 = 12, so it is 12/5. | Step 2: Convert the mixed number 2 1/2 into an improper fraction: 2 x 2 + 1 = 5, so it is 5/2. | Step 3: Multiply the numerators: 12 x 5 = 60. | Step 4: Multiply the denominators: 5 x 2 = 10. | Step 5: So the product is 60/10. | Step 6: Divide the numerator and denominator by 2: 60/10 = 30/5. | Step 7: Divide the numerator and denominator by 5: 30/5 = 6/1. | Step 8: Final Answer: 6.

Question: Review Exercise 3, Question 8 (ii): Multiply, giving the answer in the lowest terms: 1 2/3 x 2 1/4.

Answer: Step 1: Convert the mixed number 1 2/3 into an improper fraction: 1 x 3 + 2 = 5, so it is 5/3. | Step 2: Convert the mixed number 2 1/4 into an improper fraction: 2 x 4 + 1 = 9, so it is 9/4. | Step 3: Multiply

the numerators: 5 x 9 = 45. | Step 4: Multiply the denominators: 3 x 4 = 12. | Step 5: So the product is 45/12. | Step 6: Divide the numerator and denominator by 3: 45/12 = 15/4. | Step 7: Final Answer: 3 3/4.

Question: Review Exercise 3, Question 8 (iii): Multiply, giving the answer in the lowest terms: 28/35 x 12/21.

Answer: Step 1: Multiply the numerators: 28 x 12 = 336. | Step 2: Multiply the denominators: 35 x 21 = 735. | Step 3: So the product is 336/735. | Step 4: Divide the numerator and denominator by 3: 336/735 = 112/245. | Step 5: Divide the numerator and denominator by 7: 112/245 = 16/35. | Step 6: Final Answer: 16/35.

Question: Review Exercise 3, Question 8 (iv): Multiply, giving the answer in the lowest terms: 1 3/5 x 1 5/10.

Answer: Step 1: Convert the mixed number 1 3/5 into an improper fraction: 1 x 5 + 3 = 8, so it is 8/5. | Step 2: Convert the mixed number 1 5/10 into an improper fraction: 1 x 10 + 5 = 15, so it is 15/10. | Step 3: Multiply the numerators: 8 x 15 = 120. | Step 4: Multiply the denominators: 5 x 10 = 50. | Step 5: So the product is 120/50. | Step 6: Divide the numerator and denominator by 2: 120/50 = 60/25. | Step 7: Divide the numerator and denominator by 5: 60/25 = 12/5. | Step 8: Final Answer: 2 2/5.

Question: Review Exercise 3, Question 9 (i): Divide, giving the answer in the lowest terms: 5/6 / 25.

Answer: Step 1: To divide, keep the first fraction 5/6 as it is. | Step 2: Flip the second fraction: 25 becomes 1/25. | Step 3: Now multiply: 5/6 x 1/25. | Step 4: Multiply the numerators: 5 x 1 = 5. | Step 5: Multiply the denominators: 6 x 25 = 150. | Step 6: So the product is 5/150. | Step 7: Divide the numerator and denominator by 5: 5/150 = 1/30. | Step 8: Final Answer: 1/30.

Question: Review Exercise 3, Question 9 (ii): Divide, giving the answer in the lowest terms: 3/4 / 5/8.

Answer: Step 1: To divide, keep the first fraction 3/4 as it is. | Step 2: Flip the second fraction: 5/8 becomes 8/5. | Step 3: Now multiply: 3/4 x 8/5. | Step 4: Convert the mixed number 1 3/5 into an improper fraction: 1 x 5 + 3 = 8, so it is 8/5. | Step 5: Multiply the numerators: 3 x 8 = 24. | Step 6: Multiply the denominators: 4 x 5 = 20. | Step 7: So the product is 24/20. | Step 8: Divide the numerator and denominator by 2: 24/20 = 12/10. | Step 9: Divide the numerator and denominator by 2: 12/10 = 6/5. | Step 10: Final Answer: 1 1/5.

Question: Review Exercise 3, Question 9 (iii): Divide, giving the answer in the lowest terms: 5 1/2 / 1 1/4.

Answer: Step 1: To divide, keep the first fraction 5 1/2 as it is. | Step 2: Flip the second fraction: 1 1/4 becomes 4/5. | Step 3: Now multiply: 5 1/2 x 4/5. | Step 4: Convert the mixed number 5 1/2 into an improper fraction: 5 x 2 + 1 = 11, so it is 11/2. | Step 5: Multiply the numerators: 11 x 4 = 44. | Step 6: Multiply the denominators: 2 x 5 = 10. | Step 7: So the product is 44/10. | Step 8: Divide the numerator and denominator by 2: 44/10 = 22/5. | Step 9: Final Answer: 4 2/5.

Question: Review Exercise 3, Question 9 (iv): Divide, giving the answer in the lowest terms: 7 3/5 / 1 7/12.

Answer: Step 1: To divide, keep the first fraction 7 3/5 as it is. | Step 2: Flip the second fraction: 1 7/12 becomes 12/19. | Step 3: Now multiply: 7 3/5 x 12/19. | Step 4: Convert the mixed number 7 3/5 into an improper fraction: 7 x 5 + 3 = 38, so it is 38/5. | Step 5: Multiply the numerators: 38 x 12 = 456. | Step 6: Multiply the denominators: 5 x 19 = 95. | Step 7: So the product is 456/95. | Step 8: Divide the numerator and denominator by 19: 456/95 = 24/5. | Step 9: Final Answer: 4 4/5.

Question: Review Exercise 3, Question 10: Nasir jogged 7 1/2 km and his friend jogged 4 2/3 km. How much more distance did Nasir cover?

Answer: Step 1: Nasir covered more distance, so subtract his friend's distance from Nasir's distance. | Step 2: The denominators are 2, 3. Their LCM is 6. | Step 3: Convert 7 1/2: 15 x 3 = 45, so it becomes 45/6. | Step 4: Convert 4 2/3: 14 x 2 = 28, so it becomes 28/6. | Step 5: Now subtract the numerators: 45 - 28 = 17; the result is 17/6. | Step 6: Final Answer: Nasir covered 2 5/6 km more distance.

Question: Review Exercise 3, Question 11: There are 32 balloons in a room, out of which 1/4 are red. How many balloons are red?

Answer: Step 1: There are 32 balloons in all and 1/4 of them are red, so find 1/4 of 32. | Step 2: Convert the mixed number 32 into an improper fraction: 32 x 1 + 0 = 32, so it is 32/1. | Step 3: Multiply the numerators: 1 x 32 = 32. | Step 4: Multiply the denominators: 4 x 1 = 4. | Step 5: So the product is 32/4. | Step 6: Divide the numerator and denominator by 2: 32/4 = 16/2. | Step 7: Divide the numerator and denominator by 2: 16/2 = 8/1. | Step 8: Final Answer: 8 balloons are red.

Question: Review Exercise 3, Question 12: Javeria's doll dress requires 3/4 metre of fabric. How many dresses can be made from 3 3/4 metres?

Answer: Step 1: The fabric is 3 3/4 metres in all and each dress needs 3/4 metre, so divide. | Step 2: To divide, keep the first fraction 3 3/4 as it is. | Step 3: Flip the second fraction: 3/4 becomes 4/3. | Step 4: Now multiply: 3 3/4 x 4/3. | Step 5: Convert the mixed number 3 3/4 into an improper fraction: 3 x 4 + 3 = 15, so it is 15/4. | Step 6: Convert the mixed number 1 1/3 into an improper fraction: 1 x 3 + 1 = 4, so it is 4/3. | Step 7: Multiply the numerators: 15 x 4 = 60. | Step 8: Multiply the denominators: 4 x 3 = 12. | Step 9: So the product is 60/12. | Step 10: Divide the numerator and denominator by 2: 60/12 = 30/6. | Step 11: Divide the numerator and

denominator by 2: 30/6 = 15/3. | Step 12: Divide the numerator and denominator by 3: 15/3 = 5/1. | Step 13: Final Answer: 5 dresses can be made.

Question: Review Exercise 3, Question 13: A factory makes 9 1/2 litres of apple juice each hour. How many litres will it make in 14 hours?

Answer: Step 1: The factory makes 9 1/2 litres of juice each hour; multiply for 14 hours. | Step 2: Convert the mixed number 9 1/2 into an improper fraction: 9 x 2 + 1 = 19, so it is 19/2. | Step 3: Convert the mixed number 14 into an improper fraction: 14 x 1 + 0 = 14, so it is 14/1. | Step 4: Multiply the numerators: 19 x 14 = 266. | Step 5: Multiply the denominators: 2 x 1 = 2. | Step 6: So the product is 266/2. | Step 7: Divide the numerator and denominator by 2: 266/2 = 133/1. | Step 8: Final Answer: 133 litres of juice will be made in 14 hours.

4. Decimal and Fractions

Activity: Decimal Fractions

Question: Activity (a): Write the fraction and decimal fraction for the coloured portions.

Answer: Step 1: The strip is divided into 10 equal parts and 4 parts are coloured. | Step 2: The coloured portion as a fraction: 4/10. | Step 3: In decimal form it is written as 0.4. | Step 4: Final Answer: 4/10 = 0.4.

Question: Activity (b): Write the fraction and decimal fraction for the coloured portions.

Answer: Step 1: The circle is divided into 10 equal parts and 3 parts are coloured. | Step 2: The coloured portion as a fraction: 3/10. | Step 3: In decimal form it is written as 0.3. | Step 4: Final Answer: 3/10 = 0.3.

Question: Activity (c): Write the fraction and decimal fraction for the coloured portions.

Answer: Step 1: The figure shows 3 full strips (10 parts each) and 3 parts of another strip coloured. | Step 2: The coloured portion as a fraction: 33/10. | Step 3: In decimal form it is written as 3.3. | Step 4: Final Answer: 33/10 = 3.3.

Question: Activity (d): Write the fraction and decimal fraction for the coloured portions.

Answer: Step 1: The figure shows 2 full circles (10 parts each) and 7 parts of another circle coloured. | Step 2: The coloured portion as a fraction: 27/10. | Step 3: In decimal form it is written as 2.7. | Step 4: Final Answer: 27/10 = 2.7.

Question: Activity: Fill the table with the denominator and the number of

decimal places for: 0.3, 13.5, 0.84, 732.52, 0.324, 583.123.

Answer: Step 1: Look at the digits after the decimal point in each number. | Step 2: If there is 1 digit after the point, the denominator is 10; for 2 digits it is 100; for 3 digits it is 1000. | Step 3: 0.3: denominator 10, decimal places: One (1). | Step 4: 13.5: denominator 10, decimal places: One (1). | Step 5: 0.84: denominator 100, decimal places: Two (2). | Step 6: 732.52: denominator 100, decimal places: Two (2). | Step 7: 0.324: denominator 1000, decimal places: Three (3). | Step 8: 583.123: denominator 1000, decimal places: Three (3). | Step 9: Final Answer: the completed table is given in the steps above.

Exercise 4.1: Decimal Figures and Place Value

Question: Exercise 4.1, Question 1 (i): Draw the figure for 0.3.

Answer: Step 1: Draw the figure for 0.3: Divide a strip into 10 equal parts and shade 3 parts. | Step 2: Final Answer: the figure for 0.3 is described above.

Question: Exercise 4.1, Question 1 (ii): Draw the figure for 0.8.

Answer: Step 1: Draw the figure for 0.8: Divide a strip into 10 equal parts and shade 8 parts. | Step 2: Final Answer: the figure for 0.8 is described above.

Question: Exercise 4.1, Question 1 (iii): Draw the figure for 1.3.

Answer: Step 1: Draw the figure for 1.3: Shade one full strip and 3 parts (out of 10) of another strip. | Step 2: Final Answer: the figure for 1.3 is described above.

Question: Exercise 4.1, Question 1 (iv): Draw the figure for 1.7.

Answer: Step 1: Draw the figure for 1.7: Shade one full strip and 7 parts (out of 10) of another strip. | Step 2: Final Answer: the figure for 1.7 is described above.

Question: Exercise 4.1, Question 1 (v): Draw the figure for 0.34.

Answer: Step 1: Draw the figure for 0.34: In a grid of 100 squares, shade 34 small squares. | Step 2: Final Answer: the figure for 0.34 is described above.

Question: Exercise 4.1, Question 1 (vi): Draw the figure for 1.5.

Answer: Step 1: Draw the figure for 1.5: Shade one full strip and 5 parts (out of 10) of another strip. | Step 2: Final Answer: the figure for 1.5 is described above.

Question: Exercise 4.1, Question 2 (i): Write the number of decimal places in 123.1.

Answer: Step 1: Find the decimal point in 123.1. | Step 2: Count the digits after the point: there are 1 digits. | Step 3: Final Answer: 123.1 has one (1) decimal place(s).

Question: Exercise 4.1, Question 2 (ii): Write the number of decimal places in 57.322.

Answer: Step 1: Find the decimal point in 57.322. | Step 2: Count the digits after the point: there are 3 digits. | Step 3: Final Answer: 57.322 has three (3) decimal place(s).

Question: Exercise 4.1, Question 2 (iii): Write the number of decimal places in 0.87.

Answer: Step 1: Find the decimal point in 0.87. | Step 2: Count the digits after the point: there are 2 digits. | Step 3: Final Answer: 0.87 has two (2) decimal place(s).

Question: Exercise 4.1, Question 2 (iv): Write the number of decimal places in 139.45.

Answer: Step 1: Find the decimal point in 139.45. | Step 2: Count the digits after the point: there are 2 digits. | Step 3: Final Answer: 139.45 has two (2) decimal place(s).

Question: Exercise 4.1, Question 2 (v): Write the number of decimal places in 59.552.

Answer: Step 1: Find the decimal point in 59.552. | Step 2: Count the digits after the point: there are 3 digits. | Step 3: Final Answer: 59.552 has three (3) decimal place(s).

Question: Exercise 4.1, Question 2 (vi): Write the number of decimal places in 735.9.

Answer: Step 1: Find the decimal point in 735.9. | Step 2: Count the digits after the point: there are 1 digits. | Step 3: Final Answer: 735.9 has one (1) decimal place(s).

Question: Exercise 4.1, Question 3 (i): Write the place value and value of the encircled digit(s) in 725.04.

Answer: Step 1: Write 725.04 and name the place of each marked digit from the decimal point. | Step 2: The digit 4 is in the hundredths place; its value is 0.04. | Step 3: Final Answer: 4 (hundredths) = 0.04.

Question: Exercise 4.1, Question 3 (ii): Write the place value and value of the encircled digit(s) in 135.85.

Answer: Step 1: Write 135.85 and name the place of each marked digit from the decimal point. | Step 2: The digit 8 is in the tenths place; its value is 0.8. | Step 3: Final Answer: 8 (tenths) = 0.8.

Question: Exercise 4.1, Question 3 (iii): Write the place value and value of the encircled digit(s) in 5.092.

Answer: Step 1: Write 5.092 and name the place of each marked digit from the decimal point. | Step 2: The digit 2 is in the thousandths place; its value is 0.002. | Step 3: Final Answer: 2 (thousandths) = 0.002.

Question: Exercise 4.1, Question 3 (iv): Write the place value and value of the encircled digit(s) in 124.32.

Answer: Step 1: Write 124.32 and name the place of each marked digit

from the decimal point. | Step 2: The digit 1 is in the hundreds place; its value is 100. | Step 3: The digit 2 is in the hundredths place; its value is 0.02. | Step 4: Final Answer: 1 (hundreds) = 100; 2 (hundredths) = 0.02.

Question: Exercise 4.1, Question 3 (v): Write the place value and value of the encircled digit(s) in 43.59.

Answer: Step 1: Write 43.59 and name the place of each marked digit from the decimal point. | Step 2: The digit 4 is in the tens place; its value is 40. | Step 3: The digit 9 is in the hundredths place; its value is 0.09. | Step 4: Final Answer: 4 (tens) = 40; 9 (hundredths) = 0.09.

Question: Exercise 4.1, Question 3 (vi): Write the place value and value of the encircled digit(s) in 34.67.

Answer: Step 1: Write 34.67 and name the place of each marked digit from the decimal point. | Step 2: The digit 3 is in the tens place; its value is 30. | Step 3: The digit 7 is in the hundredths place; its value is 0.07. | Step 4: Final Answer: 3 (tens) = 30; 7 (hundredths) = 0.07.

Question: Exercise 4.1, Question 4 (i): Write down the place value and value of each digit in 575.1.

Answer: Step 1: Go through 575.1 digit by digit from the left. | Step 2: The digit 5 is in the hundreds place; its value is 500. | Step 3: The digit 7 is in the tens place; its value is 70. | Step 4: The digit 5 is in the ones place; its value is 5. | Step 5: The digit 1 is in the tenths place; its value is 0.1. | Step 6: Final Answer: the values of all the digits are given in the steps above.

Question: Exercise 4.1, Question 4 (ii): Write down the place value and value of each digit in 0.534.

Answer: Step 1: Go through 0.534 digit by digit from the left. | Step 2: The digit 0 is in the ones place; its value is 0. | Step 3: The digit 5 is in the tenths place; its value is 0.5. | Step 4: The digit 3 is in the hundredths place; its value is 0.03. | Step 5: The digit 4 is in the thousandths place; its

value is 0.004. | Step 6: Final Answer: the values of all the digits are given in the steps above.

Question: Exercise 4.1, Question 4 (iii): Write down the place value and value of each digit in 18.97.

Answer: Step 1: Go through 18.97 digit by digit from the left. | Step 2: The digit 1 is in the tens place; its value is 10. | Step 3: The digit 8 is in the ones place; its value is 8. | Step 4: The digit 9 is in the tenths place; its value is 0.9. | Step 5: The digit 7 is in the hundredths place; its value is 0.07. | Step 6: Final Answer: the values of all the digits are given in the steps above.

Question: Exercise 4.1, Question 4 (iv): Write down the place value and value of each digit in 9.02.

Answer: Step 1: Go through 9.02 digit by digit from the left. | Step 2: The digit 9 is in the ones place; its value is 9. | Step 3: The digit 0 is in the tenths place; its value is 0. | Step 4: The digit 2 is in the hundredths place; its value is 0.02. | Step 5: Final Answer: the values of all the digits are given in the steps above.

Question: Exercise 4.1, Question 4 (v): Write down the place value and value of each digit in 75.84.

Answer: Step 1: Go through 75.84 digit by digit from the left. | Step 2: The digit 7 is in the tens place; its value is 70. | Step 3: The digit 5 is in the ones place; its value is 5. | Step 4: The digit 8 is in the tenths place; its value is 0.8. | Step 5: The digit 4 is in the hundredths place; its value is 0.04. | Step 6: Final Answer: the values of all the digits are given in the steps above.

Question: Exercise 4.1, Question 4 (vi): Write down the place value and value of each digit in 51.15.

Answer: Step 1: Go through 51.15 digit by digit from the left. | Step 2: The

digit 5 is in the tens place; its value is 50. | Step 3: The digit 1 is in the ones place; its value is 1. | Step 4: The digit 1 is in the tenths place; its value is 0.1. | Step 5: The digit 5 is in the hundredths place; its value is 0.05. | Step 6: Final Answer: the values of all the digits are given in the steps above.

Question: Exercise 4.1, Question 5 (a): Match the coloured part of the figure with the correct decimal fraction.

Answer: Step 1: Look at the figure: 9 out of 10 vertical strips shaded. | Step 2: The coloured part in decimal form is 0.9. | Step 3: Final Answer: 0.9.

Question: Exercise 4.1, Question 5 (b): Match the coloured part of the figure with the correct decimal fraction.

Answer: Step 1: Look at the figure: 7 out of 10 horizontal strips shaded. | Step 2: The coloured part in decimal form is 0.7. | Step 3: Final Answer: 0.7.

Question: Exercise 4.1, Question 5 (c): Match the coloured part of the figure with the correct decimal fraction.

Answer: Step 1: Look at the figure: two circles of 10 parts: one full and 4 parts of the other shaded. | Step 2: The coloured part in decimal form is 1.4. | Step 3: Final Answer: 1.4.

Question: Exercise 4.1, Question 5 (d): Match the coloured part of the figure with the correct decimal fraction.

Answer: Step 1: Look at the figure: 3 strips of 10 parts: two full and 1 part of the third shaded. | Step 2: The coloured part in decimal form is 2.1. | Step 3: Final Answer: 2.1.

Question: Exercise 4.1, Question 5 (e): Match the coloured part of the figure with the correct decimal fraction.

Answer: Step 1: Look at the figure: 23 squares of a 100-square grid

shaded. | Step 2: The coloured part in decimal form is 0.23. | Step 3: Final Answer: 0.23.

Question: Exercise 4.1, Question 5 (f): Match the coloured part of the figure with the correct decimal fraction.

Answer: Step 1: Look at the figure: 55 squares of a 100-square grid shaded. | Step 2: The coloured part in decimal form is 0.55. | Step 3: Final Answer: 0.55.

Question: Exercise 4.1, Question 5 (g): Match the coloured part of the figure with the correct decimal fraction.

Answer: Step 1: Look at the figure: 2 strips of 10 parts: one full and 6 parts of the other shaded. | Step 2: The coloured part in decimal form is 1.6. | Step 3: Final Answer: 1.6.

Question: Exercise 4.1, Question 5 (h): Match the coloured part of the figure with the correct decimal fraction.

Answer: Step 1: Look at the figure: 7 squares of a 100-square grid shaded. | Step 2: The coloured part in decimal form is 0.07. | Step 3: Final Answer: 0.07.

Question: Exercise 4.1, Question 5 (i): Match the coloured part of the figure with the correct decimal fraction.

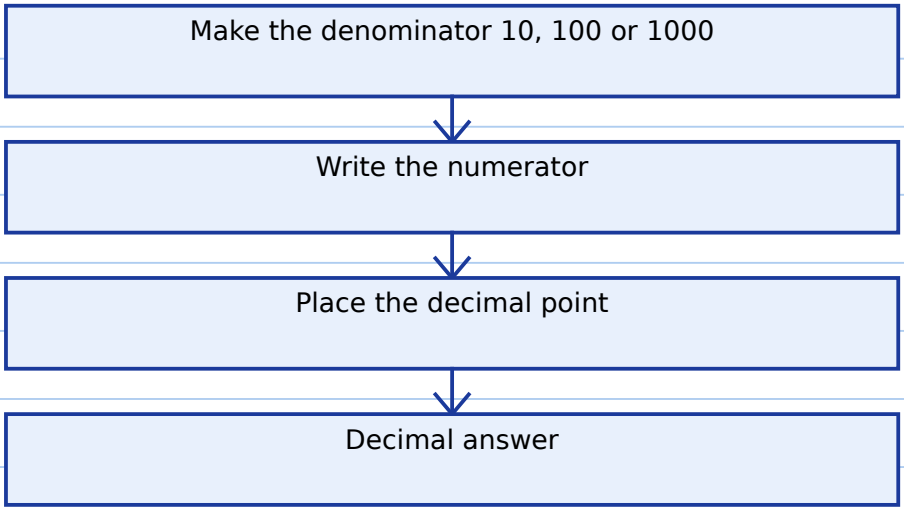
Answer: Step 1: Look at the figure: two circles of 10 parts: one full and 8 parts of the other shaded. | Step 2: The coloured part in decimal form is 1.8. | Step 3: Final Answer: 1.8.

Question: Exercise 4.1, Question 5 (j): Match the coloured part of the figure with the correct decimal fraction.

Answer: Step 1: Look at the figure: 3 strips of 10 parts: two full and 2 parts of the third shaded. | Step 2: The coloured part in decimal form is 2.2. | Step 3: Final Answer: 2.2.

Exercise 4.2: Conversion of Fractions and Decimals

Converting a Fraction into a Decimal



Question: Exercise 4.2, Question 1 (i): Convert the fraction 1/100 into a decimal.

Answer: Step 1: The denominator is 100, which has 2 zero(s). | Step 2: Write the numerator 1 and place the decimal point 2 place(s) from the right. | Step 3: Final Answer: 1/100 = 0.01.

Question: Exercise 4.2, Question 1 (ii): Convert the fraction 5/10 into a decimal.

Answer: Step 1: The denominator is 10, which has 1 zero(s). | Step 2: Write the numerator 5 and place the decimal point 1 place(s) from the right. | Step 3: Final Answer: 5/10 = 0.5.

Question: Exercise 4.2, Question 1 (iii): Convert the fraction 23/100 into a decimal.

Answer: Step 1: The denominator is 100, which has 2 zero(s). | Step 2: Write the numerator 23 and place the decimal point 2 place(s) from the right. | Step 3: Final Answer: 23/100 = 0.23.

Question: Exercise 4.2, Question 1 (iv): Convert the fraction 431/1000 into a decimal.

Answer: Step 1: The denominator is 1000, which has 3 zero(s). | Step 2: Write the numerator 431 and place the decimal point 3 place(s) from the right. | Step 3: Final Answer: 431/1000 = 0.431.

Question: Exercise 4.2, Question 1 (v): Convert the fraction 231/100 into a decimal.

Answer: Step 1: The denominator is 100, which has 2 zero(s). | Step 2: Write the numerator 231 and place the decimal point 2 place(s) from the right. | Step 3: Final Answer: 231/100 = 2.31.

Question: Exercise 4.2, Question 1 (vi): Convert the fraction 17/10 into a decimal.

Answer: Step 1: The denominator is 10, which has 1 zero(s). | Step 2: Write the numerator 17 and place the decimal point 1 place(s) from the right. | Step 3: Final Answer: 17/10 = 1.7.

Question: Exercise 4.2, Question 1 (vii): Convert the fraction 29/1000 into a decimal.

Answer: Step 1: The denominator is 1000, which has 3 zero(s). | Step 2: Write the numerator 29 and place the decimal point 3 place(s) from the right. | Step 3: Final Answer: 29/1000 = 0.029.

Question: Exercise 4.2, Question 1 (viii): Convert the fraction 11/100 into a decimal.

Answer: Step 1: The denominator is 100, which has 2 zero(s). | Step 2: Write the numerator 11 and place the decimal point 2 place(s) from the right. | Step 3: Final Answer: 11/100 = 0.11.

Question: Exercise 4.2, Question 2 (i): Convert the fraction 7/25 into a decimal.

Answer: Step 1: To make the denominator a power of ten, multiply the numerator and denominator by 4. | Step 2: 7 x 4 = 28; the new fraction is 28/100. | Step 3: Now the denominator is 100; write the numerator and

place the point 2 place(s) from the right: 0.28. | Step 4: Final Answer: 7/25 = 0.28.

Question: Exercise 4.2, Question 2 (ii): Convert the fraction 42/4 into a decimal.

Answer: Step 1: To make the denominator a power of ten, multiply the numerator and denominator by 25. | Step 2: 42 x 25 = 1050; the new fraction is 1050/100. | Step 3: Now the denominator is 100; write the numerator and place the point 2 place(s) from the right: 10.5. | Step 4: Final Answer: 42/4 = 10.5.

Question: Exercise 4.2, Question 2 (iii): Convert the fraction 9/2 into a decimal.

Answer: Step 1: To make the denominator a power of ten, multiply the numerator and denominator by 5. | Step 2: 9 x 5 = 45; the new fraction is 45/10. | Step 3: Now the denominator is 10; write the numerator and place the point 1 place(s) from the right: 4.5. | Step 4: Final Answer: 9/2 = 4.5.

Question: Exercise 4.2, Question 2 (iv): Convert the fraction 3/4 into a decimal.

Answer: Step 1: To make the denominator a power of ten, multiply the numerator and denominator by 25. | Step 2: 3 x 25 = 75; the new fraction is 75/100. | Step 3: Now the denominator is 100; write the numerator and place the point 2 place(s) from the right: 0.75. | Step 4: Final Answer: 3/4 = 0.75.

Question: Exercise 4.2, Question 2 (v): Convert the fraction 13/50 into a decimal.

Answer: Step 1: To make the denominator a power of ten, multiply the numerator and denominator by 2. | Step 2: 13 x 2 = 26; the new fraction is 26/100. | Step 3: Now the denominator is 100; write the numerator and place the point 2 place(s) from the right: 0.26. | Step 4: Final Answer:

13/50 = 0.26.

Question: Exercise 4.2, Question 2 (vi): Convert the fraction 3/8 into a decimal.

Answer: Step 1: To make the denominator a power of ten, multiply the numerator and denominator by 125. | Step 2: 3 x 125 = 375; the new fraction is 375/1000. | Step 3: Now the denominator is 1000; write the numerator and place the point 3 place(s) from the right: 0.375. | Step 4: Final Answer: 3/8 = 0.375.

Question: Exercise 4.2, Question 2 (vii): Convert the fraction 19/25 into a decimal.

Answer: Step 1: To make the denominator a power of ten, multiply the numerator and denominator by 4. | Step 2: 19 x 4 = 76; the new fraction is 76/100. | Step 3: Now the denominator is 100; write the numerator and place the point 2 place(s) from the right: 0.76. | Step 4: Final Answer: 19/25 = 0.76.

Question: Exercise 4.2, Question 2 (viii): Convert the fraction 30/50 into a decimal.

Answer: Step 1: To make the denominator a power of ten, multiply the numerator and denominator by 2. | Step 2: 30 x 2 = 60; the new fraction is 60/100. | Step 3: Now the denominator is 100; write the numerator and place the point 2 place(s) from the right: 0.6. | Step 4: Final Answer: 30/50 = 0.6.

Question: Exercise 4.2, Question 3 (i): Convert the decimal 0.3 into a fraction in the simplest form.

Answer: Step 1: Write 0.3 as a fraction: the digits without the point over 10: 3/10. | Step 2: Final Answer: 0.3 = 3/10.

Question: Exercise 4.2, Question 3 (ii): Convert the decimal 1.35 into a fraction in the simplest form.

Answer: Step 1: Write 1.35 as a fraction: the digits without the point over 100: 135/100. | Step 2: Divide the numerator and denominator by 5: 135/100 = 27/20. | Step 3: Final Answer: 1.35 = 1 7/20.

Question: Exercise 4.2, Question 3 (iii): Convert the decimal 12.37 into a fraction in the simplest form.

Answer: Step 1: Write 12.37 as a fraction: the digits without the point over 100: 1237/100. | Step 2: Final Answer: 12.37 = 12 37/100.

Question: Exercise 4.2, Question 3 (iv): Convert the decimal 0.432 into a fraction in the simplest form.

Answer: Step 1: Write 0.432 as a fraction: the digits without the point over 1000: 432/1000. | Step 2: Divide the numerator and denominator by 2: 432/1000 = 216/500. | Step 3: Divide the numerator and denominator by 2: 216/500 = 108/250. | Step 4: Divide the numerator and denominator by 2: 108/250 = 54/125. | Step 5: Final Answer: 0.432 = 54/125.

Question: Exercise 4.2, Question 3 (v): Convert the decimal 25.5 into a fraction in the simplest form.

Answer: Step 1: Write 25.5 as a fraction: the digits without the point over 10: 255/10. | Step 2: Divide the numerator and denominator by 5: 255/10 = 51/2. | Step 3: Final Answer: 25.5 = 25 1/2.

Question: Exercise 4.2, Question 3 (vi): Convert the decimal 2.32 into a fraction in the simplest form.

Answer: Step 1: Write 2.32 as a fraction: the digits without the point over 100: 232/100. | Step 2: Divide the numerator and denominator by 2: 232/100 = 116/50. | Step 3: Divide the numerator and denominator by 2: 116/50 = 58/25. | Step 4: Final Answer: 2.32 = 2 8/25.

Question: Exercise 4.2, Question 3 (vii): Convert the decimal 0.45 into a fraction in the simplest form.

Answer: Step 1: Write 0.45 as a fraction: the digits without the point over

100: 45/100. | Step 2: Divide the numerator and denominator by 5: 45/100 = 9/20. | Step 3: Final Answer: 0.45 = 9/20.

Question: Exercise 4.2, Question 3 (viii): Convert the decimal 135.2 into a fraction in the simplest form.

Answer: Step 1: Write 135.2 as a fraction: the digits without the point over 10: 1352/10. | Step 2: Divide the numerator and denominator by 2: 1352/10 = 676/5. | Step 3: Final Answer: 135.2 = 135 1/5.

Exercise 4.3: Addition and Subtraction of Decimals

Question: Exercise 4.3, Question 1 (i): Add: 0.35 + 0.42.

Answer: Step 1: Write every number with 2 decimal place(s): 0.35 = 35 hundredths, 0.42 = 42 hundredths. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 5 + 2 = 7; write 7. | Step 4: Add the tens column: 3 + 4 = 7; write 7. | Step 5: Put the decimal point 2 place(s) from the right in 77: 0.77. | Step 6: Final Answer: 0.35 + 0.42 = 0.77.

Question: Exercise 4.3, Question 1 (ii): Add: 0.21 + 0.35 + 0.11.

Answer: Step 1: Write every number with 2 decimal place(s): 0.21 = 21 hundredths, 0.35 = 35 hundredths, 0.11 = 11 hundredths. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 1 + 5 + 1 = 7; write 7. | Step 4: Add the tens column: 2 + 3 + 1 = 6; write 6. | Step 5: Put the decimal point 2 place(s) from the right in 67: 0.67. | Step 6: Final Answer: 0.21 + 0.35 + 0.11 = 0.67.

Question: Exercise 4.3, Question 1 (iii): Add: 1.27 + 3.49.

Answer: Step 1: Write every number with 2 decimal place(s): 1.27 = 127 hundredths, 3.49 = 349 hundredths. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones

column: 7 + 9 = 16; write 6 and carry 1 to the next column. | Step 4: Add the tens column: 2 + 4 + 1 carried over = 7; write 7. | Step 5: Add the hundreds column: 1 + 3 = 4; write 4. | Step 6: Put the decimal point 2 place(s) from the right in 476: 4.76. | Step 7: Final Answer: 1.27 + 3.49 = 4.76.

Question: Exercise 4.3, Question 1 (iv): Add: 13.5 + 14.2 + 7.9.

Answer: Step 1: Write every number with 1 decimal place(s): 13.5 = 135 tenths, 14.2 = 142 tenths, 7.9 = 79 tenths. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 5 + 2 + 9 = 16; write 6 and carry 1 to the next column. | Step 4: Add the tens column: 3 + 4 + 7 + 1 carried over = 15; write 5 and carry 1 to the next column. | Step 5: Add the hundreds column: 1 + 1 + 0 + 1 carried over = 3; write 3. | Step 6: Put the decimal point 1 place(s) from the right in 356: 35.6. | Step 7: Final Answer: 13.5 + 14.2 + 7.9 = 35.6.

Question: Exercise 4.3, Question 1 (v): Add: 1.34 + 0.43.

Answer: Step 1: Write every number with 2 decimal place(s): 1.34 = 134 hundredths, 0.43 = 43 hundredths. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 4 + 3 = 7; write 7. | Step 4: Add the tens column: 3 + 4 = 7; write 7. | Step 5: Add the hundreds column: 1 + 0 = 1; write 1. | Step 6: Put the decimal point 2 place(s) from the right in 177: 1.77. | Step 7: Final Answer: 1.34 + 0.43 = 1.77.

Question: Exercise 4.3, Question 1 (vi): Add: 25.32 + 15.75.

Answer: Step 1: Write every number with 2 decimal place(s): 25.32 = 2532 hundredths, 15.75 = 1575 hundredths. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 2 + 5 = 7; write 7. | Step 4: Add the tens column: 3 + 7 = 10;

write 0 and carry 1 to the next column. | Step 5: Add the hundreds column: 5 + 5 + 1 carried over = 11; write 1 and carry 1 to the next column. | Step 6: Add the thousands column: 2 + 1 + 1 carried over = 4; write 4. | Step 7: Put the decimal point 2 place(s) from the right in 4107: 41.07. | Step 8: Final Answer: 25.32 + 15.75 = 41.07.

Question: Exercise 4.3, Question 1 (vii): Add: 0.3 + 0.9 + 0.4.

Answer: Step 1: Write every number with 1 decimal place(s): 0.3 = 3 tenths, 0.9 = 9 tenths, 0.4 = 4 tenths. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 3 + 9 + 4 = 16; write 6 and carry 1 to the next column. | Step 4: Write the carried 1 in the next column. | Step 5: Put the decimal point 1 place(s) from the right in 16: 1.6. | Step 6: Final Answer: 0.3 + 0.9 + 0.4 = 1.6.

Question: Exercise 4.3, Question 1 (viii): Add: 1.5 + 3.9 + 17.2.

Answer: Step 1: Write every number with 1 decimal place(s): 1.5 = 15 tenths, 3.9 = 39 tenths, 17.2 = 172 tenths. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 5 + 9 + 2 = 16; write 6 and carry 1 to the next column. | Step 4: Add the tens column: 1 + 3 + 7 + 1 carried over = 12; write 2 and carry 1 to the next column. | Step 5: Add the hundreds column: 0 + 0 + 1 + 1 carried over = 2; write 2. | Step 6: Put the decimal point 1 place(s) from the right in 226: 22.6. | Step 7: Final Answer: 1.5 + 3.9 + 17.2 = 22.6.

Question: Exercise 4.3, Question 1 (ix): Add: 68.378 + 12.633.

Answer: Step 1: Write every number with 3 decimal place(s): 68.378 = 68378 thousandths, 12.633 = 12633 thousandths. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 8 + 3 = 11; write 1 and carry 1 to the next column. |

Step 4: Add the tens column: 7 + 3 + 1 carried over = 11; write 1 and carry 1 to the next column. | Step 5: Add the hundreds column: 3 + 6 + 1 carried over = 10; write 0 and carry 1 to the next column. | Step 6: Add the thousands column: 8 + 2 + 1 carried over = 11; write 1 and carry 1 to the next column. | Step 7: Add the ten thousands column: 6 + 1 + 1 carried over = 8; write 8. | Step 8: Put the decimal point 3 place(s) from the right in 81011: 81.011. | Step 9: Final Answer: 68.378 + 12.633 = 81.011.

Question: Exercise 4.3, Question 1 (x): Add: 42.243 + 81.483.

Answer: Step 1: Write every number with 3 decimal place(s): 42.243 = 42243 thousandths, 81.483 = 81483 thousandths. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 3 + 3 = 6; write 6. | Step 4: Add the tens column: 4 + 8 = 12; write 2 and carry 1 to the next column. | Step 5: Add the hundreds column: 2 + 4 + 1 carried over = 7; write 7. | Step 6: Add the thousands column: 2 + 1 = 3; write 3. | Step 7: Add the ten thousands column: 4 + 8 = 12; write 2 and carry 1 to the next column. | Step 8: Write the carried 1 in the next column. | Step 9: Put the decimal point 3 place(s) from the right in 123726: 123.726. | Step 10: Final Answer: 42.243 + 81.483 = 123.726.

Question: Exercise 4.3, Question 2 (i): Subtract: 0.9 - 0.3.

Answer: Step 1: Write both numbers with 1 decimal place(s): 0.9 = 9 tenths and 0.3 = 3 tenths. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 9 - 3 = 6; write 6. | Step 4: Put the decimal point 1 place(s) from the right in 6: 0.6. | Step 5: Final Answer: 0.9 - 0.3 = 0.6.

Question: Exercise 4.3, Question 2 (ii): Subtract: 0.84 - 0.61.

Answer: Step 1: Write both numbers with 2 decimal place(s): 0.84 = 84

hundredths and 0.61 = 61 hundredths. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 4 - 1 = 3; write 3. | Step 4: Subtract the tens column: 8 - 6 = 2; write 2. | Step 5: Put the decimal point 2 place(s) from the right in 23: 0.23. | Step 6: Final Answer: 0.84 - 0.61 = 0.23.

Question: Exercise 4.3, Question 2 (iii): Subtract: 10.59 - 3.27.

Answer: Step 1: Write both numbers with 2 decimal place(s): 10.59 = 1059 hundredths and 3.27 = 327 hundredths. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 9 - 7 = 2; write 2. | Step 4: Subtract the tens column: 5 - 2 = 3; write 3. | Step 5: Hundreds column: after borrowing, the top digit becomes 10; 10 - 3 = 7; write 7. | Step 6: Thousands column: after borrowing, the top digit becomes 0; 0 - 0 = 0; write 0. | Step 7: Put the decimal point 2 place(s) from the right in 732: 7.32. | Step 8: Final Answer: 10.59 - 3.27 = 7.32.

Question: Exercise 4.3, Question 2 (iv): Subtract: 13.54 - 12.87.

Answer: Step 1: Write both numbers with 2 decimal place(s): 13.54 = 1354 hundredths and 12.87 = 1287 hundredths. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Ones column: after borrowing, the top digit becomes 14; 14 - 7 = 7; write 7. | Step 4: Tens column: after borrowing, the top digit becomes 14; 14 - 8 = 6; write 6. | Step 5: Hundreds column: after borrowing, the top digit becomes 2; 2 - 2 = 0; write 0. | Step 6: Subtract the thousands column: 1 - 1 = 0; write 0. | Step 7: Put the decimal point 2 place(s) from the right in 67: 0.67. | Step 8: Final Answer: 13.54 - 12.87 = 0.67.

Question: Exercise 4.3, Question 2 (v): Subtract: 15.39 - 12.41.

Answer: Step 1: Write both numbers with 2 decimal place(s): 15.39 = 1539 hundredths and 12.41 = 1241 hundredths. | Step 2: Write the bigger

number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 9 - 1 = 8; write 8. | Step 4: Tens column: after borrowing, the top digit becomes 13; 13 - 4 = 9; write 9. | Step 5: Hundreds column: after borrowing, the top digit becomes 4; 4 - 2 = 2; write 2. | Step 6: Subtract the thousands column: 1 - 1 = 0; write 0. | Step 7: Put the decimal point 2 place(s) from the right in 298: 2.98. | Step 8: Final Answer: 15.39 - 12.41 = 2.98.

Question: Exercise 4.3, Question 2 (vi): Subtract: 87.34 - 85.56.

Answer: Step 1: Write both numbers with 2 decimal place(s): 87.34 = 8734 hundredths and 85.56 = 8556 hundredths. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Ones column: after borrowing, the top digit becomes 14; 14 - 6 = 8; write 8. | Step 4: Tens column: after borrowing, the top digit becomes 12; 12 - 5 = 7; write 7. | Step 5: Hundreds column: after borrowing, the top digit becomes 6; 6 - 5 = 1; write 1. | Step 6: Subtract the thousands column: 8 - 8 = 0; write 0. | Step 7: Put the decimal point 2 place(s) from the right in 178: 1.78. | Step 8: Final Answer: 87.34 - 85.56 = 1.78.

Question: Exercise 4.3, Question 2 (vii): Subtract: 19.31 - 10.32.

Answer: Step 1: Write both numbers with 2 decimal place(s): 19.31 = 1931 hundredths and 10.32 = 1032 hundredths. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Ones column: after borrowing, the top digit becomes 11; 11 - 2 = 9; write 9. | Step 4: Tens column: after borrowing, the top digit becomes 12; 12 - 3 = 9; write 9. | Step 5: Hundreds column: after borrowing, the top digit becomes 8; 8 - 0 = 8; write 8. | Step 6: Subtract the thousands column: 1 - 1 = 0; write 0. | Step 7: Put the decimal point 2 place(s) from the right in 899: 8.99. | Step 8: Final Answer: 19.31 - 10.32 = 8.99.

Question: Exercise 4.3, Question 2 (viii): Subtract: 54.79 - 39.84.

Answer: Step 1: Write both numbers with 2 decimal place(s): 54.79 = 5479 hundredths and 39.84 = 3984 hundredths. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 9 - 4 = 5; write 5. | Step 4: Tens column: after borrowing, the top digit becomes 17; 17 - 8 = 9; write 9. | Step 5: Hundreds column: after borrowing, the top digit becomes 13; 13 - 9 = 4; write 4. | Step 6: Thousands column: after borrowing, the top digit becomes 4; 4 - 3 = 1; write 1. | Step 7: Put the decimal point 2 place(s) from the right in 1495: 14.95. | Step 8: Final Answer: 54.79 - 39.84 = 14.95.

Question: Exercise 4.3, Question 2 (ix): Subtract: 49.243 - 17.157.

Answer: Step 1: Write both numbers with 3 decimal place(s): 49.243 = 49243 thousandths and 17.157 = 17157 thousandths. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Ones column: after borrowing, the top digit becomes 13; 13 - 7 = 6; write 6. | Step 4: Tens column: after borrowing, the top digit becomes 13; 13 - 5 = 8; write 8. | Step 5: Hundreds column: after borrowing, the top digit becomes 1; 1 - 1 = 0; write 0. | Step 6: Subtract the thousands column: 9 - 7 = 2; write 2. | Step 7: Subtract the ten thousands column: 4 - 1 = 3; write 3. | Step 8: Put the decimal point 3 place(s) from the right in 32086: 32.086. | Step 9: Final Answer: 49.243 - 17.157 = 32.086.

Question: Exercise 4.3, Question 2 (x): Subtract: 76.919 - 64.883.

Answer: Step 1: Write both numbers with 3 decimal place(s): 76.919 = 76919 thousandths and 64.883 = 64883 thousandths. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 9 - 3 = 6; write 6. | Step 4: Tens column: after borrowing, the top digit becomes 11; 11 - 8 = 3; write 3. | Step 5: Hundreds column: after borrowing, the top digit becomes 8; 8 - 8 = 0;

write 0. | Step 6: Subtract the thousands column: 6 - 4 = 2; write 2. | Step 7: Subtract the ten thousands column: 7 - 6 = 1; write 1. | Step 8: Put the decimal point 3 place(s) from the right in 12036: 12.036. | Step 9: Final Answer: 76.919 - 64.883 = 12.036.

Exercise 4.4: Multiplication of Decimals

Question: Exercise 4.4, Question 1: Multiply 6.3 by 10.

Answer: Step 1: To multiply by 10, move the decimal point one place(s) to the right. | Step 2: 6.3 x 10 = 63. | Step 3: Final Answer: 63.

Question: Exercise 4.4, Question 2: Multiply 0.25 by 10.

Answer: Step 1: To multiply by 10, move the decimal point one place(s) to the right. | Step 2: 0.25 x 10 = 2.50. | Step 3: Final Answer: 2.50.

Question: Exercise 4.4, Question 3: Multiply 1.732 by 100.

Answer: Step 1: To multiply by 100, move the decimal point two place(s) to the right. | Step 2: 1.732 x 100 = 173.200. | Step 3: Final Answer: 173.200.

Question: Exercise 4.4, Question 4: Multiply 0.327 by 100.

Answer: Step 1: To multiply by 100, move the decimal point two place(s) to the right. | Step 2: 0.327 x 100 = 32.700. | Step 3: Final Answer: 32.700.

Question: Exercise 4.4, Question 5: Multiply 22.3542 by 100.

Answer: Step 1: To multiply by 100, move the decimal point two place(s) to the right. | Step 2: 22.3542 x 100 = 2235.4200. | Step 3: Final Answer: 2235.4200.

Question: Exercise 4.4, Question 6: Multiply 1.265 by 1000.

Answer: Step 1: To multiply by 1000, move the decimal point three place(s) to the right. | Step 2: 1.265 x 1000 = 1265. | Step 3: Final Answer: 1265.

Question: Exercise 4.4, Question 7: Multiply 1.872 by 1000.

Answer: Step 1: To multiply by 1000, move the decimal point three

place(s) to the right. | Step 2: 1.872 x 1000 = 1872. | Step 3: Final Answer: 1872.

Question: Exercise 4.4, Question 8: Multiply 6.3 by 3.

Answer: Step 1: Ignore the decimal point for a while: 6.3 has 1 decimal place(s), so take 63. | Step 2: Break 63 into its parts: 60 + 3. | Step 3: 60 x 3 = 180. | Step 4: 3 x 3 = 9. | Step 5: Add the parts: 180 + 9 = 189. | Step 6: Now put the decimal point 1 place(s) from the right in 189: 18.9. | Step 7: Final Answer: 6.3 x 3 = 18.9.

Question: Exercise 4.4, Question 9: Multiply 0.85 by 25.

Answer: Step 1: Ignore the decimal point for a while: 0.85 has 2 decimal place(s), so take 85. | Step 2: Break 25 into 5 + 20. | Step 3: Multiply by the ones: 85 x 5 = 80 x 5 = 400 + 5 x 5 = 25; total 425. | Step 4: Multiply by the tens: first 85 x 2 = 80 x 2 = 160 + 5 x 2 = 10; total 170; then x 10 gives 1,700. | Step 5: Add the partial products: 425 + 1,700 = 2,125. | Step 6: Now put the decimal point 2 place(s) from the right in 2125: 21.25. | Step 7: Final Answer: 0.85 x 25 = 21.25.

Question: Exercise 4.4, Question 10: Multiply 7.9 by 13.

Answer: Step 1: Ignore the decimal point for a while: 7.9 has 1 decimal place(s), so take 79. | Step 2: Break 13 into 3 + 10. | Step 3: Multiply by the ones: 79 x 3 = 70 x 3 = 210 + 9 x 3 = 27; total 237. | Step 4: Multiply by the tens: first 79 x 1 = 70 x 1 = 70 + 9 x 1 = 9; total 79; then x 10 gives 790. | Step 5: Add the partial products: 237 + 790 = 1,027. | Step 6: Now put the decimal point 1 place(s) from the right in 1027: 102.7. | Step 7: Final Answer: 7.9 x 13 = 102.7.

Question: Exercise 4.4, Question 11: Multiply 7.21 by 33.

Answer: Step 1: Ignore the decimal point for a while: 7.21 has 2 decimal place(s), so take 721. | Step 2: Break 33 into 3 + 30. | Step 3: Multiply by the ones: 721 x 3 = 700 x 3 = 2100 + 20 x 3 = 60 + 1 x 3 = 3; total 2163.

| Step 4: Multiply by the tens: first 721 x 3 = 700 x 3 = 2100 + 20 x 3 = 60 + 1 x 3 = 3; total 2163; then x 10 gives 21,630. | Step 5: Add the partial products: 2,163 + 21,630 = 23,793. | Step 6: Now put the decimal point 2 place(s) from the right in 23793: 237.93. | Step 7: Final Answer: 7.21 x 33 = 237.93.

Question: Exercise 4.4, Question 12: Multiply 1.834 by 11.

Answer: Step 1: Ignore the decimal point for a while: 1.834 has 3 decimal place(s), so take 1834. | Step 2: Break 11 into 1 + 10. | Step 3: Multiply by the ones: 1,834 x 1 = 1000 x 1 = 1000 + 800 x 1 = 800 + 30 x 1 = 30 + 4 x 1 = 4; total 1834. | Step 4: Multiply by the tens: first 1,834 x 1 = 1000 x 1 = 1000 + 800 x 1 = 800 + 30 x 1 = 30 + 4 x 1 = 4; total 1834; then x 10 gives 18,340. | Step 5: Add the partial products: 1,834 + 18,340 = 20,174. | Step 6: Now put the decimal point 3 place(s) from the right in 20174: 20.174. | Step 7: Final Answer: 1.834 x 11 = 20.174.

Question: Exercise 4.4, Question 13: Multiply 5.42 by 24.

Answer: Step 1: Ignore the decimal point for a while: 5.42 has 2 decimal place(s), so take 542. | Step 2: Break 24 into 4 + 20. | Step 3: Multiply by the ones: 542 x 4 = 500 x 4 = 2000 + 40 x 4 = 160 + 2 x 4 = 8; total 2168. | Step 4: Multiply by the tens: first 542 x 2 = 500 x 2 = 1000 + 40 x 2 = 80 + 2 x 2 = 4; total 1084; then x 10 gives 10,840. | Step 5: Add the partial products: 2,168 + 10,840 = 13,008. | Step 6: Now put the decimal point 2 place(s) from the right in 13008: 130.08. | Step 7: Final Answer: 5.42 x 24 = 130.08.

Question: Exercise 4.4, Question 14: Multiply 0.49 by 71.

Answer: Step 1: Ignore the decimal point for a while: 0.49 has 2 decimal place(s), so take 49. | Step 2: Break 71 into 1 + 70. | Step 3: Multiply by the ones: 49 x 1 = 40 x 1 = 40 + 9 x 1 = 9; total 49. | Step 4: Multiply by the tens: first 49 x 7 = 40 x 7 = 280 + 9 x 7 = 63; total 343; then x 10

gives 3,430. | Step 5: Add the partial products: 49 + 3,430 = 3,479. | Step 6: Now put the decimal point 2 place(s) from the right in 3479: 34.79. | Step 7: Final Answer: 0.49 x 71 = 34.79.

Question: Exercise 4.4, Question 15: Multiply 3.52 by 45.

Answer: Step 1: Ignore the decimal point for a while: 3.52 has 2 decimal place(s), so take 352. | Step 2: Break 45 into 5 + 40. | Step 3: Multiply by the ones: 352 x 5 = 300 x 5 = 1500 + 50 x 5 = 250 + 2 x 5 = 10; total 1760. | Step 4: Multiply by the tens: first 352 x 4 = 300 x 4 = 1200 + 50 x 4 = 200 + 2 x 4 = 8; total 1408; then x 10 gives 14,080. | Step 5: Add the partial products: 1,760 + 14,080 = 15,840. | Step 6: Now put the decimal point 2 place(s) from the right in 15840: 158.4. | Step 7: Final Answer: 3.52 x 45 = 158.4.

Exercise 4.5: Division of Decimals

Question: Exercise 4.5, Question 1: Divide 17.1 by 6.

Answer: Step 1: Divide 17.1 by 6 digit by digit (long division). | Step 2: 17 / 6: 6 x 2 = 12; remainder 17 - 12 = 5. | Step 3: Now place the decimal point in the quotient. | Step 4: 51 / 6: 6 x 8 = 48; remainder 51 - 48 = 3. | Step 5: 30 / 6: 6 x 5 = 30; remainder 30 - 30 = 0. | Step 6: Final Answer: 17.1 / 6 = 2.85.

Question: Exercise 4.5, Question 2: Divide 1.8 by 5.

Answer: Step 1: Divide 1.8 by 5 digit by digit (long division). | Step 2: Now place the decimal point in the quotient. | Step 3: 18 / 5: 5 x 3 = 15; remainder 18 - 15 = 3. | Step 4: 30 / 5: 5 x 6 = 30; remainder 30 - 30 = 0. | Step 5: Final Answer: 1.8 / 5 = 0.36.

Question: Exercise 4.5, Question 3: Divide 9.36 by 4.

Answer: Step 1: Divide 9.36 by 4 digit by digit (long division). | Step 2: 9 / 4: 4 x 2 = 8; remainder 9 - 8 = 1. | Step 3: Now place the decimal point in

the quotient. | Step 4: 13 / 4: 4 x 3 = 12; remainder 13 - 12 = 1. | Step 5: 16 / 4: 4 x 4 = 16; remainder 16 - 16 = 0. | Step 6: Final Answer: 9.36 / 4 = 2.34.

Question: Exercise 4.5, Question 4: Divide 9.66 by 3.

Answer: Step 1: Divide 9.66 by 3 digit by digit (long division). | Step 2: 9 / 3: 3 x 3 = 9; remainder 9 - 9 = 0. | Step 3: Now place the decimal point in the quotient. | Step 4: 6 / 3: 3 x 2 = 6; remainder 6 - 6 = 0. | Step 5: 6 / 3: 3 x 2 = 6; remainder 6 - 6 = 0. | Step 6: Final Answer: 9.66 / 3 = 3.22.

Question: Exercise 4.5, Question 5: Divide 4.68 by 9.

Answer: Step 1: Divide 4.68 by 9 digit by digit (long division). | Step 2: Now place the decimal point in the quotient. | Step 3: 46 / 9: 9 x 5 = 45; remainder 46 - 45 = 1. | Step 4: 18 / 9: 9 x 2 = 18; remainder 18 - 18 = 0. | Step 5: Final Answer: 4.68 / 9 = 0.52.

Question: Exercise 4.5, Question 6: Divide 17.85 by 7.

Answer: Step 1: Divide 17.85 by 7 digit by digit (long division). | Step 2: 17 / 7: 7 x 2 = 14; remainder 17 - 14 = 3. | Step 3: Now place the decimal point in the quotient. | Step 4: 38 / 7: 7 x 5 = 35; remainder 38 - 35 = 3. | Step 5: 35 / 7: 7 x 5 = 35; remainder 35 - 35 = 0. | Step 6: Final Answer: 17.85 / 7 = 2.55.

Question: Exercise 4.5, Question 7: Divide 6.28 by 4.

Answer: Step 1: Divide 6.28 by 4 digit by digit (long division). | Step 2: 6 / 4: 4 x 1 = 4; remainder 6 - 4 = 2. | Step 3: Now place the decimal point in the quotient. | Step 4: 22 / 4: 4 x 5 = 20; remainder 22 - 20 = 2. | Step 5: 28 / 4: 4 x 7 = 28; remainder 28 - 28 = 0. | Step 6: Final Answer: 6.28 / 4 = 1.57.

Question: Exercise 4.5, Question 8: Divide 0.72 by 2.

Answer: Step 1: Divide 0.72 by 2 digit by digit (long division). | Step 2: Now place the decimal point in the quotient. | Step 3: 7 / 2: 2 x 3 = 6;

remainder 7 - 6 = 1. | Step 4: 12 / 2: 2 x 6 = 12; remainder 12 - 12 = 0. | Step 5: Final Answer: 0.72 / 2 = 0.36.

Question: Exercise 4.5, Question 9: Divide 5.62 by 2.

Answer: Step 1: Divide 5.62 by 2 digit by digit (long division). | Step 2: 5 / 2: 2 x 2 = 4; remainder 5 - 4 = 1. | Step 3: Now place the decimal point in the quotient. | Step 4: 16 / 2: 2 x 8 = 16; remainder 16 - 16 = 0. | Step 5: 2 / 2: 2 x 1 = 2; remainder 2 - 2 = 0. | Step 6: Final Answer: 5.62 / 2 = 2.81.

Question: Exercise 4.5, Question 10: Divide 34.23 by 3.

Answer: Step 1: Divide 34.23 by 3 digit by digit (long division). | Step 2: 3 / 3: 3 x 1 = 3; remainder 3 - 3 = 0. | Step 3: 4 / 3: 3 x 1 = 3; remainder 4 - 3 = 1. | Step 4: Now place the decimal point in the quotient. | Step 5: 12 / 3: 3 x 4 = 12; remainder 12 - 12 = 0. | Step 6: 3 / 3: 3 x 1 = 3; remainder 3 - 3 = 0. | Step 7: Final Answer: 34.23 / 3 = 11.41.

Question: Exercise 4.5, Question 11: Divide 67.25 by 5.

Answer: Step 1: Divide 67.25 by 5 digit by digit (long division). | Step 2: 6 / 5: 5 x 1 = 5; remainder 6 - 5 = 1. | Step 3: 17 / 5: 5 x 3 = 15; remainder 17 - 15 = 2. | Step 4: Now place the decimal point in the quotient. | Step 5: 22 / 5: 5 x 4 = 20; remainder 22 - 20 = 2. | Step 6: 25 / 5: 5 x 5 = 25; remainder 25 - 25 = 0. | Step 7: Final Answer: 67.25 / 5 = 13.45.

Question: Exercise 4.5, Question 12: Divide 32.34 by 6.

Answer: Step 1: Divide 32.34 by 6 digit by digit (long division). | Step 2: 32 / 6: 6 x 5 = 30; remainder 32 - 30 = 2. | Step 3: Now place the decimal point in the quotient. | Step 4: 23 / 6: 6 x 3 = 18; remainder 23 - 18 = 5. | Step 5: 54 / 6: 6 x 9 = 54; remainder 54 - 54 = 0. | Step 6: Final Answer: 32.34 / 6 = 5.39.

Question: Exercise 4.5, Question 13: Divide 96.20 by 2.

Answer: Step 1: Divide 96.20 by 2 digit by digit (long division). | Step 2: 9 / 2: 2 x 4 = 8; remainder 9 - 8 = 1. | Step 3: 16 / 2: 2 x 8 = 16; remainder

16 - 16 = 0. | Step 4: Now place the decimal point in the quotient. | Step 5: 2 / 2: 2 x 1 = 2; remainder 2 - 2 = 0. | Step 6: 0 / 2: 2 x 0 = 0; remainder 0 - 0 = 0. | Step 7: Final Answer: 96.20 / 2 = 48.10.

Question: Exercise 4.5, Question 14: Divide 10.11 by 3.

Answer: Step 1: Divide 10.11 by 3 digit by digit (long division). | Step 2: 10 / 3: 3 x 3 = 9; remainder 10 - 9 = 1. | Step 3: Now place the decimal point in the quotient. | Step 4: 11 / 3: 3 x 3 = 9; remainder 11 - 9 = 2. | Step 5: 21 / 3: 3 x 7 = 21; remainder 21 - 21 = 0. | Step 6: Final Answer: 10.11 / 3 = 3.37.

Question: Exercise 4.5, Question 15: Divide 40.05 by 5.

Answer: Step 1: Divide 40.05 by 5 digit by digit (long division). | Step 2: 40 / 5: 5 x 8 = 40; remainder 40 - 40 = 0. | Step 3: Now place the decimal point in the quotient. | Step 4: 0 / 5: 5 x 0 = 0; remainder 0 - 0 = 0. | Step 5: 5 / 5: 5 x 1 = 5; remainder 5 - 5 = 0. | Step 6: Final Answer: 40.05 / 5 = 8.01.

Exercise 4.6: Decimal Word Problems

Question: Exercise 4.6, Question 1: Ahmed purchased a shirt for Rs 325.80 and jeans for Rs 525.25. Find the amount spent.

Answer: Step 1: Ahmed spent on both the shirt and the jeans, so add the two prices. | Step 2: Write every number with 2 decimal place(s): 325.80 = 32580 hundredths, 525.25 = 52525 hundredths. | Step 3: Write the numbers one below the other, ones under ones, tens under tens. | Step 4: Add the ones column: 0 + 5 = 5; write 5. | Step 5: Add the tens column: 8 + 2 = 10; write 0 and carry 1 to the next column. | Step 6: Add the hundreds column: 5 + 5 + 1 carried over = 11; write 1 and carry 1 to the next column. | Step 7: Add the thousands column: 2 + 2 + 1 carried over = 5; write 5. | Step 8: Add the ten thousands column: 3 + 5 = 8; write 8. |

Step 9: Put the decimal point 2 place(s) from the right in 85105: 851.05. | Step 10: Final Answer: Ahmed spent Rs 851.05 in all.

Question: Exercise 4.6, Question 2: Javeria's weight is 10.24 kg and her sister's weight is 11.28 kg. What is the total weight of both sisters?

Answer: Step 1: The total weight of both sisters is needed, so add the two weights. | Step 2: Write every number with 2 decimal place(s): 10.24 = 1024 hundredths, 11.28 = 1128 hundredths. | Step 3: Write the numbers one below the other, ones under ones, tens under tens. | Step 4: Add the ones column: 4 + 8 = 12; write 2 and carry 1 to the next column. | Step 5: Add the tens column: 2 + 2 + 1 carried over = 5; write 5. | Step 6: Add the hundreds column: 0 + 1 = 1; write 1. | Step 7: Add the thousands column: 1 + 1 = 2; write 2. | Step 8: Put the decimal point 2 place(s) from the right in 2152: 21.52. | Step 9: Final Answer: The total weight of both sisters is 21.52 kg.

Question: Exercise 4.6, Question 3: Ansa paid a total of Rs 97.5 for a pencil case and a colour box; the pencil case costs Rs 30.25. Find the cost of the colour box.

Answer: Step 1: Subtract the cost of the pencil case from the total amount to get the cost of the colour box. | Step 2: Write both numbers with 2 decimal place(s): 97.5 = 9750 hundredths and 30.25 = 3025 hundredths. | Step 3: Write the bigger number on top and the smaller number below it, in columns. | Step 4: Ones column: after borrowing, the top digit becomes 10; 10 - 5 = 5; write 5. | Step 5: Tens column: after borrowing, the top digit becomes 4; 4 - 2 = 2; write 2. | Step 6: Subtract the hundreds column: 7 - 0 = 7; write 7. | Step 7: Subtract the thousands column: 9 - 3 = 6; write 6. | Step 8: Put the decimal point 2 place(s) from the right in 6725: 67.25. | Step 9: Final Answer: The colour box costs Rs 67.25.

Question: Exercise 4.6, Question 4: Haroon and Shafique spent Rs 95.82 in

all; Haroon spent Rs 89.75. What amount was spent by Shafique?

Answer: Step 1: Subtract Haroon's spending from the total to get Shafique's spending. | Step 2: Write both numbers with 2 decimal place(s): 95.82 = 9582 hundredths and 89.75 = 8975 hundredths. | Step 3: Write the bigger number on top and the smaller number below it, in columns. | Step 4: Ones column: after borrowing, the top digit becomes 12; 12 - 5 = 7; write 7. | Step 5: Tens column: after borrowing, the top digit becomes 7; 7 - 7 = 0; write 0. | Step 6: Hundreds column: after borrowing, the top digit becomes 15; 15 - 9 = 6; write 6. | Step 7: Thousands column: after borrowing, the top digit becomes 8; 8 - 8 = 0; write 0. | Step 8: Put the decimal point 2 place(s) from the right in 607: 6.07. | Step 9: Final Answer: Shafique spent Rs 6.07.

Question: Exercise 4.6, Question 5: Ali has Rs 50.29 pocket money; he gave Rs 15.45 to his sister Nida and spent Rs 13.84 on coffee. How much money is left?

Answer: Step 1: Ali has Rs 50.29; first subtract the amount given to his sister Nida. | Step 2: Write both numbers with 2 decimal place(s): 50.29 = 5029 hundredths and 15.45 = 1545 hundredths. | Step 3: Write the bigger number on top and the smaller number below it, in columns. | Step 4: Subtract the ones column: 9 - 5 = 4; write 4. | Step 5: Tens column: after borrowing, the top digit becomes 12; 12 - 4 = 8; write 8. | Step 6: Hundreds column: after borrowing, the top digit becomes 9; 9 - 5 = 4; write 4. | Step 7: Thousands column: after borrowing, the top digit becomes 4; 4 - 1 = 3; write 3. | Step 8: Put the decimal point 2 place(s) from the right in 3484: 34.84. | Step 9: Now subtract the amount spent on coffee from what is left. | Step 10: Write both numbers with 2 decimal place(s): 34.84 = 3484 hundredths and 13.84 = 1384 hundredths. | Step 11: Write the bigger number on top and the smaller number below it, in

columns. | Step 12: Subtract the ones column: 4 - 4 = 0; write 0. | Step 13: Subtract the tens column: 8 - 8 = 0; write 0. | Step 14: Subtract the hundreds column: 4 - 3 = 1; write 1. | Step 15: Subtract the thousands column: 3 - 1 = 2; write 2. | Step 16: Put the decimal point 2 place(s) from the right in 2100: 21. | Step 17: Final Answer: Rs 21.00 is left with Ali.

Question: Exercise 4.6, Question 6: Aslam brought 65 hens; the weight of each hen is 2.72 kg. What is the weight of all of them?

Answer: Step 1: Aslam brought 65 hens and each hen weighs 2.72 kg, so multiply. | Step 2: Ignore the decimal point for a while: 2.72 has 2 decimal place(s), so take 272. | Step 3: Break 65 into 5 + 60. | Step 4: Multiply by the ones: 272 x 5 = 200 x 5 = 1000 + 70 x 5 = 350 + 2 x 5 = 10; total 1360. | Step 5: Multiply by the tens: first 272 x 6 = 200 x 6 = 1200 + 70 x 6 = 420 + 2 x 6 = 12; total 1632; then x 10 gives 16,320. | Step 6: Add the partial products: 1,360 + 16,320 = 17,680. | Step 7: Now put the decimal point 2 place(s) from the right in 17680: 176.8. | Step 8: Final Answer: The weight of all the hens is 176.8 kg.

Question: Exercise 4.6, Question 7: One set of books weighs 3.75 kg. What is the weight of 32 such sets?

Answer: Step 1: One set of books weighs 3.75 kg; there are 32 such sets, so multiply. | Step 2: Ignore the decimal point for a while: 3.75 has 2 decimal place(s), so take 375. | Step 3: Break 32 into 2 + 30. | Step 4: Multiply by the ones: 375 x 2 = 300 x 2 = 600 + 70 x 2 = 140 + 5 x 2 = 10; total 750. | Step 5: Multiply by the tens: first 375 x 3 = 300 x 3 = 900 + 70 x 3 = 210 + 5 x 3 = 15; total 1125; then x 10 gives 11,250. | Step 6: Add the partial products: 750 + 11,250 = 12,000. | Step 7: Now put the decimal point 2 place(s) from the right in 12000: 120. | Step 8: Final Answer: 32 sets weigh 120 kg.

Question: Exercise 4.6, Question 8: Yusra cuts a ribbon 13.75 m long into 5

equal pieces. What is the length of each piece?

Answer: Step 1: Yusra cuts a 13.75 m ribbon into 5 equal pieces, so divide. | Step 2: Divide 13.75 by 5 digit by digit (long division). | Step 3: 13 / 5: 5 x 2 = 10; remainder 13 - 10 = 3. | Step 4: Now place the decimal point in the quotient. | Step 5: 37 / 5: 5 x 7 = 35; remainder 37 - 35 = 2. | Step 6: 25 / 5: 5 x 5 = 25; remainder 25 - 25 = 0. | Step 7: Final Answer: Each piece is 2.75 m long.

Question: Exercise 4.6, Question 9: The total weight of 5 sacks of flour is 58.75 kg. What is the weight of one sack?

Answer: Step 1: The total weight of 5 sacks of flour is 58.75 kg; divide to get one sack. | Step 2: Divide 58.75 by 5 digit by digit (long division). | Step 3: 5 / 5: 5 x 1 = 5; remainder 5 - 5 = 0. | Step 4: 8 / 5: 5 x 1 = 5; remainder 8 - 5 = 3. | Step 5: Now place the decimal point in the quotient. | Step 6: 37 / 5: 5 x 7 = 35; remainder 37 - 35 = 2. | Step 7: 25 / 5: 5 x 5 = 25; remainder 25 - 25 = 0. | Step 8: Final Answer: One sack weighs 11.75 kg.

Review Exercise 4

Question: Exercise Review 4, Question 1 (i): Write the place value and value of the encircled digit(s) in 2.325.

Answer: Step 1: Write 2.325 and name the place of each marked digit from the decimal point. | Step 2: The digit 5 is in the thousandths place; its value is 0.005. | Step 3: Final Answer: 5 (thousandths) = 0.005.

Question: Exercise Review 4, Question 1 (ii): Write the place value and value of the encircled digit(s) in 17.953.

Answer: Step 1: Write 17.953 and name the place of each marked digit from the decimal point. | Step 2: The digit 1 is in the tens place; its value is 10. | Step 3: The digit 9 is in the tenths place; its value is 0.9. | Step 4:

Final Answer: 1 (tens) = 10; 9 (tenths) = 0.9.

Question: Review Exercise 4, Question 1 (iii): Write the place value and value of the circled digit 7 in 37.87.

Answer: Step 1: Write 37.87 and name the place of each 7. | Step 2: The digit 7 is in the ones place; its value is 7. | Step 3: The digit 7 is in the hundredths place; its value is 0.07. | Step 4: Final Answer: 7 (ones) = 7; 7 (hundredths) = 0.07.

Question: Review Exercise 4, Question 2 (i): Convert the fraction 321/100 into a decimal.

Answer: Step 1: The denominator is 100; write 321 and place the point 2 place(s) from the right. | Step 2: Final Answer: 321/100 = 3.21.

Question: Review Exercise 4, Question 2 (ii): Convert the fraction 175/1000 into a decimal.

Answer: Step 1: The denominator is 1000; write 175 and place the point 3 place(s) from the right. | Step 2: Final Answer: 175/1000 = 0.175.

Question: Review Exercise 4, Question 2 (iii): Convert the fraction 19/4 into a decimal.

Answer: Step 1: Multiply the numerator and denominator by 25: 475/100. | Step 2: Write the numerator and place the point 2 place(s) from the right: 4.75. | Step 3: Final Answer: 19/4 = 4.75.

Question: Review Exercise 4, Question 2 (iv): Convert the fraction 27/8 into a decimal.

Answer: Step 1: Multiply the numerator and denominator by 125: 3375/1000. | Step 2: Write the numerator and place the point 3 place(s) from the right: 3.375. | Step 3: Final Answer: 27/8 = 3.375.

Question: Review Exercise 4, Question 3 (i): Convert the decimal 1.54 into a fraction.

Answer: Step 1: Write 1.54 as a fraction: 154/100. | Step 2: Divide the

numerator and denominator by 2: 154/100 = 77/50. | Step 3: Final Answer: 1.54 = 1 27/50.

Question: Review Exercise 4, Question 3 (ii): Convert the decimal 0.35 into a fraction.

Answer: Step 1: Write 0.35 as a fraction: 35/100. | Step 2: Divide the numerator and denominator by 5: 35/100 = 7/20. | Step 3: Final Answer: 0.35 = 7/20.

Question: Review Exercise 4, Question 3 (iii): Convert the decimal 13.7 into a fraction.

Answer: Step 1: Write 13.7 as a fraction: 137/10. | Step 2: Final Answer: 13.7 = 13 7/10.

Question: Review Exercise 4, Question 3 (iv): Convert the decimal 0.345 into a fraction.

Answer: Step 1: Write 0.345 as a fraction: 345/1000. | Step 2: Divide the numerator and denominator by 5: 345/1000 = 69/200. | Step 3: Final Answer: 0.345 = 69/200.

Question: Review Exercise 4, Question 4 (i): Solve: 1.57 + 3.42.

Answer: Step 1: Write every number with 2 decimal place(s): 1.57 = 157 hundredths, 3.42 = 342 hundredths. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 7 + 2 = 9; write 9. | Step 4: Add the tens column: 5 + 4 = 9; write 9. | Step 5: Add the hundreds column: 1 + 3 = 4; write 4. | Step 6: Put the decimal point 2 place(s) from the right in 499: 4.99. | Step 7: Final Answer: 1.57 + 3.42 = 4.99.

Question: Review Exercise 4, Question 4 (ii): Solve: 0.37 - 0.18.

Answer: Step 1: Write both numbers with 2 decimal place(s): 0.37 = 37 hundredths and 0.18 = 18 hundredths. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Ones

column: after borrowing, the top digit becomes 17; 17 - 8 = 9; write 9. | Step 4: Tens column: after borrowing, the top digit becomes 2; 2 - 1 = 1; write 1. | Step 5: Put the decimal point 2 place(s) from the right in 19: 0.19. | Step 6: Final Answer: 0.37 - 0.18 = 0.19.

Question: Review Exercise 4, Question 4 (iii): Solve: 17.55 + 23.42.

Answer: Step 1: Write every number with 2 decimal place(s): 17.55 = 1755 hundredths, 23.42 = 2342 hundredths. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 5 + 2 = 7; write 7. | Step 4: Add the tens column: 5 + 4 = 9; write 9. | Step 5: Add the hundreds column: 7 + 3 = 10; write 0 and carry 1 to the next column. | Step 6: Add the thousands column: 1 + 2 + 1 carried over = 4; write 4. | Step 7: Put the decimal point 2 place(s) from the right in 4097: 40.97. | Step 8: Final Answer: 17.55 + 23.42 = 40.97.

Question: Review Exercise 4, Question 4 (iv): Solve: 1.94 - 0.89.

Answer: Step 1: Write both numbers with 2 decimal place(s): 1.94 = 194 hundredths and 0.89 = 89 hundredths. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Ones column: after borrowing, the top digit becomes 14; 14 - 9 = 5; write 5. | Step 4: Tens column: after borrowing, the top digit becomes 8; 8 - 8 = 0; write 0. | Step 5: Subtract the hundreds column: 1 - 0 = 1; write 1. | Step 6: Put the decimal point 2 place(s) from the right in 105: 1.05. | Step 7: Final Answer: 1.94 - 0.89 = 1.05.

Question: Review Exercise 4, Question 5 (i): Multiply 0.325 by 100.

Answer: Step 1: To multiply by 100, move the decimal point two place(s) to the right. | Step 2: 0.325 x 100 = 32.500. | Step 3: Final Answer: 32.500.

Question: Review Exercise 4, Question 5 (ii): Multiply 17.55 by 10.

Answer: Step 1: To multiply by 10, move the decimal point one place(s) to the right. | Step 2: 17.55 x 10 = 175.50. | Step 3: Final Answer: 175.50.

Question: Review Exercise 4, Question 5 (iii): Multiply 3.5252 by 1000.

Answer: Step 1: To multiply by 1000, move the decimal point three place(s) to the right. | Step 2: 3.5252 x 1000 = 3525.2000. | Step 3: Final Answer: 3525.2000.

Question: Review Exercise 4, Question 5 (iv): Multiply 2.57 by 32.

Answer: Step 1: Ignore the decimal point for a while: 2.57 has 2 decimal place(s), so take 257. | Step 2: Break 32 into 2 + 30. | Step 3: Multiply by the ones: 257 x 2 = 200 x 2 = 400 + 50 x 2 = 100 + 7 x 2 = 14; total 514. | Step 4: Multiply by the tens: first 257 x 3 = 200 x 3 = 600 + 50 x 3 = 150 + 7 x 3 = 21; total 771; then x 10 gives 7,710. | Step 5: Add the partial products: 514 + 7,710 = 8,224. | Step 6: Now put the decimal point 2 place(s) from the right in 8224: 82.24. | Step 7: Final Answer: 2.57 x 32 = 82.24.

Question: Review Exercise 4, Question 6 (i): Divide 9.42 by 6.

Answer: Step 1: Divide 9.42 by 6 digit by digit (long division). | Step 2: 9 / 6: 6 x 1 = 6; remainder 9 - 6 = 3. | Step 3: Now place the decimal point in the quotient. | Step 4: 34 / 6: 6 x 5 = 30; remainder 34 - 30 = 4. | Step 5: 42 / 6: 6 x 7 = 42; remainder 42 - 42 = 0. | Step 6: Final Answer: 9.42 / 6 = 1.57.

Question: Review Exercise 4, Question 6 (ii): Divide 2.52 by 7.

Answer: Step 1: Divide 2.52 by 7 digit by digit (long division). | Step 2: Now place the decimal point in the quotient. | Step 3: 25 / 7: 7 x 3 = 21; remainder 25 - 21 = 4. | Step 4: 42 / 7: 7 x 6 = 42; remainder 42 - 42 = 0. | Step 5: Final Answer: 2.52 / 7 = 0.36.

Question: Review Exercise 4, Question 6 (iii): Divide 8.61 by 3.

Answer: Step 1: Divide 8.61 by 3 digit by digit (long division). | Step 2: 8 / 3: 3 x 2 = 6; remainder 8 - 6 = 2. | Step 3: Now place the decimal point in the quotient. | Step 4: 26 / 3: 3 x 8 = 24; remainder 26 - 24 = 2. | Step 5:

21 / 3: 3 x 7 = 21; remainder 21 - 21 = 0. | Step 6: Final Answer: 8.61 / 3 = 2.87.

Question: Review Exercise 4, Question 7 (a): Anas ran 110.25 m and Asma ran 97.75 m. What total distance did both run altogether?

Answer: Step 1: The total distance run by both is needed, so add the two distances. | Step 2: Write every number with 2 decimal place(s): 110.25 = 11025 hundredths, 97.75 = 9775 hundredths. | Step 3: Write the numbers one below the other, ones under ones, tens under tens. | Step 4: Add the ones column: 5 + 5 = 10; write 0 and carry 1 to the next column. | Step 5: Add the tens column: 2 + 7 + 1 carried over = 10; write 0 and carry 1 to the next column. | Step 6: Add the hundreds column: 0 + 7 + 1 carried over = 8; write 8. | Step 7: Add the thousands column: 1 + 9 = 10; write 0 and carry 1 to the next column. | Step 8: Add the ten thousands column: 1 + 0 + 1 carried over = 2; write 2. | Step 9: Put the decimal point 2 place(s) from the right in 20800: 208. | Step 10: Final Answer: Both ran 208 m altogether.

Question: Review Exercise 4, Question 7 (b): How much more distance did Anas run than Asma?

Answer: Step 1: To find how much more Anas ran, subtract Asma's distance from Anas's distance. | Step 2: Write both numbers with 2 decimal place(s): 110.25 = 11025 hundredths and 97.75 = 9775 hundredths. | Step 3: Write the bigger number on top and the smaller number below it, in columns. | Step 4: Subtract the ones column: 5 - 5 = 0; write 0. | Step 5: Tens column: after borrowing, the top digit becomes 12; 12 - 7 = 5; write 5. | Step 6: Hundreds column: after borrowing, the top digit becomes 9; 9 - 7 = 2; write 2. | Step 7: Thousands column: after borrowing, the top digit becomes 10; 10 - 9 = 1; write 1. | Step 8: Ten thousands column: after borrowing, the top digit becomes 0; 0 - 0 = 0;

write 0. | Step 9: Put the decimal point 2 place(s) from the right in 1250: 12.5. | Step 10: Final Answer: Anas ran 12.5 m more than Asma.

Question: Review Exercise 4, Question 8: Abdullah drinks 0.45 litre of juice every day. How many litres will he drink in 30 days?

Answer: Step 1: Abdullah drinks 0.45 litre of juice every day; multiply for 30 days. | Step 2: Ignore the decimal point for a while: 0.45 has 2 decimal place(s), so take 45. | Step 3: Break 30 into 30. | Step 4: Multiply by the tens: first 45 x 3 = 40 x 3 = 120 + 5 x 3 = 15; total 135; then x 10 gives 1,350. | Step 5: Now put the decimal point 2 place(s) from the right in 1350: 13.5. | Step 6: Final Answer: Abdullah will drink 13.5 litres in 30 days.

5. Measurement

Activities and Exercise 5.1: Converting Units of Length

Question: Activity 1: Convert to metres: 6 km (given), 9 km, 12 km, 25 km, 50 km, 75 km, 80 km.

Answer: Step 1: Remember: 1 kilometre = 1,000 metres. | Step 2: 6 x 1000 = 6000 metres. | Step 3: 9 x 1000 = 9000 metres. | Step 4: 12 x 1000 = 12000 metres. | Step 5: 25 x 1000 = 25000 metres. | Step 6: 50 x 1000 = 50000 metres. | Step 7: 75 x 1000 = 75000 metres. | Step 8: 80 x 1000 = 80000 metres. | Step 9: Final Answer: all the conversions above are complete.

Question: Activity 2: Convert metres into centimetres: 5 m (given), 9 m, 18 m, 45 m, 60 m, 87 m.

Answer: Step 1: Remember: 1 metre = 100 centimetres. | Step 2: 5 x 100 = 500 centimetres. | Step 3: 9 x 100 = 900 centimetres. | Step 4: 18 x 100 = 1800 centimetres. | Step 5: 45 x 100 = 4500 centimetres. | Step 6: 60 x 100 = 6000 centimetres. | Step 7: 87 x 100 = 8700 centimetres. | Step 8: Final Answer: all the conversions above are complete.

Question: Activity 3: Convert into millimetres: 7 cm (given), 10 cm, 11 cm, 31 cm, 49 cm.

Answer: Step 1: Remember: 1 centimetre = 10 millimetres. | Step 2: 7 x 10 = 70 millimetres. | Step 3: 10 x 10 = 100 millimetres. | Step 4: 11 x 10 = 110 millimetres. | Step 5: 31 x 10 = 310 millimetres. | Step 6: 49 x 10 = 490 millimetres. | Step 7: Final Answer: all the conversions above are complete.

Question: Exercise 5.1, Question 1 (i): Convert 5 km into metres.

Answer: Step 1: Remember: 1 kilometre = 1,000 metres. | Step 2: Multiply: 5 x 1000 = 5,000. | Step 3: Final Answer: 5,000 metres.

Question: Exercise 5.1, Question 1 (ii): Convert 14 km into metres.

Answer: Step 1: Remember: 1 kilometre = 1,000 metres. | Step 2: Multiply: 14 x 1000 = 14,000. | Step 3: Final Answer: 14,000 metres.

Question: Exercise 5.1, Question 1 (iii): Convert 20 km into metres.

Answer: Step 1: Remember: 1 kilometre = 1,000 metres. | Step 2: Multiply: 20 x 1000 = 20,000. | Step 3: Final Answer: 20,000 metres.

Question: Exercise 5.1, Question 2 (i): Convert 17 m into centimetres.

Answer: Step 1: Remember: 1 metre = 100 centimetres. | Step 2: Multiply: 17 x 100 = 1,700. | Step 3: Final Answer: 1,700 centimetres.

Question: Exercise 5.1, Question 2 (ii): Convert 32 m into centimetres.

Answer: Step 1: Remember: 1 metre = 100 centimetres. | Step 2: Multiply: 32 x 100 = 3,200. | Step 3: Final Answer: 3,200 centimetres.

Question: Exercise 5.1, Question 2 (iii): Convert 54 m into centimetres.

Answer: Step 1: Remember: 1 metre = 100 centimetres. | Step 2: Multiply: 54 x 100 = 5,400. | Step 3: Final Answer: 5,400 centimetres.

Question: Exercise 5.1, Question 3 (i): Convert 15 cm into millimetres.

Answer: Step 1: Remember: 1 centimetre = 10 millimetres. | Step 2: Multiply: 15 x 10 = 150. | Step 3: Final Answer: 150 millimetres.

Question: Exercise 5.1, Question 3 (ii): Convert 19 cm into millimetres.

Answer: Step 1: Remember: 1 centimetre = 10 millimetres. | Step 2: Multiply: 19 x 10 = 190. | Step 3: Final Answer: 190 millimetres.

Question: Exercise 5.1, Question 3 (iii): Convert 30 cm into millimetres.

Answer: Step 1: Remember: 1 centimetre = 10 millimetres. | Step 2: Multiply: 30 x 10 = 300. | Step 3: Final Answer: 300 millimetres.

Question: Exercise 5.1, Question 4 (i): Change 4 m into centimetres and millimetres.

Answer: Step 1: First convert into centimetres: 4 x 100 = 400 cm. | Step 2: Then convert into millimetres: 4 x 1,000 = 4,000 mm. | Step 3: Final Answer: 400 cm and 4,000 mm.

Question: Exercise 5.1, Question 4 (ii): Change 10 m into centimetres and millimetres.

Answer: Step 1: First convert into centimetres: 10 x 100 = 1000 cm. | Step 2: Then convert into millimetres: 10 x 1,000 = 10,000 mm. | Step 3: Final Answer: 1000 cm and 10,000 mm.

Question: Exercise 5.1, Question 4 (iii): Change 35 m into centimetres and millimetres.

Answer: Step 1: First convert into centimetres: 35 x 100 = 3500 cm. | Step 2: Then convert into millimetres: 35 x 1,000 = 35,000 mm. | Step 3: Final Answer: 3500 cm and 35,000 mm.

Question: Exercise 5.1, Question 4 (iv): Change 64 m into centimetres and millimetres.

Answer: Step 1: First convert into centimetres: 64 x 100 = 6400 cm. | Step 2: Then convert into millimetres: 64 x 1,000 = 64,000 mm. | Step 3: Final Answer: 6400 cm and 64,000 mm.

Question: Exercise 5.1, Question 4 (v): Change 83 m into centimetres and millimetres.

Answer: Step 1: First convert into centimetres: 83 x 100 = 8300 cm. | Step 2: Then convert into millimetres: 83 x 1,000 = 83,000 mm. | Step 3: Final Answer: 8300 cm and 83,000 mm.

Question: Exercise 5.1, Question 4 (vi): Change 98 m into centimetres and millimetres.

Answer: Step 1: First convert into centimetres: 98 x 100 = 9800 cm. | Step 2: Then convert into millimetres: 98 x 1,000 = 98,000 mm. | Step 3: Final Answer: 9800 cm and 98,000 mm.

Exercise 5.2: Addition and Subtraction of Length

Question: Exercise 5.2, Question 1 (i): Add: 4,200 m and 9,600 m.

Answer: Step 1: Both lengths are in metres, so add directly. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 0 + 0 = 0; write 0. | Step 4: Add the tens column: 0 + 0 = 0; write 0. | Step 5: Add the hundreds column: 2 + 6 = 8; write 8. | Step 6: Add the thousands column: 4 + 9 = 13; write 3 and carry 1 to the next column. | Step 7: Write the carried 1 in the next column. | Step 8: Final Answer: 13,800 m.

Question: Exercise 5.2, Question 1 (ii): Add: 25 km 520 m and 12 km 840 m.

Answer: Step 1: Add the m parts: 520 + 840 = 1360 m. | Step 2: 1360 m = 1 km and 360 m, because 1 km = 1000 m; carry 1 to the km side. | Step 3: Add the km parts: 25 + 12 + 1 carried = 38 km. | Step 4: Final Answer: 38 km 360 m.

Question: Exercise 5.2, Question 1 (iii): Add: 49 km 719 m and 32 km 103 m.

Answer: Step 1: Add the m parts: 719 + 103 = 822 m. | Step 2: Add the km parts: 49 + 32 = 81 km. | Step 3: Final Answer: 81 km 822 m.

Question: Exercise 5.2, Question 1 (iv): Add: 30 km 60 m and 29 km 29 m.

Answer: Step 1: Add the m parts: 60 + 29 = 89 m. | Step 2: Add the km parts: 30 + 29 = 59 km. | Step 3: Final Answer: 59 km 89 m.

Question: Exercise 5.2, Question 1 (v): Add: 69 m 17 cm and 99 m 32 cm.

Answer: Step 1: Add the cm parts: 17 + 32 = 49 cm. | Step 2: Add the m parts: 69 + 99 = 168 m. | Step 3: Final Answer: 168 m 49 cm.

Question: Exercise 5.2, Question 1 (vi): Add: 42 cm 3 mm and 68 cm 5 mm.

Answer: Step 1: Add the mm parts: 3 + 5 = 8 mm. | Step 2: Add the cm

parts: 42 + 68 = 110 cm. | Step 3: Final Answer: 110 cm 8 mm.

Question: Exercise 5.2, Question 1 (vii): Add: 13 m 25 cm, 40 m and 65 m 5 cm.

Answer: Step 1: Add the cm parts: 25 + 0 + 5 = 30 cm. | Step 2: Add the m parts: 13 + 40 + 65 = 118 m. | Step 3: Final Answer: 118 m 30 cm.

Question: Exercise 5.2, Question 1 (viii): Add: 90 km 820 m, 75 km 500 m and 110 km 175 m.

Answer: Step 1: Add the m parts: 820 + 500 + 175 = 1495 m. | Step 2: 1495 m = 1 km and 495 m, because 1 km = 1000 m; carry 1 to the km side. | Step 3: Add the km parts: 90 + 75 + 110 + 1 carried = 276 km. | Step 4: Final Answer: 276 km 495 m.

Question: Exercise 5.2, Question 1 (ix): Add: 45 km 340 m, 82 km 399 m and 230 km 180 m.

Answer: Step 1: Add the m parts: 340 + 399 + 180 = 919 m. | Step 2: Add the km parts: 45 + 82 + 230 = 357 km. | Step 3: Final Answer: 357 km 919 m.

Question: Exercise 5.2, Question 2 (i): Subtract 5,050 m from 7,000 m.

Answer: Step 1: Both lengths are in metres, so subtract directly. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 0 - 0 = 0; write 0. | Step 4: Tens column: after borrowing, the top digit becomes 10; 10 - 5 = 5; write 5. | Step 5: Hundreds column: after borrowing, the top digit becomes 9; 9 - 0 = 9; write 9. | Step 6: Thousands column: after borrowing, the top digit becomes 6; 6 - 5 = 1; write 1. | Step 7: Final Answer: 1,950 m.

Question: Exercise 5.2, Question 2 (ii): Subtract: 2 m 76 cm from 6 m 35 cm.

Answer: Step 1: The cm part 35 is smaller than 76, so borrow 1 m: 1 m = 100 cm; now the cm part is 135 and the m part is 5. | Step 2: Subtract the

cm parts: 135 - 76 = 59 cm. | Step 3: Subtract the m parts: 5 - 2 = 3 m. | Step 4: Final Answer: 3 m 59 cm.

Question: Exercise 5.2, Question 2 (iii): Subtract: 34 m 20 cm from 36 m 80 cm.

Answer: Step 1: Subtract the cm parts: 80 - 20 = 60 cm. | Step 2: Subtract the m parts: 36 - 34 = 2 m. | Step 3: Final Answer: 2 m 60 cm.

Question: Exercise 5.2, Question 2 (iv): Subtract: 305 m 20 cm from 862 m 60 cm.

Answer: Step 1: Subtract the cm parts: 60 - 20 = 40 cm. | Step 2: Subtract the m parts: 862 - 305 = 557 m. | Step 3: Final Answer: 557 m 40 cm.

Question: Exercise 5.2, Question 2 (v): Subtract: 36 km 500 m from 87 km 250 m.

Answer: Step 1: The m part 250 is smaller than 500, so borrow 1 km: 1 km = 1000 m; now the m part is 1250 and the km part is 86. | Step 2: Subtract the m parts: 1250 - 500 = 750 m. | Step 3: Subtract the km parts: 86 - 36 = 50 km. | Step 4: Final Answer: 50 km 750 m.

Question: Exercise 5.2, Question 2 (vi): Subtract: 18 km 352 m from 70 km 100 m.

Answer: Step 1: The m part 100 is smaller than 352, so borrow 1 km: 1 km = 1000 m; now the m part is 1100 and the km part is 69. | Step 2: Subtract the m parts: 1100 - 352 = 748 m. | Step 3: Subtract the km parts: 69 - 18 = 51 km. | Step 4: Final Answer: 51 km 748 m.

Question: Exercise 5.2, Question 2 (vii): Subtract: 106 m 18 cm from 300 m 29 cm.

Answer: Step 1: Subtract the cm parts: 29 - 18 = 11 cm. | Step 2: Subtract the m parts: 300 - 106 = 194 m. | Step 3: Final Answer: 194 m 11 cm.

Question: Exercise 5.2, Question 2 (viii): Subtract: 27 cm 8 mm from 74 cm 7 mm.

Answer: Step 1: The mm part 7 is smaller than 8, so borrow 1 cm: 1 cm = 10 mm; now the mm part is 17 and the cm part is 73. | Step 2: Subtract the mm parts: 17 - 8 = 9 mm. | Step 3: Subtract the cm parts: 73 - 27 = 46 cm. | Step 4: Final Answer: 46 cm 9 mm.

Question: Exercise 5.2, Question 2 (ix): Subtract: 37 cm 5 mm from 64 cm 3 mm.

Answer: Step 1: The mm part 3 is smaller than 5, so borrow 1 cm: 1 cm = 10 mm; now the mm part is 13 and the cm part is 63. | Step 2: Subtract the mm parts: 13 - 5 = 8 mm. | Step 3: Subtract the cm parts: 63 - 37 = 26 cm. | Step 4: Final Answer: 26 cm 8 mm.

Question: Activity: Use the correct unit of cm, m or km to fill each box: (1) The length of a pen (2) The length of a bed (3) The length of a hockey ground (4) The width of your geometry box (5) The distance from Karachi to Larkana.

Answer: Step 1: Think of the length of each thing and choose the correct unit. | Step 2: The length of a pen: centimetre (cm) (given). | Step 3: The length of a bed: metre (m), because a bed is long. | Step 4: The length of a hockey ground: metre (m). | Step 5: The width of your geometry box: centimetre (cm). | Step 6: The distance from Karachi to Larkana: kilometre (km), because the distance between cities is very large. | Step 7: Final Answer: pen cm, bed m, hockey ground m, geometry box cm, Karachi-Larkana km.

Exercise 5.3: Length Word Problems

Exercise 5.3, Question 1 (i): Choose the best unit of length. The thickness of an eraser is:

(a) 2 mm

✓ (b) 2 cm

(c) 2 m

(d) 2 km

Exercise 5.3, Question 1 (ii): Choose the best unit of length. The length of a football ground is:

(a) 30 mm

(b) 30 cm

✓ (c) 30 m

(d) 30 km

Exercise 5.3, Question 1 (iii): Choose the best unit of length. The distance between Karachi and Hyderabad by road is:

(a) 165 mm

(b) 165 cm

(c) 165 m

✓ (d) 165 km

Question: Exercise 5.3, Question 2: The length of an iron rod is 2 m 86 cm; how much is left if 1 m 38 cm has been cut off?

Answer: Step 1: Subtract the cm parts: 86 - 38 = 48 cm. | Step 2: Subtract the m parts: 2 - 1 = 1 m. | Step 3: Final Answer: 1 m 48 cm.

Question: Exercise 5.3, Question 3: Ali covered 789 m from his house to Jamia Masjid and then 368 m from Jamia Masjid to School. What is the total distance covered?

Answer: Step 1: Ali went from home to the mosque and then to school, so add the two distances. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 9 + 8 = 17; write 7 and carry 1 to the next column. | Step 4: Add the tens column: 8 + 6 + 1 carried over = 15; write 5 and carry 1 to the next column. | Step 5: Add the hundreds column: 7 + 3 + 1 carried over = 11; write 1 and carry 1 to the next column. | Step 6: Write the carried 1 in the next

column. | Step 7: Final Answer: Ali covered 1,157 m in all.

Question: Exercise 5.3, Question 4: A car is 1 m 62 cm wide; a garage is 2 m 41 cm wide. How much space is left when the car is in the garage?

Answer: Step 1: The cm part 41 is smaller than 62, so borrow 1 m: 1 m = 100 cm; now the cm part is 141 and the m part is 1. | Step 2: Subtract the cm parts: 141 - 62 = 79 cm. | Step 3: Subtract the m parts: 1 - 1 = 0 m. | Step 4: Final Answer: 79 cm.

Question: Exercise 5.3, Question 5: The red part of a colour pencil is 65 mm long; the blue part is 52 mm long. What is the length of the full pencil in millimetres and centimetres?

Answer: Step 1: The red part is 65 mm long and the blue part is 52 mm long, so add them. | Step 2: 65 + 52 = 117 mm. | Step 3: Convert millimetres into centimetres: 10 mm make 1 cm; 117 mm makes 11 cm and 7 mm. | Step 4: Final Answer: The full pencil is 117 mm long, that is, 11 cm 7 mm.

Question: Exercise 5.3, Question 6: In a walking race, Tariq ran 9 km 200 m and Sajjad ran only 8 km 850 m. How far ahead of Sajjad was Tariq?

Answer: Step 1: The m part 200 is smaller than 850, so borrow 1 km: 1 km = 1000 m; now the m part is 1200 and the km part is 8. | Step 2: Subtract the m parts: 1200 - 850 = 350 m. | Step 3: Subtract the km parts: 8 - 8 = 0 km. | Step 4: Final Answer: 350 m.

Question: Exercise 5.3, Question 7: In a 10 kilometre race, a horse fell down at a distance of 245 m from the winning point. What distance had the horse run before it fell down?

Answer: Step 1: First change the total race length into metres: 10 km = 10,000 m. Then subtract the 245 m left to the winning point. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Ones column: after borrowing, the top digit becomes

10; 10 - 5 = 5; write 5. | Step 4: Tens column: after borrowing, the top digit becomes 9; 9 - 4 = 5; write 5. | Step 5: Hundreds column: after borrowing, the top digit becomes 9; 9 - 2 = 7; write 7. | Step 6: Thousands column: after borrowing, the top digit becomes 9; 9 - 0 = 9; write 9. | Step 7: Ten thousands column: after borrowing, the top digit becomes 0; 0 - 0 = 0; write 0. | Step 8: Final Answer: The horse had run 9,755 m, that is, 9 km 755 m, before it fell down.

Question: Exercise 5.3, Question 8: Nasir is 142 cm tall; his friend is 8 cm taller than Nasir. How tall is his friend?

Answer: Step 1: The friend is 8 cm taller than Nasir, so add 8 to Nasir's height. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 2 + 8 = 10; write 0 and carry 1 to the next column. | Step 4: Add the tens column: 4 + 0 + 1 carried over = 5; write 5. | Step 5: Add the hundreds column: 1 + 0 = 1; write 1. | Step 6: Final Answer: His friend is 150 cm tall.

Question: Exercise 5.3, Question 9: Saba's house is at a distance of 375 m from school and 505 m from the railway station. What is the difference between the distances of school and railway station from Saba's home?

Answer: Step 1: The difference between the two distances is needed, so subtract the smaller distance from the bigger one. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 5 - 5 = 0; write 0. | Step 4: Tens column: after borrowing, the top digit becomes 10; 10 - 7 = 3; write 3. | Step 5: Hundreds column: after borrowing, the top digit becomes 4; 4 - 3 = 1; write 1. | Step 6: Final Answer: The difference between the distances is 130 m.

Activity and Exercise 5.4: Mass

Question: Activity: Convert into grams: (1) 18 kg (given) (2) 25 kg (3) 4 kg 80 g (4) 5 kg 125 g.

Answer: Step 1: Remember: 1 kilogram = 1,000 grams. | Step 2: 18 x 1,000 = 18,000 grams. | Step 3: 25 x 1,000 = 25,000 grams. | Step 4: For 4 kg 80 g: 4 x 1,000 = 4,000; add 80 to get 4,080 grams. | Step 5: For 5 kg 125 g: 5 x 1,000 = 5,000; add 125 to get 5,125 grams. | Step 6: Final Answer: 18 kg = 18,000 g, 25 kg = 25,000 g, 4 kg 80 g = 4,080 g and 5 kg 125 g = 5,125 g.

Question: Exercise 5.4, Question 1 (i): Add: 3,705 g, 8,536 g and 4,000 g.

Answer: Step 1: All three weights are in grams, so add directly. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 5 + 6 + 0 = 11; write 1 and carry 1 to the next column. | Step 4: Add the tens column: 0 + 3 + 0 + 1 carried over = 4; write 4. | Step 5: Add the hundreds column: 7 + 5 + 0 = 12; write 2 and carry 1 to the next column. | Step 6: Add the thousands column: 3 + 8 + 4 + 1 carried over = 16; write 6 and carry 1 to the next column. | Step 7: Write the carried 1 in the next column. | Step 8: Final Answer: 16,241 g.

Question: Exercise 5.4, Question 1 (ii): Add: 4 kg 485 g, 2 kg 390 g and 4 kg 425 g.

Answer: Step 1: Add the g parts: 485 + 390 + 425 = 1300 g. | Step 2: 1300 g = 1 kg and 300 g, because 1 kg = 1000 g; carry 1 to the kg side. | Step 3: Add the kg parts: 4 + 2 + 4 + 1 carried = 11 kg. | Step 4: Final Answer: 11 kg 300 g.

Question: Exercise 5.4, Question 1 (iii): Add: 8 kg 75 g and 9 kg 46 g.

Answer: Step 1: Add the g parts: 75 + 46 = 121 g. | Step 2: Add the kg parts: 8 + 9 = 17 kg. | Step 3: Final Answer: 17 kg 121 g.

Question: Exercise 5.4, Question 1 (iv): Add: 4 kg 32 g and 3 kg 85 g.

Answer: Step 1: Add the g parts: 32 + 85 = 117 g. | Step 2: Add the kg

parts: 4 + 3 = 7 kg. | Step 3: Final Answer: 7 kg 117 g.

Question: Exercise 5.4, Question 1 (v): Add: 16 kg 860 g, 23 kg 545 g and 49 kg 360 g.

Answer: Step 1: Add the g parts: 860 + 545 + 360 = 1765 g. | Step 2: 1765 g = 1 kg and 765 g, because 1 kg = 1000 g; carry 1 to the kg side. | Step 3: Add the kg parts: 16 + 23 + 49 + 1 carried = 89 kg. | Step 4: Final Answer: 89 kg 765 g.

Question: Exercise 5.4, Question 2 (i): Subtract: 1 kg 250 g from 5 kg.

Answer: Step 1: The g part 0 is smaller than 250, so borrow 1 kg: 1 kg = 1000 g; now the g part is 1000 and the kg part is 4. | Step 2: Subtract the g parts: 1000 - 250 = 750 g. | Step 3: Subtract the kg parts: 4 - 1 = 3 kg. | Step 4: Final Answer: 3 kg 750 g.

Question: Exercise 5.4, Question 2 (ii): Subtract: 3 kg 33 g from 6 kg 86 g.

Answer: Step 1: Subtract the g parts: 86 - 33 = 53 g. | Step 2: Subtract the kg parts: 6 - 3 = 3 kg. | Step 3: Final Answer: 3 kg 53 g.

Question: Exercise 5.4, Question 2 (iii): Subtract: 4,505 g from 9,007 g.

Answer: Step 1: Both weights are in grams, so subtract directly. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 7 - 5 = 2; write 2. | Step 4: Subtract the tens column: 0 - 0 = 0; write 0. | Step 5: Hundreds column: after borrowing, the top digit becomes 10; 10 - 5 = 5; write 5. | Step 6: Thousands column: after borrowing, the top digit becomes 8; 8 - 4 = 4; write 4. | Step 7: Final Answer: 4,502 g.

Question: Exercise 5.4, Question 2 (iv): Subtract: 36 kg 740 g from 59 kg 960 g.

Answer: Step 1: Subtract the g parts: 960 - 740 = 220 g. | Step 2: Subtract the kg parts: 59 - 36 = 23 kg. | Step 3: Final Answer: 23 kg 220 g.

Question: Exercise 5.4, Question 2 (v): Subtract: 14 kg 72 g from 20 kg 40

g.

Answer: Step 1: The g part 40 is smaller than 72, so borrow 1 kg: 1 kg = 1000 g; now the g part is 1040 and the kg part is 19. | Step 2: Subtract the g parts: 1040 - 72 = 968 g. | Step 3: Subtract the kg parts: 19 - 14 = 5 kg. | Step 4: Final Answer: 5 kg 968 g.

Question: Activity: Use the appropriate unit 'kg' or 'g' in the blanks: (1) The mass of a silver ring (2) The mass of bags of flour (3) The mass of a sugar bag (4) The mass of one soap (5) The mass of potatoes and onions (6) The mass of a salt bag.

Answer: Step 1: Think of the weight of each thing and choose the correct unit (kg or g). | Step 2: The mass of a silver ring: gram (g) (given), because a ring is light. | Step 3: The mass of bags of flour: kilogram (kg). | Step 4: The mass of a sugar bag: kilogram (kg). | Step 5: The mass of one soap: gram (g). | Step 6: The mass of potatoes and onions: kilogram (kg). | Step 7: The mass of a salt bag: kilogram (kg). | Step 8: Final Answer: ring g, flour bags kg, sugar kg, soap g, potatoes and onions kg, salt kg.

Exercise 5.5: Mass Word Problems

Exercise 5.5, Question 1 (i): Choose the answer in the best unit of mass. The mass of a paper clip:

- ✓ (a) 1 g
- (b) 1 kg
- (c) 100 g
- (d) 100 kg

Exercise 5.5, Question 1 (ii): Choose the answer in the best unit of mass. The mass of a 13 years old boy:

- (a) 4 g
- (b) 4 kg

(c) 40 kg

(d) 400 g

Exercise 5.5, Question 1 (iii): Choose the answer in the best unit of mass.

The mass of a box of tea:

(a) 40 g

(b) 400 g

(c) 4 g

(d) 4 kg

Exercise 5.5, Question 1 (iv): Choose the answer in the best unit of mass.

The weight of a watermelon:

(a) 5 kg

(b) 50 g

(c) 50 kg

(d) 500 g

Question: Exercise 5.5, Question 2: A bale of rubber weighs 75 kg 700 g; another weighs 86 kg 400 g. Find the total weight.

Answer: Step 1: Add the g parts: 700 + 400 = 1100 g. | Step 2: 1100 g = 1 kg and 100 g, because 1 kg = 1000 g; carry 1 to the kg side. | Step 3: Add the kg parts: 75 + 86 + 1 carried = 162 kg. | Step 4: Final Answer: 162 kg 100 g.

Question: Exercise 5.5, Question 3: Hussain weighs 28 kg 750 g and his father weighs 63 kg 500 g. How much lighter is Hussain than his father?

Answer: Step 1: The g part 500 is smaller than 750, so borrow 1 kg: 1 kg = 1000 g; now the g part is 1500 and the kg part is 62. | Step 2: Subtract the g parts: 1500 - 750 = 750 g. | Step 3: Subtract the kg parts: 62 - 28 = 34 kg. | Step 4: Final Answer: 34 kg 750 g.

Question: Exercise 5.5, Question 4: Fozia bought 21 kg 350 g of sweet from one shop and 1 kg 200 g from another shop. What is the total weight

of sweet purchased in all?

Answer: Step 1: Add the g parts: 350 + 200 = 550 g. | Step 2: Add the kg parts: 21 + 1 = 22 kg. | Step 3: Final Answer: 22 kg 550 g.

Question: Exercise 5.5, Question 5: A grain merchant had 3,000 kg of peas; he sold 1,856 kg 750 g of it. What weight of peas was left with him?

Answer: Step 1: The g part 0 is smaller than 750, so borrow 1 kg: 1 kg = 1000 g; now the g part is 1000 and the kg part is 2999. | Step 2: Subtract the g parts: 1000 - 750 = 250 g. | Step 3: Subtract the kg parts: 2999 - 1856 = 1143 kg. | Step 4: Final Answer: 1143 kg 250 g.

Activity and Exercise 5.6: Volume

Question: Activity: Convert into millilitres: (1) 15 l (given) (2) 40 l (3) 75 l (4) 66 l.

Answer: Step 1: Remember: 1 litre = 1,000 millilitres. | Step 2: 15 x 1000 = 15000 millilitres. | Step 3: 40 x 1000 = 40000 millilitres. | Step 4: 75 x 1000 = 75000 millilitres. | Step 5: 66 x 1000 = 66000 millilitres. | Step 6: Final Answer: all the conversions above are complete.

Question: Exercise 5.6, Question 1 (i): Add: 7 l 420 ml and 10 l 500 ml.

Answer: Step 1: Add the ml parts: 420 + 500 = 920 ml. | Step 2: Add the l parts: 7 + 10 = 17 l. | Step 3: Final Answer: 17 l 920 ml.

Question: Exercise 5.6, Question 1 (ii): Add: 2,100 ml, 4,960 ml and 3,755 ml.

Answer: Step 1: All three quantities are in millilitres, so add directly. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 0 + 0 + 5 = 5; write 5. | Step 4: Add the tens column: 0 + 6 + 5 = 11; write 1 and carry 1 to the next column. | Step 5: Add the hundreds column: 1 + 9 + 7 + 1 carried over = 18; write 8 and carry 1 to the next column. | Step 6: Add the thousands column: 2 + 4

+ 3 + 1 carried over = 10; write 0 and carry 1 to the next column. | Step 7: Write the carried 1 in the next column. | Step 8: Final Answer: 10,815 ml, that is, 10 l 815 ml.

Question: Exercise 5.6, Question 1 (iii): Add: 7 l 25 ml, 16 l 400 ml and 31 l 251 ml.

Answer: Step 1: Add the ml parts: 25 + 400 + 251 = 676 ml. | Step 2: Add the l parts: 7 + 16 + 31 = 54 l. | Step 3: Final Answer: 54 l 676 ml.

Question: Exercise 5.6, Question 1 (iv): Add: 705 ml, 820 ml and 695 ml.

Answer: Step 1: All three quantities are in millilitres, so add directly. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 5 + 0 + 5 = 10; write 0 and carry 1 to the next column. | Step 4: Add the tens column: 0 + 2 + 9 + 1 carried over = 12; write 2 and carry 1 to the next column. | Step 5: Add the hundreds column: 7 + 8 + 6 + 1 carried over = 22; write 2 and carry 2 to the next column. | Step 6: Write the carried 2 in the next column. | Step 7: Final Answer: 2,220 ml, that is, 2 l 220 ml.

Question: Exercise 5.6, Question 1 (v): Add: 14 l 782 ml, 17 l 300 ml and 26 l 450 ml.

Answer: Step 1: Add the ml parts: 782 + 300 + 450 = 1532 ml. | Step 2: 1532 ml = 1 l and 532 ml, because 1 l = 1000 ml; carry 1 to the l side. | Step 3: Add the l parts: 14 + 17 + 26 + 1 carried = 58 l. | Step 4: Final Answer: 58 l 532 ml.

Question: Exercise 5.6, Question 2 (i): Subtract: 719 l from 825 l.

Answer: Step 1: Both quantities are in litres, so subtract directly. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Ones column: after borrowing, the top digit becomes 15; 15 - 9 = 6; write 6. | Step 4: Tens column: after borrowing, the top digit becomes 1; 1 - 1 = 0; write 0. | Step 5: Subtract the hundreds column: 8 -

7 = 1; write 1. | Step 6: Final Answer: 106 l.

Question: Exercise 5.6, Question 2 (ii): Subtract: 16 l 415 ml from 60 l 600 ml.

Answer: Step 1: Subtract the ml parts: 600 - 415 = 185 ml. | Step 2: Subtract the l parts: 60 - 16 = 44 l. | Step 3: Final Answer: 44 l 185 ml.

Question: Exercise 5.6, Question 2 (iv): Subtract: 8 l 205 ml from 11 l 150 ml.

Answer: Step 1: The ml part 150 is smaller than 205, so borrow 1 l: 1 l = 1000 ml; now the ml part is 1150 and the l part is 10. | Step 2: Subtract the ml parts: 1150 - 205 = 945 ml. | Step 3: Subtract the l parts: 10 - 8 = 2 l. | Step 4: Final Answer: 2 l 945 ml.

Question: Exercise 5.6, Question 2 (v): Subtract: 76 l 223 ml from 97 l 660 ml.

Answer: Step 1: Subtract the ml parts: 660 - 223 = 437 ml. | Step 2: Subtract the l parts: 97 - 76 = 21 l. | Step 3: Final Answer: 21 l 437 ml.

Question: Exercise 5.6, Question 2 (iii): Subtract: 640 ml from 905 ml.

Answer: Step 1: Both quantities are in millilitres, so subtract directly. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 5 - 0 = 5; write 5. | Step 4: Tens column: after borrowing, the top digit becomes 10; 10 - 4 = 6; write 6. | Step 5: Hundreds column: after borrowing, the top digit becomes 8; 8 - 6 = 2; write 2. | Step 6: Final Answer: 265 ml.

Question: Activity: Choose the appropriate measure: (1) Cold drink bottle: 25 ml / 250 ml / 2500 ml (2) Drink can: 36 ml / 360 ml / 3600 ml (3) Milk pack: 1 l / 10 l / 100 l.

Answer: Step 1: Think of the usual capacity of each thing and choose the appropriate measure. | Step 2: Cold drink bottle: 250 ml (given). | Step 3: Drink can: 360 ml, because a can is bigger than the small bottle but less

than litres. | Step 4: Milk pack: 1 litre, because milk packs are commonly of one litre. | Step 5: Final Answer: bottle 250 ml, can 360 ml, milk pack 1 litre.

Exercise 5.7: Volume Word Problems

Exercise 5.7, Question 1 (i): Choose the best unit of volume. Oil tin:

✓ (a) 4 l

(b) 40 l

(c) 400 l

Exercise 5.7, Question 1 (ii): Choose the best unit of volume. Juice carton:

✓ (a) 1 l

(b) 1 ml

(c) 2 ml

Exercise 5.7, Question 1 (iii): Choose the best unit of volume. Ice-cream tub:

(a) 500 l

✓ (b) 500 ml

(c) 800 ml

Exercise 5.7, Question 1 (iv): Choose the best unit of volume. Milk jug:

✓ (a) 4 l

(b) 4 ml

Exercise 5.7, Question 1 (v): Choose the best unit of volume. Thermos flask:

(a) 140 l

✓ (b) 140 ml

Exercise 5.7, Question 1 (vi): Choose the best unit of volume. Small medicine bottle:

(a) 5 l

✓ (b) 5 ml

Question: Exercise 5.7, Question 2: How much water is left if 19 ml are taken from a cup holding 28 ml?

Answer: Step 1: The cup holds 28 ml and 19 ml are taken out, so subtract. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Ones column: after borrowing, the top digit becomes 18; 18 - 9 = 9; write 9. | Step 4: Tens column: after borrowing, the top digit becomes 1; 1 - 1 = 0; write 0. | Step 5: Final Answer: 9 ml of water is left.

Question: Exercise 5.7, Question 3: The bath tub in Sara's house requires 850 l of water to fill; it now holds 552 l. How many more litres are needed?

Answer: Step 1: The tub needs 850 litres to fill and already holds 552 litres, so subtract. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Ones column: after borrowing, the top digit becomes 10; 10 - 2 = 8; write 8. | Step 4: Tens column: after borrowing, the top digit becomes 14; 14 - 5 = 9; write 9. | Step 5: Hundreds column: after borrowing, the top digit becomes 7; 7 - 5 = 2; write 2. | Step 6: Final Answer: 298 more litres are needed.

Question: Exercise 5.7, Question 4: A container contains 98 l 300 ml of oil; if 51 l 700 ml more is added, how much oil will there be altogether?

Answer: Step 1: Add the ml parts: 300 + 700 = 1000 ml. | Step 2: 1000 ml = 1 l and 0 ml, because 1 l = 1000 ml; carry 1 to the l side. | Step 3: Add the l parts: 98 + 51 + 1 carried = 150 l. | Step 4: Final Answer: 150 l.

Question: Exercise 5.7, Question 5: A milk van carried 272 l of milk; 35 l 875 ml of the milk was spilt in an accident. How much milk was left?

Answer: Step 1: The ml part 0 is smaller than 875, so borrow 1 l: 1 l = 1000 ml; now the ml part is 1000 and the l part is 271. | Step 2: Subtract the ml parts: 1000 - 875 = 125 ml. | Step 3: Subtract the l parts: 271 - 35 = 236 l. | Step 4: Final Answer: 236 l 125 ml.

Question: Exercise 5.7, Question 6: A water drum contains 500 l of water; after watering the flowers, 260 ml of water is left. How much water has been used for watering?

Answer: Step 1: The ml part 0 is smaller than 260, so borrow 1 l: 1 l = 1000 ml; now the ml part is 1000 and the l part is 499. | Step 2: Subtract the ml parts: 1000 - 260 = 740 ml. | Step 3: Subtract the l parts: 499 - 0 = 499 l. | Step 4: Final Answer: 499 l 740 ml.

Question: Exercise 5.7, Question 7: There is 1 l 10 ml of syrup in one bottle and another bottle contains 2 l 75 ml. Find the total quantity of syrup in both bottles.

Answer: Step 1: Add the ml parts: 10 + 75 = 85 ml. | Step 2: Add the l parts: 1 + 2 = 3 l. | Step 3: Final Answer: 3 l 85 ml.

Question: Exercise 5.7, Question 8: There is 80 l 750 ml of diesel in the tank of a bus. How much diesel must be added to make it 100 l?

Answer: Step 1: The ml part 0 is smaller than 750, so borrow 1 l: 1 l = 1000 ml; now the ml part is 1000 and the l part is 99. | Step 2: Subtract the ml parts: 1000 - 750 = 250 ml. | Step 3: Subtract the l parts: 99 - 80 = 19 l. | Step 4: Final Answer: 19 l 250 ml.

Time Activities and Exercises 5.8-5.10: Measurement of Time

Question: Activity Time-1: Use a.m. or p.m.: (1) 8 o'clock in the morning (2) 5 o'clock in the evening (3) 1 o'clock in the morning (4) 9 o'clock at night (5) 2 hours before midnight.

Answer: Step 1: Remember: the time before midnight (12 o'clock) in the morning is called a.m. and the time after noon is called p.m. | Step 2: 8 o'clock in the morning: 8 a.m. (given). | Step 3: 5 o'clock in the evening: 5 p.m., because it is after noon. | Step 4: 1 o'clock in the morning: 1 a.m. | Step 5: 9 o'clock at night: 9 p.m. | Step 6: 2 hours before midnight: 10

o'clock at night, that is, 10 p.m. | Step 7: Final Answer: 8 a.m., 5 p.m., 1 a.m., 9 p.m., 10 p.m.

Question: Activity Time-2 (i): Write the time of the clock in words and in numbers.

Answer: Step 1: In the clock, the short hand shows the hour and the long hand shows the minutes. | Step 2: The time in words: 5 minutes to 3. | Step 3: Final Answer: 2:55 (5 minutes to 3).

Question: Activity Time-2 (ii): Write the time of the clock in words and in numbers.

Answer: Step 1: In the clock, the short hand shows the hour and the long hand shows the minutes. | Step 2: The time in words: 10 minutes past 8. | Step 3: Final Answer: 8:10 (10 minutes past 8).

Question: Activity Time-2 (iii): Write the time of the clock in words and in numbers.

Answer: Step 1: In the clock, the short hand shows the hour and the long hand shows the minutes. | Step 2: The time in words: 40 minutes past 1. | Step 3: Final Answer: 1:40 (40 minutes past 1).

Question: Activity Time-2 (iv): Write the time of the clock in words and in numbers.

Answer: Step 1: In the clock, the short hand shows the hour and the long hand shows the minutes. | Step 2: The time in words: quarter to 5. | Step 3: Final Answer: 4:45 (quarter to 5).

Question: Activity Time-2 (v): Write the time of the clock in words and in numbers.

Answer: Step 1: In the clock, the short hand shows the hour and the long hand shows the minutes. | Step 2: The time in words: quarter past 7. | Step 3: Final Answer: 7:15 (quarter past 7).

Question: Activity Time-2 (vi): Write the time of the clock in words and in

numbers.

Answer: Step 1: In the clock, the short hand shows the hour and the long hand shows the minutes. | Step 2: The time in words: half past 9. | Step 3: Final Answer: 9:30 (half past 9).

Question: Activity Time-3: Convert into minutes: (1) 2 h 10 min (2) 3 h 32 min (3) 1 h 45 min (4) 5 hours.

Answer: Step 1: Remember: 1 hour = 60 minutes, so multiply the hours by 60 and add the minutes. | Step 2: 2 h 10 min: 2 x 60 = 120; add 10 to get 130 minutes (given). | Step 3: 3 h 32 min: 3 x 60 = 180; add 32 to get 212 minutes. | Step 4: 1 h 45 min: 1 x 60 = 60; add 45 to get 105 minutes. | Step 5: 5 hours: 5 x 60 = 300 minutes. | Step 6: Final Answer: 130 minutes, 212 minutes, 105 minutes, 300 minutes.

Question: Activity Time-4: Convert into seconds: (1) 47 min (2) 1 min 5 sec (3) 12 min 15 sec (4) 45 min 10 sec (5) 59 min.

Answer: Step 1: Remember: 1 minute = 60 seconds, so multiply the minutes by 60 and add the seconds. | Step 2: 47 min: 47 x 60 = 2,820 seconds (given). | Step 3: 1 min 5 sec: 1 x 60 = 60; add 5 to get 65 seconds. | Step 4: 12 min 15 sec: 12 x 60 = 720; add 15 to get 735 seconds. | Step 5: 45 min 10 sec: 45 x 60 = 2,700; add 10 to get 2,710 seconds. | Step 6: 59 min: 59 x 60 = 3,540 seconds. | Step 7: Final Answer: 2,820 seconds, 65 seconds, 735 seconds, 2,710 seconds, 3,540 seconds.

Question: Activity Time-5: Convert into months: (1) 2 years (2) 5 years 2 months (3) 10 years 8 months (4) 7 years 6 months.

Answer: Step 1: Remember: 1 year = 12 months, so multiply the years by 12 and add the months. | Step 2: 2 years: 2 x 12 = 24 months (given). | Step 3: 5 years 2 months: 5 x 12 = 60; add 2 to get 62 months. | Step 4: 10 years 8 months: 10 x 12 = 120; add 8 to get 128 months. | Step 5: 7

years 6 months: 7 x 12 = 84; add 6 to get 90 months. | Step 6: Final Answer: 24 months, 62 months, 128 months, 90 months.

Question: Activity Time-6: Convert into days: (1) 4 months 10 days (2) 6 months 4 days (3) 9 months 20 days (4) 18 months 15 days.

Answer: Step 1: Remember: 1 month is taken as 30 days, so multiply the months by 30 and add the days. | Step 2: 4 months 10 days: 4 x 30 = 120; add 10 to get 130 days (given). | Step 3: 6 months 4 days: 6 x 30 = 180; add 4 to get 184 days. | Step 4: 9 months 20 days: 9 x 30 = 270; add 20 to get 290 days. | Step 5: 18 months 15 days: 18 x 30 = 540; add 15 to get 555 days. | Step 6: Final Answer: 130 days, 184 days, 290 days, 555 days.

Question: Exercise 5.8, Question 1: Convert into months: (i) 5 years (ii) 8 years 6 months (iii) 4 years 9 months (iv) 10 years 2 months (v) 15 years 8 months (vi) 20 years 10 months.

Answer: Step 1: Remember: 1 year = 12 months. | Step 2: 5 years: 5 x 12 = 60. | Step 3: 8 years 6 months: 8 x 12 = 96; add 6 to get 102 months. | Step 4: 4 years 9 months: 4 x 12 = 48; add 9 to get 57 months. | Step 5: 10 years 2 months: 10 x 12 = 120; add 2 to get 122 months. | Step 6: 15 years 8 months: 15 x 12 = 180; add 8 to get 188 months. | Step 7: 20 years 10 months: 20 x 12 = 240; add 10 to get 250 months. | Step 8: Final Answer: 60 months, 102 months, 57 months, 122 months, 188 months, 250 months.

Question: Exercise 5.8, Question 2: Convert into days: (i) 3 months (ii) 3 months 12 days (iii) 8 months 20 days (iv) 4 months 25 days (v) 10 months 28 days (vi) 2 months 15 days.

Answer: Step 1: Remember: 1 month is taken as 30 days. | Step 2: 3 months: 3 x 30 = 90. | Step 3: 3 months 12 days: 3 x 30 = 90; add 12 to get 102 days. | Step 4: 8 months 20 days: 8 x 30 = 240; add 20 to get 260

days. | Step 5: 4 months 25 days: 4 x 30 = 120; add 25 to get 145 days. | Step 6: 10 months 28 days: 10 x 30 = 300; add 28 to get 328 days. | Step 7: 2 months 15 days: 2 x 30 = 60; add 15 to get 75 days. | Step 8: Final Answer: 90 days, 102 days, 260 days, 145 days, 328 days, 75 days.

Question: Exercise 5.8, Question 3: Convert into days: (i) 8 weeks (ii) 25 weeks 3 days (iii) 17 weeks 6 days (iv) 30 weeks (v) 35 weeks 5 days (vi) 41 weeks 2 days.

Answer: Step 1: Remember: 1 week = 7 days. | Step 2: 8 weeks: 8 x 7 = 56. | Step 3: 25 weeks 3 days: 25 x 7 = 175; add 3 to get 178 days. | Step 4: 17 weeks 6 days: 17 x 7 = 119; add 6 to get 125 days. | Step 5: 30 weeks: 30 x 7 = 210. | Step 6: 35 weeks 5 days: 35 x 7 = 245; add 5 to get 250 days. | Step 7: 41 weeks 2 days: 41 x 7 = 287; add 2 to get 289 days. | Step 8: Final Answer: 56 days, 178 days, 125 days, 210 days, 250 days, 289 days.

Question: Exercise 5.9, Question 1 (i): Add: 45 minutes 38 seconds and 30 minutes 40 seconds.

Answer: Step 1: Add the seconds parts: 38 + 40 = 78 seconds. | Step 2: 78 seconds = 1 minutes and 18 seconds, because 1 minutes = 60 seconds; carry 1 to the minutes side. | Step 3: Add the minutes parts: 45 + 30 + 1 carried = 76 minutes. | Step 4: Final Answer: 76 minutes 18 seconds.

Question: Exercise 5.9, Question 1 (ii): Add: 48 minutes 39 seconds and 37 minutes 20 seconds.

Answer: Step 1: Add the seconds parts: 39 + 20 = 59 seconds. | Step 2: Add the minutes parts: 48 + 37 = 85 minutes. | Step 3: Final Answer: 85 minutes 59 seconds.

Question: Exercise 5.9, Question 1 (iii): Add: 28 hours 10 minutes and 31 hours 20 minutes.

Answer: Step 1: Add the minutes parts: 10 + 20 = 30 minutes. | Step 2:

Add the hours parts: 28 + 31 = 59 hours. | Step 3: Final Answer: 59 hours 30 minutes.

Question: Exercise 5.9, Question 1 (iv): Add: 25 hours 20 minutes and 34 hours 15 minutes.

Answer: Step 1: Add the minutes parts: 20 + 15 = 35 minutes. | Step 2: Add the hours parts: 25 + 34 = 59 hours. | Step 3: Final Answer: 59 hours 35 minutes.

Question: Exercise 5.9, Question 2 (i): Subtract: 48 minutes 39 seconds from 59 minutes 49 seconds.

Answer: Step 1: Subtract the seconds parts: 49 - 39 = 10 seconds. | Step 2: Subtract the minutes parts: 59 - 48 = 11 minutes. | Step 3: Final Answer: 11 minutes 10 seconds.

Question: Exercise 5.9, Question 2 (ii): Subtract: 35 minutes 25 seconds from 55 minutes 35 seconds.

Answer: Step 1: Subtract the seconds parts: 35 - 25 = 10 seconds. | Step 2: Subtract the minutes parts: 55 - 35 = 20 minutes. | Step 3: Final Answer: 20 minutes 10 seconds.

Question: Exercise 5.9, Question 2 (iii): Subtract: 32 hours 12 minutes from 45 hours 46 minutes.

Answer: Step 1: Subtract the minutes parts: 46 - 12 = 34 minutes. | Step 2: Subtract the hours parts: 45 - 32 = 13 hours. | Step 3: Final Answer: 13 hours 34 minutes.

Question: Exercise 5.9, Question 2 (iv): Subtract: 23 hours 10 minutes from 54 hours 30 minutes.

Answer: Step 1: Subtract the minutes parts: 30 - 10 = 20 minutes. | Step 2: Subtract the hours parts: 54 - 23 = 31 hours. | Step 3: Final Answer: 31 hours 20 minutes.

Question: Exercise 5.10, Question 1: Rafique took 25 minutes 30 seconds

to reach school while 23 minutes 25 seconds to come back. How much total time did he take to go and come back?

Answer: Step 1: Add the seconds parts: 30 + 25 = 55 seconds. | Step 2: Add the minutes parts: 25 + 23 = 48 minutes. | Step 3: Final Answer: 48 minutes 55 seconds.

Question: Exercise 5.10, Question 2: The Pakistani cricket team took 4 hours 25 minutes to complete their innings, the Indian team took 3 hours 20 minutes. How much total time did both teams take?

Answer: Step 1: Add the minutes parts: 25 + 20 = 45 minutes. | Step 2: Add the hours parts: 4 + 3 = 7 hours. | Step 3: Final Answer: 7 hours 45 minutes.

Question: Exercise 5.10, Question 3: Ahsan takes 42 minutes 54 seconds to complete a job, his friend takes 32 minutes 12 seconds. How much more time does Ahsan take?

Answer: Step 1: Subtract the seconds parts: 54 - 12 = 42 seconds. | Step 2: Subtract the minutes parts: 42 - 32 = 10 minutes. | Step 3: Final Answer: 10 minutes 42 seconds.

Question: Exercise 5.10, Question 4: A train takes 19 hours 48 minutes from Lahore to Karachi, another train takes 17 hours 23 minutes. Find the difference between the times.

Answer: Step 1: Subtract the minutes parts: 48 - 23 = 25 minutes. | Step 2: Subtract the hours parts: 19 - 17 = 2 hours. | Step 3: Final Answer: 2 hours 25 minutes.

Question: Exercise 5.10, Question 5: Momal spent 53 minutes 47 seconds to watch television while Zahid spent 39 minutes 23 seconds. Find the difference in their time.

Answer: Step 1: Subtract the seconds parts: 47 - 23 = 24 seconds. | Step 2: Subtract the minutes parts: 53 - 39 = 14 minutes. | Step 3: Final

Answer: 14 minutes 24 seconds.

Review Exercise 5

Review Exercise 5, Question 1 (i): Tick the correct answer. 1 centimetre is equal to:

(a) 100 mm

✓ (b) 10 mm

(c) 1000 mm

Review Exercise 5, Question 1 (ii): Tick the correct answer. 1 kilometre is equal to:

✓ (a) 1000 m

(b) 100 m

(c) 10 m

Review Exercise 5, Question 1 (iii): Tick the correct answer. 1 litre is equal to:

(a) 100 ml

(b) 10 ml

✓ (c) 1000 ml

Review Exercise 5, Question 1 (iv): Tick the correct answer. The number of hours in a day is:

(a) 12

✓ (b) 24

(c) 30

Question: Review Exercise 5, Question 2: List any four lengths from your surrounding that would be measured by using kilometres.

Answer: Step 1: Think of lengths around you that are so large that they can only be measured in kilometres. | Step 2: For example: the distance between my home and school (in kilometres). | Step 3: The distance between two cities, for example Karachi to Hyderabad. | Step 4: The

length of a long road or motorway. | Step 5: The distance between two villages. | Step 6: Final Answer: home-to-school distance, distance between cities, length of a motorway, distance between villages — all four are measured in kilometres.

Question: Review Exercise 5, Question 3 (i): Convert this length into kilometres: 6,000 m

Answer: Step 1: To change into kilometres, divide the metres by 1,000 (because 1 km = 1,000 m). | Step 2: Multiply: 6000 x 1 = 6. | Step 3: Final Answer: 6 km.

Question: Review Exercise 5, Question 3 (ii): Convert this length into kilometres: 3,500 m

Answer: Step 1: To change into kilometres, divide the metres by 1,000. | Step 2: 3,500 / 1,000 = 3.5, that is, 3 km and 500 m. | Step 3: Final Answer: 3.5 km

Question: Review Exercise 5, Question 4 (i): Convert this length into metres: 15 km

Answer: Step 1: Remember: 1 kilometre = 1,000 metres. | Step 2: Multiply: 15 x 1000 = 15,000. | Step 3: Final Answer: 15,000 metres.

Question: Review Exercise 5, Question 4 (ii): Convert this length into metres: 3 km

Answer: Step 1: Remember: 1 kilometre = 1,000 metres. | Step 2: Multiply: 3 x 1000 = 3,000. | Step 3: Final Answer: 3,000 metres.

Question: Review Exercise 5, Question 5 (i): Add: 30 km 43 m and 18 km 84 m.

Answer: Step 1: Add the m parts: 43 + 84 = 127 m. | Step 2: Add the km parts: 30 + 18 = 48 km. | Step 3: Final Answer: 48 km 127 m.

Question: Review Exercise 5, Question 5 (ii): Add: 48 m 65 cm and 38 m 76 cm.

Answer: Step 1: Add the cm parts: 65 + 76 = 141 cm. | Step 2: 141 cm = 1 m and 41 cm, because 1 m = 100 cm; carry 1 to the m side. | Step 3: Add the m parts: 48 + 38 + 1 carried = 87 m. | Step 4: Final Answer: 87 m 41 cm.

Question: Review Exercise 5, Question 5 (iii): Add: 13 l 800 ml and 12 l 700 ml.

Answer: Step 1: Add the ml parts: 800 + 700 = 1500 ml. | Step 2: 1500 ml = 1 l and 500 ml, because 1 l = 1000 ml; carry 1 to the l side. | Step 3: Add the l parts: 13 + 12 + 1 carried = 26 l. | Step 4: Final Answer: 26 l 500 ml.

Question: Review Exercise 5, Question 5 (iv): Add: 44 kg 380 g and 38 kg 960 g.

Answer: Step 1: Add the g parts: 380 + 960 = 1340 g. | Step 2: 1340 g = 1 kg and 340 g, because 1 kg = 1000 g; carry 1 to the kg side. | Step 3: Add the kg parts: 44 + 38 + 1 carried = 83 kg. | Step 4: Final Answer: 83 kg 340 g.

Question: Review Exercise 5, Question 6 (i): Subtract: 40 km 65 m from 76 km 72 m.

Answer: Step 1: Subtract the m parts: 72 - 65 = 7 m. | Step 2: Subtract the km parts: 76 - 40 = 36 km. | Step 3: Final Answer: 36 km 7 m.

Question: Review Exercise 5, Question 6 (ii): Subtract: 43 m 81 cm from 72 m 34 cm.

Answer: Step 1: The cm part 34 is smaller than 81, so borrow 1 m: 1 m = 100 cm; now the cm part is 134 and the m part is 71. | Step 2: Subtract the cm parts: 134 - 81 = 53 cm. | Step 3: Subtract the m parts: 71 - 43 = 28 m. | Step 4: Final Answer: 28 m 53 cm.

Question: Review Exercise 5, Question 6 (iii): Subtract: 4 l from 12 l 250 ml.

Answer: Step 1: Subtract the ml parts: 250 - 0 = 250 ml. | Step 2: Subtract the l parts: 12 - 4 = 8 l. | Step 3: Final Answer: 8 l 250 ml.

Question: Review Exercise 5, Question 6 (iv): Subtract: 45 kg 325 g from 86 kg 638 g.

Answer: Step 1: Subtract the g parts: 638 - 325 = 313 g. | Step 2: Subtract the kg parts: 86 - 45 = 41 kg. | Step 3: Final Answer: 41 kg 313 g.

6. Geometry

Activities: Drawing Figures and Points

Question: Activity Geo-1: Draw the following diagrams with H and HB grade pencils: (i) Line segment (ii) Square (iii) Circle.

Answer: Step 1: For a line segment: mark two points and join them with a straight edge. | Step 2: For a square: draw four equal sides and four right angles (use a set square). | Step 3: For a circle: put the compass needle on one point and turn the pencil round to complete the circle. | Step 4: Final Answer: All three diagrams are drawn by the steps above.

Question: Activity Geo-2: Take a point A. From point A draw AB with a straight edge. Draw another AC from point A. How many lines can be drawn from point A?

Answer: Step 1: A line from one point can go in any direction. | Step 2: Besides AB and AC, many more lines can be drawn — a new line in every direction. | Step 3: Final Answer: Countless (unlimited) lines can be drawn from one point.

Question: Activity Geo-3: Take a pair of dividers. Penetrate the pointed ends on a paper. Name the two marks as A and B. Now draw lines with the help of a straightedge so that they may pass through both points A and B. How many such lines can be drawn?

Answer: Step 1: Try to join both the points A and B. | Step 2: You will see that however you place the straight edge, only one straight line passes through both points. | Step 3: Final Answer: Only one straight line can be drawn through two given points.

Exercise 6.1: Measuring Sides

Question: Exercise 6.1, Question 1 (i): Measure the sides of triangle ABC with a ruler and write their lengths: m AC, m AB, m CB.

Answer: Step 1: Place the zero of the ruler at one end and read up to the other end. | Step 2: Measure each side separately. | Step 3: Final Answer: m AC = 3 cm, m AB = 5.6 cm and m CB = 3.5 cm.

Question: Exercise 6.1, Question 1 (ii): Measure the sides of quadrilateral PQRS and write: m PQ, m QR, m RS, m SP.

Answer: Step 1: Place the ruler on each side and read the length. | Step 2: Final Answer: m PQ = 5.4 cm, m QR = 5.4 cm, m RS = 2.6 cm and m SP = 3.3 cm.

Question: Exercise 6.1, Question 1 (iii): Measure the sides of rectangle WXYZ and write: m WX, m YZ, m XY, m ZW.

Answer: Step 1: The opposite sides of a rectangle are equal, so two measurements tell all. | Step 2: Final Answer: m WX and m YZ are both 2.7 cm; m XY and m ZW are both 3.5 cm.

Question: Exercise 6.1, Question 2 (i): Join the pair of points to draw the line segment PQ, measure its length with a ruler and verify by dividers.

Answer: Step 1: Join the two points with a straight line. | Step 2: Measure with the ruler: the length of PQ is 8.5 cm (given). | Step 3: Put the divider points on both ends and check the same length on the ruler — it matches. | Step 4: Final Answer: m PQ = 8.5 cm.

Question: Exercise 6.1, Question 2 (ii): Join the pair of points to draw the line segment ST, measure its length with a ruler and verify by dividers.

Answer: Step 1: Join the two points with a straight line. | Step 2: Measure with the ruler: the length of ST is 7.9 cm. | Step 3: Put the divider points on both ends and check the same length on the ruler — it matches. | Step 4: Final Answer: m ST = 7.9 cm.

Question: Exercise 6.1, Question 2 (iii): Join the pair of points to draw the line segment FG, measure its length with a ruler and verify by dividers.

Answer: Step 1: Join the two points with a straight line. | Step 2: Measure with the ruler: the length of FG is 9.1 cm. | Step 3: Put the divider points on both ends and check the same length on the ruler — it matches. | Step 4: Final Answer: m FG = 9.1 cm.

Question: Exercise 6.1, Question 2 (iv): Join the pair of points to draw the line segment YZ, measure its length with a ruler and verify by dividers.

Answer: Step 1: Join the two points with a straight line. | Step 2: Measure with the ruler: the length of YZ is 10 cm. | Step 3: Put the divider points on both ends and check the same length on the ruler — it matches. | Step 4: Final Answer: m YZ = 10 cm.

Question: Exercise 6.1, Question 2 (v): Join the pair of points to draw the line segment MN, measure its length with a ruler and verify by dividers.

Answer: Step 1: Join the two points with a straight line. | Step 2: Measure with the ruler: the length of MN is 10.6 cm. | Step 3: Put the divider points on both ends and check the same length on the ruler — it matches. | Step 4: Final Answer: m MN = 10.6 cm.

Exercise 6.2: Drawing Segments of Given Lengths and Measuring Curved Lines

Question: Exercise 6.2, Question 1 (i): Draw a line segment of the following length using (a) a ruler (b) a ruler and dividers: mAB = 7.4 cm.

Answer: Step 1: (a) With a ruler: mark a point, place the zero of the ruler there and mark a second point at 7.4 cm; join them. | Step 2: (b) With dividers: open the dividers to 7.4 cm on the ruler, then transfer this distance to the paper, mark two points and join them. | Step 3: Final Answer: A segment AB of 7.4 cm has been drawn.

Question: Exercise 6.2, Question 1 (ii): Draw a line segment of the following length using (a) a ruler (b) a ruler and dividers: mBC = 6.6 cm.

Answer: Step 1: (a) With a ruler: mark a point, place the zero of the ruler there and mark a second point at 6.6 cm; join them. | Step 2: (b) With dividers: open the dividers to 6.6 cm on the ruler, then transfer this distance to the paper, mark two points and join them. | Step 3: Final Answer: A segment BC of 6.6 cm has been drawn.

Question: Exercise 6.2, Question 1 (iii): Draw a line segment of the following length using (a) a ruler (b) a ruler and dividers: mCD = 5.7 cm.

Answer: Step 1: (a) With a ruler: mark a point, place the zero of the ruler there and mark a second point at 5.7 cm; join them. | Step 2: (b) With dividers: open the dividers to 5.7 cm on the ruler, then transfer this distance to the paper, mark two points and join them. | Step 3: Final Answer: A segment CD of 5.7 cm has been drawn.

Question: Exercise 6.2, Question 1 (iv): Draw a line segment of the following length using (a) a ruler (b) a ruler and dividers: mDE = 3.8 cm.

Answer: Step 1: (a) With a ruler: mark a point, place the zero of the ruler there and mark a second point at 3.8 cm; join them. | Step 2: (b) With dividers: open the dividers to 3.8 cm on the ruler, then transfer this distance to the paper, mark two points and join them. | Step 3: Final Answer: A segment DE of 3.8 cm has been drawn.

Question: Exercise 6.2, Question 1 (v): Draw a line segment of the following length using (a) a ruler (b) a ruler and dividers: mEF = 4.9 cm.

Answer: Step 1: (a) With a ruler: mark a point, place the zero of the ruler there and mark a second point at 4.9 cm; join them. | Step 2: (b) With dividers: open the dividers to 4.9 cm on the ruler, then transfer this distance to the paper, mark two points and join them. | Step 3: Final Answer: A segment EF of 4.9 cm has been drawn.

Question: Exercise 6.2, Question 1 (vi): Draw a line segment of the following length using (a) a ruler (b) a ruler and dividers: mPQ = 6.0 cm.

Answer: Step 1: (a) With a ruler: mark a point, place the zero of the ruler there and mark a second point at 6.0 cm; join them. | Step 2: (b) With dividers: open the dividers to 6.0 cm on the ruler, then transfer this distance to the paper, mark two points and join them. | Step 3: Final Answer: A segment PQ of 6.0 cm has been drawn.

Question: Activity: Measure the curved line ABC of Fig (i) with thread and ruler.

Answer: Step 1: Lay the thread carefully along the curved line ABC and mark its ends. | Step 2: Straighten the thread and measure it with the ruler. | Step 3: Final Answer: The length of the curved line ABC is 9.6 cm (book result).

Question: Activity: Measure the curved line PQRS of fig (a) with dividers and ruler.

Answer: Step 1: Open the dividers a little and walk small steps along the curve to cover the whole line; count the steps. | Step 2: Measure the divider opening with the ruler and multiply by the number of steps. | Step 3: Final Answer: The length of the curved line PQRS is 10.5 cm (book result).

Question: Exercise 6.2, Question 2 (i): Measure the curved line CD with thread, ruler and dividers and write its length.

Answer: Step 1: Lay the thread along the curved line CD and mark its ends. | Step 2: Straighten the thread and measure it with the ruler; also verify with dividers. | Step 3: Final Answer: The length of the curved line CD is 4.2 cm.

Question: Exercise 6.2, Question 2 (ii): Measure the curved line EF with thread, ruler and dividers and write its length.

Answer: Step 1: Lay the thread along the curved line EF and mark its ends. | Step 2: Straighten the thread and measure it with the ruler; also verify with dividers. | Step 3: Final Answer: The length of the curved line EF is 4.7 cm.

Question: Exercise 6.2, Question 2 (iii): Measure the curved line PK with thread, ruler and dividers and write its length.

Answer: Step 1: Lay the thread along the curved line PK and mark its ends. | Step 2: Straighten the thread and measure it with the ruler; also verify with dividers. | Step 3: Final Answer: The length of the curved line PK is 4.3 cm.

Exercise 6.3: Directions and Horizontal/Vertical Lines

Exercise 6.3, Question (1): Fill in the blank: Horizontal line shows _____ directions.

Answer: West and East

Exercise 6.3, Question (2): Fill in the blank: Vertical line shows _____ directions.

Answer: North and South

Exercise 6.3, Question (3): Fill in the blank: North to South direction represents _____ line.

Answer: Vertical

Exercise 6.3, Question (4): Fill in the blank: West to East direction represents _____ line.

Answer: Horizontal

Exercise 6.4: Parallel and Non-Parallel Lines

Question: Activity: Cross the shapes which represent parallel lines and tick the shapes which represent non-parallel lines: rectangle, square, triangle, trapezium, rhombus (diamond), right triangle.

Answer: Step 1: Look at the opposite sides of each shape: those that never meet are parallel. | Step 2: The opposite sides of a rectangle and a square are parallel; the trapezium and rhombus also have parallel opposite sides — these are crossed. | Step 3: The sides of a triangle and a right triangle cut each other, they are not parallel — these are ticked. | Step 4: Final Answer: Cross — rectangle, square, trapezium, rhombus; Tick — triangle, right triangle.

Question: Exercise 6.4, Question 1 (i): Identify whether line AB and line CD are parallel or non-parallel lines.

Answer: Step 1: Look at the direction of both lines: both are horizontal, going in the same direction. | Step 2: Parallel lines never meet and the distance between them stays the same everywhere. | Step 3: Final Answer: line AB and line CD are parallel lines.

Question: Exercise 6.4, Question 1 (ii): Identify whether line EF and line GH are parallel or non-parallel lines.

Answer: Step 1: Look at the direction of both lines: EF is horizontal but GH is slanting, their directions are different. | Step 2: Parallel lines never meet and the distance between them stays the same everywhere. | Step 3: Final Answer: line EF and line GH are non-parallel lines.

Question: Exercise 6.4, Question 1 (iii): Identify whether line IJ and line KL are parallel or non-parallel lines.

Answer: Step 1: Look at the direction of both lines: IJ is slanting and KL is vertical. | Step 2: Parallel lines never meet and the distance between them stays the same everywhere. | Step 3: Final Answer: line IJ and line KL are non-parallel lines.

Question: Exercise 6.4, Question 1 (iv): Identify whether line MN and line OP are parallel or non-parallel lines.

Answer: Step 1: Look at the direction of both lines: both are slanting the

same way. | Step 2: Parallel lines never meet and the distance between them stays the same everywhere. | Step 3: Final Answer: line MN and line OP are parallel lines.

Question: Exercise 6.4, Question 1 (v): Identify whether line QS and line RT are parallel or non-parallel lines.

Answer: Step 1: Look at the direction of both lines: QS is horizontal and RT is vertical, they would cut each other. | Step 2: Parallel lines never meet and the distance between them stays the same everywhere. | Step 3: Final Answer: line QS and line RT are non-parallel lines.

Question: Exercise 6.4, Question 1 (vi): Identify whether line WU and line VX are parallel or non-parallel lines.

Answer: Step 1: Look at the direction of both lines: both are vertical. | Step 2: Parallel lines never meet and the distance between them stays the same everywhere. | Step 3: Final Answer: line WU and line VX are parallel lines.

Question: Exercise 6.4, Question 1 (vii): Identify whether line YZ and line AB are parallel or non-parallel lines.

Answer: Step 1: Look at the direction of both lines: YZ is vertical and AB is horizontal. | Step 2: Parallel lines never meet and the distance between them stays the same everywhere. | Step 3: Final Answer: line YZ and line AB are non-parallel lines.

Question: Exercise 6.4, Question 1 (viii): Identify whether line BC and line DE are parallel or non-parallel lines.

Answer: Step 1: Look at the direction of both lines: both are slanting the same way. | Step 2: Parallel lines never meet and the distance between them stays the same everywhere. | Step 3: Final Answer: line BC and line DE are parallel lines.

Question: Exercise 6.4, Question 2: Tick the lines which are parallel to line

XY: lines AB, CD and WV are shown with XY.

Answer: Step 1: XY is a horizontal line, so only a line that is also horizontal can be parallel to it. | Step 2: AB and CD are slanting, they would cut XY; WV is horizontal, like XY. | Step 3: Final Answer: Only WV is parallel to XY, so tick only WV.

Question: Exercise 6.4, Question 3: Cross the lines which are non-parallel to line LM: lines ZA, DE and MN are shown with LM.

Answer: Step 1: LM is horizontal; ZA and MN are also horizontal, so they are parallel to LM. | Step 2: DE is slanting, it would cut LM, so it is non-parallel. | Step 3: Final Answer: Cross only DE.

Question: Activity: Draw a line parallel to line AB using set squares.

Answer: Step 1: Place the longest side of one set square exactly on line AB. | Step 2: Place the second set square against another side of the first and hold it firm. | Step 3: Slide the first set square along the second one to the required place. | Step 4: Draw the new line along its longest side — it is parallel to AB. | Step 5: Final Answer: The parallel line is drawn by the steps above.

Exercise 6.5: Drawing Lines with Set Squares

Question: Exercise 6.5, Question 1: Draw a vertical line PQ on a given horizontal line XY, using set squares.

Answer: Step 1: Place the right-angle corner of a set square on XY so that one side lies along XY. | Step 2: Draw a line upward along the other (vertical) side and name it PQ. | Step 3: Check: PQ and XY make a right angle (90 degrees) between them. | Step 4: Final Answer: The vertical line PQ has been drawn.

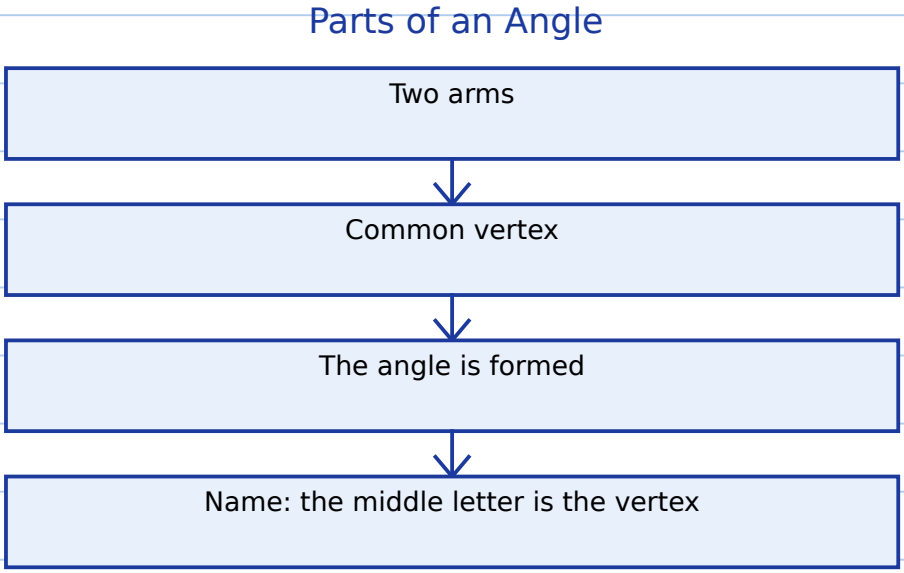
Question: Exercise 6.5, Question 2: Draw a parallel line YZ to a given line PQ using set squares.

Answer: Step 1: Place the long side of one set square on PQ. | Step 2: Put the second set square against it, hold it, and slide the first one to the new place. | Step 3: Draw the line along the long side and name it YZ. | Step 4: Final Answer: YZ has been drawn parallel to PQ.

Question: Exercise 6.5, Question 3: Draw AB which passes through a given point E and is parallel to a given CD (using set squares).

Answer: Step 1: Place the long side of a set square on CD and support it with the second set square. | Step 2: Slide the first set square until its long side reaches the point E. | Step 3: Draw the line through E and name it AB. | Step 4: Final Answer: AB passes through E and is parallel to CD.

Exercise 6.6: Vertex and Arms of Angles



Question: Exercise 6.6, Question 1 (i): Write the names of the vertex and arms of the angle ∠ABC.

Answer: Step 1: In the name of an angle, the middle letter is the vertex: it is B. | Step 2: The two arms starting from the vertex are BA and BC. | Step 3: Final Answer: The vertex is B and the arms are BA and BC.

Question: Exercise 6.6, Question 1 (ii): Write the names of the vertex and arms of the angle ∠PQR.

Answer: Step 1: In the name of an angle, the middle letter is the vertex: it is Q. | Step 2: The two arms starting from the vertex are QP and QR. | Step 3: Final Answer: The vertex is Q and the arms are QP and QR.

Question: Exercise 6.6, Question 1 (iii): Write the names of the vertex and arms of the angle ∠XWY.

Answer: Step 1: In the name of an angle, the middle letter is the vertex: it is W. | Step 2: The two arms starting from the vertex are WX and WY. | Step 3: Final Answer: The vertex is W and the arms are WX and WY.

Question: Exercise 6.6, Question 2 (i): Write the following angle in symbols: the vertex is O and the points on its arms are A and B.

Answer: Step 1: Write the sign ∠, then three letters: first a point on one arm, the vertex O in the middle, and a point on the other arm at the end. | Step 2: Final Answer: ∠AOB

Question: Exercise 6.6, Question 2 (ii): Write the following angle in symbols: the vertex is E and the points on its arms are A and F.

Answer: Step 1: Write the sign ∠, then three letters: first a point on one arm, the vertex E in the middle, and a point on the other arm at the end. | Step 2: Final Answer: ∠AEF

Question: Exercise 6.6, Question 2 (iii): Write the following angle in symbols: the vertex is K and the points on its arms are J and L.

Answer: Step 1: Write the sign ∠, then three letters: first a point on one arm, the vertex K in the middle, and a point on the other arm at the end. | Step 2: Final Answer: ∠JKL

Exercise 6.7: Identifying Right Angles

Question: Exercise 6.7, Question 1 (i): Look and tell whether angle ACD is a right angle.

Answer: Step 1: Check by placing the right-angle corner of a set square on

the angle. | Step 2: The angle does not match the corner of the set square (it is smaller), so it is not a right angle. | Step 3: Final Answer: No, it is not a right angle.

Question: Exercise 6.7, Question 1 (ii): Look and tell whether angle EFG is a right angle.

Answer: Step 1: Check by placing the right-angle corner of a set square on the angle. | Step 2: The angle does not match the corner of the set square (it is bigger), so it is not a right angle. | Step 3: Final Answer: No, it is not a right angle.

Question: Exercise 6.7, Question 1 (iii): Look and tell whether angle MNO is a right angle.

Answer: Step 1: Check by placing the right-angle corner of a set square on the angle. | Step 2: The angle matches the corner of the set square exactly, so it is a right angle (90 degrees). | Step 3: Final Answer: Yes, it is a right angle, so tick it.

Question: Exercise 6.7, Question 1 (iv): Look and tell whether angle PKL is a right angle.

Answer: Step 1: Check by placing the right-angle corner of a set square on the angle. | Step 2: The angle matches the corner of the set square exactly, so it is a right angle (90 degrees). | Step 3: Final Answer: Yes, it is a right angle, so tick it.

Question: Exercise 6.7, Question 1 (v): Look and tell whether angle ONM is a right angle.

Answer: Step 1: Check by placing the right-angle corner of a set square on the angle. | Step 2: The angle matches the corner of the set square exactly, so it is a right angle (90 degrees). | Step 3: Final Answer: Yes, it is a right angle, so tick it.

Question: Exercise 6.7, Question 1 (vi): Look and tell whether the acute

angle shown without labels is a right angle.

Answer: Step 1: Check by placing the right-angle corner of a set square on the angle. | Step 2: The angle does not match the corner of the set square (it is smaller), so it is not a right angle. | Step 3: Final Answer: No, it is not a right angle.

Question: Exercise 6.7, Question 2 (i): Does this figure show a right angle? Figure: horizontal line PQ and vertical line QR meeting at Q.

Answer: Step 1: Check the angles formed in the figure with the corner of a set square. | Step 2: Final Answer: Yes, this figure shows a right angle.

Question: Exercise 6.7, Question 2 (ii): Does this figure show a right angle? Figure: two lines crossing at O.

Answer: Step 1: Check the angles formed in the figure with the corner of a set square. | Step 2: Final Answer: Yes, this figure shows a right angle.

Question: Exercise 6.7, Question 2 (iii): Does this figure show a right angle? Figure: straight line AP with ray OB from O.

Answer: Step 1: Check the angles formed in the figure with the corner of a set square. | Step 2: Final Answer: No, this figure does not show a right angle.

Question: Exercise 6.7, Question 2 (iv): Does this figure show a right angle? Figure: horizontal line PR with line MN crossing at Q.

Answer: Step 1: Check the angles formed in the figure with the corner of a set square. | Step 2: Final Answer: No, this figure does not show a right angle.

Question: Exercise 6.7, Question 2 (v): Does this figure show a right angle? Figure: lines EG and MN crossing at F.

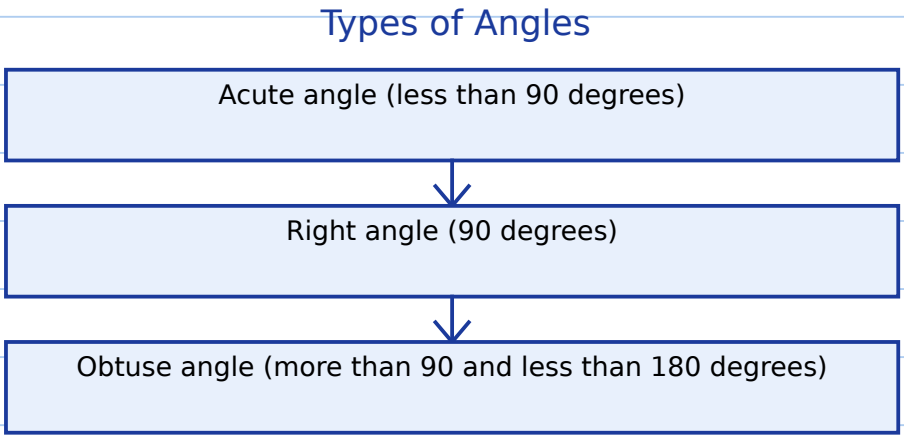
Answer: Step 1: Check the angles formed in the figure with the corner of a set square. | Step 2: Final Answer: Yes, this figure shows a right angle.

Question: Exercise 6.7, Question 2 (vi): Does this figure show a right

angle? Figure: horizontal line AD with vertical line CK at C.

Answer: Step 1: Check the angles formed in the figure with the corner of a set square. | Step 2: Final Answer: Yes, this figure shows a right angle.

Exercise 6.8: Types of Angles



Exercise 6.8, Question 1 (i): Complete the sentence: Angle BAC is _____ angle.

Answer: right

Exercise 6.8, Question 1 (ii): Complete the sentence: Angle BAD is _____ angle.

Answer: acute

Exercise 6.8, Question 1 (iii): Complete the sentence: Angle BAF is _____ angle.

Answer: obtuse

Exercise 6.8, Question 1 (iv): Complete the sentence: Angle DAB is _____ angle.

Answer: acute

Exercise 6.8, Question 1 (v): Complete the sentence: Angle FAB is _____ angle.

Answer: obtuse

Exercise 6.8, Question 1 (vi): Complete the sentence: Angle CAB is

_____ angle.

Answer: right

Question: Exercise 6.8, Question 2 (i): Draw the following angle: angle ABC (obtuse angle).

Answer: Step 1: Choose a vertex and draw one arm from it. | Step 2: Using a set square or protractor, make a obtuse angle and draw the second arm. | Step 3: Name the angle ABC (the middle letter is the vertex). | Step 4: Final Answer: The obtuse angle ABC has been drawn.

Question: Exercise 6.8, Question 2 (ii): Draw the following angle: angle PQR (acute angle).

Answer: Step 1: Choose a vertex and draw one arm from it. | Step 2: Using a set square or protractor, make a acute angle and draw the second arm. | Step 3: Name the angle PQR (the middle letter is the vertex). | Step 4: Final Answer: The acute angle PQR has been drawn.

Question: Exercise 6.8, Question 2 (iii): Draw the following angle: angle ABC (right angle).

Answer: Step 1: Choose a vertex and draw one arm from it. | Step 2: Using a set square or protractor, make a right angle and draw the second arm. | Step 3: Name the angle ABC (the middle letter is the vertex). | Step 4: Final Answer: The right angle ABC has been drawn.

Question: Exercise 6.8, Question 2 (iv): Draw the following angle: angle DEF (obtuse angle).

Answer: Step 1: Choose a vertex and draw one arm from it. | Step 2: Using a set square or protractor, make a obtuse angle and draw the second arm. | Step 3: Name the angle DEF (the middle letter is the vertex). | Step 4: Final Answer: The obtuse angle DEF has been drawn.

Question: Exercise 6.8, Question 2 (v): Draw the following angle: angle WXY (right angle).

Answer: Step 1: Choose a vertex and draw one arm from it. | Step 2: Using a set square or protractor, make a right angle and draw the second arm. | Step 3: Name the angle WXY (the middle letter is the vertex). | Step 4: Final Answer: The right angle WXY has been drawn.

Exercise 6.9: Measuring Angles with a Protractor

Question: Exercise 6.9: Using the protractor, measure the following angles: (1) angle FEG (2) angle RST (3) angle XYZ (4) angle ABC (5) angle JLK (6) angle DEF (7) angle PQR (8) angle STU (9) angle XYZ.

Answer: Step 1: Place the centre of the protractor on the vertex of the angle and its base line exactly on one arm. | Step 2: Read the degrees at the other arm where it cuts the protractor. | Step 3: Angle FEG: the measure is 30 degrees. | Step 4: Angle RST: the measure is 40 degrees. | Step 5: Angle XYZ: the measure is 40 degrees. | Step 6: Angle ABC: the measure is 28 degrees. | Step 7: Angle JLK: the measure is 43 degrees. | Step 8: Angle DEF: the measure is 43 degrees. | Step 9: Angle PQR: the measure is 110 degrees. | Step 10: Angle STU: the measure is 110 degrees. | Step 11: Angle XYZ: the measure is 142 degrees. | Step 12: Final Answer: The measures of all the angles are given above.

Exercise 6.10: Measuring, Naming the Type and Drawing Angles

Question: Exercise 6.10, Question 1 (i): Measure the angle PQR and then tell the type of the angle.

Answer: Step 1: Place the centre of the protractor on the vertex and read the degrees. | Step 2: The measure is 43 degrees. | Step 3: Compare 43 degrees with 90 degrees: it is an acute angle. | Step 4: Final Answer: 43 degrees, acute angle.

Question: Exercise 6.10, Question 1 (ii): Measure the angle ABC and then tell the type of the angle.

Answer: Step 1: Place the centre of the protractor on the vertex and read the degrees. | Step 2: The measure is 43 degrees. | Step 3: Compare 43 degrees with 90 degrees: it is an acute angle. | Step 4: Final Answer: 43 degrees, acute angle.

Question: Exercise 6.10, Question 1 (iii): Measure the angle XYZ and then tell the type of the angle.

Answer: Step 1: Place the centre of the protractor on the vertex and read the degrees. | Step 2: The measure is 70 degrees. | Step 3: Compare 70 degrees with 90 degrees: it is an acute angle. | Step 4: Final Answer: 70 degrees, acute angle.

Question: Exercise 6.10, Question 1 (iv): Measure the angle ABC and then tell the type of the angle.

Answer: Step 1: Place the centre of the protractor on the vertex and read the degrees. | Step 2: The measure is 155 degrees. | Step 3: Compare 155 degrees with 90 degrees: it is an obtuse angle. | Step 4: Final Answer: 155 degrees, obtuse angle.

Question: Exercise 6.10, Question 1 (v): Measure the angle TLM and then tell the type of the angle.

Answer: Step 1: Place the centre of the protractor on the vertex and read the degrees. | Step 2: The measure is 90 degrees. | Step 3: Compare 90 degrees with 90 degrees: it is an right angle. | Step 4: Final Answer: 90 degrees, right angle.

Question: Exercise 6.10, Question 1 (vi): Measure the angle COT and then tell the type of the angle.

Answer: Step 1: Place the centre of the protractor on the vertex and read the degrees. | Step 2: The measure is 155 degrees. | Step 3: Compare 155 degrees with 90 degrees: it is an obtuse angle. | Step 4: Final Answer: 155 degrees, obtuse angle.

Question: Exercise 6.10, Question 2: Draw the following angles with the help of a protractor: (i) 10 degrees (ii) 40 degrees (iii) 20 degrees (iv) 60 degrees (v) 30 degrees (vi) 80 degrees (vii) 90 degrees (viii) 120 degrees (ix) 145 degrees (x) 45 degrees.

Answer: Step 1: Draw a straight line and choose the vertex point on it. | Step 2: Place the centre of the protractor on that point and its base line exactly on the line. | Step 3: (i) Mark at 10 degrees, remove the protractor and join the point to the vertex — an angle of 10 degrees is made. | Step 4: (ii) Mark at 40 degrees, remove the protractor and join the point to the vertex — an angle of 40 degrees is made. | Step 5: (iii) Mark at 20 degrees, remove the protractor and join the point to the vertex — an angle of 20 degrees is made. | Step 6: (iv) Mark at 60 degrees, remove the protractor and join the point to the vertex — an angle of 60 degrees is made. | Step 7: (v) Mark at 30 degrees, remove the protractor and join the point to the vertex — an angle of 30 degrees is made. | Step 8: (vi) Mark at 80 degrees, remove the protractor and join the point to the vertex — an angle of 80 degrees is made. | Step 9: (vii) Mark at 90 degrees, remove the protractor and join the point to the vertex — an angle of 90 degrees is made. | Step 10: (viii) Mark at 120 degrees, remove the protractor and join the point to the vertex — an angle of 120 degrees is made. | Step 11: (ix) Mark at 145 degrees, remove the protractor and join the point to the vertex — an angle of 145 degrees is made. | Step 12: (x) Mark at 45 degrees, remove the protractor and join the point to the vertex — an angle of 45 degrees is made. | Step 13: Final Answer: All ten angles have been drawn.

Exercise 6.11: Drawing Equal, Double and Sum Angles

Question: Exercise 6.11, Question 1 (i): Draw an angle with a protractor

equal in measure to the given angle ABC.

Answer: Step 1: First measure the given angle ABC with the protractor and note its measure. | Step 2: On a new line, choose a vertex and draw exactly the same measure with the protractor. | Step 3: Final Answer: The new angle is drawn equal to the given angle ABC.

Question: Exercise 6.11, Question 1 (ii): Draw an angle with a protractor equal in measure to the given angle PQR.

Answer: Step 1: First measure the given angle PQR with the protractor and note its measure. | Step 2: On a new line, choose a vertex and draw exactly the same measure with the protractor. | Step 3: Final Answer: The new angle is drawn equal to the given angle PQR.

Question: Exercise 6.11, Question 1 (iii): Draw an angle with a protractor equal in measure to the given angle DEF.

Answer: Step 1: First measure the given angle DEF with the protractor and note its measure. | Step 2: On a new line, choose a vertex and draw exactly the same measure with the protractor. | Step 3: Final Answer: The new angle is drawn equal to the given angle DEF.

Question: Exercise 6.11, Question 1 (iv): Draw an angle with a protractor equal in measure to the given angle XYZ.

Answer: Step 1: First measure the given angle XYZ with the protractor and note its measure. | Step 2: On a new line, choose a vertex and draw exactly the same measure with the protractor. | Step 3: Final Answer: The new angle is drawn equal to the given angle XYZ.

Question: Exercise 6.11, Question 2 (i): Draw an angle twice the measure of the given angle ABC.

Answer: Step 1: First measure the given angle ABC. | Step 2: Double its measure (multiply by 2). | Step 3: Draw an angle of that doubled measure with the protractor. | Step 4: Final Answer: The angle of twice the measure

has been drawn.

Question: Exercise 6.11, Question 2 (ii): Draw an angle twice the measure of the given angle ABC.

Answer: Step 1: First measure the given angle ABC. | Step 2: Double its measure (multiply by 2). | Step 3: Draw an angle of that doubled measure with the protractor. | Step 4: Final Answer: The angle of twice the measure has been drawn.

Question: Exercise 6.11, Question 2 (iii): Draw an angle twice the measure of the given angle PQR.

Answer: Step 1: First measure the given angle PQR. | Step 2: Double its measure (multiply by 2). | Step 3: Draw an angle of that doubled measure with the protractor. | Step 4: Final Answer: The angle of twice the measure has been drawn.

Question: Exercise 6.11, Question 2 (iv): Draw an angle twice the measure of the given angle XYZ.

Answer: Step 1: First measure the given angle XYZ. | Step 2: Double its measure (multiply by 2). | Step 3: Draw an angle of that doubled measure with the protractor. | Step 4: Final Answer: The angle of twice the measure has been drawn.

Question: Exercise 6.11, Question 3 (i): Draw an angle equal in measure to the sum of two angles: angle ABC and angle DEF.

Answer: Step 1: Measure both given angles separately. | Step 2: Add the two measures. | Step 3: Draw an angle equal to that sum with the protractor. | Step 4: Final Answer: The angle equal to the sum has been drawn.

Question: Exercise 6.11, Question 3 (ii): Draw an angle equal in measure to the sum of two angles: angle ABC and angle PQR.

Answer: Step 1: Measure both given angles separately. | Step 2: Add the

two measures. | Step 3: Draw an angle equal to that sum with the protractor. | Step 4: Final Answer: The angle equal to the sum has been drawn.

Exercise 6.12: Drawing Circles with Given Radii

Question: Exercise 6.12, Question 1: By using compass and ruler, draw a circle whose radius is 3.2 cm.

Answer: Step 1: Open the compass to 3.2 cm on the ruler. | Step 2: Put the compass needle on the centre point and turn the pencil all the way round. | Step 3: Final Answer: A circle of radius 3.2 cm has been drawn.

Question: Exercise 6.12, Question 2: By using compass and ruler, draw a circle whose radius is 4.1 cm.

Answer: Step 1: Open the compass to 4.1 cm on the ruler. | Step 2: Put the compass needle on the centre point and turn the pencil all the way round. | Step 3: Final Answer: A circle of radius 4.1 cm has been drawn.

Question: Exercise 6.12, Question 3: By using compass and ruler, draw a circle whose radius is 4.4 cm.

Answer: Step 1: Open the compass to 4.4 cm on the ruler. | Step 2: Put the compass needle on the centre point and turn the pencil all the way round. | Step 3: Final Answer: A circle of radius 4.4 cm has been drawn.

Question: Exercise 6.12, Question 4: By using compass and ruler, draw a circle whose radius is 5.5 cm.

Answer: Step 1: Open the compass to 5.5 cm on the ruler. | Step 2: Put the compass needle on the centre point and turn the pencil all the way round. | Step 3: Final Answer: A circle of radius 5.5 cm has been drawn.

Question: Exercise 6.12, Question 5: By using compass and ruler, draw a circle whose radius is 6 cm.

Answer: Step 1: Open the compass to 6 cm on the ruler. | Step 2: Put the

compass needle on the centre point and turn the pencil all the way round. | Step 3: Final Answer: A circle of radius 6 cm has been drawn.

Question: Exercise 6.12, Question 6: By using compass and ruler, draw a circle whose radius is 4.8 cm.

Answer: Step 1: Open the compass to 4.8 cm on the ruler. | Step 2: Put the compass needle on the centre point and turn the pencil all the way round. | Step 3: Final Answer: A circle of radius 4.8 cm has been drawn.

Exercise 6.13: Drawing Squares and Rectangles

Question: Exercise 6.13, Question 1 (i): Draw a square with sides of 6.5 cm using a protractor or set square.

Answer: Step 1: First draw one side 6.5 cm long. | Step 2: With a set square, make right angles at both ends and draw sides of 6.5 cm upward. | Step 3: Join the top ends — all four sides are 6.5 cm and all four angles are right angles. | Step 4: Final Answer: A square with sides of 6.5 cm has been drawn.

Question: Exercise 6.13, Question 1 (ii): Draw a square with sides of 4.9 cm using a protractor or set square.

Answer: Step 1: First draw one side 4.9 cm long. | Step 2: With a set square, make right angles at both ends and draw sides of 4.9 cm upward. | Step 3: Join the top ends — all four sides are 4.9 cm and all four angles are right angles. | Step 4: Final Answer: A square with sides of 4.9 cm has been drawn.

Question: Exercise 6.13, Question 1 (iii): Draw a square with sides of 5.8 cm using a protractor or set square.

Answer: Step 1: First draw one side 5.8 cm long. | Step 2: With a set square, make right angles at both ends and draw sides of 5.8 cm upward. | Step 3: Join the top ends — all four sides are 5.8 cm and all four angles are

right angles. | Step 4: Final Answer: A square with sides of 5.8 cm has been drawn.

Question: Exercise 6.13, Question 1 (iv): Draw a square with sides of 6.3 cm using a protractor or set square.

Answer: Step 1: First draw one side 6.3 cm long. | Step 2: With a set square, make right angles at both ends and draw sides of 6.3 cm upward. | Step 3: Join the top ends — all four sides are 6.3 cm and all four angles are right angles. | Step 4: Final Answer: A square with sides of 6.3 cm has been drawn.

Question: Exercise 6.13, Question 1 (v): Draw a square with sides of 7.3 cm using a protractor or set square.

Answer: Step 1: First draw one side 7.3 cm long. | Step 2: With a set square, make right angles at both ends and draw sides of 7.3 cm upward. | Step 3: Join the top ends — all four sides are 7.3 cm and all four angles are right angles. | Step 4: Final Answer: A square with sides of 7.3 cm has been drawn.

Question: Exercise 6.13, Question 1 (vi): Draw a square with sides of 4.1 cm using a protractor or set square.

Answer: Step 1: First draw one side 4.1 cm long. | Step 2: With a set square, make right angles at both ends and draw sides of 4.1 cm upward. | Step 3: Join the top ends — all four sides are 4.1 cm and all four angles are right angles. | Step 4: Final Answer: A square with sides of 4.1 cm has been drawn.

Question: Exercise 6.13, Question 1 (vii): Draw a square with sides of 8.7 cm using a protractor or set square.

Answer: Step 1: First draw one side 8.7 cm long. | Step 2: With a set square, make right angles at both ends and draw sides of 8.7 cm upward. | Step 3: Join the top ends — all four sides are 8.7 cm and all four angles are

right angles. | Step 4: Final Answer: A square with sides of 8.7 cm has been drawn.

Question: Exercise 6.13, Question 1 (viii): Draw a square with sides of 8.5 cm using a protractor or set square.

Answer: Step 1: First draw one side 8.5 cm long. | Step 2: With a set square, make right angles at both ends and draw sides of 8.5 cm upward. | Step 3: Join the top ends — all four sides are 8.5 cm and all four angles are right angles. | Step 4: Final Answer: A square with sides of 8.5 cm has been drawn.

Question: Exercise 6.13, Question 2 (i): Draw a rectangle with sides 8 cm and 7 cm.

Answer: Step 1: First draw the length side of 8 cm. | Step 2: Make right angles at both ends and draw the width sides of 7 cm. | Step 3: Join the top ends with a line of 8 cm. | Step 4: Final Answer: A rectangle with sides 8 cm and 7 cm has been drawn.

Question: Exercise 6.13, Question 2 (ii): Draw a rectangle with sides 9.5 cm and 4.7 cm.

Answer: Step 1: First draw the length side of 9.5 cm. | Step 2: Make right angles at both ends and draw the width sides of 4.7 cm. | Step 3: Join the top ends with a line of 9.5 cm. | Step 4: Final Answer: A rectangle with sides 9.5 cm and 4.7 cm has been drawn.

Question: Exercise 6.13, Question 2 (iii): Draw a rectangle with sides 12.3 cm and 5.2 cm.

Answer: Step 1: First draw the length side of 12.3 cm. | Step 2: Make right angles at both ends and draw the width sides of 5.2 cm. | Step 3: Join the top ends with a line of 12.3 cm. | Step 4: Final Answer: A rectangle with sides 12.3 cm and 5.2 cm has been drawn.

Question: Exercise 6.13, Question 2 (iv): Draw a rectangle with sides 9.6

cm and 3.7 cm.

Answer: Step 1: First draw the length side of 9.6 cm. | Step 2: Make right angles at both ends and draw the width sides of 3.7 cm. | Step 3: Join the top ends with a line of 9.6 cm. | Step 4: Final Answer: A rectangle with sides 9.6 cm and 3.7 cm has been drawn.

Question: Exercise 6.13, Question 2 (v): Draw a rectangle with sides 6.1 cm and 2.9 cm.

Answer: Step 1: First draw the length side of 6.1 cm. | Step 2: Make right angles at both ends and draw the width sides of 2.9 cm. | Step 3: Join the top ends with a line of 6.1 cm. | Step 4: Final Answer: A rectangle with sides 6.1 cm and 2.9 cm has been drawn.

Question: Exercise 6.13, Question 2 (vi): Draw a rectangle with sides 2.9 cm and 7.6 cm.

Answer: Step 1: First draw the length side of 2.9 cm. | Step 2: Make right angles at both ends and draw the width sides of 7.6 cm. | Step 3: Join the top ends with a line of 2.9 cm. | Step 4: Final Answer: A rectangle with sides 2.9 cm and 7.6 cm has been drawn.

Review Exercise 6

Question: Review Exercise 6, Question 1: Write down the names of the following instruments: (i) [picture of a scale] (ii) [set square] (iii) [set square] (iv) [protractor] (v) [compasses] (vi) [dividers].

Answer: Step 1: Look at each picture carefully and remember the work of the instrument. | Step 2: (i) Ruler (scale). | Step 3: (ii) Set square. | Step 4: (iii) Set square. | Step 5: (iv) Protractor. | Step 6: (v) Compasses. | Step 7: (vi) Dividers. | Step 8: Final Answer: The names of the instruments are given above.

Question: Review Exercise 6, Question 2: If you have to draw a thin line

segment, which grade of pencil will you use?

Answer: Step 1: A thin line needs a hard pencil with a thin point. | Step 2: Final Answer: An H-grade pencil.

Question: Review Exercise 6, Question 3: Measure the length and width of the following objects with a ruler in millimetres: (i) A post card (ii) Cover of Maths-book (iii) Geometry Box.

Answer: Step 1: Place each object at the zero of the ruler and read the length, then the width. | Step 2: Write the measurements in millimetres (remember: 1 cm = 10 mm). | Step 3: Final Answer: Each object has been measured with the ruler and written in millimetres (the measurements may differ slightly for each student's objects).

Question: Review Exercise 6, Question 4: Draw a pair of: (a) Parallel and non-parallel lines (b) Horizontal and vertical lines.

Answer: Step 1: (a) Draw two lines in the same direction that never meet (parallel), and two lines that cut each other (non-parallel). | Step 2: (b) Draw one lying line (horizontal) and one standing line (vertical). | Step 3: Final Answer: Both pairs have been drawn.

Question: Review Exercise 6, Question 5: Draw a curved line and measure its length by using thread and ruler.

Answer: Step 1: Draw a gently curved line. | Step 2: Lay the thread along the curve, mark the ends, then straighten it and measure with the ruler. | Step 3: Final Answer: The length of the curved line has been measured with thread and ruler.

Question: Review Exercise 6, Question 6: Draw a line which passes through a given point and is parallel to a given line using a set square.

Answer: Step 1: Place the long side of a set square on the given line and support it with the second set square. | Step 2: Slide the first set square until the long side reaches the given point. | Step 3: Draw the line through

the point — it is parallel to the given line. | Step 4: Final Answer: The required line has been drawn.

Question: Review Exercise 6, Question 7: Draw the figure to go with each sentence: (i) line PQ is parallel to line RS (ii) line UV intersects line PQ at point A (iii) line UV intersects line RS at point B.

Answer: Step 1: (i) Draw two lines PQ and RS in the same direction. | Step 2: (ii) Draw a slanting line UV that cuts PQ; name the crossing point A. | Step 3: (iii) Extend the same UV so that it also cuts RS; name that point B. | Step 4: Final Answer: All three figures have been drawn.

Question: Review Exercise 6, Question 8: List some objects in your class room that represent right angles.

Answer: Step 1: Look around the class where two sides meet at a right angle. | Step 2: Examples: the corner of a book, the corner of the blackboard, the corner of the door, the corner of a table, the corner of a window, the corner of a notebook. | Step 3: Final Answer: The corners of a book, blackboard, door, table, window and notebook are right angles.

Question: Review Exercise 6, Question 9 (i): How many angles can you find in the figure? Name and mention the type of the angle: rays BA (up), BD (diagonal) and BC from point B.

Answer: Step 1: An angle is formed between every two rays: from three rays, 3 angles are formed. | Step 2: Angle ABD: acute; angle CBD: acute; the whole angle ABC: right. | Step 3: Final Answer: There are 3 angles in all — ABD acute, CBD acute and ABC right.

Question: Review Exercise 6, Question 9 (ii): How many angles can you find in the figure? Name and mention the type of the angle: rays DG, DP, DE and DC from point D.

Answer: Step 1: Count all the angles formed by four rays: neighbouring and combined ones both. | Step 2: The angles are: GDP acute, PDE acute,

GDE right, CDE obtuse, CDG obtuse, CDP obtuse. | Step 3: Final Answer: There are 6 angles in all.

Question: Review Exercise 6, Question 9 (iii): How many angles can you find in the figure? Name and mention the type of the angle: semicircle ABC with centre B and rays BF, BE, BD.

Answer: Step 1: Count the angles between every two rays from centre B. | Step 2: Acute angles: ABF, FBE, EBD, DBC; right angles: ABE, FBD, EBC; obtuse angles: ABD, FBC. | Step 3: Final Answer: There are 9 angles in all — 4 acute, 3 right and 2 obtuse.

Question: Review Exercise 6, Question 10 (i): Count the number of squares in the figure. Also name them: two squares side by side.

Answer: Step 1: Look at every closed figure whose four sides are equal and angles are right angles. | Step 2: Final Answer: There are 2 squares in this figure (the two side-by-side squares; the combined big figure is a rectangle, not a square).

Question: Review Exercise 6, Question 10 (ii): Count the number of squares in the figure. Also name them: a big square divided into four small squares.

Answer: Step 1: First count the small squares: there are 4. | Step 2: Then look at the big outer figure: it is also a square. | Step 3: Final Answer: There are 5 squares in all — 4 small and 1 big.

Question: Review Exercise 6, Question 11: How many triangles are in the figure? Figure: triangle ADC with DB drawn from D to B on AC.

Answer: Step 1: Start with the small triangles: ADB and CDB — there are 2. | Step 2: Then the whole big figure ADC is also a triangle. | Step 3: Final Answer: There are 3 triangles in all.

Question: Review Exercise 6, Question 12: Get a compass and practice drawing same circles; make some designs of your own.

Answer: Step 1: Keep the compass open at one measure and draw many circles — all will be of the same size. | Step 2: Make designs by overlapping circles or by taking different radii from one centre. | Step 3: Final Answer: The practice is complete and designs have been made.

Question: Review Exercise 6, Question 13: (i) Draw a circle with 25 mm radius. (ii) Name its centre, radius, diameter and circumference. (iii) Measure its circumference with thread and ruler.

Answer: Step 1: (i) Open the compass to 25 mm (2.5 cm) on the ruler and draw the circle. | Step 2: (ii) The middle point is the centre; the length from the centre to the circumference is the radius (25 mm); the longest chord through the centre is the diameter, which is twice the radius, that is, 50 mm; the full length around the circle is the circumference. | Step 3: (iii) Lay the thread all the way round the circumference, then straighten it and measure with the ruler: about 157 mm is obtained. | Step 4: Final Answer: Radius 25 mm, diameter 50 mm and circumference about 157 mm.

7. Information Handling

Activity: Bar Graph (Runs in a Cricket Match)

Question: Activity: The bar graph shows the runs made by three players Ali, Anwar and Rashid in a cricket match: (i) What are the runs made by Ali? (ii) What are the runs made by Anwar? (iii) What are the runs made by Rashid? (iv) Whose runs are the highest? (v) Whose runs are the least?

Answer: Step 1: Read the height of each bar of the bar graph and write the runs. | Step 2: (i) Ali's runs are 50 (given in the book). | Step 3: (ii) Anwar's runs are 40. | Step 4: (iii) Rashid's runs are 30. | Step 5: (iv) 50 is the biggest number, so Ali's runs are the highest. | Step 6: (v) 30 is the smallest number, so Rashid's runs are the least. | Step 7: Final Answer: Ali 50, Anwar 40, Rashid 30; highest Ali and least Rashid.

Activity: Bar Graph (Saima's Marks)

Question: Activity: The bar graph shows Saima's marks in her final examination (Sindhi 70, Social Studies 50, English 60, Urdu 40, Science 80, Maths 50). (1) In which subject did she secure the highest marks? (2) In which subject did she secure the lowest marks? (3) In how many subjects did she appear? (4) In how many subjects did she secure equal marks? (5) In which subject did she secure more than 60 marks? (6) What were the total marks of Saima?

Answer: Step 1: Read the height of each bar of the bar graph and write the marks: Sindhi 70, Social Studies 50, English 60, Urdu 40, Science 80, Maths 50. | Step 2: (1) The tallest bar is of Science, so the highest marks are in Science (80) — (given in the book). | Step 3: (2) The shortest bar is of Urdu, so the lowest marks are in Urdu (40). | Step 4: (3) Count the

subjects: there are 6 subjects, so she appeared in 6 subjects. | Step 5: (4) Social Studies and Maths both have 50 marks, so she secured equal marks in 2 subjects. | Step 6: (5) She secured more than 60 marks in Science (80 marks). | Step 7: (6) For the total, add all six marks (working below). | Step 8: Final Answer: (1) Science (2) Urdu (3) 6 subjects (4) 2 subjects (5) Science (6) Total marks 340.

Question: Activity (Saima), Question 6 working

Answer: Step 1: The marks of all six subjects of Saima are added. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 0 + 0 + 0 + 0 + 0 + 0 = 0; write 0. | Step 4: Add the tens column: 7 + 5 + 6 + 4 + 8 + 5 = 35; write 5 and carry 3 to the next column. | Step 5: Write the carried 3 in the next column. | Step 6: Final Answer: Saima's total marks are 340.

Exercise 7.1: Reading Bar Graphs

Question: Exercise 7.1, Question 1: The bar graph shows a survey of favourite colours of students of class IV (blue 15, yellow 5, pink 30, red 15, black 10, green 20 students). Answer: (i) How many students like yellow colour? (ii) Which colour is liked most? (iii) Which colour is liked least? (iv) How many students like green colour? (v) How many students like red colour? (vi) How many colours are included in this survey? (vii) How many students are included in this survey?

Answer: Step 1: Read from each bar of the bar graph the students who like each colour. | Step 2: (i) 5 students like yellow colour. | Step 3: (ii) The tallest bar is of pink colour (30 students), so pink is liked most. | Step 4: (iii) The shortest bar is of yellow colour (5 students), so yellow is liked least. | Step 5: (iv) 20 students like green colour. | Step 6: (v) 15 students like red colour. | Step 7: (vi) Count the colours: blue, yellow, pink, red,

black, green — 6 colours in all. | Step 8: (vii) For the total students, add the numbers of all the colours (working below). | Step 9: Final Answer: (i) 5 (ii) Pink (iii) Yellow (iv) 20 (v) 15 (vi) 6 colours (vii) 95 students.

Question: Exercise 7.1, Question 1 (vii) working

Answer: Step 1: The students of all the colours are added. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 5 + 5 + 0 + 5 + 0 + 0 = 15; write 5 and carry 1 to the next column. | Step 4: Add the tens column: 1 + 0 + 3 + 1 + 1 + 2 + 1 carried over = 9; write 9. | Step 5: Final Answer: 95 students are included in the survey.

Question: Exercise 7.1, Question 2: The bar graph shows the favourite games of students of Class IV (Football 25, Swimming 30, Hockey 20, Cricket 35 students). Answer: (i) How many students like Hockey? (ii) Which game is liked most? (iii) How many students like Football? (iv) How many games are liked by students? (v) Which game is liked least? (vi) How many students like cricket? (vii) How many total students are involved in games?

Answer: Step 1: Read from each bar of the bar graph the students who like each game. | Step 2: (i) 20 students like Hockey. | Step 3: (ii) The tallest bar is of Cricket (35 students), so Cricket is liked most. | Step 4: (iii) 25 students like Football. | Step 5: (iv) Count the games: Football, Swimming, Hockey, Cricket — 4 games in all. | Step 6: (v) The shortest bar is of Hockey (20 students), so Hockey is liked least. | Step 7: (vi) 35 students like cricket. | Step 8: (vii) For the total, add the numbers of all four games (working below). | Step 9: Final Answer: (i) 20 (ii) Cricket (iii) 25 (iv) 4 games (v) Hockey (vi) 35 (vii) Total 110 students.

Question: Exercise 7.1, Question 2 (vii) working

Answer: Step 1: The students of all the games are added. | Step 2: Write

the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 5 + 0 + 0 + 5 = 10; write 0 and carry 1 to the next column. | Step 4: Add the tens column: 2 + 3 + 2 + 3 + 1 carried over = 11; write 1 and carry 1 to the next column. | Step 5: Write the carried 1 in the next column. | Step 6: Final Answer: 110 students are involved in games in total.

Activity: Line Graph (Temperature of Jacobabad)

Question: Activity: The line graph shows the temperature of Jacobabad town at various times (5 a.m. 35 C, 7 a.m. 37 C, 9 a.m. 38 C, 11 a.m. 39 C, 1 p.m. 41 C, 3 p.m. 39 C): (1) What was the temperature at 9 a.m.? (2) At what time the temperature was the lowest? (3) Were the temperatures same at 7 a.m. and 3 p.m.? (4) What was the temperature at 1 p.m.? (5) What was the lowest temperature?

Answer: Step 1: Read the value of each time point on the line graph: 5 a.m. 35, 7 a.m. 37, 9 a.m. 38, 11 a.m. 39, 1 p.m. 41, 3 p.m. 39 degrees Celsius. | Step 2: (1) At 9 a.m. the temperature was 38 degrees Celsius (given in the book). | Step 3: (2) The smallest value is at 5 a.m., so the temperature was the lowest at 5 a.m. | Step 4: (3) At 7 a.m. it was 37 degrees and at 3 p.m. 39 degrees — both are different, so no, they were not the same. | Step 5: (4) At 1 p.m. the temperature was 41 degrees Celsius. | Step 6: (5) The lowest temperature was 35 degrees Celsius. | Step 7: Final Answer: (1) 38 degrees (2) At 5 a.m. (3) No (4) 41 degrees (5) 35 degrees Celsius.

Exercise 7.2: Reading Line Graphs

Question: Exercise 7.2, Question 1: The line graph shows the time in minutes taken by each of five teams (A, B, C, D, E) to complete a task: (i) Which team completed the work in the shortest time? (ii) Which team took

the longest time? (iii) Which teams took the same length of time? (iv) What was the shortest time? (v) What was the longest time? (vi) How long did team C take to complete the task?

Answer: Step 1: Read the value of each team's point on the line graph in minutes. | Step 2: (i) The lowest point is of team D — 10 minutes, so D completed the work in the shortest time. | Step 3: (ii) The highest point is of team E — 40 minutes, so E took the longest time. | Step 4: (iii) According to the book's answer key, teams A and B took the same length of time. | Step 5: (iv) The shortest time was 10 minutes. | Step 6: (v) The longest time was 40 minutes. | Step 7: (vi) Team C's point is at 15 minutes, so C took 15 minutes. | Step 8: Final Answer: (i) Team D (ii) Team E (iii) Teams A and B (iv) 10 minutes (v) 40 minutes (vi) 15 minutes.

Question: Exercise 7.2, Question 2: The line graph shows the progress of one week of a farmer's tractor taken on rent to plough his fields (Monday 5, Tuesday 10, Wednesday 10, Thursday 15, Friday 5, Saturday 10, Sunday 10 acres): (i) On which day the tractor ploughed the maximum number of acres? (ii) How many acres were ploughed on Wednesday? (iii) For how many days the land was ploughed? (iv) What was his progress on Tuesday? (v) On which days the progress was minimum? (vi) How many acres were ploughed during the whole week? (vii) Give the separate output for Thursday, Friday and Saturday. (viii) On which days the output was same?

Answer: Step 1: Read the value of each day's point on the line graph in acres: Monday 5, Tuesday 10, Wednesday 10, Thursday 15, Friday 5, Saturday 10, Sunday 10. | Step 2: (i) The greatest value, 15 acres, is on Thursday, so the maximum acres were ploughed on Thursday. | Step 3: (ii) On Wednesday 10 acres were ploughed. | Step 4: (iii) Count the days: from Monday to Sunday the land was ploughed for 7 days. | Step 5: (iv) The

progress on Tuesday was 10 acres. | Step 6: (v) The smallest value is 5 acres, which is on Monday and Friday, so the progress was minimum on these days. | Step 7: (vi) For the total, add the acres of all the days (working below). | Step 8: (vii) Thursday 15 acres, Friday 5 acres, Saturday 10 acres. | Step 9: (viii) Tuesday, Wednesday, Saturday and Sunday — all four days had the same output of 10 acres; Monday and Friday also had the same (5 acres). | Step 10: Final Answer: (i) Thursday (ii) 10 acres (iii) 7 days (iv) 10 acres (v) Monday and Friday (vi) Total 65 acres (vii) Thursday 15, Friday 5, Saturday 10 (viii) Tuesday, Wednesday, Saturday, Sunday are equal.

Question: Exercise 7.2, Question 2 (vi) working

Answer: Step 1: The acres of all the days of the week are added. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 5 + 0 + 0 + 5 + 5 + 0 + 0 = 15; write 5 and carry 1 to the next column. | Step 4: Add the tens column: 0 + 1 + 1 + 1 + 0 + 1 + 1 + 1 carried over = 6; write 6. | Step 5: Final Answer: 65 acres were ploughed during the whole week.

Question: Exercise 7.2, Question 3: The line graph shows Samina's progress in quarterly tests (Maths 80, English 70, Science 60, Islamiyat 80, Sindhi 70): (i) In which subject(s) Samina secured the highest marks? (ii) What were her total marks obtained? (iii) How many marks did she get in English? (iv) How many marks did she get in Maths? (v) In which subject did she get the lowest marks? (vi) How many marks did she get in Sindhi?

Answer: Step 1: Read the value of each subject's point on the line graph: Maths 80, English 70, Science 60, Islamiyat 80, Sindhi 70. | Step 2: (i) The highest points are of Maths and Islamiyat — both 80 marks, so the highest marks are in these two subjects. | Step 3: (ii) For the total, add all five marks (working below). | Step 4: (iii) She got 70 marks in English. | Step 5:

(iv) She got 80 marks in Maths. | Step 6: (v) The lowest point is of Science — 60 marks, so she got the lowest marks in Science. | Step 7: (vi) She got 70 marks in Sindhi. | Step 8: Final Answer: (i) Maths and Islamiyat (80, 80) (ii) Total 360 (iii) 70 (iv) 80 (v) Science (vi) 70.

Question: Exercise 7.2, Question 3 (ii) working

Answer: Step 1: The marks of all five subjects of Samina are added. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 0 + 0 + 0 + 0 + 0 = 0; write 0. | Step 4: Add the tens column: 8 + 7 + 6 + 8 + 7 = 36; write 6 and carry 3 to the next column. | Step 5: Write the carried 3 in the next column. | Step 6: Final Answer: Samina's total marks are 360.

Review Exercise 7

Question: Review Exercise 7, Question 1: The bar graph shows the cost of mango squash (1 litre Rs 40, 2 litres Rs 80, 3 litres Rs 100, 4 litres Rs 120, 5 litres Rs 160, 6 litres Rs 180): (i) What is the cost of 2 litres of mango squash? (ii) What is the cost of 4 litres of mango squash? (iii) If we have six notes of Rs 20, can we buy 6 litres of mango squash? (iv) How much change will we have from Rs 100 when we buy 2 litres of mango squash? (v) How many litres of squash can we buy from Rs 180?

Answer: Step 1: Read the cost of each quantity from the bar graph. | Step 2: (i) The cost of 2 litres is Rs 80. | Step 3: (ii) The cost of 4 litres is Rs 120. | Step 4: (iii) We have six notes of Rs 20 each — write its total in the working below; the cost of 6 litres is Rs 180, which is more than the money we have. | Step 5: (iv) The cost of 2 litres is Rs 80; subtracting from Rs 100 gives the change (working below). | Step 6: (v) Rs 180 is exactly the cost of 6 litres. | Step 7: Final Answer: (i) Rs 80 (ii) Rs 120 (iii) No, we cannot buy 6 litres (iv) Rs 20 change (v) 6 litres can be bought.

Question: Review Exercise 7, Question 1 (iii) working: What is the total of six notes?

Answer: Step 1: There are six notes of Rs 20 each, so multiply. | Step 2: Break 20 into its parts: 20. | Step 3: 20 x 6 = 120. | Step 4: Final Answer: The total of six notes is Rs 120, which is less than the cost of 6 litres, Rs 180.

Question: Review Exercise 7, Question 1 (iv) working: How much change will we get?

Answer: Step 1: The cost of 2 litres is subtracted from Rs 100. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 0 - 0 = 0; write 0. | Step 4: Tens column: after borrowing, the top digit becomes 10; 10 - 8 = 2; write 2. | Step 5: Hundreds column: after borrowing, the top digit becomes 0; 0 - 0 = 0; write 0. | Step 6: Final Answer: We will get Rs 20 back.

Question: Review Exercise 7, Question 2: The bar graph shows Rashid's family expenditure on food for six months (January Rs 5500, February Rs 3000, March Rs 2000, April Rs 4000, May Rs 2500, June Rs 1000): (i) How much amount is spent on food during February? (ii) How much during April? (iii) In which month the expenditure was the lowest? (iv) In which month the expenditure was the highest? (v) What are the total expenditures from January to June? (vi) What was the difference in expenditure spent for April and May?

Answer: Step 1: Read the height of each bar of the bar graph and write each month's expenditure: January 5500, February 3000, March 2000, April 4000, May 2500, June 1000 rupees. | Step 2: (i) In February Rs 3000 were spent. | Step 3: (ii) In April Rs 4000 were spent. | Step 4: (iii) The shortest bar is of June (Rs 1000), so the expenditure was the lowest in June. | Step 5: (iv) The tallest bar is of January (Rs 5500), so the

expenditure was the highest in January. | Step 6: (v) For the total, add the expenditure of all six months (working below). | Step 7: (vi) Find the difference between April's and May's expenditure (working below). | Step 8: Final Answer: (i) Rs 3000 (ii) Rs 4000 (iii) June (iv) January (v) Total Rs 18,000 (vi) Difference Rs 1500.

Question: Review Exercise 7, Question 2 (v) working

Answer: Step 1: The expenditure of all six months is added. | Step 2: Write the numbers one below the other, ones under ones, tens under tens. | Step 3: Add the ones column: 0 + 0 + 0 + 0 + 0 + 0 = 0; write 0. | Step 4: Add the tens column: 0 + 0 + 0 + 0 + 0 + 0 = 0; write 0. | Step 5: Add the hundreds column: 5 + 0 + 0 + 0 + 5 + 0 = 10; write 0 and carry 1 to the next column. | Step 6: Add the thousands column: 5 + 3 + 2 + 4 + 2 + 1 + 1 carried over = 18; write 8 and carry 1 to the next column. | Step 7: Write the carried 1 in the next column. | Step 8: Final Answer: The total expenditure from January to June is Rs 18,000.

Question: Review Exercise 7, Question 2 (vi) working

Answer: Step 1: May's expenditure is subtracted from April's expenditure. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 0 - 0 = 0; write 0. | Step 4: Subtract the tens column: 0 - 0 = 0; write 0. | Step 5: Hundreds column: after borrowing, the top digit becomes 10; 10 - 5 = 5; write 5. | Step 6: Thousands column: after borrowing, the top digit becomes 3; 3 - 2 = 1; write 1. | Step 7: Final Answer: The difference in expenditure for April and May is Rs 1500.

Question: Review Exercise 7, Question 3: The line graph shows the production of an oil plant over 5 days (Day 1: 2 million litres, Day 2: 4, Day 3: 6, Day 4: 7, Day 5: 10 million litres): (i) What was the production of the plant on the 1st day? (ii) What was its production on the 3rd day? (iii)

What production was made between the 2nd and 3rd day? (iv) How much production was made between the 2nd and 5th days? (v) On which day the plant growth production was the highest? (vi) How many times the plant production was measured?

Answer: Step 1: Read the value of each day's point on the line graph: day 1 is 2, day 2 is 4, day 3 is 6, day 4 is 7, day 5 is 10 — all in million litres. | Step 2: (i) The production on the 1st day was 2 million litres. | Step 3: (ii) The production on the 3rd day was 6 million litres. | Step 4: (iii) For the production made between the 2nd and 3rd day, subtract the 2nd day's production from the 3rd day's (working below). | Step 5: (iv) For the production made between the 2nd and 5th days, subtract the 2nd day's production from the 5th day's (working below). | Step 6: (v) Look at the daily increases: the increase from day 4 to day 5 is the greatest, so the plant growth production was the highest on the 5th day (and the production was also the most on the 5th day, 10 million litres). | Step 7: (vi) The production was measured for 5 days, so it was measured 5 times. | Step 8: Final Answer: (i) 2 million litres (ii) 6 million litres (iii) 2 million litres (iv) 6 million litres (v) 5th day (vi) 5 times.

Question: Review Exercise 7, Question 3 (iii) working

Answer: Step 1: The 2nd day's production is subtracted from the 3rd day's production. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Subtract the ones column: 6 - 4 = 2; write 2. | Step 4: Final Answer: Between the 2nd and 3rd day, 2 million litres were produced.

Question: Review Exercise 7, Question 3 (iv) working

Answer: Step 1: The 2nd day's production is subtracted from the 5th day's production. | Step 2: Write the bigger number on top and the smaller number below it, in columns. | Step 3: Ones column: after borrowing, the

top digit becomes 10; 10 - 4 = 6; write 6. | Step 4: Tens column: after borrowing, the top digit becomes 0; 0 - 0 = 0; write 0. | Step 5: Final Answer: Between the 2nd and 5th days, 6 million litres were produced.

8. Glossary

Glossary: Definitions of Terms

Question: Glossary: Define the term 'Addition'.

Answer: Step 1: Definition: The process of putting two or more numbers together to find their total is called addition.

Question: Glossary: Define the term 'Associative Property of Addition'.

Answer: Step 1: Definition: While adding three numbers, changing their grouping does not change the sum; this is the associative property of addition. For example, (2 + 3) + 4 = 2 + (3 + 4).

Question: Glossary: Define the term 'Associative Property of Multiplication'.

Answer: Step 1: Definition: While multiplying three numbers, changing their grouping does not change the product; this is the associative property of multiplication.

Question: Glossary: Define the term 'Angle'.

Answer: Step 1: Definition: Two rays starting from a common endpoint (vertex) form an angle.

Question: Glossary: Define the term 'Arc'.

Answer: Step 1: Definition: Any part of the circumference of a circle is called an arc.

Question: Glossary: Define the term 'Acute Angle'.

Answer: Step 1: Definition: An angle which is smaller than a right angle (90 degrees) is called an acute angle.

Question: Glossary: Define the term 'Acute Angled Triangle'.

Answer: Step 1: Definition: A triangle whose all three angles are acute (less than 90 degrees) is called an acute angled triangle.

Question: Glossary: Define the term 'Capacity'.

Answer: Step 1: Definition: The amount that a container can hold is called its capacity; it is measured in litres and millilitres.

Question: Glossary: Define the term 'Centimetre'.

Answer: Step 1: Definition: A small unit for measuring length; there are 100 centimetres in one metre.

Question: Glossary: Define the term 'Circle'.

Answer: Step 1: Definition: A closed curved figure whose all points are at an equal distance from a middle point (centre) is called a circle.

Question: Glossary: Define the term 'Commutative Property of Multiplication'.

Answer: Step 1: Definition: While multiplying two numbers, changing their order does not change the product; this is the commutative property of multiplication.

Question: Glossary: Define the term 'Commutative Property of Addition'.

Answer: Step 1: Definition: While adding two numbers, changing their order does not change the sum; this is the commutative property of addition.

Question: Glossary: Define the term 'Common Multiples'.

Answer: Step 1: Definition: The multiples which are multiples of each of two or more numbers are called their common multiples.

Question: Glossary: Define the term 'Composite Number'.

Answer: Step 1: Definition: A number which has more than two factors is called a composite number; for example 4, 6 and 8.

Question: Glossary: Define the term 'Division'.

Answer: Step 1: Definition: The process of sharing a number into equal parts is called division; it is the opposite process of multiplication.

Question: Glossary: Define the term 'Denominator'.

Answer: Step 1: Definition: In a fraction, the number below the line is the denominator; it tells into how many equal parts the whole is divided.

Question: Glossary: Define the term 'Diameter'.

Answer: Step 1: Definition: The longest chord passing through the centre of a circle is called the diameter; it is twice the radius.

Question: Glossary: Define the term 'Divisibility'.

Answer: Step 1: Definition: If one number is divided exactly by another number with no remainder, the first number is divisible by the second.

Question: Glossary: Define the term 'Dividend'.

Answer: Step 1: Definition: In division, the number which is being divided is called the dividend.

Question: Glossary: Define the term 'Divisor'.

Answer: Step 1: Definition: In division, the number by which we divide is called the divisor.

Question: Glossary: Define the term 'Decimal Fraction'.

Answer: Step 1: Definition: A fraction whose denominator is 10, 100 or 1000 and which is written with a decimal point is called a decimal fraction.

Question: Glossary: Define the term 'Even Numbers'.

Answer: Step 1: Definition: Numbers which are exactly divisible by two (their ones digit is 0, 2, 4, 6 or 8) are called even numbers.

Question: Glossary: Define the term 'Edge'.

Answer: Step 1: Definition: In a solid shape, the line segment where two faces meet is called an edge.

Question: Glossary: Define the term 'Equivalent Fractions'.

Answer: Step 1: Definition: Fractions which show the same amount, even though their numbers are different, are called equivalent fractions.

Question: Glossary: Define the term 'Fraction'.

Answer: Step 1: Definition: A part or some parts of a whole thing are

written as a fraction; it is written with a numerator and a denominator.

Question: Glossary: Define the term 'Factors'.

Answer: Step 1: Definition: The numbers which divide a given number exactly, with no remainder, are called its factors.

Question: Glossary: Define the term 'Factorization'.

Answer: Step 1: Definition: Writing a number as a product of its factors is called factorization.

Question: Glossary: Define the term 'Gram'.

Answer: Step 1: Definition: A small unit for measuring mass; there are 1000 grams in one kilogram.

Question: Glossary: Define the term 'Graph'.

Answer: Step 1: Definition: A diagram which shows information with bars, lines or pictures is called a graph.

Question: Glossary: Define the term 'GCD'.

Answer: Step 1: Definition: The greatest common factor of two or more numbers, which divides them all exactly, is called GCD or HCF.

Question: Glossary: Define the term 'Hours'.

Answer: Step 1: Definition: A unit for measuring time; there are 60 minutes in one hour.

Question: Glossary: Define the term 'HCF'.

Answer: Step 1: Definition: The largest factor common to two or more numbers is called their HCF (Highest Common Factor).

Question: Glossary: Define the term 'Improper Fraction'.

Answer: Step 1: Definition: A fraction whose numerator is greater than or equal to its denominator is called an improper fraction.

Question: Glossary: Define the term 'Kilogram'.

Answer: Step 1: Definition: A bigger unit for measuring mass; there are 1000 grams in one kilogram.

Question: Glossary: Define the term 'Litre'.

Answer: Step 1: Definition: A unit for measuring capacity; there are 1000 millilitres in one litre.

Question: Glossary: Define the term 'Line'.

Answer: Step 1: Definition: A straight path which goes on endlessly in both directions is called a line; it has no endpoint.

Question: Glossary: Define the term 'Line Segment'.

Answer: Step 1: Definition: The part of a line which has two fixed endpoints is called a line segment; its length can be measured.

Question: Glossary: Define the term 'Lunar Calendar'.

Answer: Step 1: Definition: A calendar based on the cycle of the moon is called a lunar calendar; the Islamic calendar is lunar.

Question: Glossary: Define the term 'L.C.M'.

Answer: Step 1: Definition: The smallest multiple common to two or more numbers is called their L.C.M (Least Common Multiple).

Question: Glossary: Define the term 'Like Fractions'.

Answer: Step 1: Definition: Fractions which have the same denominator are called like fractions.

Question: Glossary: Define the term 'Multiplication'.

Answer: Step 1: Definition: The short way of repeated addition of the same number is multiplication; it gives the product.

Question: Glossary: Define the term 'Mass'.

Answer: Step 1: Definition: The amount of matter in an object is called its mass; it is measured in grams and kilograms.

Question: Glossary: Define the term 'Millilitre'.

Answer: Step 1: Definition: A small unit for measuring capacity; there are 1000 millilitres in one litre.

Question: Glossary: Define the term 'Millimetre'.

Answer: Step 1: Definition: A very small unit for measuring length; there are 10 millimetres in one centimetre.

Question: Glossary: Define the term 'Minute'.

Answer: Step 1: Definition: A unit for measuring time; there are 60 seconds in one minute.

Question: Glossary: Define the term 'Month'.

Answer: Step 1: Definition: A unit of time; according to the book, one month has 30 days and one year has 12 months.

Question: Glossary: Define the term 'Million'.

Answer: Step 1: Definition: A million is a big number: there are ten hundred-thousands (ten lakh) in one million and it is written as 1,000,000.

Question: Glossary: Define the term 'Mixed Number'.

Answer: Step 1: Definition: A number made of a whole number together with a fraction is called a mixed number.

Question: Glossary: Define the term 'Numerator'.

Answer: Step 1: Definition: In a fraction, the number above the line is the numerator; it tells how many parts are taken.

Question: Glossary: Define the term 'Obtuse Angle'.

Answer: Step 1: Definition: An angle which is greater than a right angle (90 degrees) but smaller than a straight angle (180 degrees) is called an obtuse angle.

Question: Glossary: Define the term 'Obtuse Angled Triangle'.

Answer: Step 1: Definition: A triangle which has one obtuse angle (more than 90 degrees) is called an obtuse angled triangle.

Question: Glossary: Define the term 'Place Value'.

Answer: Step 1: Definition: The value of a digit according to its position is called its place value; for example the digit in the hundreds place.

Question: Glossary: Define the term 'Proper Fraction'.

Answer: Step 1: Definition: A fraction whose numerator is smaller than its denominator is called a proper fraction.

Question: Glossary: Define the term 'Paisa'.

Answer: Step 1: Definition: A small unit of money; there are 100 paisas in one rupee.

Question: Glossary: Define the term 'Point'.

Answer: Step 1: Definition: A point shows an exact position; it has neither length nor width.

Question: Glossary: Define the term 'Prime Factorisation'.

Answer: Step 1: Definition: Writing a number as a product of prime factors only is called prime factorisation.

Question: Glossary: Define the term 'Protractor'.

Answer: Step 1: Definition: The half-round instrument used for measuring and drawing angles is called a protractor (printed as Protector in the book).

Question: Glossary: Define the term 'Quadrilateral'.

Answer: Step 1: Definition: A closed figure which has four sides is called a quadrilateral; square and rectangle are quadrilaterals.

Question: Glossary: Define the term 'Quotient'.

Answer: Step 1: Definition: The answer obtained on dividing is called the quotient.

Question: Glossary: Define the term 'Remainder'.

Answer: Step 1: Definition: The amount left over after division is called the remainder.

Question: Glossary: Define the term 'Ray'.

Answer: Step 1: Definition: The part of a line which has one fixed endpoint and goes on endlessly in one direction is called a ray.

Question: Glossary: Define the term 'Rectangle'.

Answer: Step 1: Definition: A quadrilateral whose opposite sides are equal and all four angles are right angles is called a rectangle.

Question: Glossary: Define the term 'Radius'.

Answer: Step 1: Definition: The distance from the centre of a circle to its circumference is the radius; it is half of the diameter.

Question: Glossary: Define the term 'Rupee'.

Answer: Step 1: Definition: The unit of money used in Pakistan; there are 100 paisas in one rupee.

Question: Glossary: Define the term 'Ruler'.

Answer: Step 1: Definition: The instrument used for measuring lengths and drawing straight lines is called a ruler (scale).

Question: Glossary: Define the term 'Right Angle'.

Answer: Step 1: Definition: An angle which is exactly 90 degrees is called a right angle.

Question: Glossary: Define the term 'Right Triangle'.

Answer: Step 1: Definition: A triangle which has one right angle (90 degrees) is called a right triangle.

Question: Glossary: Define the term 'Symbol'.

Answer: Step 1: Definition: Any sign used in mathematics is called a symbol; for example the signs of addition, subtraction, multiplication and division.

Question: Glossary: Define the term 'Square'.

Answer: Step 1: Definition: A quadrilateral whose all four sides are equal and all four angles are right angles is called a square.

Question: Glossary: Define the term 'Subtraction'.

Answer: Step 1: Definition: The process of taking one number away from another to find the difference is called subtraction.

Question: Glossary: Define the term 'Second'.

Answer: Step 1: Definition: A small unit for measuring time; there are 60 seconds in one minute.

Question: Glossary: Define the term 'Solar Calendar'.

Answer: Step 1: Definition: A calendar based on the cycle of the sun is called a solar calendar; its one year has 365 days.

Question: Glossary: Define the term 'Scalene Triangle'.

Answer: Step 1: Definition: A triangle whose all three sides are of different lengths is called a scalene triangle.

Question: Glossary: Define the term 'Triangle'.

Answer: Step 1: Definition: A closed figure which has three sides and three angles is called a triangle.

Question: Glossary: Define the term 'Unlike Fractions'.

Answer: Step 1: Definition: Fractions which have different denominators are called unlike fractions.

Question: Glossary: Define the term 'Unit Fraction'.

Answer: Step 1: Definition: A fraction whose numerator is one (1) is called a unit fraction.

Question: Glossary: Define the term 'Vertex'.

Answer: Step 1: Definition: The common endpoint of the two arms of an angle is the vertex; the corner point of a figure is also called a vertex.

Question: Glossary: Define the term 'Week'.

Answer: Step 1: Definition: A unit of time; there are 7 days in one week.

Question: Glossary: Define the term 'Year'.

Answer: Step 1: Definition: A big unit of time; according to the book, one year has 365 days and one year has 12 months.